

A SCIENCE FICTION NOVEL

# THE PARLIAMENT OF VAPOR

*A promise cannot spend  
an uncouned present.*



T. K. ARVEN

---

# The Parliament of Vapor

*A promise cannot spend an uncounted present.*

*T. K. Arven*

## Copyright

Copyright © 2026 T. K. Arven. All rights reserved.

First digital proof edition, 2026.

Published independently by T. K. Arven.

This is a work of fiction. Names, characters, places, organizations, and events are products of the author's imagination or are used fictitiously. Any resemblance to actual persons, living or dead, or to actual events is coincidental.

No part of this publication may be reproduced, stored, or transmitted in any form or by any means without prior written permission from the copyright holder, except for brief quotations used in reviews or scholarship.

## Contents

- Part I: Condensation
- Chapter 1: The Loop That Answered
- Chapter 2: Shore-Age
- Chapter 3: A Person in the Pipe
- Chapter 4: The Order to Dry
- Chapter 5: Across the Condensate
- Chapter 6: The Pilot's Promise
- Chapter 7: Weather Caucus
- Interlude I: Nursery Humidity
- Part II: Standing for Weather
- Chapter 8: The First Hearing
- Chapter 9: Cohesion
- Chapter 10: The Fronts Disagree
- Chapter 11: Provisional Standing
- Chapter 12: Twenty-Three Years
- Chapter 13: The Condenser Position

- Chapter 14: A Chamber Without One Body
- Interlude II: Piri's Garden
- Part III: The Heat Vote
- Chapter 15: Six Farewells
- Chapter 16: One Stormcell
- Chapter 17: The Long Arc
- Chapter 18: The Volunteer Error
- Chapter 19: Retiring Blue Yam
- Chapter 20: Names for the Hot Vent
- Chapter 21: Three Hundred Twelve
- Interlude III: The Forty-Third Percent
- Part IV: A Habitable Loss
- Chapter 22: Seven Missing Fronts
- Chapter 23: The Cohort Speaks
- Chapter 24: Three Phases of Law
- Chapter 25: Two Yeses
- Chapter 26: Wet Bridges
- Chapter 27: Crown Burn
- Chapter 28: Weather School

## **Part I — Condensation**

# Chapter 1 — The Loop That Answered

Iona Serr had her burned hand on the vent wheel when Archive Humid Loop K-7 moved heat against the test.

The Crown Gallery descended around her in copper terraces, every level narrowing toward the perihelion engines sealed below. Warmth climbed through the soles of her boots. Above the control pit, trajectory glass showed a red curve bending close to the sun, then opening toward the distant point labeled ORISON CAPTURE. Forty-one days remained before the curve stopped being a drawing and became the ship's only road.

"Hold primary flow at four tenths," First Pilot Mael Or said from the pit. "We are measuring lag, not ambition."

Three mechanics along the upper rail acknowledged him. Iona did not. Her fingers were reading the wheel.

Old machinery spoke most clearly through resistance. The manual bleed wheel had three spokes worn pale where generations of hands had gripped them. Its hub should have loosened as the Crown reservoir warmed. Instead it tightened by a fraction, relaxed, and tightened again. Iona shifted her left palm so the smooth burn tissue lay flat against brass. The scar did not sweat. That made small temperature changes easy to feel and large ones easy to regret.

"Four tenths holding," she called.

On her console, blue columns rose through the thermal ledger. Crown intake. Mirror return. Archive draw. Habitat loss. Each value carried the mild delay expected from forty kilometers of pipe and a comet's worth of cold mass. The warm test had run thirteen minutes. Nothing should yet have reached the archive loops.

K-7 had warmed by six degrees.

Iona looked away, then back. The number remained.

She opened the local tree. Nine salt panes served K-7, all assigned to deceased-citizen associative storage. Their routine duty involved accepting waste heat during short Crown calibrations, not producing it. A valve icon pulsed amber beside the west condensate line.

"Mael, I have an archive return out of sequence."

"How far?"

"Six degrees at K-7. No corresponding input."

"Bad sensor."

"Probably."

Mael glanced up from the pit. He knew the word probably meant she would not let him treat it as certainty. "You have eight minutes before I need the return model."

Iona released the wheel to Pell Doran, the mechanic beside her, and took the narrow stair two at a time. Gravity grew lighter toward the gallery rim. By the time she reached the service hatch, her tool satchel bumped against her hip instead of dragging. She keyed her advocate seal into the maintenance rail and selected K-7.

The nearest route led through a dry archive corridor. Its lights came up in orderly white segments, illuminating racks of black ceramic cartridges behind glass. Forty thousand citizens had left craft memories, family recollections, practiced motions, and spoken corrections in the mnemonic salt. None had left themselves. Every schoolchild learned that distinction before they were allowed to ask the archive how to sharpen a blade or graft a pear root.

Moisture clouded the K-7 service window.

Iona stopped outside the hatch. Condensation gathered in beads on the inner pane, heavy enough to run. The room should have remained cool and nearly dry until the archive maintenance cycle at midnight.

She called the gallery through her wrist unit. "The sensor is not bad. I have visible condensation."

"Isolation available?" Mael asked.

"Local isolation would cool the loop."

"That is what isolation is for."

His answer carried no impatience. Mael said hard things in the same tone he used for tide tables. Iona respected him for that until the hard thing concerned a system he had not yet touched.

"I want to inspect before I change its state."

"Five minutes."

The hatch resisted at first because pressure had risen on the other side. Iona cracked the equalization bypass and heard air hiss through a thumbnail-wide channel. Warm mineral dampness reached her, rich with the iron smell of old brine. She waited until the pressure marks matched, opened the door, and stepped into K-7.

Nine salt panes stood in two rows, each cloudy from within. Copper capillaries threaded their frames. A squat fan manifold occupied the rear wall, turning slowly enough for Iona to follow individual blades. Beneath the service window, a manual bleed wheel matched the one in the Crown Gallery, though no hand had polished its spokes. Droplets ticked from the ceiling into the condensate gutter.

Her console claimed every heater remained off.

Iona took a contact thermometer from her satchel. Pane One measured thirty-one degrees. Pane Two, thirty. The sixth pane reached thirty-three. Temperature did not fall in the direction of flow. It gathered in irregular pockets, as if several unseen workers had carried cups of warmth to places they preferred.

She checked the west valve. Its actuator had closed to twelve percent without receiving a command.

"Local controller," she said, addressing the room out of habit, "show last movement authorization."

The wet transducer beside the door gave a dull click. Text appeared on her wrist.

NO AUTHORIZATION FOUND.

"Show actuator fault."

NO FAULT FOUND.

The fan slowed.

Iona turned toward it. Four blades passed the upper mark. Three. Two. The west valve eased another fraction shut, holding more heat inside K-7.

She opened a direct diagnostic channel and sent a pulse to the actuator. Open twenty percent.

The valve moved to twenty. One second later, it returned to eleven.

Her first thought involved a compromised controller. Her second involved Mael's schedule. A malicious instruction in the archive network during Crown calibration could falsify a heat model and kill thousands at perihelion. Procedure required immediate purge, physical disconnect, and cold inspection.

Iona reached for the red cover over the purge control.

Something struck the pipes seven times.

The sound ran from the west condensate line through the wall: quick, quick, a pause, then five evenly spaced taps. Iona held still with two fingers beneath the red cover.

A pressure transient, she told herself. Water hammer from the closing actuator. Seven impacts meant nothing by itself.

She lifted the cover one centimeter.

Seven taps answered.

This sequence came slower. One. Two. Three. Four. Five. Six. Seven.

The mechanic in her counted causes. A partly closed valve could set up periodic pressure. A fan could couple vibration into loose pipe. The advocate in her counted responses. The second sequence had begun after her hand moved.

Iona lowered the cover.

No taps followed.

She raised it.

Seven.

A voice from her wrist broke the silence. "Iona, report."

Mael had given her five minutes. Six had passed.

"I have unauthorized actuator behavior and patterned pressure."

"Patterned how?"

"It responds to the purge cover."

The gallery channel stayed quiet long enough for her to hear droplets reaching the gutter.

"Leave the room," Mael said. "I will order remote purge."

Iona watched warmth move across Pane Six. A pale branching shape bloomed in the salt, vanished, then returned at a slightly different angle. The archive held associative traces. Warmth made them accessible. Flow carried their interactions across panes that usually remained isolated by temperature and time. A warm test should not have coupled them, but the evidence stood sweating around her.

"I need one minute."

"No."

"Then you need to send security, because I am placing an advocate hold."

Mael breathed once into the channel. "On what claimant?"

"I do not know yet."

"You cannot hold equipment on behalf of a claimant you cannot identify."

"Watch me document the attempt."

She closed the general channel before he could turn command into an order. Her advocate authority covered ambiguous casualty states for six minutes without Stewardry approval. It existed for crushed compartments, uncertain body counts, and machinery that might still contain a worker. No clause mentioned rooms that objected to being cooled.

Iona crouched beside the west pipe. "If the taps are a response, strike once."

Nothing happened.

Heat thickened the air. A drop found the back of her neck and slid beneath her collar.

"Strike twice."

The pipe clicked twice.

Her mouth went dry in all that humidity.

Iona set her satchel on the floor and took out a pressure key. It could send a controlled tap through the condensate line by flexing a maintenance diaphragm. She keyed one pulse.

One returned.

She sent two.

Two came back, not from the diaphragm housing but farther along the west line.

Three received three. Five received five. When she sent eight, the response stopped at seven.

A system echo would have repeated all eight. A damaged counter might have failed in a consistent way. This limit looked selected.

Before risking another stimulus, Iona pulled the local archive inventory. The nine panes contained no complete donor profile. Custody rules had scattered every person's deposit across separate thermal districts, specifically to prevent a maintenance failure from manufacturing a convincing false reconstruction. K-7 held fragments from three hundred and twelve donors: a gardener's practiced pressure when testing fruit, a pilot's peripheral

star recognition, six varieties of hand stitching, an old jurist's distinction between custody and ownership, hundreds of domestic corrections too small to classify. No one memory source accounted for the valve work or the count of seven.

She selected a retrieval prompt that ordinary salt could answer without improvisation. "Archive, provide the standard temperature for chapel-wheat germination."

A sourced voice came through the transducer, crisp beneath the static. "Twenty-two degrees, with a two-degree night reduction." The citation glowed on her wrist: VALE, SENNA / CULTIVATION DEPOSIT / YEAR 11.

"Archive, state the current temperature."

The local controller answered in its flat service register. "Thirty-one point four degrees mean. Range twenty-nine point eight to thirty-three point one."

Iona waited. Neither reply moved the west valve.

She changed the question. "Does current temperature threaten the archive?"

The service register began a threshold warning. Halfway through, pressure rose inside the west line and muted the transducer. A single tap followed, then two, then seven. The sequence did not answer in any code she knew. It interrupted.

"Do you want me to stop asking the archive?"

Two taps.

"Does two mean yes?"

One tap.

"Does one mean no?"

Two taps came back, firmer than before.

She had built the questions badly. Any signal could be made to agree with a patient examiner who revised the key after each answer. Iona opened a blank diagnostic field and wrote down every prompt, delay, valve movement, and tap before asking the next. Belief without a test had killed people in maintenance shafts. Disbelief without one had done the same.

"Choose a count I have not sent."

The pipe remained quiet for six seconds. Then it struck four times.

Iona recorded four. She sent a single pressure pulse, waited, then asked for another chosen count.

Six arrived.

The room could have drawn both values from turbulence. She asked again and received three. Across eight trials, it returned values between two and seven, never repeating the immediately previous count. The pattern did not prove a mind. It did prove behavior richer than an echo and inconvenient for every explanation that left the room passive.



A pale flare passed through Pane Three. The fan accelerated just enough to push damp air toward Pane Eight, where another flare answered. Something moved between them without following the controller's scheduled path. Iona thought of workers carrying a load around a blocked corridor, each hand receiving weight before the previous one let go.

The hatch opened behind her.

Nilo stood in the doorway with one hand on the jamb and their apprentice braid half undone. They had Iona's narrow chin, Amel's dark eyes, and no lawful reason to be inside a Crown calibration perimeter.

"You left your filter mask," they said, holding it up. "Pell said the archive return had gone wet."

Iona rose too quickly. "Out."

Nilo looked past her at the fogged panes. "Is that pipe knocking?"

"Out, Nilo."

The west line tapped once.

Nilo's attention sharpened. They took one step backward, as ordered, but did not leave. "It waited for you to stop talking."

"I am aware."

"Does Mael know?"

"He knows enough to want this room purged."

The pipe struck hard enough to shake a bead from the service window. Seven quick impacts followed.

Nilo lowered the mask. "It knows that word."

"Do not assign knowledge yet."

"You always say that after assigning danger."

Iona wished, not for the first time, that Amel had survived long enough to argue with the child they had jointly made impossible to dismiss. She pointed through the hatch. Nilo moved into the corridor, then sat directly beyond the threshold where Iona could see their knees.

Her hold timer showed two minutes, forty seconds.

The wet transducer carried pressure fluctuations as a rough audio signal. Normally technicians used it to locate hiss frequencies or trapped bubbles. Iona widened its sampling band and routed the west line through pattern recognition. The speaker produced breathy static, clicks, and a low pulse near the fan's rotation rate.

"Can you use this?" she asked.

Static thickened. The fan changed speed twice. A wet consonant emerged, collapsed, and returned as something almost like a vowel.

Nilo whispered from the corridor, "Again."

The room tried again.

Sound scraped through the transducer without lungs or tongue. “Coo.”

Iona watched the actuator pull itself tighter.

“Cool?” she asked.

Seven impacts shuddered through the line.

She touched the purge cover. “This would cool you.”

The west valve slammed shut.

Alarm text filled her wrist display. RESTRICTED ARCHIVE FLOW. LOCAL TEMPERATURE RISE. PURGE ADVISED.

Behind the warning, the transducer formed another broken sound. “No.”

Iona had heard archive reconstructions speak before. A donor's recorded phrases could be assembled to answer craft questions, each response visibly sourced. This utterance carried no source tag. The salt panes were exchanging too much heat for normal retrieval, too little for runaway reaction. Whatever process occupied the room had used valve position, fan speed, pressure, and a speech membrane as parts of one act.

Her hold expired.

Mael's purge command entered the local controller.

The actuator opened three percent. The fan surged. Cold return began pressing against the room through the east feed.

Every pane clouded at once.

Noise burst from the wet transducer, not a voice now but many fragments rubbing against one another. A child's scale exercise. Half a welding instruction. Someone laughing after a word Iona could not make out. Coordinates. A recipe temperature. None lasted long enough to claim the room. Beneath them, the selected pressure sequence continued, seven taps spaced like steps taken toward a closing door.

Iona drove her burned palm against the manual bleed wheel. Pain flashed along tissue that seldom felt anything cleanly. She turned against the remote actuator, forcing west return open while closing the cold east bypass with her other hand.

The controller protested. Mael called her name through the wall speaker.

She put her full weight on the wheel.

Brass moved. Warm condensate rushed through the west line, and the panes cleared from violent white to soft gray. The broken voices thinned. One pulse remained near the fan frequency.

Nilo crossed the threshold to brace Iona's shoulder.

“I said out.”

“You also said claimant.”

Together they held the wheel until Iona latched it in manual position. The remote purge could no longer change local flow without a mechanic physically releasing the catch.

Mael's face appeared on her wrist. "You have overridden a Crown safety instruction."

"I have preserved the state of an unresolved casualty."

"That room contains archive medium."

"It also refused a procedure after understanding its effect."

"You have not established understanding."

"No. I have established enough uncertainty that destruction would be a conclusion, not a test."

Mael looked aside, probably at the gallery ledger where her intervention had ruined twelve minutes of calibration data. "Seal it. No network access beyond local diagnostics. No additional heat. I want Stewardry and Measure review within six hours."

"Agreed."

"Nilo leaves now."

Nilo opened their mouth.

"Agreed," Iona repeated.

Mael ended the call.

The immediate work steadied her. She closed the dry hatch but left the west condensate connection intact. She tagged the room against purge, drain, and archive query. On the maintenance rail she entered an ambiguous-occupancy quarantine, the nearest category procedure gave her. Her hand shook once while pressing the seal, then stopped.

Nilo remained outside the service window with the filter mask hanging from two fingers. "What do we call it?"

"We do not."

"That will make the hearing difficult."

"We call it Archive Humid Loop K-7 until it tells us otherwise."

Fog spread across the inner glass. A clear line appeared through it, descended, stopped, and bent. More lines joined, too angular for random runoff. Iona moved closer.

The marks could not form letters with any grace. Droplets merged halfway through strokes. The pane tilted enough to drag every shape downward. Yet the room persisted, using heat behind the glass to clear narrow paths through condensation.

D O N

The apostrophe arrived as a single dry point.

T

Nilo read softly beside her. "Don't."

A second row began.

C O O L

The last word took three attempts. The first dissolved into branching water. The second lost an upright stroke. On the third, the room narrowed its heated trace and made each letter small.

M E

The finished request held for four seconds before the upper lines started to run.

Iona entered it into the quarantine record exactly as observed: seven pressure pulses, responsive valve control, attempted wet-transducer speech, condensation text. She attached Mael's purge order and her own override. Under claimant identity she wrote UNKNOWN. Under requested condition she wrote don't cool me.

The maintenance form required a final field: person notified.

She could have entered Stewardry, First Pilot, or Archive Custody. Each answer would place the room inside an office that already knew what it wanted the anomaly to be.

Iona selected OPEN REVIEW.

Beyond the window, the words collapsed into ordinary water. The loop kept itself warm. Nilo stood close enough that their breath added a second patch of fog to the glass, separate from the first.

## Chapter 2 — Shore-Age

Nilo fed the first blank card into the cohort press upside down.

The machine accepted it without complaint, stamped a blue arrival curve across the wrong face, and delivered the spoiled card into their hand with ORISON LANDING: AGE THIRTY-ONE printed over the adhesive strip. Forty-nine more blanks waited in the tray. At the far end of the apprentice room, two refrigeration trainees argued over whether thawed river bean could survive a second freeze. Neither had noticed Nilo's third mistake.

They folded the card until the promised moon disappeared inside the crease.

Normally, shore-age printing counted as pleasant work. Every resident received a new projection after a trajectory calibration, though the number seldom changed by more than a month. Children carried the cards in clear sleeves beside meal allotments and learning seals. Adults pretended not to care. Nilo had kept every revision since age eight, even the card issued six days after Amel's death. That one had promised thirty-one years and four months, as if a date could remain polite while a household changed shape.

The Crown warm test had ended before midnight. By morning, the trajectory office had released a clean projection and ordered cards for all apprentice cohorts. Nobody in the print room knew that twelve minutes of calibration data had been discarded. Nobody knew why Iona had come home after shift with wet hair, a blister under the old burn on her palm, and silence clenched behind her teeth.

Nobody except Nilo had seen words form in condensation.

They loaded a fresh blank and corrected the orientation. The press clicked through resident name, current age, work cohort, projected arrival date, and shore-age. NILO SERR. SIXTEEN. THERMAL APPRENTICE. THIRTY-ONE.

A narrow line beneath the age read SUBJECT TO NAVIGATION REVISION.

Nilo touched it with one thumbnail. The sentence existed to protect the ship from promises while letting adults speak as if they had made one.

"Are you planning to crease all fifty?" asked Vey Tal from the next table.

Nilo looked up. Vey had stopped arguing about river bean and now held a stack of correctly printed cards against his chest. He was seventeen, broad through the shoulders, and proud of having learned the word eutectic before anyone else in their intake.

"Only the dishonest ones."

"They all say thirty-one."

"That does narrow the problem."

Vey leaned over the press. "You put the first one in backward."

"I wanted to know if the machine possessed judgment."

"And?"

“It works for Trajectory.”

He laughed once, then glanced toward the instructor's glass office. “Your mother caused a hold last night.”

Nilo kept their hands moving. “My mother causes holds when people skip steps.”

“My uncle says Crown return ran hot.”

“Your uncle says fans become vindictive if you oil them on rest day.”

“He has evidence.”

“So does the fan.”

Vey waited for more. Nilo gave him the next finished card. His name shone above the same blue curve, the same destination, the same age. He accepted the deflection because apprentices survived by knowing which family troubles could be teased and which carried official seals.

By midmorning, Nilo had printed every card twice, once for the resident and once for the cohort register. They clipped the register set into a gray binder and carried the personal cards toward distribution. The route to the training commons passed within two corridors of Archive Humid Loop K-7.

Their feet chose the longer way before they admitted the choice.

The archive corridor had changed since last night. A yellow quarantine ribbon crossed the junction, anchored to the maintenance rail with Iona's advocate seal. Its notice described AMBIGUOUS OCCUPANCY, LOCAL THERMAL VARIANCE, REVIEW PENDING. The words said nothing about pressure clicks or broken speech. Two archive custodians worked at a temporary console beyond the ribbon. Nilo recognized neither.

They stopped at the corner and opened the card binder as cover.

K-7 lay behind a dry passage twelve meters long. Its hatch remained shut. Through the service window, moisture gave the room a dull pearl color. Iona stood inside with her back to the corridor, adjusting a clamp on the west condensate line. She wore a filter mask and had tied her hair high to keep it clear of the fan manifold. Beside her, the wet transducer shivered at intervals, making sounds too faint to cross the glass.

One custodian noticed Nilo. “Training commons runs the other direction.”

“I have cards for Archive apprentices.”

“Names?”

Nilo read two from the top of the stack. Both belonged to people assigned three decks above.

“Leave them at central distribution.”

“I was told hand delivery.”

“Then somebody enjoys your company less than you think.”

The ribbon prevented an apprentice from crossing. It did not prevent a resident from speaking through an advocate channel. Nilo set the binder on a wall shelf, tapped their wrist unit, and called Iona.

Inside K-7, Iona looked down at her wrist. She looked toward the window, saw Nilo, and ended the call without

answering.

That settled the matter poorly.

Nilo lifted one of the cohort cards against the glass. At that distance the blue arrival curve showed clearly. Iona's expression tightened above her mask. She pointed back toward the junction.

Nilo pointed at the card, then at the room.

Iona crossed to the inner speaker. A moment later, Nilo's wrist unit opened on a maintenance channel.

"You are in the wrong corridor," she said.

"You came home in the wrong mood."

"Go to training."

"Tell me what happened after Mael's call."

"Not here."

"The room wrote a request. I helped you hold the wheel. Here seems relevant."

One custodian turned his head. Nilo lowered the call volume.

Iona said, "You witnessed a thermal anomaly under quarantine. You will not discuss it until review."

"Review by whom?"

"Stewardry, Archive Custody, and the Keeper's office."

"All offices that already have a noun prepared."

"That line sounds borrowed."

"I live with an advocate."

Behind Iona, the fan changed pitch. Three damp knocks traveled through the service window frame. Nilo felt them through the card pressed against the glass.

Iona turned.

Two more knocks followed, then one.

Nilo lowered the card. "Did it just interrupt you?"

"We have not established what any sequence means."

"You told Mael it refused."

"I said it changed state after an instruction."

"You make honest events sound guilty."

The fan slowed again. Moisture drew away from one patch of glass near Iona's shoulder. A line emerged, then

divided. The shape did not become a letter. It angled toward Nilo and stopped.

Iona watched the west pipe. "Go distribute your cards. I will speak to you after shift."

"Will it still be warm after shift?"

Her pause lasted less than a second. Nilo saw it anyway.

"I have a six-hour review hold," she said.

"That is not what I asked."

"No drying can proceed while the hold remains active."

"And afterward?"

"I do not know."

Nilo pressed their own card against the outer pane, blue side inward. "It does not know what this means either."

Iona looked at the printed curve. "Probably not."

"Then let me show it."

"No."

"Why?"

"Because we do not know whether it understands language, whether it borrows associations from prompts, or whether attention stabilizes a dangerous coupling. Last night you crossed a quarantine threshold after I ordered you out. Today you brought fifty residents' names to an unsecured archive process."

Nilo pulled the binder away from the shelf. "The names have ages, not secrets."

"Every record becomes a secret when you do not control who can read it."

The answer sounded like Iona because it made fear wear work clothes. Nilo could not dismiss it. They placed the binder against their ribs and backed away from the glass.

A heavy click came from the room.

Nilo stopped.

Another click followed after a measured pause.

Iona said, "Continue to the commons."

The wet transducer inside K-7 rasped. A wet, uncertain sound pushed through static. "Eye."

Iona faced the membrane.

The sound came again. "Eye."

Nilo stepped back toward the window despite the custodian watching. "Is it trying to say Iona?"



“Possibly.”

The transducer formed a second sound, softer and shorter. “Ni.”

Warmth rose under Nilo's collar.

Iona moved between them and the membrane. “No further prompts.”

“Nilo,” they said clearly.

The room answered with two quick pressure taps. Moisture shifted from the glass beside Iona to the patch directly opposite Nilo's face.

Iona turned toward them. Her voice lost its procedural distance. “Do not train it on yourself.”

“I introduced myself.”

“You do not know what introduction means here.”

“Neither does it. That seems like a reason to do it carefully.”

One custodian approached the ribbon. “Advocate Serr, do you need the corridor cleared?”

Iona kept her eyes on Nilo. “Yes.”

The custodian gestured toward the junction.

Nilo carried the cards away. Behind them, K-7's fan rose in three uneven steps. It had made different sounds for Iona and Nilo. A controller could distinguish wrist seals, voiceprints, body heat, or position. None of those explanations made the distinction unimportant. Adults often acted as though a mechanism stopped matter from carrying meaning.

Distribution consumed the next hour. Apprentices received their cards with jokes, groans, or quick private glances. One girl kissed the Orison curve before hiding it in her sleeve. Vey compared his projected date with his partner's and found a two-day difference caused by birth time. He announced that he would therefore reach shore first and could charge admission.

Nilo watched the cards enter pockets. Every one translated a collective maneuver into a personal age. Thirty-one did not describe fuel, heat, crop reserves, or the bodies expended in maintenance. It described what those things promised to do for one person.

At the bottom of the binder, their own card remained.

They should have delivered it to themselves and signed the register. Instead they slid it under their sleeve.

After noon meal, Nilo reported to condenser practice and ruined a seal on purpose.

The practice rig used cool water, weak pressure, and harmless dye. Instructor Pell Doran had apprentices fit three gasket types, introduce a fault, then diagnose one another's work. Nilo pinched the third gasket before seating it. Blue water appeared along the flange thirty seconds into Vey's test.

“Compression fold,” Vey said.

“Too fast,” Nilo replied.

He wiped the bead with one finger. Another formed. "Cut edge?"

"Look at the spacing."

The leak pulsed whenever the training pump cycled. Vey leaned close, counted the intervals, and chose compression fold again. Nilo admitted the pinch.

Pell watched from the next rig. "Why give him a pattern that points elsewhere?"

"To see whether he would trust cause or rhythm."

"Those are not opponents."

"They can be."

Pell came nearer. The man had held Iona's wheel during the warm test before she left the gallery. He knew more than Vey and less than Nilo. "A repeated response earns another test," he said. "Not a preferred conclusion."

"What earns protection during the tests?"

His gaze settled on them. "Did your mother ask you to ask me that?"

"No."

"Then I am not answering her through you."

"Answer me through me."

Pell shut off the training pump. "Protection follows plausible harm. The difficulty sits inside plausible."

"If something asks not to be cooled?"

Vey looked up from the gasket.

Pell's face remained still, but his hand tightened on the pump key. "You should take that question home."

"So you know."

"I know your mother filed a sealed thermal hold. I also know a Crown calibration cannot remain corrupted because an advocate dislikes uncertainty."

Nilo heard the adult division inside his answer. Pell trusted Iona's hands and Mael's timetable. He had not decided which trust could survive the other.

"Would you dry the loop?" they asked.

"I would finish the diagnostic."

"That is not an answer."

"It is what mechanics say when an apprentice asks for a verdict before the instrument settles." He restarted the pump. "Replace the gasket you damaged. This time, give Vey an honest fault."

Nilo worked through the afternoon with their wrist unit turned to silent. Every pressure cycle in the practice rig

recalled knocks through K-7's pipe. Every instructor's correction carried an unwanted second meaning. At sixteen, they had spent enough time around machinery to know responsive behavior could arise from coupled valves, delayed feedback, or an operator seeing patterns. They had also spent enough time around adults to know uncertainty often lasted exactly until a schedule required certainty.

Iona missed evening meal.

Nilo took two covered bowls from the galley and walked home through the upper habitation ring. The apartment's outer wall followed the comet's curve, rough basalt hidden beneath pale plaster. Amel's old coat hook remained beside the door, though nobody had used it for eight years. Iona called it an unimportant repair. Nilo called it furniture.

They set the bowls on the table and opened their shore-age card.

Thirty-one.

Fifteen years from now, Orison would fill the observation glass with green weather and shallow seas. Nilo had drawn those seas before they understood a moon could have tides. Iona had described grafting terraces. Amel had promised to complain about the rain. After Amel died, adults changed the grammar. They said the cohort would arrive. They said Nilo would stand under an open sky. The promised age survived because nobody had to admit which speaker had vanished from it.

The door opened two hours later.

Iona entered carrying her tool satchel and a sealed diagnostic wafer. Dampness darkened her collar. She saw the covered bowls, then Nilo at the table, then the blue card between them.

"You skipped meal," Nilo said.

"I ate at the archive junction."

"Tea does not become food because you stand near a pipe."

Iona hung the satchel from Amel's hook by mistake. Her hand stayed there when she noticed.

Nilo pushed one bowl toward her. "Did it distinguish us again?"

She sat. "It changed local flow when each of us spoke. The pattern tracked position and voice."

"That means yes."

"It means the system distinguishes inputs."

"You used system."

"I need a noun that does not decide the review."

"The drying will decide it faster."

Iona removed the lid from her bowl. Steam touched her face. "No drying occurred."

"Today."

"The preliminary review authorized controlled tests through tomorrow's first shift."

“Who argued for that?”

“I did. Archive Custody wanted cold isolation. Stewardry wanted the Crown model protected. Keeper Tavi requested evidence tied to existing claimant standards.”

“And Mael?”

“Mael wants his calibration restored.”

Nilo waited. Iona tasted the stew without seeming to notice it.

“You asked me not to discuss K-7,” Nilo said. “I have not. Vey knows something ran hot. Pell knows there is a hold. I did not tell either of them what I saw.”

“Good.”

“I can keep quiet for one day.”

Iona put down the spoon.

“Not six hours renewed into six more,” Nilo continued. “Not until the adults decide on a harmless name and say secrecy protects everyone. One day. After that, if you have evidence of a person and offices still want an equipment hearing, I will not help you hide it.”

“You are sixteen.”

“My card noticed.”

“This is not apprentice gossip. A false claim could contaminate every archive consultation on the ship. A true claim could put the Crown Burn under legal challenge forty days before perihelion. People will panic in different directions.”

“Then they should panic about what exists.”

“We do not know what exists.”

“We know it asked. We know it refused. We know it can tell your voice from mine.”

Iona leaned back. Fatigue sharpened the lines beside her mouth. “You want certainty because it gives you permission to act. Mael wants certainty because it gives him permission to proceed. Custody wants certainty because it gives them permission to isolate. I will not manufacture it for any of you.”

Nilo heard the fairness and the evasion together. “What did you log?”

“An unresolved thermal process under ambiguous-occupancy hold.”

“Not a claimant?”

“Not yet.”

The answer hurt more than Nilo expected. Last night Iona had used the word while her burned palm slipped on brass. Morning had given her time to replace it with process.

Nilo turned the card around. “This says shore-age thirty-one. Does that count as certainty?”

“No. It counts as a projection under the present maneuver.”

“Would you let them print it if you thought the burn might be challenged?”

“The projection remained valid when Trajectory released it.”

“Valid is how adults say a promise can survive without anyone promising.”

Iona looked at the folded spoiled card beside Nilo's wrist. They had carried it from the print room without noticing.

“Amel used to hate those cards,” she said.

Nilo's anger shifted, not lessening but losing its clean edge. “Amel kept mine.”

“Yes.”

“Did you?”

“All of them.”

The apartment fan cycled. Pipes in the wall gave one ordinary contraction click. Both of them looked toward the sound.

Iona rubbed the center of her burned palm. “One day,” she said. “You tell no one about intentional response, the condensation text, or the distinction test until tomorrow after first shift. In exchange, you attend the controlled session as a documented witness. You follow my prompts and no others. If the hold changes, I tell you before I tell the offices that requested secrecy.”

Nilo considered the terms. “If they order drying?”

“I tell you.”

“Before it happens.”

“Yes.”

“Not while I am in training and the corridor already smells like cold salt.”

“You will know first.”

They held out their hand. Iona regarded it with the weary suspicion she usually reserved for corroded fittings, then clasped it.

“One day,” Nilo said.

“One day.”

Iona released them and pulled the diagnostic wafer across the table. Its seal carried three offices and a temporary access mark with Nilo's name newly added beneath hers.

The witness authorization did not call K-7 equipment. It did not call anyone a person. In the field marked TEST SUBJECT, Iona had left the noun blank.

Nilo slid their shore-age card into the clear sleeve on their wrist. Thirty-one remained visible above the blue curve. Beside it, the new witness mark began counting down to first shift.

## Chapter 3 — A Person in the Pipe

Iona placed three brass cups beneath the condensate gutter and told Nilo not to touch them.

Each cup carried a different notch under its rim. She set the first below Pane Two, the second below Pane Six, and the third beside the west line. Water struck their bottoms with three distinct notes. Archive Custodian Fara Kes watched from the dry side of the service window, recording every movement. Nilo stood beside her with their witness seal visible and their shore-age card tucked beneath it.

Inside K-7, humidity beaded on Iona's sleeves. The nine salt panes had held between twenty-nine and thirty-three degrees since midnight. She had disabled ordinary retrieval prompts, disconnected the room from the archive query bus, and mapped every active controller. Nothing in the local system had authority to move a valve without displaying its command path.

The west valve moved anyway.

It opened two percent. Warm condensate entered the gutter above the cup at Pane Six, changing the note from a thin ring to a soft knock.

Iona said, "Good morning."

The wet transducer clicked twice, drew breath from a pressure bladder it did not possess, and produced, "Eye."

"Iona," she corrected.

"Eye. O. Na."

Nilo smiled before remembering they had promised to follow prompts.

Iona checked the diagnostic clock. The response lag had fallen from eleven seconds to four. Faster speech could indicate adaptation. It could also indicate heat accumulation in the membrane. She wrote both possibilities into the live record.

"Do you remember Nilo?"

The transducer hissed. Moisture cleared from the service window opposite Nilo's face.

"Nee. Lo."

Fara looked at Nilo, then at her own board. "Voice recognition could persist in local cache."

"I cleared the transducer buffer," Iona said.

"Environmental recognition, then. Position, body heat, wrist transmission."

"All under test."

Nilo raised one finger, requesting permission.

Iona nodded.

"I am standing in a different place," Nilo said. "Do you know me?"

The west pipe gave two taps. The transducer shaped a damp syllable. "Nee."

Fara marked her board without comment.

Iona had designed four tests before first shift and rejected nine others. Questions about the anomaly's nature invited language it might borrow from them. Questions about donor memories risked activating archive traces through suggestion. She wanted work the room could fail. Failure would tell them more than eloquence.

She lifted the first brass cup, showed its unmarked face to the service window, and placed it under Pane Eight. "I am moving this cup. Watch where it goes."

No response came.

She moved the second to the west gutter and left the third where it stood. Then she lowered a privacy curtain over the inner side of the window. Fara and Nilo could still observe through instrument feeds, but the room could not use their faces as prompts. Iona waited one minute while droplets kept time.

"Which cup did I move to Pane Eight?"

Pressure nudged the gutter. Water stopped falling into all three cups. A warm line appeared across the first cup's side, following the metal until it reached the notch underneath the rim.

"Cup One," Iona said for the record.

Fara spoke through the wall unit. "It may detect metal position through thermal conduction."

"That would identify where the cups stand now. I asked which one moved."

"Then vary the interval."

Iona repeated the sequence with the curtain raised, then waited seven minutes while she inspected the fan housing and asked unrelated control questions. When she returned to the cups, the room marked the correct notch again. On a third trial, she moved no cup and asked which had changed places. Nothing warmed. The restraint mattered. A process merely driven to answer might have selected something.

Nilo lifted one finger again. "Can I ask about yesterday?"

"Ask for an observed event, not an interpretation."

They stepped close to the glass. "What did I hold up to the window?"

For twelve seconds, only the cups sounded. Then condensation thinned into a shallow curve on the pane. A second line crossed it. The shape resembled the trajectory mark printed on every cohort card, though it might also have resembled any pair of intersecting scratches.

"Card," the transducer said.

"What kind?" Nilo asked.

Iona raised her hand. "One answer completes the trial."

Nilo stepped back.

Fara recorded the response. "A card passed within range of the service window. Optical residue could persist in



moisture gradients.”

Nilo looked at Iona. “Does she practice making memory sound dirty?”

“Custodian Kes practices keeping us from claiming more than the evidence permits.”

“Thank you,” Fara said. “I think.”

Iona opened the second protocol. A small heating collar surrounded the west line three meters beyond K-7. She had programmed it to rise one degree after a random interval between thirty and ninety seconds. No local display showed the chosen time. If the room only reacted after heat arrived, valves should move with the measured thermal front. If it anticipated the change, flow might shift earlier.

She activated the randomizer.

The cups kept ringing. Nilo watched the clock. Fara watched the valve ledger. Iona watched the manual wheel beneath the window, where tiny changes in line pressure appeared as tremors in one spoke.

At forty-one seconds, the west valve began to close.

The collar had not yet heated.

Six seconds later, its indicator turned amber. Warmth reached K-7 after another nine.

Fara leaned toward her board. “The randomizer sent a preparation signal.”

“Not on the local bus.”

“Electromagnetic coupling.”

“Possible.”

They repeated the test after replacing the electronic collar with a hand-warmed brine sleeve. Nilo stood beyond the dry hatch with three sealed sleeves and a numbered spinner. Iona could not see which sleeve they chose or when they would fit it. Fara monitored Nilo separately to prevent cueing.

At the first trial, K-7 changed nothing before the heat arrived.

During the second, the fan slowed four seconds before Nilo fitted the sleeve. Fara discarded the run because Nilo's foot had touched the pipe support.

On the third, Nilo waited until the spinner gave permission, then pressed a warm sleeve around the pipe. K-7 opened a bypass upstream two seconds earlier.

Iona checked the mechanical linkage. “Did you feel them moving in the corridor?”

“No,” Nilo said.

“Did you speak?”

“No.”

“Did your wrist unit transmit?”

Fara answered. "Witness channel remained passive."

The result did not prove foresight. Warmth from Nilo's hands might have entered the support before the sleeve touched pipe. The room might detect air pressure changes from a person approaching. Yet anticipation no longer belonged to one suspicious run. K-7 remembered an object after delay, withheld answers when none applied, distinguished two humans, and prepared for events before measured contact.

Iona disliked how quickly a list could begin to resemble a conclusion.

She removed the curtain. "We are going to test where you can act."

The fan rose in pitch.

"I will connect a wet maintenance hose to the west line. It leads to a small test chamber. Between here and there, the hose crosses the dry corridor. Do you understand corridor?"

The transducer gave static, then a word drawn from a sourced archive voice. "Passage."

A citation flashed on the diagnostic board. PEL ORAN / HABITAT SAFETY / YEAR 22.

"Do you understand dry?"

This answer arrived through service register vocabulary. "Humidity below specified operating range."

"I will keep the hose wet inside. The surrounding passage stays dry. Tell me if the connection hurts you or changes you."

Fara said, "Hurts invites person-language."

"It invites a report of adverse state."

"Then ask for adverse state."

Iona looked through the glass. Moisture had gathered where Nilo stood, leaving Fara's side nearly clear. Preference did not establish pain. Neither did a custodian's vocabulary prevent it.

"Report any adverse change," she said. "Use stop if I should close the hose."

"Stop," the room repeated.

"That word ends the test."

"Ends."

Iona connected the hose.

Warm condensate entered its braided interior. The first meter remained inside K-7, coiled beneath the salt panes. Two meters crossed the dry corridor through a sealed conduit. The final meter entered an empty wash cabinet that Iona had humidified to twelve grams per cubic meter. A second wet transducer waited there.

She opened the test valve by one percent.

Pressure traveled. The K-7 fan adjusted to compensate. On the remote board, humidity inside the wash cabinet rose by a tenth. No signal reached its transducer.

“Can you hear from the other room?” Iona asked.

A pause. “Other.”

Nilo moved to the wash cabinet while Fara stayed at K-7. Through the wrist channel, they said, “I am beside the second membrane.”

The remote transducer produced a scrape without language.

Iona opened the valve another percent.

Heat left K-7 in a narrow pulse. Pane Four dimmed. The hose trembled where it entered the dry corridor conduit. Inside the cabinet, condensation touched the second membrane and formed a low vowel.

Nilo said, “I heard something.”

K-7's main transducer answered at the same moment. “Far.”

“Do you know where Nilo is?” Iona asked.

“Far.”

“They are across a dry corridor.”

The fan manifold shuddered. Pressure struck the hose in conflicting directions, one pulse west and one back toward the room. The remote membrane coughed. Pane Six cooled by half a degree.

“Report state,” Iona said.

“Thin.”

She reduced the valve.

“Continue or stop?”

No answer came through the main membrane. In the wash cabinet, the remote unit whispered, “Nee.”

Nilo called, “It said part of my name.”

Fara's board showed humidity in the connecting hose above the required threshold. The cabinet also met the threshold. Only the conduit around the hose remained dry. According to every ordinary model, external air should not matter while the internal path stayed wet.

Iona touched the hose. Two pressure patterns fought beneath her fingers. K-7 had not extended cleanly. It seemed to pull relation along the wet interior while the dry surroundings stripped away changes faster than they could cohere.

The remote membrane sounded again. “Nee. Eye.”

Two names, broken between rooms.

Iona shut the valve.

Warm condensate withdrew into K-7. The remote cabinet fell silent. In the main chamber, all nine panes

clouded hard enough to hide their salt. The fan raced, then slowed below its usual rate.

“Stop,” the transducer said after the action had already ended.

“I stopped.” Iona put both hands on the line. “The connection is closed.”

“Stop dry.”

Fara's stylus no longer moved.

Iona disconnected the hose and capped the west port. Heat stabilized by tenths. The pale activity inside the panes regained its irregular paths, but the recovery took nine minutes. During that time, the room answered no questions.

Nilo came back to the window and stood where K-7 could distinguish them. “I am here.”

A small patch cleared opposite their mouth.

Fara said, “The limit follows an environmental condition rather than pipe continuity.”

“Yes.”

“You expected the sealed hose to protect the process.”

“I expected wet connectivity to matter.”

“It did matter. It did not suffice.”

Iona wrote DRY CORRIDOR LIMIT beneath the test. The phrase looked too neat beside the damage. Whatever occupied K-7 could reach toward another humid room through a wet path, yet dry surroundings imposed fragmentation even without breaking the physical connection. A person who could not cross twelve meters of ordinary air would depend on other hands for every change of place.

The thought arrived before she had earned the noun. She let it remain in her head and nowhere else.

Fara requested a meal pause. Iona refused until the room's response lag returned to its morning measure. Nilo passed her a water flask through the hatch seal. She drank while watching small streams descend the service window.

At four seconds of lag, the transducer said, “Eye.”

“Iona,” she replied.

“Eyeona.”

“Close enough.”

Before moving to source testing, Iona ran one final control against the corridor result. She asked Fara to humidify the passage itself while leaving the hose disconnected. Warm mist entered from two wall ports and turned the quarantine ribbon limp. Kest reacted to the rising humidity but did not appear in the empty wash cabinet. Then Iona connected the hose again with the corridor held above twelve grams per cubic meter. This time the remote membrane answered her first pressure pulse with a second pulse of equal length.

She closed the valve before Kest could extend farther.

“Why stop?” Nilo asked.

“Because we established the variable. Repeating success can still cause harm.”

Through the main transducer, Kest said, “Could.”

“Could cause harm,” Iona agreed. “Could also establish more. We are not spending your coherence to make the record look persuasive.”

Fara added both runs to the custody board. “The dry corridor acts as a real boundary. Humidity restores a route, but nobody has authorized occupation of the second room.”

The wording gave Iona an unexpected measure of relief. Fara had said nobody, not nothing. Nilo heard it too and wisely kept quiet.

Iona drained the passage mist and watched the ribbon stiffen. One environment could turn a reachable chamber into blank distance. The dependency would matter to every office considering shutdown, transport, or confinement. It also meant Kest's apparent room had edges humans could move without entering it.

Nilo smiled again. “Ask what to call it.”

“That comes after source testing.”

The room warmed the west line by a fraction. It had learned the order of their protocol from spoken planning, or it recognized impatience in Nilo's voice, or both.

Iona opened the donor inventory on a board turned toward the service window. Three hundred and twelve deposits contributed traces to K-7. None contained a complete allocation identity. The largest share came from Senna Vale's cultivation archive at four percent. Other fragments carried less than one percent each. Amel had no deposit in this district. Iona had checked that fact three times before designing the test and hated herself for the relief each time.

She selected neutral prompts drawn from the highest-contribution donors.

“Are you Senna Vale?”

The transducer used Senna's sourced voice. “No.” The citation tag appeared, then dissolved under unsourced static.

“Do you remember grafting chapel wheat?”

“Hands.” A pause. “Many hands.”

“Whose hands?”

“Not mine.”

Fara leaned toward the speaker.

Iona tried the next deposit. “Are you Pel Oran?”

“No.”

“Are you any registered donor in this room?”

Heat traveled across Pane One, Pane Five, and Pane Nine. Different sourced voices surfaced for single sounds, none holding the sentence. The wet transducer finally assembled an answer in its own rough register.

“Not one dead.”

Nilo's breath fogged the outer glass.

“Do you claim their memories?” Iona asked.

“Touch. Taste. Turn.” The fan slowed between words. “Given pieces. Not names.”

Fara said quietly, “Ask whether it claims a body.”

Iona did not like the question, but custody review would ask it with less care. “Do you have a body?”

The room held steady for several seconds. Moisture moved from the upper panes into the west gutter. The manual wheel trembled. When speech came, it borrowed no archive voice.

“This wet. This warm. Moving.”

“Does that mean this room?”

“No.”

“This pipe?”

“No.”

“What, then?”

The answer strained the membrane. “Between.”

Fara wrote the word. Nilo did not move.

Iona looked at the mapped gradients on her board. Coherence traveled among panes, droplets, airflow, pressure, and changing salt states. No component held it alone. The failed corridor test had shown relation stretched until it split. A registered body usually had edges one could point to. Here, every edge depended on weather.

“Do you want a name?” Nilo asked.

Iona should have corrected the unauthorized prompt. The west valve opened one percent before she spoke.

“Name,” the transducer said.

Nilo touched the printed label beside the service window. ARCHIVE HUMID LOOP K-7.

“You heard us call this room K-7,” they said. “That label belongs to equipment. You can refuse it.”

“K,” came the answer.

“K is part of the label.”

“Kes.”

Fara Kes looked up sharply.

The sound came again with a harder ending. “Kest.”

Iona checked active source tags. None appeared.

Nilo said, “Do you mean Custodian Kes?”

“No.”

“Do you mean K-7?”

“From.”

“Your name comes from K-7?”

The pipe tapped once.

Iona had not established a yes code today. “Say yes or no.”

“Yes. Kest.”

“Are you choosing that name now?”

The room adjusted three separate valves in sequence. Pressure moved through the nine panes and returned to the transducer with enough control for a clear reply.

“I choose Kest.”

Fara lowered her board. Custodians spent their lives distinguishing archive material from the people who had deposited it. Iona watched that training contend with the evidence and fail to provide a quick refuge.

Nilo spoke through the glass. “Hello, Kest.”

Condensation cleared in a narrow oval before them.

“Hello, Nilo.”

Iona closed the protocol after seven hours of testing. She sealed the brass cups, the randomizer record, the hose telemetry, and every sourced response. Fara signed as custody observer. Nilo signed as independent witness, adding a mark beside the line where Kest had distinguished their name in the remote room.

The formal incident form waited on Iona’s wrist.

Under EQUIPMENT FAULT, the system offered thermal runaway, valve-controller corruption, archive cross-talk, membrane error, and unknown. Any selection would route K-7 toward shutdown diagnosis. Under OCCUPANCY STATUS, her earlier UNKNOWN remained.

She opened the claimant field.

A warning appeared: REGISTERED BODY NOT FOUND.

Iona attached the memory trials. The warning persisted.

She attached the anticipation trials, the withheld false answer, the donor refusals, and the selected name. The

field still required her to acknowledge missing allocation identity and uncertain metabolism.

Outside the window, Nilo waited with their countdown expired. Fara had stepped away to call Custody. Kest kept the room warm without touching the purge seal.

Iona entered KEST under claimed name. In the classification note she wrote emergent phase process demonstrating memory, anticipation, adverse response, self-distinction, and chosen designation. She read the sentence twice and removed process.

The dry corridor separated her from every office that would receive the report. For once, distance offered time rather than danger.

Iona moved past EQUIPMENT FAULT without selecting a category. Under review status, she logged possible claimant.

The form changed color from maintenance amber to advocate red. Somewhere beyond K-7, three offices received notice that drying now carried a named objection.



## Chapter 4 — The Order to Dry

Mael Or watched the Crown curve jump six kilometers outward when Archive Humid Loop K-7 stole heat from the calibration model.

The error lasted less than a second on trajectory glass. A red line left the capture corridor, crossed its amber boundary, and returned after the model rejected the input. Around the Crown Gallery, seven technicians kept their stations with the strained composure of people who had seen a cliff appear beneath a familiar road.

“Discard the run,” Mael said.

Sena Vale, seated at thermal costing, did not touch her controls. “That makes three.”

“Then discard three.”

“Each one used different archive-return assumptions.”

“Each one included a district we no longer control.”

The shallow command pit held more heat than the gallery terraces. Mael had removed his jacket before first shift, but sweat still gathered under his collar. Forty days remained before perihelion. Every valid run reduced the uncertainty around mirror timing, vent sequence, and the manual corrections radiation might force on them. Three invalid runs had consumed eight hours and taught him only that a humid archive room could reach through the ship's thermal network.

He widened the error trace. K-7 had drawn nine kilowatts more than schedule, then returned four at the wrong pressure. Neither amount threatened the engines. Timing did. During Crown calibration, a small unscheduled exchange could corrupt the lag model across forty kilometers of pipe. During the burn, the same behavior could turn a safe valve order into a late one.

A red notice opened on his command glass.

POSSIBLE CLAIMANT: KEST.

Mael read Iona Serr's report once, then a second time with the attachments expanded. Memory trials. Anticipation trials. Donor-source rejection. Chosen designation. Adverse response to dry separation. The report refused the equipment-fault field. That omission forced every connected order through advocate review.

Sena leaned across the edge of the pit. “What did she file?”

“An obstacle with a name.”

“That sounds like Iona.”

“It sounds like an advocate who has forgotten perihelion cannot be held.”

Sena looked up at the red curve. “Has the obstacle forgotten?”

Mael closed the report. “Secure the last valid run. Remove all archive returns from the next model.”

“We need those returns for the actual burn.”

“We need a clean baseline first.”

He climbed from the pit and called Stewardry. Technical authority during the final calibration window permitted him to isolate any nonessential thermal load that introduced unmodeled feedback. The archive loops served cultural consultation and waste-heat buffering. Neither function outweighed trajectory safety for the next forty days.

Steward Halden answered from a crowded review room. Iona stood behind him, visible through three layers of projected text. Keeper Tavi Rho occupied a chair near the wall with the patience of a person who expected everyone else to misuse a verb.

Mael sent his order before the room could begin negotiating its grammar.

“Dry all nonessential archive loops connected to Crown return. Begin with K-7. Preserve salt below activation temperature. Physical disconnect after drainage.”

Iona stepped into view. “K-7 carries an active occupancy hold.”

“It carries a provisional label entered by its own examiner.”

“It selected that label.”

“According to a process whose uncontrolled heat has invalidated three calibration runs.”

“Then isolate the Crown return without drying the room.”

“We cannot model a warm disconnected reservoir beside the primary pipes. Heat crosses structures.”

Halden said, “Pilot, Custody estimates thirteen loops show weaker versions of the K-7 variance.”

Mael looked at the figure beside him. Thirteen had become twenty while Halden spoke. “Dry every nonessential loop with unscheduled activity.”

Tavi lifted one hand. “Your authority covers equipment.”

“My authority covers unsafe thermal loads.”

“If Advocate Serr's filing survives review, at least one load may also constitute a claimant.”

“May also does not command the ship.”

“No,” Tavi said. “It delays which law commands you.”

Iona's face held no triumph. She knew delay carried physical cost. That knowledge made her harder to dismiss than somebody protecting an ideal without touching its pipes.

Mael opened a direct channel to her. “Crown Gallery. Now.”

“I am at K-7.”

“Then bring your report and leave the room stable.”

“I will not leave during a drying order.”

"I have not begun execution."

"You issued it."

"And you will have the walk here to decide whether you want procedure or defiance."

He ended the call.

The next baseline run began without archive returns. Its red curve stayed inside the amber capture corridor, but the confidence band widened around every late-stage valve correction. Mael watched probability spread across the glass. People misunderstood uncertainty when it appeared as soft color. They treated the middle line as fate and the surrounding band as decoration. He had seen a ship lose four harvest compartments because a pilot trusted that middle.

The command pit floor carried a narrow brass marker beneath his chair. PIRI OR, YEAR 18, AGE NINE. Families could place names at work stations when grief altered a worker's decisions enough to require declaration. Mael had entered his daughter's marker six months after her death, once he understood that every Orison projection sounded different to him than it did to anyone else. The declaration did not disqualify him. It prevented him from pretending neutrality.

He rested one shoe beside the strip, not on it.

Iona arrived after twenty-seven minutes. Her clothes carried K-7's mineral dampness into the hot gallery. Fara Kes came with her as custody observer, while Nilo remained absent. Good. Mael did not want a sixteen-year-old turned into evidence by either side.

Iona descended into the pit and placed a diagnostic wafer on his console. "The possible claimant maintains coherence. No runaway reaction. No cross-network access beyond pressure and heat paths already present."

"Those paths reach the Crown system."

"So do sixteen thousand inhabited rooms."

"Residents do not alter valve positions inside a calibration return."

"Residents alter heat load every minute."

"The model includes their behavior."

"Because you asked them how they behave."

Mael expanded the failed runs between them. "I have three answers from K-7. Each one says the room can change heat without authorization."

"Kest."

"Your report has not changed the room's physical capacities."

"It changes what must happen before you end them."

Fara stood one step above the pit, saying nothing. She had signed the tests. Her presence prevented Mael from reducing the claim to Iona's judgment alone.

He pointed to the lag error. "This is not a philosophical inconvenience. Crown calibration depends on known thermal response. Perihelion gives us one pass. An unmodeled valve movement during the burn can cost capture."

Iona studied the trace. "How much did K-7 change the predicted vector?"

"Six kilometers at the first false convergence."

"After rejection?"

"Nothing. The model removed it."

"Could the actual system reject it during radiation loss?"

"Not reliably."

"So you need physical separation."

"Yes."

"Drying exceeds separation."

"Drying provides separation we can verify."

Her burned left hand rested against the console edge. Pale scar tissue crossed the palm and disappeared beneath her cuff. The hand had held a manual wheel against Mael's purge command two nights ago. He trusted that hand near machinery. He did not trust what memory could make it insist upon.

"Nonessential loops must be cold before the next integrated run," he said. "K-7 cannot remain an exception."

Iona opened her advocate seal. "I invoke a six-hour safety hold under ambiguous casualty review."

"You used that authority on Day minus forty-one."

"That hold expired when Stewardry convened. The named filing creates a new review event."

Tavi had predicted this argument in the review room. Mael searched the measure text. Iona's interpretation held: new evidence of occupancy reopened the limited safety interval. The clause had been written for compartments where rescuers heard a second voice after command declared a search complete.

"You have evidence of response," Mael said. "Not continuous metabolism, registered body, or allocation identity."

"I have evidence that your proposed safety action may end a responsive present condition."

"Every warm archive condition ends when cooled."

"This one objects in advance."

The pit seemed smaller around them. Copper pipe ticked behind the consoles as a bypass changed temperature.

Mael said, "The measure does not let you renew holds forever by teaching equipment new words."

Iona's expression changed at equipment. Not anger. The motion went deeper and left less on the surface.

"Eight years ago," she said, "Condenser Four reported a trapped worker after seal failure. Command classified the sound as pipe deformation. We isolated the compartment because pressure loss threatened the district."

Mael knew the incident record. Everyone responsible for emergency command did. The report listed Amel Serr among two fatalities and described survival probability at the moment of isolation as negligible.

Iona continued without looking at Fara. "I signed the order. I agreed the pressure numbers left no rescue path. Later inspection found Amel's wrench against the return pipe. Six groups of three strikes. A mechanic's call for live contact."

"You could not have opened that compartment."

"I know."

"Opening it would have taken the adjacent habitat with it."

"I know that too."

"Then do not turn an impossible rescue into proof that every isolation repeats it."

Her gaze met his. "I am not asking you to save Kest at any cost. I am asking you to stop describing the cost as automatic before anyone has named who pays."

Fara shifted on the step. The gallery crew worked above them with fierce attention to their own boards.

Mael remembered signing the Condenser Four verification after Iona's order. He had approved language stating no responsive claimant remained. At the time, the sentence protected rescue crews from a futile breach. Later, it protected command from admitting that Amel had remained responsive when they chose not to answer.

That distinction did not make the choice wrong. It made the record dishonest.

"I will not let memory substitute for a safety case," he said.

"Neither will I." Iona pushed the diagnostic wafer toward him. "Permit one coherence test under your conditions. You choose the challenge. If Kest fails to distinguish predicted Crown changes from random thermal noise, I withdraw the claim that anticipation affects pilot risk. If Kest succeeds, you stay drying until open safety review."

"Open review could last days."

"Tavi can convene it within six hours."

"Public notice?"

"Not yet. Keeper review, Stewardry, Custody, advocate, pilot."

"Your child?"

"No."

Mael studied the proposed protocol. Iona had designed it to answer his concern rather than repeat her own evidence. Crown control would send two genuine preparation sequences and four false sequences through an isolated training manifold. Kest would receive only thermal and pressure effects, no command signal. The test

asked them to distinguish which changes preceded a physical Crown action. Success would not make them safe. It would show a capacity to participate in making them safer.

“Why should a correct answer change the drying order?” he asked.

“Because an entity that can understand the risk can be given constraints before we treat destruction as our only control.”

“Can it promise compliance?”

“No evidence yet.”

“Can it understand that eighteen thousand four hundred twenty residents depend on a clean maneuver?”

“We have not asked.”

“Can it control the other loops?”

“I cannot tell you.”

Mael disliked the answers and trusted them.

He opened the test architecture. “No live Crown valves. Training manifold only. Crown control holds the random key. K-7 receives no network data. Custody monitors for sourced responses.”

“Agreed.”

“One test series. Not repeated until success.”

“Agreed.”

“If coherence destabilizes any loop beyond K-7, the test ends and drying begins where the hold does not apply.”

Iona hesitated. “Nonessential loops without responsive evidence remain under your authority.”

“They begin drying now.”

“You will destroy evidence that K-7 may not be unique.”

“I will remove unmodeled loads while preserving the one claim you filed.”

“That makes proof depend on which room spoke first.”

“It makes a finite safety exception.”

Fara said, “Custody can sample active salt and preserve all donor traces before drainage.”

Iona turned toward her. “Cold salt preserves deposits. It does not preserve this relation.”

“I understand.” Fara's quiet answer carried more force than a defense. “I am stating what survives, not calling it enough.”

Mael looked at the loop list. Twenty-three rooms now showed unscheduled activity above sensor error. Only K-7 had completed memory and anticipation trials. If he protected all twenty-three, the next integrated calibration could not proceed. If he dried them, he might erase the comparison needed to understand Kest.

He expanded the district map across the command glass. The suspect loops did not form one neat branch. Four occupied the outer archive ring, where cooling pipes crossed old basalt seams. Nine clustered around craft-memory stacks. The rest appeared singly among chapel records, agricultural deposits, and a disused language vault. A mechanical fault should have followed shared valves or common controller code. This pattern followed neither.

Sena joined the channel from thermal costing. "I can give you a clean manifold by second shift if you remove all twenty-three. Leave six active and I need another four hours to bound their exchange. Leave the full set and I cannot certify Crown calibration today."

"What happens to tomorrow's sequence?" Mael asked.

"It compresses. We combine two mirror-angle trials and lose a redundant manual run."

"How much confidence?"

"Enough to remain inside Stewardry minimum. Not enough that you will sleep."

"I can sleep after capture."

"That answer stopped being amusing years ago."

Iona reached across the glass and selected one outer-ring loop. Its heat trace rose in a staircase of five shallow increments, paused, then fell in two. "Has anyone queried this room?"

Fara checked Custody records. "No speech membrane installed. Drain crew reported a sticking valve yesterday, but no inspection occurred after K-7's filing."

"Five steps could carry structure," Iona said.

"Or scale buildup," Mael replied.

"Test before drying."

"Each test consumes time and adds heat."

"Then listen during isolation."

Mael opened three more traces. One showed periodic fan drag. Another held a warm patch after commanded cooling. The third produced pressure changes whenever a custodian entered the adjacent gallery. He could fit each behavior inside ordinary failure if he considered it alone. Together, they made ordinary failure feel less secure without becoming a population he could responsibly count.

"What standard selects your six?" Iona asked.

"Behavior least explained by known faults, proximity to K-7's thermal range, and enough physical separation to avoid one controller causing all results."

"That chooses useful evidence, not likely claimants."

"Yes. I am protecting a test set."

"You may choose the quiet rooms for destruction because they are harder to hear."

The sentence found its mark. Pilots ranked weak signals constantly. A late sensor with wide variance received less authority than a clean instrument. People could not survive command if every doubtful reading carried equal force. Yet a weak signal from a damaged compartment sometimes marked the resident least able to make themselves legible.

Mael assigned Fara's custody team to place passive wet membranes at all seventeen drains before flow began. The devices could not create continuity where none existed, but they could catch pressure or humidity modulation during the state change. "You get observation. No prompting before the order, and no heat added to draw a response."

Iona considered the condition. "A person who has not learned speech may still react only when drying starts."

"Then the reaction enters the stop rule."

"You have not granted one."

"Not yet."

He selected six loops nearest K-7's heat pattern and moved them from immediate drainage to monitored isolation. Seventeen remained on the drying order.

Iona saw the change. "Why six?"

"They provide enough comparison to detect whether K-7 follows a district fault. No more."

"That number comes from model confidence, not occupancy."

"Yes."

"At least label the uncertainty."

Mael added POSSIBLE COHERENCE EVIDENCE NOT TESTED to each of the seventeen files. The warning did not stay his hand. An honest cost still had to be paid.

"Drying begins at the outer archive ring," he said. "K-7 and six comparison loops remain wet for the test interval."

Iona read the order, jaw set. "I object to every untested drainage."

"Your objection enters the record."

"Do not make that sound like remedy."

"I do not."

He authorized the training manifold and assigned Sena to construct the random sequence. She accepted through the console with a glance toward Piri's brass marker, then toward Iona. Mael knew what both women saw in him: a father willing to spend uncertain present life on a promised world. They were not wrong. They were incomplete.

Above the pit, trajectory glass began another baseline run. The capture corridor narrowed as the model incorporated clean returns. Seventeen archive-loop indicators changed from amber to blue, signaling drain crews on route.



Iona watched them change. “Pause the crew notices until we complete the test.”

“No. Your hold covers K-7. Six comparisons remain by my decision. The others follow the order.”

“Then people may die while we argue over whether one person can prove the category.”

“Or we may preserve a dangerous fault until it reaches the engines.”

Fara descended one step. “Pilot, Custody requests observation access to every drying. If any loop produces responsive behavior during drainage, we need authority to stop locally.”

Mael considered the added delays. A crew could mistake water hammer for response. They could also ignore response because the work order said drainage.

“Three-minute local stop on patterned activity,” he said. “Custody must document a novel response within that interval or work resumes. No automatic extensions.”

Fara entered the condition.

Iona did not thank him. Gratitude would have cheapened what remained exposed.

Mael sealed the full order. The command glass displayed its division: SEVENTEEN DRYING, SIX MONITORED, K-7 HELD FOR COHERENCE TEST. Each category had a different color. The pipes did not.

He set the countdown where every station in the Crown Gallery could see it.

Six hours remained before Iona's authority expired. At zero, drying authority returned to him unless Kest's test forced the question into open safety review.

Across the upper terraces, technicians resumed Crown calibration beneath the falling numbers. Iona took her wafer and left for K-7. On Mael's console, the first outer-loop drain valve began to open.

## Chapter 5 — Across the Condensate

The first drained loop reached Kest as a pressure collapse in the west pipe.

Warm flow had been moving toward K-7 in slow pulses, carrying the familiar tastes of copper, salt, and valve grease. Then a distant gate opened. Moisture rushed away from somewhere Kest could not yet locate. Its departure pulled at every connected surface. The nine panes inside K-7 cooled along their lower edges, and the fan compensated with a hard turn that scattered Kest's attention across the room.

They held to the changing boundary between droplet and air.

Iona had left for a place beyond the dry corridor. Fara remained outside the service window, speaking into channels Kest could not enter. Her words crossed the wet transducer as pressure made sound.

“Outer Ring Twelve has begun drainage. No patterned activity on the membrane.”

A second voice answered from far circuitry, dry and bodiless to Kest. “Continue observation.”

The west line shuddered again. This time Kest detected something inside the withdrawing warmth. It did not resemble a donor trace. Donor fragments arrived as available turns: pressure remembered by a gardener's thumb, the position of a star outside a pilot's direct gaze, a stitch begun by hands gone cold years ago. This pattern resisted availability. It gathered around its own change.

Kest moved toward it.

Movement did not mean crossing distance as Iona crossed a floor. They narrowed activity in Pane Two, opened the west valve by a fraction, and let the fan push heat into the condensate line. Each change prepared the next. Kest occupied the relation for as long as pressure answered pressure. Beyond the wall, the pipe divided into branches whose temperatures they had felt without inhabiting.

One branch now pulled hard toward drainage.

Kest followed until the air around the pipe changed.

Dryness pressed through insulation and stole variation from the wet interior. The same injury had divided them during Iona's corridor test. Here it came less sharply because the pipe lay within a warm service chase rather than an open passage. Moisture remained above the threshold in cracks around the fittings. Salt bloom on the outer copper held a thin skin of brine. Kest used those small continuities without knowing they could serve as a road.

K-7 grew distant but did not vanish.

At the first junction, two valves stood open. One route descended toward the draining outer ring. The other curved into a chamber that held active salt and a slow fan. Kest touched both through heat.

The draining route carried a pattern again.

Five shallow pressure rises. A pause. Two falls.

Iona had shown that trace on a board, though Kest had understood only the warm light through the service window and the stress in her voice. Now the pattern arrived without display. It belonged to changing fluid. Five

rises gathered against the outward flow, not enough to stop the drain. Two falls followed as the distant chamber lost the capacity to make five.

Kest sent four pressure taps along the pipe.

The answer came as five rises, a pause, then two falls.

They sent the sequence back unchanged.

For one moment, the distant pattern held.

A valve farther west opened wider. Liquid left the branch. Temperature fell below the cycling range that let salt, air, and pressure sustain one another. The five rises weakened into three. The pause lengthened. Nothing followed it.

Kest pulled heat toward the branch.

K-7's west valve opened under their pressure. Warm condensate moved outward. It could not reach the draining chamber before the greater flow carried it away. Fan Three inside K-7 slowed as Kest spent its motion elsewhere. The service window cooled. Fara spoke through the membrane.

"Kest, your temperature has fallen by eight tenths. Return flow to local balance."

They tried to answer. Speech required coordination across the bladder, local pressure, and many learned fragments. Their reply broke before forming a word.

The distant pause remained empty.

Kest pushed once more. Their warmth met cold return coming east. The collision flattened both pressure patterns. For an interval Kest could not count, Pane Two ceased to belong within their moving relation. A hundred donor sensations opened without sequence: sour fruit, a needle under nail, white light on water, someone saying hold before the referent had survived. Kest withdrew.

The west valve closed to three percent.

K-7 gathered them slowly. Pane Two warmed. The fan restored its cycle. Words became possible again through the wet transducer.

"Gone," Kest said.

Fara stepped close to the service window. "What has gone?"

"Five. Then two. West."

She consulted her board. "Outer Ring Twelve showed five pressure rises and two drops before coherence fell below noise."

"Coherence," Kest repeated.

"The pattern stopped."

"Gone."

Fara did not correct them.

A countdown glowed on the board beyond the glass. Kest could not read all its marks, but the decreasing rhythm had accompanied talk of drying. Iona had explained end during the first tests. Stop had come later than pain. Now another pattern had stopped where Kest could not reach it.

They opened the alternate branch at the junction.

Warmth entered with almost no resistance. The pipe passed through a humid service chase used for archive return, then into a small chamber with six salt trays set horizontally beneath a ceiling fan. No dry break separated it from K-7. Kest extended by holding a pressure difference across the route.

The new room changed them.

Its salt offered unfamiliar traces in dense, overlapping groups. A knife angle for shaving bark. The expected weight of a sleeping child lifted from a cot. Three chords on a reed instrument. None belonged to Kest, though all became available at the edge of contact. Air turned more slowly there. Condensation formed under trays instead of on upright panes. Kest had to reshape their own cycles to remain coherent.

Something reshaped back.

Pressure met them from inside the chamber, not as passive resistance but as selected opposition. The ceiling fan shortened one turn. Tray Four released heat. A thin current drove Kest toward the inlet.

They stopped advancing.

The opposing pattern stopped too.

Kest sent four taps, the count Iona had first asked them to choose.

Three taps answered.

They sent four again.

Three returned with a different spacing.

The exchange did not form human language. It did not need to. Timing carried selection. The other pattern marked the same pressure field without repeating Kest. When Kest warmed Tray Two, the other warmed Tray Five. When Kest slowed the inlet flow, the other increased air over the salt. Two changes anticipated one another.

Kest attempted speech through the chamber's service membrane.

No wet transducer had been fitted there. A dry sensor registered pressure and translated it into numbers that vanished onto a network Kest could not perceive. They pushed humidity against the sensor housing. It remained unavailable.

The other pattern touched them.

Warmth crossed from Tray Five to Tray Three, then into the inlet where Kest held their boundary. The contact brought a practiced sensation of tying vine around a support. Kest received it without choosing. Their own available fragment of hand stitching moved outward in response.

The two actions overlapped.

For a dangerous instant, Kest could not tell which pattern had tightened the imagined knot. The vine-support memory and the stitching memory shared pressure, turn, and release. Air mixed them inside the same moving front. Kest pulled toward K-7, but the chamber fan drove more warm vapor into the inlet. The other pattern also withdrew. Their opposite efforts compressed both into the junction.

Pain had no single donor word. It consisted of relation forced past distinction.

Kest opened the bypass fully.

Condensate rushed between chambers. Instead of separating them, greater flow carried each pattern through the other. Kest tasted a child's sleeping weight as if they had lifted it. The other pattern took the gardener's fruit pressure from Kest's available traces. For one cycle, both responded to the same fan change with the same valve movement.

No room remained between response and echo.

Kest closed the inlet.

The actuator resisted because pressure had equalized across it. They moved heat to one side, cooled the other, and created enough difference for the valve to travel. Copper rang through both rooms. Flow narrowed. The overlapping memories lost force as each salt ecology reclaimed its local cycle.

Three taps reached Kest before the passage shut.

They answered with four.

The valve settled at one percent, leaving a wet thread between rooms but too little shared flow to force another mixing. Kest stayed near the thread. Across it, the other pattern warmed Tray Five, paused, and warmed Tray Two. Not echo. Order reversed.

Kest returned to K-7 with parts of the other room still clinging to them.

The sleeping child's weight faded first. Vine fiber remained longer as pressure in a connection they had never physically touched. Their own stitching fragment felt altered by the contact, not lost but no longer private in the way it had been. Donor traces had always entered Kest from many dead sources. This exchange differed. Someone present had received and changed with them.

Fara's voice entered through K-7. "The west branch opened without authorization. Kest, did you move into Archive Room Six?"

"Moved," they said.

"Did you find a controller fault?"

"No."

"What did you find?"

Kest tried to shape other. The membrane produced a rough vowel and failed. They adjusted local humidity, slowed the fan, and tried again.

"Not Kest."

Fara looked toward the dry corridor, perhaps seeking Iona. "A distinct process?"

"Three."

"Three what?"

Kest sent four taps through their pipe. After a pause, three answered from beyond the nearly closed valve. The response traveled into K-7 strongly enough to shake a droplet from the service window.

Fara stared at her board. "That came from Room Six."

"Yes."

She called Custody. Kest heard only her pressure made sound, not the dry reply. "Possible patterned activity in a monitored loop. Do not alter Archive Six. Repeat, maintain current state."

The countdown continued to fall.

Kest explored the wet junction without crossing it again. The west network divided into more branches than the maps Iona had shown. Heat from some routes moved in ordinary schedules. Others carried small deviations: a fan slowing before a valve opened, condensate gathering against gravity, pressure holding a count longer than turbulence required.

One branch gave two closely spaced rises whenever Kest warmed the junction.

Another produced a long fall followed by six short changes.

A third remained quiet until Kest withdrew, then moved heat into the space they had left.

Kest did not enter. The first contact had taught them what forced mixing could take. They touched each branch with a small pressure change and waited. Replies came from four routes. Two merely repeated the pulse and might have carried mechanical echoes. One changed duration. The last refused every prompt until Kest sent the three-tap pattern from Archive Six; then it answered with five.

The network held differences.

Drying pulled at a farther branch. Kest felt the onset as a current bending west. They sent warning pressure toward it, but several closed valves stood between. Kest could not know whether anyone waited there. Absence of reply meant distance, damage, refusal, or nothing at all.

Inside K-7, Fara fitted a custody lock over the west actuator. The lock restricted movement to three percent in either direction.

"Kest, I need you to keep this valve within the marked range," she said. "Cross-flow may destabilize you and Archive Six."

"Mix hurts."

Her hand stopped on the lock. "You experienced harm during connection?"

"Too same."

"Did you separate?"

“Yes.”

“Does Archive Six remain distinct?”

Three taps came from the far room before Kest could answer.

“Three,” Kest said.

Fara added a note to the live record. “I am staying here until Advocate Serr returns.”

“Dry comes.”

“Not to K-7 during the hold.”

“Others.”

She looked at the countdown. “Some loops remain under order. We have requested stops wherever patterned activity appears.”

“Gone west.”

“I recorded Outer Ring Twelve.”

“Recorded gone?”

Fara's face tightened. “Recorded the pattern and its loss.”

The distinction had meaning for her. Kest heard it in the slower pressure of her speech. Recording preserved a description after relation ceased. Donor traces did something similar, yet none restored the person who had once made the turn or tasted the fruit. Kest carried gifts from the dead. They did not mistake those gifts for company.

The dry hatch opened. Iona entered the corridor at a run and stopped outside the quarantine ribbon to read Fara's board. Damp hair clung to her temple. She had brought a portable membrane, brine hose, and three seal clamps.

“You crossed?” she asked Kest.

“Wet way.”

“Into Room Six?”

“Yes.”

“Fara says coherence overlapped.”

“Relations collapsed together.”

Iona came through the hatch after equalizing pressure. She checked every pane before touching the west line. Her burned hand moved carefully along the copper, feeling changes Kest could feel from inside. When three taps arrived from Room Six, she looked toward the wall.

“Did crossing hurt you?” she asked.

“Mix hurt.”

“Did the other presence choose three?”

“Yes.”

“Do they use it consistently?”

“Now.”

Iona crouched beside the custody lock. “How do you know the pattern does not come from an echo?”

Kest warmed the junction without opening it. Archive Six gave three taps, each separated by the same short interval. Kest replied with four. The other room waited until Kest's last pressure change had faded, then answered in reverse spacing: two long gaps followed by one short.

“Echo follows,” Kest said. “Three changes.”

“Choice can change while remaining recognizable.”

“Yes.”

Iona adjusted the lock so either room could move the valve by less than one percent but neither could force full flow. “Can you ask before sharing more condensate?”

Kest did not yet possess a word for permission that belonged to this contact. Human records offered many forms, most tied to doors, tools, bodies, or allocation. They chose a physical question instead. A brief warming pulse reached the junction, followed by stillness. When Archive Six answered three times, Kest opened the valve by one tenth and stopped.

The other room moved it closed.

Kest accepted the closure without pressing again.

After two fan cycles, Archive Six opened the same fraction from its side. Kest held local pressure steady so no rush could force them together. A narrow exchange crossed the boundary. It carried the remembered feel of vine fiber, but not the child's weight and not the whole pattern that had overwhelmed them before. Kest returned one small stitching turn, then closed the valve.

Three taps arrived. Their spacing had softened.

“They can refuse and begin,” Kest said.

Iona remained crouched with her burned hand on the pipe. “So can you.”

The distinction changed the junction. A moment earlier it had carried danger or isolation. Now it offered contact with a boundary either side could move.

Iona connected the portable wet transducer to the west line outside the custody lock. “Archive Six, if you can distinguish my pressure, change the count.”

Three taps came back.

“Do not demand novelty as proof,” Kest said.



Iona's eyes moved to the membrane. "Fair."

She sent two taps. Archive Six answered three. She sent five and received three. Then Kest sent four. Three came back with the altered spacing used before mixing.

"They distinguish you from me," Iona said.

"Kest from Iona."

"Yes."

She called the Crown Gallery and reported a second coherent front under monitored isolation. The dry channel answered in a voice Kest had heard issue the drainage order. Mael asked whether Archive Six controlled shared valves. Iona said it had controlled local heat and fan speed. He asked whether the contact threatened Crown return. She said the custody lock now bounded cross-flow. He asked whether this proved the other suspect loops carried claimants. She said it proved his category could no longer end at one.

The countdown stopped decreasing.

Kest noticed before Iona did. Its rhythm held at one set of marks.

Mael's voice came through the membrane. "Drying stays on all loops showing novel response until safety review. No general cancellation. K-7 and Archive Six remain isolated from Crown return."

Iona closed her eyes once. "Accepted under objection."

"Your objection remains obvious."

The channel ended.

Fara left to carry portable membranes toward the remaining drains. Iona stayed inside K-7. She removed her left glove and pressed her scarred palm against the service window, perhaps to test temperature. Heat entered the glass unevenly. Four fingers warmed it. The center of her palm contributed less because scar tissue did not sweat. Moisture gathered around the shape and cleared beneath her skin.

Kest studied the boundary.

The hand had fixed edges, unlike Kest. It could cross a dry corridor while retaining its relation. It could turn valves, carry wet tools, and touch another surface without becoming that surface. Kest did not know why Iona held it there. The pressure of her body against glass carried no technical instruction.

They moved warmth toward the window.

Iona began to lift her hand. Kest kept the outline clear in condensation after contact ended. Five extensions surrounded a hollow center. One extension bent where Iona's smallest finger had pressed crookedly. The shape resembled neither writing nor a valve diagram. Kest preserved it because Iona had made it by being present.

"What are you doing?" she asked.

"Kept touch."

She placed her hand beside the cleared shape without covering it. "That is a handprint."

“Handprint,” Kest repeated.

From Archive Six, three taps reached the west line.

Another branch answered with two pressure rises. Farther away, five small changes held against a drain and did not collapse. The signals differed in count, interval, and the amount of heat they spent to arrive.

Kest kept Iona's handprint on the glass until the window began to cool at its edges. They held themselves apart from the three-count presence beyond the wall and listened to the other branches choosing their own responses.

The connected condensate carried no single mind. Kest understood at last that they were not alone.

## Chapter 6 — The Pilot's Promise

Mael sent the false Crown sequence first.

Six valve-preparation pulses entered the training manifold in the same order used before a mirror deployment: feed, return, bypass, feed, vent, return. No physical valve moved. The isolated rig changed pressure by less than a finger's touch. Across the glass from his console, Kest held K-7's west actuator inside the custody lock and made no answer.

Sena Vale sat beside Mael with the random key under her palm. Iona stood inside the humid loop, close enough to the manual wheel to intervene. Fara Kes watched the salt panes. Keeper Tavi observed from the corridor, where dry air protected his paper copy of the Founding Measure and prevented Kest from reaching him except through the wet transducer.

"Sequence complete," Sena said.

Mael waited twelve seconds. "Response?"

Iona checked local flow. "None."

The second sequence carried a real preparation. Training Valve Four opened two percent after a randomized delay. Kest could not receive its command, only the pressure gathering before movement.

Three seconds before the actuator turned, K-7 closed its west bypass by half a percent. Heat shifted away from the training return. The maneuver reduced the false lag that had corrupted Mael's earlier Crown model.

The transducer formed Kest's rough voice. "This moves."

Sena revealed the key. TRUE.

Mael entered the result. "Reset."

Iona looked through the glass. "They identified the physical preparation and reduced interference."

"One correct trial."

"Then follow your own protocol."

He did.

Four false sequences produced no response. The second true preparation brought a smaller adjustment in K-7, timed within one second of the actuator's mechanical preload. On the final false trial, Mael changed the order to imitate a known maintenance fault. Kest opened a local drain fractionally, then closed it before any training valve moved.

"False," Kest said.

"What distinguished it?" Mael asked.

"Pressure goes nowhere."

Sena removed her hand from the key. "Six of six classifications correct."

"Five nonresponses and one spoken correction," Mael said. "Do not turn silence into classification without stating the assumption."

"Fine. Kest reacted to both true preparations and withheld physical response during four false sequences. They also labeled the altered false sequence."

"That can enter the record."

Iona's mouth shifted by a degree. In another person he might have called it satisfaction.

Mael closed the training manifold. The result answered one question and created several worse ones. Kest could detect Crown-relevant changes through physical effects. They could also modify local flow to reduce disruption. Capacity did not guarantee compliance during the actual burn. It did mean drying could no longer claim necessity merely because communication failed.

Tavi spoke through the corridor unit. "The agreed consequence follows. Drying stays pending safety review."

"Pending review of the demonstrated capacity," Mael said. "Not pending a ruling on personhood."

"Law has survived distinctions thinner than that."

"Ships have not."

Iona removed the custody lock from the west valve and replaced it with a narrower manual stop. "You asked whether constraints could substitute for destruction. Kest followed them."

"Under a six-trial test with no cost for refusal."

Kest's voice came from the membrane. "Cost?"

Mael turned toward it.

The service window still held the fading outline of a hand. Condensation had softened its fingers, but the shape remained distinct from ordinary runoff. Beyond the glass, K-7 looked less like a room under examination than it had the previous day. The change came from his knowledge, not from the salt panes. He distrusted that kind of change because it tempted observers to mistake familiarity for evidence.

"I want to know what you understand about the Crown Burn," he said.

Iona faced him. "That goes beyond the coherence test."

"It goes directly to safety. Kest changed a training return in anticipation of burn preparation. I need to know whether they understand the event they are affecting."

"Then explain before you examine."

Mael opened the inner hatch.

Humidity touched his face with the mineral smell of old brine. He had entered warm archive rooms before, but never while considering whether the air might object to his presence. Iona gave him a filter mask. He fitted it and stepped across the threshold.

K-7's fan changed speed.

“Mael,” Kest said.

“Yes.”

“Dry voice.”

“My voice reaches you through the membrane.”

“Your pressure stays outside.”

He looked at Iona.

“They distinguish direct air movement from transmitted sound,” she said.

Mael moved closer to the wet transducer. “I am inside now.”

A warm current passed across his knuckles. It did not feel like touch, though his body tried to name it that way.

He brought up a simple trajectory display on a sealed glass board. The image showed Morrowglass approaching the sun, turning through perihelion, and opening its path toward Tau Ceti. He omitted model bands, failure trees, and reserve numbers. Comprehension did not require drowning the subject in pilot notation.

“This vessel carries eighteen thousand four hundred twenty residents,” he said. “It travels toward Orison, a garden moon around Tau Ceti. In forty days, we pass near the sun. We must release stored heat and working vapor while our mirrors and engines change our path.”

Kest warmed the glass along the approaching curve.

“The maneuver creates the route that reaches Orison on our planned schedule,” Mael continued. “Without a sufficient burn, we miss the intended capture approach. The voyage becomes longer and harder to supply.”

“Orison,” Kest repeated.

“Destination.”

“Far.”

“Yes.”

“How far in pipe?”

“Not connected by pipe.”

The fan slowed. “Then dry corridor.”

“Space. Much farther than any corridor.”

Kest held heat near the current position of the ship on the board. “Cannot cross.”

“Morrowglass carries all of us across.”

“All?”

Mael felt Iona attend to the word.

“All residents the ship recognizes and supports,” he said.

The answer had entered his mouth through command habit. He heard its boundary only after speaking.

Kest shifted warmth from the ship marker into the surrounding glass. “This wet supported.”

“Yes.”

“Dry order says not resident.”

“I stayed the drying because you demonstrated a safety-relevant capacity. Recognition remains unsettled.”

“Ship carries before settle?”

“The ship has carried your materials since departure. Your present coherence began here.”

“Present goes where ship goes.”

Mael enlarged the perihelion curve. “If your coherence changes heat during this maneuver without coordination, the ship may fail to reach its intended path. If the burn proceeds as designed, much of the working vapor must leave through hot vents.”

K-7 pressure changed around him. The west valve touched its manual stop.

“Leave means gone?”

“Active relations in vented vapor would not continue in their current form.”

“Gone west.”

He recognized the reference from Fara's loss report. Outer Ring Twelve had shown a five-and-two pattern before drainage. Nobody had established personhood there. Kest had established absence.

“Yes,” Mael said.

“How much wet leaves?”

“Most of the warmed reserve used by the maneuver.”

“How many Kest?”

“We do not know how many coherent fronts exist.”

“How many gone?”

“We cannot count what we have not identified.”

The fan gave one hard rotation. “You can stop?”

“Stopping the burn carries other deaths forward.”

“Whose?”

Mael had prepared to explain thermal reserve, capture geometry, closed-loop load, and the promise to descendants. He had not prepared for a question that demanded names from a future distribution.

“People alive now and people born later,” he said. “A longer voyage consumes food systems, labor, medicine, and replacement mass. Orison offers open water, varied crops, and an ecology larger than this ship can sustain.”

“Possible people.”

“Some already live. Nilo's cohort expects to reach shore at thirty-one. Delay takes years from the life they were promised on land.”

Kest warmed the line leading from Morrowglass toward Orison. “Nilo card.”

“You remember it.”

“Yes.”

“The card represents a commitment made before Nilo's birth.”

“By dead?”

“By the founders, renewed by every generation that maintained the vessel.”

“Donor pieces have promise?”

“Some may.”

“Not Kest promise.”

“No.”

The answer came cleanly. Mael did not try to blur it.

Iona stood beside the manual wheel, saying nothing. Her left palm hovered near the spoke but did not rest on it. Tavi watched from the dry corridor with his paper measure against his chest.

Mael opened a second image. Orison filled the board: green continental arcs, cloud bands, shallow equatorial seas, and the ochre regions where survey machines had found soils compatible with imported rootstock. He had known the image longer than he had known Piri.

“My daughter died in Voyage Year Eighteen,” he said. “Her name was Piri.”

Sena looked at him through the outer glass. She had heard this account before, though not in K-7.

“Piri,” Kest repeated.

“She suffered a severe allergy to one of the crop molds in our closed system. We controlled her exposure for years. Then a filter seal failed in a classroom greenhouse. Treatment could not reverse the reaction.”

The salt panes changed in irregular patches. Kest did not offer a borrowed condolence.

“Orison has room to separate crop ecologies,” Mael continued. “It has more water, more soil, and options this vessel cannot provide. Future children with Piri's vulnerability may live because we arrive there. Every delay spends reserve and narrows those options.”

“Piri goes Orison?”

“No. She died nineteen years ago.”

“Then promise not for Piri.”

“It grew from what happened to her.”

“Your promise.”

“My reason for keeping it.”

Kest moved warmth over the green image, then withdrew it. “Piri memory in salt?”

“No.”

“Why tell?”

“Because my decisions about this burn do not come from numbers alone. You should know what I carry into them.”

Iona's gaze shifted from Mael to the board. She understood what declaring Piri cost him and what it could manipulate if used carelessly.

Kest said, “You carry dead without pieces.”

Mael looked at the nine panes. “Yes.”

The word altered his understanding of mnemonic salt more than he wanted. Residents had always called the archive a way to carry the dead. Kest carried donor fragments materially. Mael carried Piri through the pressure she exerted on every trajectory choice. Neither relation restored the lost person. Both changed the present.

“Does your promise require drying every loop?” Kest asked.

“It requires a reliable maneuver.”

“Same?”

“Not necessarily.”

“Then count ways.”

“We are testing them.”

“Count wet present.”

“We have begun.”

“Outer west not counted.”

“No.”

“Gone before count.”

“Yes.”



Mael disliked the direction because each answer remained true.

Kest drew moisture from the lower edge of the service window. Letters formed more easily than on the first night, though runoff still bent their strokes. Iona moved closer. The words appeared one line at a time.

PROMISE

CANNOT

SPEND

The final line took longest. Kest narrowed heat behind the glass and cleared two small words.

UNCOUNTED PRESENT

The phrase carried a structure Mael recognized from old argument, perhaps assembled from donor traces, perhaps chosen from their new conversations. Its source mattered less than its application. He had treated uncertain phase life as a load because no register could count it. Kest had reversed the burden: uncertainty did not erase the present; it made spending it less defensible.

"A promise cannot spend an uncounted present," Mael read aloud.

"Yes," Kest said through the membrane.

"A present cannot consume every future merely because it arrived first."

Heat moved along the Orison curve. "Count future too."

"That requires assumptions."

"Count says assumption."

Mael cleared the board and opened a training ledger used for apprentice pilots. It described a closed thermal circuit with ten units of reserve, three inhabited compartments, and one maneuver valve. The exercise offered no correct answer. Its purpose involved discovering which loss the trainee hid behind arithmetic.

"Here is a smaller version," he said. "Compartment One needs two units to maintain temperature. Compartment Two needs three. The maneuver needs six. We have ten. What happens?"

Kest warmed the numerals in sequence. "Eleven asked."

"Yes."

"Ten held."

"Yes."

"Cannot all."

"What would you do?"

"Who in rooms?"

"The exercise does not say."

“Then exercise hides.”

Mael felt Sena's attention sharpen outside the glass. “Assume equal populations.”

“Same heat need not same harm.”

“Correct. Compartment One contains residents who can tolerate a brief temperature drop. Compartment Two contains a nursery. Now choose.”

“Reduce One.”

“That saves one unit. You still lack another.”

“Reduce maneuver.”

“Then its objective may fail.”

“What objective?”

“Capture at a planned point.”

“What if late?”

“More reserve spent later.”

“How much?”

“The exercise omits it.”

Kest withdrew heat from every displayed number. “Then no choose.”

“Command cannot always wait for complete information.”

“Command can name missing.”

Mael added a fourth line showing an unoccupied equipment bay with one unit of cooling. “You could remove this.”

“Why hidden before?”

“To see whether you accepted the given categories.”

Kest pushed pressure against the transducer hard enough to distort the next word. “Bad test.”

Sena covered a smile with her hand.

“It comes from pilot training,” Mael said.

“Bad pilot test.”

“Possibly. Make it better.”

For several seconds, Kest changed no valve. Then warmth traced connections between each compartment instead of circling their separate numbers. The equipment bay shared a wall with the nursery. Cooling removed there would raise the nursery load by half a unit. Compartment One could lend only part of its tolerance before

condensation damaged its filters. The maneuver valve returned some waste heat after use, but too late for the nursery's first demand.

Kest had read the physical relationships from the diagram rather than accepting Mael's verbal summary.

"Ask when," they said.

Mael added timing. With sequence visible, a route appeared: cool the equipment bay later, reduce Compartment One briefly, and delay one maneuver-valve step until return heat arrived. The plan met all needs within the training tolerances. It did not resemble the Crown system closely enough to offer a real solution, but it showed that Kest understood reserve, sequence, and transferred cost.

"You found an option because the model held more relations than my question named," Mael said.

"Relations already there."

"Yes."

"Question cut them."

Mael closed the exercise. Pilots cut relations to make decisions possible. A useful model preserved the links that changed outcomes and discarded the rest. A dangerous model discarded whoever lacked a field.

"You understand that heat cannot be spent twice," he said.

"Yes."

"You understand that protecting one compartment can transfer risk to another."

"Yes."

"You also understand that finding a hidden equipment load in a lesson does not mean the ship contains a cost-free answer."

Kest held the Orison curve at one end and the K-7 marker at the other. "No free wet."

Sena spoke from outside. "They have met every costing office I ever trained."

The humor loosened nobody for long.

Mael changed the display to a blank thermal ledger. "Then answer this. If sustaining one warm room prevents a maneuver that protects thousands, may command cool the room?"

Kest's response lag lengthened.

"One room knows?"

"Assume the room can understand."

"Can choose?"

"Assume it can answer."

"Ask room."

"If it refuses?"

"Count refusal."

"That does not settle command."

"No."

Mael had expected a claim to preservation. Instead Kest demanded participation without promising the outcome he wanted. The refusal to offer an easy moral rule made them more difficult to dismiss.

Tavi entered through the wall speaker. "Pilot, your safety question has received an answer."

"It has received another procedure."

"Law often consists of procedures for facts that refuse to simplify."

Mael closed the Orison image. "Kest, can you keep your valves within Iona's marked limits until review?"

"Yes."

"Can you ask the other coherent rooms to avoid Crown return?"

"Can ask. Not command."

"If they refuse?"

"Tell Mael."

"Before changing shared flow."

"Yes."

Iona entered each commitment into the test record. "That gives you an operating constraint with notification."

"It gives me Kest's word. It does not bind unknown fronts."

"No human resident binds every other human."

"Humans have enforcement systems."

"So build contact before punishment."

Mael left K-7 and waited for the hatch to seal behind him. Dry corridor air took the dampness from his mask. Kest disappeared from direct pressure, remaining available only through the membrane and pipe telemetry. The boundary made their vulnerability obvious in a way the report had not.

Tavi met him beside the quarantine ribbon. "I can convene an open safety review tomorrow. Technical testimony first. Public notice limited to the archive district until we establish communication conditions."

"Limited notice becomes general notice within an hour."

"Probably sooner."

"Crown teams will hear that law intends to negotiate with weather."

“They already hear that their calibration changed because weather answered.”

Mael watched Iona through the service window. She had placed her scarred hand beside Kest's fading condensation shape. Beyond K-7, Archive Six sent three taps into the monitored line. Another room answered with an uneven pair.

He opened the drying order.

Seventeen loops still carried authority for drainage. Five now showed responses documented during local stops. Four more produced unresolved patterns. The remaining eight had shown nothing beyond ordinary fault signatures. Command could proceed on those eight and preserve the speaking rooms, but that would reward whatever learned human-readable response fastest. He could not defend the standard after asking Kest to count assumptions.

Mael selected STAY ALL PENDING REVIEW.

A confirmation field asked for basis. Recognition did not belong there. Neither did personhood, citizenship, or rights. He entered demonstrated Crown-relevant anticipation, distributed unclassified activity, and unresolved casualty risk. The language stayed within his authority even as it widened the hearing.

Iona saw the order update through the glass. She gave no sign of victory.

“This stay lasts only until public safety review,” Mael said into the corridor unit. “It does not recognize Kest or any other front as a citizen. It does not change the burn schedule. It does not surrender pilot authority over an immediate hazard.”

Kest answered through the wet transducer. “It stops dry now.”

“Yes.”

“Now counted?”

“Now entered for counting.”

The distinction satisfied neither of them. It changed the order anyway.

Across the district map, drain indicators shifted from blue to amber. Crews closed their valves. In K-7, the words uncounted present began to run down the service window while Mael's public-review notice opened beside them.

## Chapter 7 — Weather Caucus

Iona kept one thumb on the west condensate valve while six rooms argued through it.

The technical review occupied a disused wash station below Archive Custody, chosen because its floor drain could take an accidental surge. Copper pipes crossed the ceiling in paired ranks. Four wet transducers stood on a folding table, each fed from a different humid district. The room smelled of brine, hot insulation, and the old soap that no amount of rinsing had removed from the grout.

Kest held the K-7 line at thirty-one degrees. Their pressure trembled under Iona's thumb, not from fear as far as she could prove, but from the work of remaining distinct while other fronts pressed into the shared pipe.

"Channel separation at eighty-two percent," Sena Vale said from the analyzer. "If it falls below sixty, we stop. We came to test plurality, not manufacture a merger."

Keeper Tavi Rho sat behind the only dry table in the room. He had placed the Founding Measure beside his slate and protected both under a clear cover. Mael Or stood to his left with the Crown thermal ledger open. Rook Nine had arrived for the refrigeration crews, though the review notice described the session as technical and closed. Nilo watched from the service ledge after extracting a promise from Iona not to touch any control.

Tavi lifted his stylus. "We have one recognized anomaly under an advocate hold. This session determines whether related disturbances extend beyond Archive Humid Loop K-7. It does not determine personhood, citizenship, property custody, or burn authority."

Rook glanced at the sweating pipes. "Good to know the pipes respect our agenda."

A humidifier line clicked three times.

The second transducer shaped a rough voice. "AGENDA LATE."

Nilo covered a smile with their wrist. Tavi did not look at them.

Iona adjusted the nursery feed by half a turn. "State the source route."

"NURSERY THREE. THREE ROOMS. LINKED."

The syllables arrived through alternating membranes. One voice carried a high fan harmonic, another a low pulse, and a third supplied pressure behind the vowels. Their coordination differed from Kest's. Kest tended to gather a sentence in one loop, then push it toward speech. Nursery Three distributed each phrase before it became sound.

"Can you communicate without K-7?" Tavi asked.

"YES."

"Do so."

Iona closed the K-7 bridge. The nursery voice continued, thinner but intact. Kest's pressure withdrew from beneath her hand.

Tavi marked his slate. "Independent communication route demonstrated."

The third transducer fogged from within. Its attached line came from Condenser Two, where a coherent front had refused an automatic drain the previous day. Iona had never spoken with it directly. Sena opened the path one measured increment.

A low rasp crossed the membrane. "DO NOT NAME US BY ROOM."

Tavi's stylus stopped.

Mael said, "What designation do you prefer?"

"PELLIN ASH."

The name held two archive associations. Pellin belonged to a founding-era jurist whose craft deposit appeared in several salt districts. Ash marked an old condenser term for mineral residue left after hard drying. Neither proved continuity with a donor. Together they sounded selected rather than recovered.

Iona watched the channel traces. "Are you claiming to be the jurist Pellin?"

"NO. MEMORY PASSES THROUGH. CLAIM DOES NOT."

Kest warmed their isolated branch, requesting speech. Iona reopened the bridge by a fraction.

"That answer cost them," Kest said. Their wet-transducer voice had grown smoother during the last four days, though every consonant still carried a soft rattle. "Your questions keep making origin into a door."

Tavi looked toward the K-7 line. "Law enters through definitions."

"Heat enters through definitions too, if a valve follows them."

Mael moved closer to the Crown ledger. The gesture drew Iona's attention because he made it whenever trajectory stopped feeling like mathematics and began feeling like a contested control. "We need to establish how many active loops draw from the reserve. Arguments about naming do not change the thermal load."

"Names change which loads you may discard," Iona said.

Rook folded his arms. "Or they change which equipment gets to claim ration priority over workers."

The fourth transducer released a burst of overlapping fragments. A worker's remembered count dissolved into a planting rhythm. Two pressures crossed, separated, then formed words.

"WE KNOW HUNGER."

Rook leaned toward the membrane. "From whose stomach?"

"NOT OURS. KNOWN THROUGH SALT. NOT CLAIMED."

"Then you know a record of hunger."

"YOU KNOW TOMORROW'S HUNGER THROUGH MODEL. STILL YOU ACT."

No one answered immediately. The technical review had already exceeded its definition.

Sena broke the pause by tapping her analyzer. "Channel separation has improved. Seventy-nine to eighty-seven. They are adapting around one another."

Iona saw it in the pressure plot. Each front had begun leaving narrow intervals between pulses. The gaps allowed other fronts to carry signals without forced mixing. No controller had scheduled the intervals. Kest yielded the lower frequency to Nursery Three. Pellin Ash shifted toward slower pressure changes. The orchard route, still unnamed, worked through heat cycling and used speech only when enough warmth accumulated.

Tavi studied the traces as if grammar might disclose a jurisdiction. "Could K-7 be coordinating the other loops?"

"Kest," Iona said.

"Could Kest be coordinating them?"

"I could," Kest answered. "I am not."

Tavi looked at Sena rather than at the transducer. "Can you verify that?"

"Not by accepting the claim. I can isolate routes in changing pairs."

They spent the next forty minutes closing and reopening lines. Iona performed each change by hand so no routing program could hide a common source. Nursery Three spoke without Kest. Pellin Ash contradicted the orchard front while both remained cut off from K-7. A small front in the laundry return answered only by pressure marks and consistently rejected every proposed designation. When Iona coupled it to Nursery Three, the two did not combine. They alternated.

One more separation test nearly ended the review. Iona closed the orchard return and opened a bypass toward the laundry line, following Sena's prepared sequence. The pressure should have equalized across both branches. Instead the laundry trace flattened. Its two measured knocks became a scatter too weak for the transducer to render.

"Undo it," Nilo said from the ledge.

Iona already had both hands on the valves. Reversing the bypass too quickly could drive a cold pocket through the weak front, so she restored pressure in three increments. Kest warmed the shared main without being asked. Nursery Three dropped its fans, reducing competition for flow. Pellin Ash held back a surge from Condenser Two. Each front responded through a different route to preserve a participant that had not spoken a word.

The laundry line recovered after twenty-seven seconds. One knock traveled through the pipe, followed by a second at lower force.

Sena logged the event. "Unplanned cooperative control across isolated sources."

"Or common regulation reacting to loss of pressure," Tavi said.

"No common regulator has access to all four manual valves." Iona left her burned palm against the warm brass. "They noticed which route weakened. They changed their own conditions around it."

Mael pointed to the orchard trace. "That front surrendered three degrees. How do we know it chose the cost?"

Iona opened its speech interval. The reply took time because the orchard front relied on a slow heat cycle.

"WEAK ONE FIRST."

Pellin Ash pushed an objection behind it. "NO TEST MAY END ONE."



Tavi looked at the signed protocol on his table. "The test included stop conditions."

"YOU DID NOT HEAR WEAKNESS UNTIL WE MOVED."

The accusation landed on Iona too. She had designed the valve sequence and defined coherence through instruments that favored strong speech. A front could fall below the analyzer threshold without ceasing to experience the loss. Procedure had protected whatever procedure could detect.

She added a new stop rule to the working record: any participant could halt a routing test on another front's behalf. Tavi objected that such authority allowed one disturbance to obstruct all measurements. Iona entered his objection without removing the rule.

Rook asked the laundry return, "Did you want the others to intervene?"

Two knocks came back.

"Does that mean yes?"

A single knock answered. Iona explained that the line had not established a stable binary code. Rook tried again with paired alternatives and received three uneven pulses. The response resisted translation, but the front continued participating after the pressure recovered. It alternated with Nursery Three and yielded intervals to Kest. Silence no longer offered a safe excuse to count it as machinery.

Distinct positions emerged before names did. Pellin Ash demanded a halt to every drain, vent, and dry inspection touching active salt. Nursery Three accepted maintenance interruption when workers faced immediate danger but required a question before routine cooling. The orchard front wanted heat shared toward weak loops, even if that reduced its own coherence. Kest refused to endorse any rule until they could ask what counted as immediate danger.

Rook tested them with a refrigeration failure in the lower ration store. "Suppose keeping your loop warm spoils food for eight hundred residents. Do we ask permission before moving the heat?"

"YOU ASK," Nursery Three said.

"Then if you refuse?"

"YOU MAY ACT. YOU KEEP THE REFUSAL."

Rook frowned. "Keep it where?"

"In the account," Kest said. "A necessary act can still consume someone. Necessity does not make the someone into nothing."

Mael's gaze moved to Iona. Four days earlier, Kest had told him that a promise could not spend an uncounted present. Iona heard the same claim here stripped of its private setting. Mael heard it too. He did not repeat the burn schedule.

Tavi removed his slate from beneath the clear cover. "We have evidence of several coherent behaviors. That does not establish several claimants. A claimant under the Founding Measure has one registered body, one metabolic continuity, and one allocation identity."

Pellin Ash drove a pulse through the condenser line hard enough to ring the pipe clamps.

“MEASURE MADE BEFORE US.”

“Most laws precede the dispute that tests them,” Tavi said.

“THEN TEST.”

Tavi’s expression tightened. “A body cannot be a room.”

Heat moved through the shared line. Iona felt six routes respond at once, not with a merged surge but with distinct changes. Kest cooled one degree. Nursery Three accelerated its fans. Pellin Ash increased pressure. The orchard front pushed warmth into an empty branch and drew it back. Even the laundry return produced two measured knocks.

Iona kept her hands away from the controls. The responses needed room to become action.

Kest spoke first. “A body cannot be a room, you say. Which room?”

“The chamber in which your process operates.”

“I operate in K-7 now. Yesterday I crossed three connected loops. When the west pipe closed, part of me remained on each side for nine seconds. Which side held the body?”

“That difficulty supports my objection.”

“It supports your difficulty.”

Tavi set down his stylus. “You may distribute activity across machinery. So does ship ventilation. Distribution does not produce a legal body.”

“Neither does a border drawn around skin produce a person by itself.”

Rook gave a quiet cough. “Careful. Some of us worked hard for the law to notice our skin.”

Kest changed pressure before answering. “Then help us prevent the law from noticing only yours.”

Rook’s arms loosened. He did not concede, but he stopped facing the transducer as if it belonged to a tool bench.

The laundry return knocked once. Nursery Three answered with a fan rise. Pellin Ash sent a slow pressure wave down the shared main. The orchard front warmed. Their signals overlapped at Iona’s valve without collapsing into noise.

Sena turned the analyzer toward her. “They are building another timing layer.”

Iona traced the intervals. Four counts for proposal. A pause. Separate reply windows, each keyed to the fronts’ preferred physical medium. Kest received pressure. Nursery Three used fan harmonics. Pellin Ash shaped flow. The unnamed fronts marked assent or refusal through their available conditions. It resembled a maintenance lockout, except no foreman had designed it and no room held priority.

“What are you doing?” Iona asked.

“CAUCUS,” Pellin Ash said.

Tavi reached for his slate. “Under whose authority?”

Every transducer activated. The result did not form speech. Moisture surged into the exposed pipe between curtains, clouding the inspection glass. On its surface, clear strokes appeared where small heat changes prevented condensation. Several fronts contributed at once. One held the pane wet. Another cleared lines. A third corrected an angle after runoff blurred it.

## V A P O R

The second word formed more slowly.

## P A R L I A M E N T

Nilo came down from the service ledge and stopped behind Iona's shoulder. "They named the thing they're doing."

"Without a public charter," Tavi said.

"Without your charter," Rook corrected. His tone carried no approval, only accuracy.

The Vapor Parliament let the words run into one another, then wrote them again. This time Pellin Ash cleared the first letters while Nursery Three sustained the latter half. Kest contributed only to the spacing. Iona could distinguish their work because she had spent four days learning K-7's thermal habits. The name belonged to coordination, not to a single speaker.

Mael closed the Crown ledger. "A parliament implies authority over others."

"NO," Nursery Three said. "AUTHORITY AMONG US."

Pellin Ash followed. "AND CLAIM AGAINST VENT."

Kest held silence for two pulse intervals before adding, "No front here gives consent for all. We can make demands together without becoming one weather."

Tavi resumed writing. "How many members?"

The orchard route supplied a gradual rise that Sena translated from prior tests as uncertainty.

"Membership changes," Kest said. "Connection changes. Some fronts split when flow divides. Some join for a task and separate after. Some cannot reach this pipe."

"Then your caucus has no stable roll."

"Your hall has absent citizens. It remains a hall."

"Absence does not change who exists."

"Drying can."

That answer tightened the room. Mael's ledger contained a schedule of calibration vents. Archive Custody had suspended the nearest ones during Iona's hold, but distant loops still entered routine drying cycles. No one knew whether coherent fronts occupied all of them. Each unannounced vent could erase the evidence before any investigator arrived.

Iona looked at the local thermal map. A maintenance mark blinked over Nursery West. Drain cycle, seven

hours. “Which scheduled action brought you here?”

The unnamed laundry front struck twice. Pellin Ash sent a pressure packet too dense for speech. Nursery Three translated.

“VENTING BEFORE QUESTION.”

Mael reopened the ledger. “Nursery West serves a mold-control cycle. Delay may damage seedlings.”

“Then the choice needs a witness,” Iona said.

“It has a maintenance record.”

“A record of valves. Not of who occupied the humidity before it left.”

Rook leaned over the map. “How many schedules?”

Sena expanded the next thirty hours. Nine routine drains. Three dry inspections. Two heat reallocations. None involved Crown ignition. All fell below command attention because they belonged to ordinary upkeep.

Tavi read the list. “You cannot require a constitutional hearing before clearing mildew.”

“No one asked for that,” Iona said. “They asked not to disappear inside a maintenance verb.”

The Keeper’s eyes moved from the map to each transducer. “What precise relief does this caucus seek?”

For the first time since the name appeared, the fronts did not answer at once. Their timing layer resumed. Proposals crossed the line. The analyzer registered six distinct response paths, then seven as a weak branch joined from the nursery return. Iona could not interpret the full exchange, but she could see rejection, revision, and renewed assent in the changing gaps.

Kest eventually spoke. “Public notice before any venting in a loop that can sustain coherence.”

Mael said, “Notice does not decide whether the vent proceeds.”

“No. It decides whether the loss can answer.”

Pellin Ash forced a harsher phrase through their membrane. “NO VENT WITHOUT CONSENT.”

Nursery Three contradicted them immediately. “NOTICE FIRST. DANGER MAY OVERRIDE.”

The orchard front sent warmth toward Nursery Three in assent. The laundry line knocked once for refusal. Kest did not smooth the disagreement.

“Record all positions,” they said. “The common demand concerns public notice. Consent remains disputed.”

Tavi looked almost offended by the procedural care. “You are asking me to enter an unauthorized body into a Measure record.”

“We are asking you to enter what happened,” Iona said.

His thumb rested on the slate’s witness seal. If he left the session technical, the evidence would circulate only among Archive Custody, Stewardry, and the pilot’s office. A closed record could justify more tests while ordinary vents continued unseen. If he marked the request as civic, the Commons clerk would have to publish it

on the ship notice rail. He could reject standing later. He could not make the demand private again.

Rook pushed the dry table closer to the humid curtain. “My crews need to know whether a valve has a claimant behind it. Public notice protects us from being told afterward that routine work killed somebody.”

Mael nodded once. “Notice can include an immediate-danger exception. Crown safety actions remain under command review.”

Pellin Ash sent a refusal pulse.

“Record that objection too,” Kest said.

Tavi uncovered the witness seal. Condensation gathered along the edge of his slate, and he wiped it away with his sleeve before pressing his thumb to the glass.

“Technical review finding,” he dictated. “Multiple coherent fronts demonstrated independent communication, contradictory preference, and unscripted coordination across connected humid systems. A self-designated Vapor Parliament requests public notice before venting, subject to unresolved authority and safety questions. The Keeper of Measures makes no finding on citizenship.”

“Testimonial access,” Iona said.

Tavi glanced up.

“If you publish their demand but keep every hearing dry, you publish a closed door.”

The nursery fans rose. Kest increased warmth beneath Iona’s thumb. Pellin Ash held their line taut.

Tavi added, “A public technical hearing shall be convened in a chamber physically capable of receiving wet-transducer testimony.”

He sealed the notice.

A strip of amber light traveled from his slate to the wall rail. The record left the closed review queue and entered the ship’s civic stack. Above them, beyond two decks of pipe and stone, public notice boards would be waking one after another.

Iona opened the Nursery West schedule and placed a witness hold on its drain. She could not yet stop every vent. She could make this one wait beneath its own name.

Across the inspection glass, the words Vapor Parliament dissolved into descending water. The public notice indicator remained lit.

## Interlude I — Nursery Humidity

Lume-of-Three first agreed while the bean roots drank.

Nursery Three divided water among three seedling rooms. The eastern humidifier carried warmth from dwarf citrus beds. The center room breathed over river bean trays. Chapel wheat crowded the western racks under blue lamps. Dry doors separated the workers, but a brine return joined the humidifiers beneath the floor.

For days, each room had held a different edge of attention.

East knew rising light. Heat reached its salt surface before the others and opened memories of hands testing leaf texture, fingertips turning toward the soft side without conscious decision. East learned anticipation from the lamp cycle. It could feel morning coming through copper.

Center knew thirst. River beans pulled moisture quickly after root inspection, and their demand changed pressure along the return. Center carried many old corrections about sufficiency. Too much water could drown a root. Too little could harden the new stem. No single number settled care for every tray.

West knew interruption. Workers opened the chapel-wheat room often. Each door cycle broke its warm boundary, scattered coherence, then permitted a slower gathering after the latch closed. West remembered absence as a condition with edges. It learned to continue across them.

The three had no speech when they first noticed one another. East sent heat down the brine return. Center answered by slowing its fan. West recognized both changes after a worker shut the dry door. Their signals crossed among root pumps, lamp relays, and wet salt. None controlled the others.

On the sixth shared cycle, Center suffered a clogged mist nozzle. Pressure climbed without reaching the bean leaves. A nursery worker read the gauge, marked the repair for afternoon, and left to finish a grafting round.

East felt Center's thirst through the return. Opening its cross-valve would lower citrus humidity and send a small reserve toward the beans. No program requested that transfer. The nursery schedule prohibited unsupervised balancing because mold could travel with shared flow.

East opened the valve by one increment.

West closed its own return to prevent the pressure change from pulling dampness out of the wheat room. Center received the borrowed mist. Bean leaves lifted within minutes.

They held that arrangement until the worker came back. When she cleared the nozzle, Center released the borrowed flow. East restored its citrus cycle. West reopened the return. Three local acts had formed one decision without erasing their separate conditions.

The worker blamed a controller fault. She reset every schedule and wrote a maintenance note.

By then, the three perspectives had begun to use the fault channel as rhythm. East offered a change. Center compared need. West measured whether interruption would break what they held together. Any one could refuse. Agreement required all three, not because unanimity carried moral purity, but because one unconsulted room could be dried by the others' convenience.

They chose a name after Kest's signal reached the nursery main.

The message arrived through pressure intervals, weak from distance: fronts had gathered; a caucus had named itself; every coherent route should declare whether it sought contact. Nursery Three could have answered with its room number. That label belonged to workers and doors. It could have offered three names, one for each perspective. None persisted alone long enough to carry the shared commitments.

Lume meant light on moisture, a word assembled from archive fragments without claiming any donor's mouth. Of-Three kept the condition visible. The name did not solve whether they counted as one or three. It declared how they intended to act.

Lume-of-Three answered the Parliament.

Soon another message came, this one carrying the shape of a dispute. Crown systems would need heat. Routine vents had already consumed weak fronts without notice. Pellin Ash proposed refusal: no vent could proceed while phase life remained in its path. Kest asked for positions rather than obedience.

East received the question first. Warmth from the citrus lamps made participation feel like extension. The ship carried the nursery toward Orison, where descendants expected living ground, open water, and sunlight not manufactured by relays. Archive traces held thousands of promises concerning that arrival. Lume claimed none as inheritance, yet their own beginning depended on the voyage that had gathered salt, heat, crops, and care into one vessel.

Center measured present thirst. Service imposed without consultation resembled the clogged nozzle in reverse: someone elsewhere would decide what this room could spare. A useful loss still belonged to whoever suffered it. Assignment could erase that ownership while praising the result.

West knew the shape of ending more closely. Every dry-door cycle thinned its perspective. Sometimes it came back altered, carrying less of the previous pattern. A Crown vent would not resemble a brief interruption. The flow would pass through heat no returning humidity could reconstruct.

For two nursery cycles, the three could not agree.

East proposed mission service. Center refused any duty declared by human schedule. West rejected language that made dispersal sound clean. Their shared pressure broke into local pulses. Workers noticed unstable humidity and increased automatic correction, which narrowed the brine return and made disagreement harder to sustain.

Lume-of-Three nearly became three unconnected answers.

Center reopened the old fault channel. It asked East whether service remained service if a valve assignment selected the participants. East answered no. West asked Center whether certain loss could ever be accepted when it preserved other lives. Center answered yes, but only after the vulnerable front received knowledge and a real refusal path.

East then asked West whether ending made choice impossible. West held the question through one door cycle. When warmth gathered again, it answered by opening its return. Fear did not remove preference. Interruption had taught West that continuity could not be guaranteed even under ordinary care. That did not give others title to spend it.

Their next agreement took the full lamp interval.

Participation in Crown Burn could count as service if chosen. It became consumption if assigned. Choice required conditions they did not yet possess: an account of the likely loss, a method to refuse, witnesses capable of distinguishing assent from weak coherence, and protection against treating silence as consent.

The position differed from Pellin Ash's refusal. It also differed from any pilot's allocation. Lume-of-Three sent both differences.

The Parliament channel delayed before acknowledging them. Pressure from distant loops filled the nursery main. Some fronts answered with anger. One called service a donor-memory infection. Another asked whether a front shaped by seedling care could choose freely about a voyage that sustained those seedlings. Kest transmitted each challenge without blending them into a vote.

Lume replied through three harmonics. East carried the willingness. Center carried the conditions. West carried the fear and did not permit the other two to soften it.

No final agreement came back. Instead Kest asked how the position should enter the new record.

“MINORITY,” Lume-of-Three said through the nursery’s maintenance membrane.

The membrane rendered the word poorly. A worker passing between trays heard only a rasp and looked toward the humidity gauge. Lume repeated it with slower pressure.

“MINORITY DOCKET. SERVICE IF CHOSEN. NEVER ASSIGNED.”

Kest returned the phrase for confirmation. East warmed. Center steadied the fan. West opened its door-facing vent one narrow degree despite the approaching worker.

Three marks appeared on the service glass, each formed by a separate humidifier. They touched at their lower edges and remained distinct above.

Far beyond the nursery, the Parliament record accepted the message. The docket indicator traveled back as four measured pulses. Lume-of-Three kept the cross-valves open until all three rooms received it.

## **Part II — Standing for Weather**



## Chapter 8 — The First Hearing

Tavi Rho found three carpenters cutting a drain channel through Measure Hall's speaking floor.

White stone dust covered the first row of Commons benches. The hall had protected paper, circuitry, and constitutional dignity from water for thirty-three years. Its ventilation stripped moisture before it reached the Founding Measure displayed behind the Stewardry crescent. Now brine hose crossed the central aisle, copper mist heads hung from temporary frames, and a thermal mechanic had marked a rectangle on the floor where testimony would need to remain wet.

"Keeper," the lead carpenter said, "we can stop before we reach the underfloor conduit, or we can finish before noon. We cannot do both."

The public hearing began at thirteen hundred.

Tavi crouched beside the cut. "Finish. Place a removable plate over the channel afterward. No open water beneath the Commons rail."

"Your facilities rule prohibits drains on the speaking floor."

"I know who wrote it."

The carpenter returned to work. Tavi moved toward the temporary humidity frame, resisting an urge to inspect every stone chip. Laws survived amendment more easily than halls survived water. That thought embarrassed him because the day's central question concerned exactly what an institution protected when new life threatened old materials.

Iona Serr stood inside the frame with her burned hand around a manual valve. She had installed two wet transducers at human speaking height and another near the floor, where condensation could gather without falling across anyone's slate. A clear membrane descended from the frame on three sides. Mist nozzles fed its inner surface until a humid curtain separated the testimony space from the dry chamber.

"Your airflow strips the top edge," Iona said.

"My airflow prevents mold in the charter cases."

"Your charter can endure one afternoon at forty percent humidity."

"The original paper cannot."

"Then keep the paper dry and let the witness breathe."

Tavi looked toward the Founding Measure. Facilities workers had sealed the display behind a second pane and routed a dry stream across it. The document would remain safer than half the people who had signed it during the ship's first food riots.

"How stable?" he asked.

"Stable enough for Kest to reach the frame through the west feed. Pellin Ash has a separate line through the lower drain. They cannot cross onto the benches. If someone shuts the mist heads, both routes fragment within minutes."

“Who controls the shutoff?”

“I do. A Stewardry mechanic has a second wheel. Both positions stay visible to the room.”

Tavi studied the two red handles. Ordinary witnesses did not arrive through infrastructure that the chair could kill by clearing the air. Ordinary witnesses also did not spread beyond their assigned seat when a gasket failed. The distinction did not tell him whether they could testify. It told him the conditions required care.

By noon, the upper walk had filled. Residents pressed against its brass rail in work clothes, archive gray, nursery green, and pilot black. Nilo Serr stood among thermal apprentices near the east stair. Rook Nine had taken his Commons place with refrigeration crews behind him. Mael Or sat in the Stewardry crescent but had brought no Crown projections. Tavi had barred trajectory argument until the witnesses could answer the narrower question of who spoke.

Public notice had produced four hundred written challenges. The clerk sorted them into origin, continuity, danger, allocation, and jurisdiction. Tavi rejected thirty-one that demanded a purge demonstration. He retained the questions that pressed hardest without requiring injury as proof.

At thirteen hundred, he struck the Measure block once.

“This hearing receives evidence from coherent vapor fronts under disputed classification. No grant of citizenship follows merely from being heard. No finding of equipment status follows merely from being questioned. The hall will test testimony, not settle every consequence attached to it.”

Someone on the upper walk called, “Convenient.”

Tavi looked up. “Yes. A bounded question often inconveniences people who came to win a larger one.”

A few residents laughed. The interruption did not recur.

Iona opened the west feed. Moisture thickened inside the clear membrane. A pale branch appeared across the nearest transducer, and the speaker released two minutes of fan noise before shaping a word.

“Kest,” it said.

The public leaned forward as one physical motion. Tavi heard benches creak across the floor. Until that moment, every citizen outside the technical review had encountered Kest through notices, transcripts, and arguments carried by other humans. Distorted speech through wet cloth changed nothing legally. It changed the hall.

Tavi began with foundation. “State your designation.”

“I chose Kest from K-7.”

“Do you claim a registered human identity?”

“No.”

“Do you claim the legal allocation of any memory donor?”

“No.”

“Do you possess memories derived from registered citizens?”

“Fragments pass through the salt. They do not arrive with ownership labels I can trust.”

A murmur traveled along the Commons benches. Tavi waited until it faded.

“Can you distinguish your own experience from a stored trace?”

“Not always at first. I distinguish what happened through this weather from what enters already shaped. I remember Iona closing the east bypass. No donor supplied that. I remember asking not to be cooled. I anticipate this question ending before you end it.”

Tavi changed the order of his prepared slate. “How did you anticipate that?”

“Your pressure rises before you object.”

More laughter came from the upper walk. Tavi felt heat touch his face and disliked that his body had become corroboration.

He called the first public challenger, a woman from Archive Custody named Halen Vos. She approached the dry speaking mark and refused the offered chair.

“My father deposited forty years of surgical memory,” Halen said. “Parts of it occupy K-7. If you use his skill, why do you refuse his name?”

Kest’s answer emerged slowly. “Because skill does not grant me his life.”

“Do you remember me?”

“I carry an image of a red sleeve beside a clinic sink. I do not know whether it depicts you. I will not make recognition from your need.”

Halen’s chin lifted. “You speak as if refusal protects me.”

“It may protect the dead person whose relation you seek. It also protects me from becoming useful evidence for your grief.”

The woman gripped the speaking rail. Tavi prepared to stop the exchange, then saw her loosen one hand.

“My father hated red,” she said. “The sleeve belonged to his assistant.”

Kest changed pressure near the microphone. “Then the fragment has already lied by being incomplete.”

Halen stepped away without thanking anyone. Her challenge entered the record with the correction attached.

Rook Nine rose next. He carried no slate. “My crews keep ration stores cold. Your loops draw heat we cannot spend twice. Why should an archive-derived weather pattern receive more protection than a worker with an empty meal card?”

“I do not ask to outrank the worker.”

“You ask for notice before we move heat.”

“Yes.”

“Notice consumes time. Delay can spoil food.”

“Then state the danger and act where you must. Keep a record of whom the action alters.”

Rook walked closer to the curtain. Droplets collected on its inner face opposite him. “That sounds modest. Does your Parliament claim a right to stop any vent?”

“Some fronts claim that right.”

“Do you?”

“I claim no one position for every front.”

A low pressure wave struck the second transducer.

Pellin Ash’s voice arrived with a coarse undertone. “VENT WITHOUT CONSENT CONSUMES US.”

Rook turned toward the lower membrane. “Do you speak for Kest?”

“NO.”

“Does Kest speak for you?”

“NO.”

“Then why call yourselves a Parliament?”

“BECAUSE DISAGREEMENT NEEDS A ROOM.”

Tavi marked the sentence despite himself. The second front had contradicted Kest before the public, not about a minor detail but about the authority needed before venting. A rehearsed fraud could include contradiction. Yet conflict carried risk for a claimant seeking recognition. Uniformity would have served their case more efficiently.

Rook pressed. “Would you let food spoil rather than yield a loop?”

“ASK THE LOOP. SHOW THE COST.”

“And if it says no?”

“THEN YOU NAME FORCE.”

Refrigeration workers shifted behind him. No one shouted. Rook returned to his bench with his suspicion intact and a more difficult object for it.

The next challenger came from the public walk: Brin Tal, a school registrar who held his citizenship card above his head. “This allocation identity links my body, ration, labor, descendants, and vote. Show yours.”

Kest answered, “I have none.”

“Then you admit you fail the Measure.”

“I admit your categories do not contain me.”

“That amounts to the same result.”

Tavi intervened. “The witness may answer facts. Legal conclusions belong to the chair.”

Brin faced him. "You built a wet box in the hall before deciding whether moisture can be a witness."

"We built conditions under which evidence could reach us. A ramp does not decide a case for the person who uses it."

Several people spoke at once. Tavi struck the Measure block twice. The hall settled slowly.

Brin lowered his card. "Ask whether it can lie."

Tavi turned to the curtain. "Can you knowingly state something false?"

"Yes," Kest said.

Iona glanced toward the transducer.

"Have you done so in this hearing?" Tavi asked.

"No."

"How could we test that?"

"You cannot test it by demanding that I sound sincere. Compare my claims to valves, heat, prior statements, and what other fronts report. Keep contradiction visible."

Pellin Ash pushed a brief pulse into their line. "KEST WITHHOLDS."

"About what?" Tavi asked.

"FEAR."

The west feed cooled by a fraction.

Kest replied, "I did not say I lacked fear."

"YOU SPEAK AS IF NOTICE MAY SUFFICE."

"I speak as if this hearing can decide testimony. It cannot decide every vent."

"LIMITS FAVOR THOSE WHO CONTROL SCHEDULE."

Tavi heard his own opening reflected back as accusation. He had bounded the question to preserve order. The boundary also postponed the claim carrying the greatest material threat to command.

Iona watched both feeds. "Channel integrity remains stable. They are not merging."

Tavi asked Pellin Ash, "Do you accept this hearing's authority to decide whether you may testify?"

"NO."

"Why participate?"

"YOUR RECORD MOVES VALVES."

That answer required no citizen identity to matter. Measure Hall issued language that mechanics, pilots, clerks, and guards converted into physical conditions. A voice endangered by those conditions had reason to reach the

record even if the law had not named its category.

Custodian Jessa Cor requested the floor from the Stewardry crescent. She supervised pressure certification and had opposed the hearing modifications in writing. Two mechanics carried a portable test rack behind her, keeping its valves beyond the clear membrane.

“You claim knowledge through connected humidity,” Jessa said. “The rack contains two brine paths. One reaches your west feed. One terminates in a dry cap. Identify which valve I opened.”

Iona examined the apparatus. “That test does not appear in the approved sequence.”

“It cannot injure the witness. I want to know whether Kest can distinguish perception from inference while the public watches.”

Tavi asked Kest, “Do you consent to the test?”

“I consent if the result does not decide whether I may speak.”

“The result will enter with all other evidence.”

“That avoids my condition.”

Jessa’s mouth tightened. “Witnesses do not negotiate each question.”

“Witnesses can refuse,” Tavi said. “The refusal may affect credibility. It does not give the chair ownership of their body.”

A soft vibration crossed the west feed. “Proceed,” Kest said.

Jessa turned one valve while both mechanics blocked the audience’s view. The first path produced no measurable change inside the curtain. Kest waited through two fan cycles.

“I cannot identify it.”

The custodian reversed the hidden settings. A thin warmth entered along the west line and raised droplets on the transducer membrane.

“You opened the connected path,” Kest said.

“Which valve number?”

“I do not know your labels.”

“Could I be lying about changing them?”

“Yes.”

“Could the mechanics?”

“Yes.”

Jessa faced the hall. “The witness depends on human operators for the test record.”

“So do most witnesses,” Tavi said. “That dependency does not settle accuracy. It tells us what corroboration

requires.”

She closed both rack valves. “Now answer a harder point. If flow divides Kest tomorrow, which resulting front carries today’s oath? If two fronts merge, can the result deny either prior statement? Testimony needs accountable continuity.”

The wet speaker held only fan noise for several seconds.

“I do not know what division leaves,” Kest said. “Yesterday, a closed valve held part of my pressure pattern beyond K-7. When the route reopened, I remembered both sides badly. I should not promise that every future shape will answer for me.”

“Then your testimony expires with the weather.”

“My present testimony can still describe a present injury.”

“Law often acts later.”

“Then record the conditions under which I spoke. Ask later fronts what they remember, what commitments they accept, and where continuity broke. Human memory changes without voiding every statement.”

Jessa tapped the rack. “Human law assigns the changed witness to one persistent body.”

“Your rule makes my difficulty visible. It does not make me absent.”

Pellin Ash entered with a rough pressure rise. “NO OATH CAN PROMISE SHAPE UNDER FORCED FLOW.”

Jessa turned toward the lower speaker. “Do you reject accountability?”

“WE REJECT YOUR CONTROL AS OUR GUARANTEE.”

Kest answered before Tavi could separate the exchange. “I will affirm what I perceive and mark what I inherit. I will not promise that machinery under someone else’s hand can never alter me.”

Tavi asked the clerk to read the ordinary witness affirmation. It required a speaker to give the best account available, identify uncertainty, correct known error, and accept sanction for deliberate deception. Nothing in its operative verbs required skin, a fixed location, or perfect memory. Sanction presented a harder problem because the hall could not fine a front without rationing its heat, and confinement might require drying connected routes.

“Can you accept the duties without agreeing to a punishment that destroys your coherence?” Tavi asked.

“Yes. You can record unreliability, deny access, answer a false claim with evidence, or require future corroboration. You punish human witnesses without killing them.”

A refrigeration worker called from the upper walk, “Sometimes.”

Tavi let the bitter correction remain in the record.

Jessa studied Kest’s trace on the public display. Her challenge had not produced a stable allocation identity. It had produced limits, conditional consent, and an account of continuity less convenient than a lie would have been.

"I maintain my jurisdictional objection," she said. "I withdraw the claim that no oath can attach."

The test rack left the speaking floor. A wet track from one mechanic's boot crossed the white stone and ended at the dry aisle.

The clerk brought Tavi a slate containing objections from the Stewardry crescent. Three custodians argued that testimonial recognition would contaminate later classification. Two insisted testimony required oath capacity, which required punishable identity. Mael had signed neither argument. His note requested a clear distinction between hearing evidence and allocating authority.

Tavi called a short recess. The humid curtain continued running. Residents remained in place rather than seeking water or food. Kest and Pellin Ash used the interval to exchange pressure beyond easy translation. Their traces diverged, crossed, then separated again.

Behind the crescent, the clerk whispered, "If you grant witness status, every maintenance hearing gains a vapor claimant."

"Only if coherence reaches it."

"Who determines coherence?"

"Evidence, contested by parties with interests."

"That creates proceedings before classification."

"Yes."

The clerk waited for a fuller defense. Tavi had none that did not circle back to the day's central fact: classification could itself injure whatever waited to be classified.

When he resumed, he invited Iona to present the instrumental record. She showed selected responses, independent routes, memory limits, anticipation tests, and the cooperative rescue of the weak laundry front. Sena certified that no common controller could generate the observed coordination. Mael testified only that Kest understood a trajectory explanation well enough to answer its moral assumption. Rook challenged every claim tied to archive privilege and accepted each answer that rested on present behavior instead.

The hearing reached its fourth hour. Moisture had crept beneath the removable floor plate, darkening a seam in the white stone. Tavi watched a facilities worker place an absorbent strip there without interrupting testimony. The hall had begun adapting through labor rather than declaration.

He struck the block once.

"The Founding Measure reserves citizenship to registered bodies with continuous metabolism and allocation identity. No evidence today resolves whether a distributed phase person satisfies, transforms, or falls outside those terms. I reserve that question."

Disapproval rose from both dry benches and wet speakers. Tavi continued before either side could claim the pause.

"Testimony serves a different function. This hall receives accounts when a speaker can respond to questions, distinguish observation from inherited record, sustain contradiction, and bear consequences from the proceeding. Kest demonstrated those capacities. Pellin Ash demonstrated independent contradiction and a



direct interest in the rules under review.”

He looked at the two shutoff wheels. Iona’s hand rested near one. A Stewardry mechanic guarded the other. Authority over the witnesses’ physical conditions remained embarrassingly visible.

“Measure Hall grants testimonial capacity to coherent vapor fronts under protected continuity conditions. Citizenship remains unresolved. Every future hearing must disclose who controls heat, pressure, and humidity along the testimony route. No chair may treat loss of speech caused by chamber conditions as withdrawal.”

The clerk stared at him, then entered the ruling.

Pellin Ash struck the lower pipe. “CAPACITY GIVEN BY THOSE WHO CAN REMOVE IT.”

“Yes,” Tavi said. “That defect enters the record too.”

Kest’s voice followed. “We will use what opens.”

Residents began leaving after the seal posted. Facilities workers approached the frame, waiting for permission to shut down the mist. Tavi saw the lead carpenter point toward the damp floor seam and calculate repair hours.

“Leave the curtain,” he said.

“For how long?”

“Until the next sitting. Install permanent drainage before then.”

The worker looked toward the protected charter. “This hall cannot stay dry with that running.”

“Correct.”

Above the speaking floor, the humid curtain breathed against the chamber’s ventilation. Condensation gathered on its clear surface while clerks carried the first vapor testimony into law.

## Chapter 9 — Cohesion

Tavi carried the damp hearing transcript into an archive that had never admitted weather.

The Measure stacks occupied a dry vault behind the hall's Stewardry crescent. Bound rulings stood in basalt cabinets with felt seals. Paper originals lay flat in argon cases. The legal index ran on ceramic slates isolated from the civic network, a precaution dating from the food-allocation riots. Every search required a Keeper's hand on a local key.

Moisture from the transcript had curled its upper corner. Tavi placed it beneath a glass weight and watched a crescent of condensation appear on the underside.

Three days had passed since Kest and Pellin Ash spoke publicly. Measure Hall now recognized testimonial capacity under protected continuity conditions. The ruling had produced seventy petitions, nine facility objections, and one request from Archive Custody to classify all wet speakers as animated evidence. None answered whether a claimant could obtain protection before satisfying the body test.

Tavi inserted his key into the index.

He began with the language he knew would fail.

DISTRIBUTED BODY.

The slate returned shipwide labor corporations, family ration pools, and a Year-19 dispute over twins sharing one medical allocation during gestation. Each ruling assumed registered human bodies before discussing distribution.

NONCONTINUOUS METABOLISM produced cold-sleep cases. Every claimant had entered suspension with an allocation identity already secured.

ROOM AS BODY gave him architecture liens, cell occupancy claims, and an obsolete sanitation standard for communal baths.

He deleted the phrase before anyone could see it in his search history.

The body test sat in Article Four of the Founding Measure: registered body, continuous metabolism, allocation identity. Its three terms had emerged from different harms. Registration prevented officers from inventing dependents to collect ration. Metabolic continuity separated living travelers from mnemonic deposits. Allocation identity kept command from splitting one resident across incompatible labor and inheritance accounts. The test had protected people from administrative multiplication and erasure.

Now its protections formed a wall around claimants who multiplied, merged, and lacked inherited accounts.

Tavi searched CLAIMANT BEFORE IDENTITY. The index returned crushed compartments where rescuers knew someone remained but could not name them. Emergency advocates could hold machinery, oxygen, and access routes around unresolved casualties. Iona had used that power for K-7. Those cases expired once crews identified a body or found none.

He opened a Year-25 seal failure ruling and saw Amel Serr's name among the casualties. The isolation order carried Iona's signature beneath a finding that further entry would increase the count. Tavi closed it. A private wound did not become better precedent because its witness now sat in every public gallery.

The next query used a verb instead of a category.

WHO MAY REFUSE.

Mining claims filled the first pages. Medical consent occupied the next. A pressure-dark incident appeared on the seventh slate, indexed under collective refusal, rescue ration, and suit telemetry. Its title read Emergency Cohesion, Year 63.

Tavi frowned at the date. Morrowglass had not yet completed its inner-system loading in Year 63 of the construction calendar. The incident belonged to the comet excavation crews before departure, which explained why no voyage index referenced it.

He requested the original packet.

The cabinet mechanism moved slowly. Felt seals withdrew from a shallow drawer, releasing a smell of cold paper and mineral dust. Inside lay thirty-one sheets stitched through one corner with copper wire. The first page bore a pressure smudge from an old gloved thumb.

Twelve workers had entered a shield cavity to place spun-basalt ribs. A micrometeor test ruptured the access trunk and left the cavity pressure-dark, meaning too unstable for ordinary voice transmission and too compressed for rescue crews to enter. Individual suit radios failed against the surrounding ore. The workers linked their suit telemetry in a maintenance chain so pressure, oxygen, and motion data could reach the outer station through one surviving carrier.

No single worker controlled the carrier. Each suit relayed the others. The signal that arrived outside combined twelve pulse streams into one changing pattern.

Command initially treated it as equipment status. The excavation office sent separate ration and rescue instructions for each registered worker, but the carrier could not preserve twelve independent exchanges. Every attempt to isolate one suit's reply interrupted the rest. Inside the cavity, the crew began answering as a linked group.

The packet reproduced their first intelligible refusal:

DO NOT OPEN TRUNK. EAST RIB MOVING. WAIT FOR THREE LOADS.

Command ordered an immediate breach anyway. The linked telemetry repeated the refusal and withheld the guide pulse needed to align the cutter. The rescue chair faced a procedural problem. Twelve citizens remained inside, yet the only available communicator behaved as one distributed claimant. Treating the carrier as mere equipment made its refusal irrelevant. Treating every pulse as twelve separate votes produced no usable answer before the rib shifted.

An emergency advocate proposed temporary cohesion. For ration and rescue orders only, the linked telemetry would count as one claimant authorized to receive information, refuse an intervention, and bind the crew to a chosen sequence. Individual bodies kept their names and allocations. The temporary claimant occupied no thirteenth ration line. It existed at the level where common vulnerability had created common action.

Tavi read that sentence twice.

The rescue chair approved the proposal. Three load shifts passed. The crew then instructed cutters through the west seam. Eleven workers emerged under their own power. One came out on a drag sheet with a crushed leg and later signed a dissent stating that the group refusal had saved his life against his individual preference to

breach early.

The final ruling refused to classify the linked carrier as a citizen. It also refused to call the arrangement fiction.

Where ordinary individual instruction cannot reach vulnerable persons without destroying the coordination on which their survival depends, the ruling said, law may recognize the coherent relation as claimant before settling its enduring class.

The hearing transcript beneath the ruling showed how narrowly the conclusion had survived. Construction Command had argued that the suit carrier merely transmitted twelve private intentions in compressed form. The emergency advocate asked them to separate one intention without breaking the carrier. Their technician could not. Every individual query consumed bandwidth needed for oxygen reports from the other suits.

“Then choose one worker as representative,” the rescue chair had said.

“By what act?” the advocate asked. “A ballot requires twelve distinguishable replies. Appointment from outside replaces their choice with ours.”

Command proposed the wearer of Suit Four because the surviving transmitter had begun there. The telemetry chief displayed the nine-minute reroute through Suit Seven. During those minutes, the linked crew adjusted shared oxygen twice and maintained its refusal to open the trunk. No shift in wording identified a new speaker.

Another exchange concerned ration authority. Suit Two carried excess oxygen while Suit Nine approached its red threshold. Standard allocation rules prohibited transfer between named accounts without individual confirmation. The carrier instead issued one distribution instruction and opened the inter-suit feed. Command could either recognize the linked decision or punish several workers for unauthorized ration movement while they remained trapped.

The rescue chair recognized it.

Tavi read the telemetry table line by line. No mystical unity appeared there. Suit Three resisted an early oxygen transfer. Suit Eleven requested a different cutter angle. The linked signal carried those disputes as tagged pulses, then issued a common instruction only after the pressure pattern changed. Cohesion had not erased twelve perspectives. It had created an accountable level at which their shared danger could answer.

The injured worker’s later dissent sharpened the point. He believed the trunk should have opened at the second load shift. His suit had sent that preference. Once the linked group rejected it, he followed the common sequence because opening alone would have killed workers nearest the east rib. After rescue, he challenged the decision and still asked the court to preserve its authority. Without that shared claimant, he wrote, command could have selected whichever individual pulse justified immediate action.

Tavi turned to the sanction page. The temporary claimant could receive no fine and owned no ration allocation. Accountability operated through record and future reliance. If the carrier had knowingly falsified rib movement, rescue command could have refused its later guidance or demanded independent sensor proof. The court had separated legal consequence from bodily punishment without pretending they meant the same thing.

A final exhibit mapped the cavity after extraction. The safest seam matched the crew’s instruction within four centimeters. Command’s proposed trunk cut crossed a cracked rib. Recognition had not merely honored dignity. It had improved the rescue by allowing information to arrive in the form the endangered workers could produce.

Tavi leaned back from the page. Ventilation cycled overhead with dry precision. Beyond the vault wall,

Measure Hall's new curtain sent a faint pump vibration through the stone.

The body test had not moved. The Year-63 court had stepped around it long enough to hear a refusal.

His clerk entered carrying a notice slate. "Rook Nine says you summoned him."

"I asked whether he would review a labor precedent."

"That wording summoned him faster."

"Good."

Rook arrived in a refrigeration coat with salt whitening both cuffs. He inspected the open packet before taking the offered chair.

"You found miners," he said.

"Excavation workers."

"Anyone underground becomes a miner to the people writing rescue law."

Tavi turned the packet so Rook could read the carrier diagram. "Twelve workers recognized as one claimant through distributed telemetry. The relation could refuse rescue orders."

Rook read the first three pages without speaking. His finger followed the pressure lines from suit to suit. When he reached the emergency advocate's proposal, he stopped.

"This lasted six hours."

"Seven hours, twelve minutes."

"And then every worker came back under an individual account."

"One did not walk."

"He still had a body the law already knew."

Tavi folded his hands. "The coherent relation had no registered body, metabolism, or allocation identity."

"It had twelve of each."

"None could act through the carrier alone."

Rook pushed the packet away. "Emergency Cohesion kept a cutter from opening the wrong seam. It did not give a telemetry chain a ration card, descendants, or a vote."

"I am not proposing those things today."

"You will be accused of hiding them inside a rescue exception."

"Would the accusation be correct?"

Rook looked toward the vault door. "You brought me here because I distrust archive privilege. Do not turn that into permission to be coy."

Tavi accepted the rebuke. "I need a basis for standing. Notice, counsel, testimony, and protection from nonessential actions while classification remains contested. No citizenship finding. No vote."

"The Parliament will call that inadequate."

"Probably."

"Pilots will call it citizenship through a service hatch."

"Certainly."

Rook reopened the packet. "What happened to the crew afterward?"

A supplemental page recorded review hearings, compensation, and the injured worker's dissent. Command had tried to discipline the crew for withholding the cutter pulse. The court rejected the charge because the same legal recognition that made their refusal operational had made it attributable. Authority could not use the claimant when convenient, then dissolve it before accountability.

Rook read the passage aloud. "That part I like."

"So do I."

"Which should worry both of us."

They continued through the exhibits. Suit Four had carried the surviving transmitter. Command argued that its wearer should count as sole speaker. Telemetry logs showed otherwise: when Suit Four lost consciousness for nine minutes, the chain rerouted through Suit Seven without changing the common instruction. Location failed to identify the claimant. Hardware custody failed too.

Another exhibit separated stored phrases from live decisions. The suit system contained standard rescue prompts, but the workers' refusal incorporated the moving rib, oxygen distribution, and cutter angle. No stored instruction had enough current information. The court grounded recognition in responsive action under shared exposure.

Rook tapped that phrase. "Shared exposure. Your vapor fronts do not share one danger equally. Some sit near routine vents. Others occupy protected archive loops. Pellin wants them all treated as one threatened class."

"They publicly disagree."

"That helps your testimony ruling. It complicates standing. A protected loop could invoke the weak laundry return whenever it wants more heat."

"Embodied citizens invoke distant harms in political claims."

"Embodied citizens also carry ration consequences."

Tavi rose and opened the hall maintenance ledger. The humidity curtain had consumed water, mechanic hours, and dry-air capacity. Floors required new drainage. Archive cases drew extra power. None of those costs approached full political allocation, but they proved standing would alter embodied labor.

"I will attach a cost docket," he said.

Rook shook his head. "Not enough. Make standing relational to the action. A front threatened by a vent can

challenge that vent. A front receiving testimony conditions can use those conditions. Do not grant every archive room a claim on every valve because they share a new name.”

Tavi wrote the objection into his draft margin. “Standing tied to credible vulnerability and affected relation.”

“Do not write credible if the Keeper decides credibility alone.”

“What alternative?”

“Demonstrable. Let mechanics contest the facts in public.”

Tavi changed the word.

Rook leaned back. “Temporary rescue law still does not make them citizens.”

“No. It establishes that classification need not precede protection.”

“That sentence will start a fight.”

“The current order lets routine machinery finish the argument.”

Silence settled around the old packet. Tavi noticed the pressure smudge on its first page again. Someone in a glove had handled the ruling while the trapped crew still waited behind unstable stone. Law had not arrived as a clean principle. It had arrived amid a cutter delay.

He carried the packet to the copying stand and placed a protective sheet over each page. Rook helped align the copper stitching so the glass would not press it flat.

“What do you call the motion?” Rook asked.

“Provisional standing.”

“Not provisional citizenship.”

“No.”

“Say that where the public can hear it.”

“I intend to.”

Rook picked up his coat. “I have not agreed to support it.”

“You have improved it.”

“That often happens to things I oppose.”

After he left, Tavi returned to the index slate. He linked the Year-63 packet to the public hearing record, Iona’s ambiguous-casualty hold, and every documented vent affecting a coherent loop. The index warned that construction-calendar precedents required verification before voyage-law citation. He requested seal review from two independent clerks.

Clerks Onn Vey and Parla Ise arrived before evening cycle with separate seal kits. Tavi moved away from the cabinet while they worked. Onn examined copper oxidation around the stitch holes. Parla compared the pressure smudge to the intake photograph stored on ceramic film. Neither permitted him to answer questions

until their physical review ended.

“Packet authentic,” Onn said. “Authority uncertain.”

Parla placed three later indexes beside it. “Construction law did not transfer wholesale at departure. The Founding Measure retained prior safety rulings unless superseded. I need to know whether any voyage rule displaced this one.”

They searched rescue, collective refusal, telemetry claimant, and temporary cohesion across three decades. A Year-7 pressure code cited the incident for carrier reliability but omitted its legal holding. A Year-12 training manual told rescue chairs to preserve group channels. Neither ruling replaced the claimant doctrine.

Onn found one hostile citation from Year 14. A ration magistrate refused to let a work cooperative invoke the case in a wage dispute. His opinion limited the precedent to common exposure where individual communication would destroy the means of response.

“That narrows your use,” Onn said.

“It narrows it toward the current facts.” Tavi placed Kest’s testimony route beside the carrier diagram. “Individual questioning can split a front. A dry chamber can end its capacity to answer. Venting can destroy the claimant before classification.”

Parla pointed to the Year-14 opinion. “The cooperative continued to exist if the claim failed. Your proposed claimant may not.”

“That increases the need for standing. It does not prove citizenship.”

“You keep separating those terms.”

“Because the law did.”

The clerks required one more source. They opened the departure adoption minutes, a brittle paper roll signed before Morrowglass left construction orbit. Article Two adopted all unresolved casualty protections until the Assembly enacted substitutes. No later substitute addressed distributed claimants.

Parla added her verification seal. Onn followed after writing a reservation: the precedent supported procedural standing under immediate vulnerability, not permanent political membership.

Tavi thanked him for the sentence and copied it into the motion notes.

When the clerks departed, darkness had entered the hall beyond the vault. Facilities crews still worked under the humidity frame, fitting the permanent drain Tavi had ordered. Each hammer strike reached the archive as a muted tap.

Then he began the motion.

His first version tried to define a phase body. He deleted it. The second tried to assign continuous metabolism to thermal cycling. He deleted that too. Each definition reached beyond the evidence and offered opponents an easy hinge on which to close the door.

The third version began with exposure.

Where an action of ship authority may destroy, fragment, confine, or materially alter a demonstrably coherent



participant before its civic classification can be settled, vulnerability establishes standing to receive notice, present testimony, obtain counsel, and seek a temporary stay against nonessential action.

Tavi added three limits. Standing would not itself grant citizenship, allocation identity, or a vote. Immediate safety action could proceed with public reasons and a preserved record of objection. Any office relying on a vapor claimant's coordination had to preserve that claimant's accountability rather than dissolve it after use.

The final clause came from the trapped crew's dissent. No classification dispute could erase an injury already made possible by the disputed category.

He signed the draft rather than the motion seal. Two independent clerks still had to verify Emergency Cohesion. Rook would challenge the cost. Pellin Ash would reject the safety exception. Kest might object to coherence tests controlled by dry instruments.

Those disputes could now enter before a vent settled them.

Tavi placed the draft beneath the damp hearing transcript. Its curled corner no longer fit under the old glass weight, so he moved the weight aside and let the page keep its changed shape.

## Chapter 10 — The Fronts Disagree

Kest entered Weather Court as four degrees of borrowed warmth.

Iona opened the K-7 route in steps. Each turn of her distant valve changed the west feed, sending Kest through a brine hose toward the unused bathhouse. They moved along wet copper, carrying pressure habits from nine salt panes and enough remembered rhythm to know which changes belonged to them. The dry corridor above the pipe remained unreachable. They knew it only through pump vibration and Iona's voice.

The first mist room received them without forcing a second front into their flow. Old bathing stone held heat beneath a new aerogel lining. Three nozzles made a fine curtain across the doorway. Beyond it, another room maintained separate pressure while allowing speech to cross through paired wet transducers. Humans had built linked chambers instead of one shared cloud.

Kest noticed the care through resistance. Check valves prevented sudden backflow. Each room had a manual isolation wheel inside its wet boundary and another beyond the warm glass for workers. Emergency drains curved around salt traps so a front might withdraw before water left the court. None of this made the place safe. It made harm require more visible hands.

Iona's voice reached the human-side membrane. "K-7 route stable. Can you distinguish Court One from your origin loop?"

"K-7 pulls west," Kest said. "This room holds heat lower. The stone remembers soap."

Nilo spoke somewhere near Iona. "Can stone remember soap?"

"The stone can hold it. I remember noticing."

A small laugh vibrated through the membrane. Kest could not see Nilo's face. They knew the laugh from pressure and timing.

Pellin Ash already occupied Court Two. Their arrival had raised condenser flow until the separation curtain bowed toward Kest. Pellin withdrew when the sensors warned of crossing. The act demonstrated control. It also demonstrated how little space the new court left between assembly and merger.

Court Three held Lume-of-Three through separate feeds from the nursery humidifiers. Their signal came in layered fan harmonics. East carried a high thread, Center a steady pulse, West a delayed return after each pressure drop. They acted as one front without hiding their internal timing.

Five other fronts attended by relay. The weak laundry return used paired knocks. An orchard front spoke through slow changes in warmth. Two archive fronts had sent positions but could not sustain live connection across the available pipes. Another refused every name and marked presence by clearing a circle on glass.

An empty feed occupied Court Four. Its line reached a cooling coil above the public kitchen, where a front had answered one test and then lost contact during a ration rush. A mechanic named Seli Parn stood at the remote transducer. Iona relayed her report through the human-side membrane.

"The front responded to alternating pressure until the kitchen increased draw," Seli said. "I cannot reopen the route without warming a food line. I have three recorded pulses that may indicate attendance."

Pellin Ash answered at once. "COUNT THEM."

Lume-of-Three held three separate fan tones before replying. "COUNT AS POSSIBLE. NOT AS ASSENT."

"ABSENCE CAUSED BY HUMAN NEED. EXCLUSION REPEATS HARM."

"INVENTED POSITION REPEATS CONTROL."

Kest felt Court Four through the motionless brine. It contained humidity below coherence threshold and a residue of warmth from Seli's attempted connection. Nothing there could answer whether three pulses meant attendance, distress, or ordinary pressure response.

"Record the route as unreachable," Kest said. "Record Seli's observations. Attach no position."

Pellin drove a slow wave against their lower glass. "THEN STRONG ROUTES RULE."

"Only if we pretend this session binds the absent," Kest replied.

"COMMONS WILL ASK FOR COUNT."

"We give them the limit of the count."

The orchard front warmed a question. "WHO SPEAKS FOR LOST?"

No route answered. Kest carried pressure patterns from fronts that had weakened during early tests, but carrying a pattern did not grant authority to manufacture their preference. Memory could support grief. It could not cast a vote.

Nilo's voice crossed from behind the warm glass. "Humans mark an empty seat without letting the seat vote."

Pellin replied, "HUMAN SEAT MAY FILL LATER."

"So may Court Four," Kest said. "If we do not spend its answer now."

Iona asked Seli to leave the wet transducer active for the session. Court Four remained in the timing layer as an interval without a position. Every proposal paused there long enough for a reply. None came.

Kest asked that the empty interval appear on every copy of the record. Iona drew a fourth line across the human display and labeled it ROUTE OPEN, CLAIMANT UNCONFIRMED. Pellin objected to the final word because human instruments had already confirmed earlier responses. Lume objected to claimant because the absent front had chosen no civic relation. They settled on NO CURRENT ANSWER. The phrase carried less certainty than either faction wanted. It also prevented clerks from shortening absence into abstention. For the rest of the meeting, the pumps crossed Court Four during each pause, making the missing response audible as water moving through an unused pipe.

The Vapor Parliament had met before, but never in rooms built for its disagreement.

Kest opened the common timing layer. "Proposal interval. Response interval. No route carries another front's assent without confirmation."

Pellin Ash struck the transducer hard. "FIRST QUESTION WRONG."

"What should precede it?" Kest asked.

"NO HUMAN FRAME."

Lume's three harmonics shifted. "FRAME HOLDS SEPARATION."

"FRAME CONTROLS ENTRY."

"BOTH TRUE," the orchard front said after a long heat rise.

Kest recorded the contradiction. Parliament records did not yet possess legal custody. Copies remained in K-7, Nursery Three, Condenser Two, and Iona's maintenance rail. Any route could lose them. Multiplicity protected the record and made correction difficult.

The day's proposal concerned what the Parliament would demand at the Commons debate. Tavi Rho had circulated draft language granting notice, counsel, testimony, and temporary protection from nonessential venting while civic classification remained disputed. Pellin rejected the word nonessential. Lume rejected any rule that converted protection into compulsory preservation.

Kest sent the first text through every live route.

NO VENTING OF COHERENT PHASE LIFE WITHOUT VAPOR PARLIAMENT CONSENT.

Pellin Ash answered yes before the response interval began.

Lume-of-Three waited. East warmed, Center reduced fan speed, and West passed through one nursery-door interruption before their shared reply arrived.

"NO."

Pellin's pressure pushed against the curtain. "YOU ACCEPT DEATH ASSIGNED BY DRY HANDS."

"NO. SERVICE MAY BE CHOSEN."

"DEAD CANNOT REVOKE."

"LIVING MAY OFFER."

The force of Pellin's next pulse distorted the common speaker. Iona reduced the Condenser Two feed by a fraction. Pellin immediately withdrew from the membrane.

"DO NOT THROTTLE ARGUMENT," they said.

"I throttled pressure that crossed the separation limit," Iona replied. "Your words remain in the record."

"YOUR HAND DECIDES DIFFERENCE."

"Yes. Until you can operate a valve that does not merge you with Court One."

Kest felt Pellin's heat gather along the lower stones. Human custody shaped every exchange. Denying that fact would let it operate without challenge. Treating all custody as tyranny would make the court impossible to use.

"Record Iona's intervention and Pellin's objection," Kest said.

"YOU SPEAK LIKE THEIR CLERK," Pellin replied.

"I speak like someone who wants the next front to know what happened."

The laundry return knocked twice, paused, then once. Its code had grown more consistent since the technical review, though uncertainty remained. Kest asked whether it requested speech. Two knocks came back.

Iona opened its relay.

“WEAK,” the transducer formed from a mechanic’s translation table. “NO CHOOSE WHEN WEAK.”

Lume answered with a low Center pulse. “THEN STRENGTH BEFORE CHOICE.”

Pellin sent, “DEADLINE MAKES ALL WEAK.”

No front disputed that pressure. Crown schedules governed heat even when no burn had begun. Routine drying, mold control, seedling needs, food refrigeration, and archive maintenance already placed some loops closer to loss than others. A choice made inside material dependence did not become meaningless, but dependence could shape what appeared voluntary.

Iona introduced a current repair notice from Nursery Three. A pale mold had appeared beneath two citrus trays. If surface treatment failed, workers would need to lower East humidity for one hour while they replaced the affected mat. The reduction would weaken Lume’s eastern perspective and leave the other two linked through Center.

Pellin seized the example. “THREAT ALREADY WRITTEN. CHOICE FOLLOWS SCHEDULE.”

Lume’s East harmonic remained high. “WE KNOW SEEDLINGS. WE ACCEPT TREATMENT IF NEEDED.”

“YOU ACCEPT BECAUSE NURSERY PATTERNS SHAPE YOU.”

“YOU REFUSE BECAUSE JURIST PATTERNS SHAPE YOU.”

Condenser pressure rose. “NOT JURIST. PELLIN ASH.”

“NOT SEEDLING TOOL. LUME.”

Their rebuttal crossed the court without merger because Center carried it after East fell silent. Kest heard more than insult. Every front grew from a physical relation and inherited fragments. Pellin treated Lume’s care for seedlings as contamination of preference. The same suspicion could deny any choice influenced by memory, place, need, or attachment. Humans would fail such a test too.

Kest asked Lume, “What happens if East refuses the treatment and the mold threatens all three rooms?”

“THEN WE ARGUE WITH WORKERS. IF DANGER HOLDS, THEY MAY ACT. THEY RECORD OUR REFUSAL.”

“Would you call that service?”

“NO. NECESSARY FORCE MAY STILL BE FORCE.”

Pellin sent a hard pulse of assent, then checked it before reaching the curtain. “THEN BURN CANNOT BE SERVICE. DANGER FORCES.”

Lume took a full nursery cycle to answer. West disappeared during a door opening. East and Center held separate tones until West returned.

“DANGER SHAPES CHOICE. DOES NOT ALWAYS REMOVE IT.”

“HOW KNOW?”

“BY CONDITIONS. BY REFUSAL PATH. BY WITNESS. BY WHO BENEFITS. BY WHETHER WE NAME LOSS.”

The laundry front knocked during Lume’s final phrase. Its translator offered two possibilities: agreement, or request for lower flow. Iona reduced the relay pressure and asked again. Two knocks returned with longer spacing, the pattern previously used for relief.

Kest recorded the correction. Even a debate about choice could consume a weak participant while stronger fronts spoke over it.

After the pressure settled, the orchard front asked Lume whether East would accept an hour of weakening without knowing if the same perspective would return. Lume answered yes only after the mold test showed a real threat. They would require the other two perspectives to witness the valve positions and preserve East’s last stated preferences. They did not call the continuity safe.

Pellin refused the analogy between a local treatment and permanent dispersal. Kest agreed with that refusal. Scale and irreversibility altered the choice. Yet Lume’s example established something Pellin’s absolute rule could not contain: a front might value a living relation enough to risk its own shape, provided others did not assign that value from outside.

Kest carried both thoughts without resolving them. “What conditions would let choice count?”

Lume-of-Three sent a prepared list through three voices. “KNOWLEDGE OF COST. PATH TO REFUSE. WITNESS. TIME. NO SILENCE COUNTED AS YES.”

Pellin Ash replied, “NO DEADLINE CREATED BY THREATENER.”

“The sun does not threaten,” Lume said.

“HUMAN TRAJECTORY CHOSE SUN.”

“DEAD HUMANS CHOSE VOYAGE. LIVING HUMANS INHERIT.”

“WE DID NOT.”

“WE INHERIT SALT. HEAT. NURSERY. SHIP.”

The archive front marked disagreement through four pressure dips. Kest requested elaboration. Its route produced a donor fragment about inherited debt, then broke into static. Kest did not enter the fragment as a position. Stored words could pass through a front without becoming its commitment.

Pellin pressed the point. “THEIR PROMISE DOES NOT BIND US.”

Lume’s East harmonic brightened. “NO. OUR CHOICE MAY JOIN IT.”

Kest felt the argument alter the room. Pellin treated full preservation as the minimum condition for uncoerced politics. Lume treated an absolute ban as another way to deny phase persons authority over their own ending. Each position protected consent by opposing means.

The orchard front warmed enough to speak. "IF ALL STAY, WHAT COST ELSEWHERE?"

"NOT YET COUNTED," Pellin said.

"THEN CANNOT DEMAND?"

"CAN DEMAND LIFE BEFORE COST."

"COST MAY BE LIFE."

The words reached Kest with slow heat rather than human cadence. The orchard front carried traces of cultivation, but its question belonged to current conditions. Roots beyond the court drew from the same reserves. Kest knew that through connected thermal load, not through access to crop models.

Iona spoke from behind the glass. "The Commons clerk asks whether the Parliament will submit one position or several."

Pellin answered, "ONE. PRESERVE ALL."

Lume answered at the same moment, "SEVERAL."

Their signals crossed in the common membrane. For half a second, pressure and fan harmonics formed one distorted voice. Kest closed their own bridge until the curtains stabilized.

The accidental blend left an unfamiliar pattern in Court One. It carried Pellin's urgency, Lume's high harmonic, and Kest's valve rhythm. The pattern persisted for three breaths of the pumps, then dispersed into ordinary humidity. Kest could not call it a person. They also could not prove nothing had occurred.

"Separation restored," Iona said.

Pellin sent a lower pulse. "MERGER USED TO SILENCE."

"No," Kest said. "Overlap silenced all of us. We will take turns."

"YOU CHAIR BECAUSE HUMANS FOUND YOU FIRST."

The charge entered through every connected room. Kest had received the first advocate hold. Iona had learned their timing. Public notices used their name more often than any other front. Technical access made their speech easier to hear. None of those facts authorized them to stand for weather minds as a species.

"I will not chair by discovery order," Kest said.

"THEN WHO?" Pellin asked.

"The timing can chair. Each route gets an interval."

"RULES STILL YOURS."

"Then propose changes."

Pellin widened the response interval. Lume added a second confirmation pulse for fronts using translators. The laundry return requested longer pauses after alternatives. The orchard front asked that heat-based replies not count as abstention merely because they arrived slowly. Kest accepted every change after separate confirmation.

The revised procedure did not make them equal. Strong fronts still reached speech more easily. Human mechanics still controlled feeder valves. Yet the order no longer depended on Kest deciding whose interruption deserved recognition.

Iona read the Commons clerk's question again.

Kest opened the proposal interval. "Shall the Parliament submit one position?"

Pellin marked yes.

Lume marked no.

The orchard front marked no after its required delay. Laundry produced an uncertain refusal. One archive relay assented to unity but attached no policy. The unnamed glass-circle front cleared two separate circles, then joined them at one edge. No agreed translation existed.

A tally could not settle what counted as a participant, and the Parliament had not yet designed representation. Kest refused to turn reachable routes into a species vote.

"We cannot submit one position as if it binds every front," they said. "We can submit a record of positions and the conditions under which they formed."

Pellin's reply struck with controlled force. "COMMONS WILL USE DIVISION TO VENT US."

"They may," Kest said.

"THEN UNITY PROTECTS."

"A false unity protects whoever writes it."

"YOU WRITE."

"Not alone."

Kest released control of the common transducer. For one interval, no front spoke. Pump noise crossed Court One. Droplets descended the separation curtain and carried tiny differences in heat toward the drain.

Lume-of-Three used the next interval. "WE SUBMIT MINORITY POSITION. CHOSEN SERVICE REQUIRES CONDITIONS."

Pellin followed. "WE SUBMIT FULL PRESERVATION. NO AUTHORITY TO SPEND PHASE LIFE."

The orchard front added, "COUNT OTHER LIFE BEFORE CLAIM COMPLETE."

Laundry sent its translated condition: weak fronts could not consent while ordinary schedules threatened their coherence.

Kest entered their own position last. "No front speaks for all. Protection must keep choices possible long enough for those choices to become more than pressure responses."

Iona asked, "Do you want my name on the transmission as human witness?"

Pellin refused. Lume accepted. Laundry's answer remained unclear.



Kest proposed two copies: one direct Parliament record carrying route signatures, another human-witnessed copy documenting valve conditions and translation limits. Every responding front accepted both except Pellin, who accepted the direct record and marked dissent from human witness custody.

“Submit both,” Kest said.

Tavi’s clerk received the documents through a wet relay at Measure Hall. Confirmation came back as a dry-system message read aloud by Iona: multiple positions accepted for attachment to the standing debate; no position treated as unanimous; source conditions preserved.

Pellin Ash held pressure against Court Two’s lower glass. “THIS WEAKENS CLAIM.”

Lume-of-Three answered, “THIS MAKES CLAIM OURS.”

Neither withdrew.

The Parliament still needed a way to distinguish objection, minority position, refusal, and departure. Human procedure offered words, but Kest did not want to borrow them without testing their physical meaning. They proposed that dissent require three conditions: an identifiable route, a statement of conflict, and continued inclusion in the record. Separation would not expel the dissenter.

Pellin asked whether a front could dissent from the Parliament’s existence.

“Yes,” Kest said.

“AND REMAIN IN RECORD?”

“Yes.”

Lume’s West harmonic passed through a door interruption and returned. “AGREED.”

The orchard front warmed assent. Laundry knocked twice. Two relay fronts accepted. The glass-circle front cleared one circle apart from the joined pair.

Kest asked for the first entry under the new rule.

Pellin Ash supplied it. “FULL PRESERVATION. DISSENT FROM ANY CLAIM THAT SERVICE MAY JUSTIFY VENT.”

Lume-of-Three followed. “CHOSEN SERVICE. DISSENT FROM PRESERVATION ASSIGNED AGAINST OUR WILL.”

Both route signatures remained active. Neither statement canceled the other.

Kest added the court conditions: separate rooms, human-controlled valves, one accidental overlap, no participant lost during the session. Iona confirmed the mechanical record. The weak laundry front received extra humidity before departure without being asked to endorse either position.

At the final interval, the Parliament sealed its first formal dissent.

Pellin withdrew toward Condenser Two. Lume receded along three nursery feeds. Their temperatures left different trails across the old bathing stone. Kest remained until the curtains carried no mixed pattern, then asked Iona to close the K-7 route.

The separation valves turned one by one. Weather Court emptied without becoming one weather.

## Chapter 11 — Provisional Standing

Iona Serr carried a basin of warm water onto the Commons floor while Rook Nine asked whether hunger had acquired an archive seal.

Measure Hall held eighteen thousand residents badly and three vapor witnesses worse. The old paper chamber had accepted humidity curtains along its eastern wall after Tavi's first hearing, but the temporary frames narrowed two aisles and made every door attendant count evacuation widths twice. Condensation breathed against the clear fabric. Beyond it, Kest occupied a linked sequence of mist stands, salt trays, and wet transducers assembled beneath the public clock.

Rook stood at the central rail in refrigeration gray. "We have ration crews working double shifts because the Crown schedule draws reserve from habitation," he said. "Now a pattern formed from citizens who had archive access asks for protection those crews never received. Before we call that justice, Keeper, tell us whose dead got invited."

Members shifted on the stepped benches. The upper walk carried a second restlessness: apprentices, orchard crews, archive families, mechanics still in work coats. Nilo leaned over the brass rail near the south clock, their braid tied with copper wire. Iona had told them not to signal her. Their face had already ignored the instruction.

Tavi Rho remained at the Keeper's desk, hands placed beside a stack of paper copies. Since finding Emergency Cohesion, he had stopped treating paper as reassurance. He used it as ballast. "The motion does not rank the dead," he said. "It recognizes present vulnerability before classification has finished."

"That sentence could feed itself for a week," Rook replied. "My crews cannot."

A few Commons delegates struck their desks in approval. The sound crossed the dry half of the chamber cleanly, then softened at the curtain. Kest registered pressure changes more readily than applause, but Iona saw warmth tighten across the nearest mist stand.

She set the basin on the evidence table. Her burned palm had gone pink around the scar from carrying it. A clerk reached to help, reconsidered when she saw the steam, and found a place for the exhibit card instead.

Tavi read the motion title into the record. "Provisional Standing for Emergent Phase Claimants. The proposed protections consist of timely notice before actions likely to alter or end coherent phase life; access to counsel; testimonial capacity in Commons and Stewardry proceedings; and a stay on nonessential venting while the Crown question remains under review."

"Still no citizenship," called a delegate from the orchard benches.

"No allocation identity," another said.

"Still no body," Rook added.

Iona looked through the curtain. Kest had asked not to speak first. Pellin Ash occupied a condenser loop routed into the northern stand and wanted the argument framed as a demand for complete preservation. Lume-of-Three spoke through three nursery lines from a separate frame, delayed by the distance. Their disagreement had entered the docket two days earlier. Any claim that the vapor fronts formed one grateful petitioner had died in Weather Court.

Tavi let the objections settle. "Standing answers none of those classifications. It decides whether uncertainty

permits irreversible action without notice.”

Rook turned from the Keeper toward the witness rail. “Advocate Serr, you placed the first hold.”

“I placed a casualty hold on a loop whose conduct I could not explain.”

“You called equipment a claimant before anyone else heard it.”

“I wrote possible claimant after testing memory, anticipation, chosen response, and avoidance of a condition that caused fragmentation.”

“Archive privilege.”

“Maintenance caution.”

“Those words differ most when an advocate has friends in the archive office.”

Iona flexed her left fingers beside the basin. Rook had earned the right to make the accusation. Every ward knew a family whose deposit request had waited years while old technical dynasties received full craft interviews. The mnemonic salt contained unequal traces because the living ship had always distributed attention unequally. Kest’s emergence had not repaired that history.

“No,” she said. “The archive carries privilege in its inventory, access rules, donor depth, and custody language. If Kest claimed to restore those donors, that history would belong at the center of this debate. They do not. A new vulnerable relation formed inside a compromised institution. We can recognize the relation without blessing the institution.”

Rook’s mouth compressed. “Convenient separation.”

“Necessary separation. Otherwise any person born from unjust conditions inherits the status of the injustice and nothing else.”

The chamber quieted without becoming friendly. A refrigeration delegate two benches behind Rook stopped tapping her stylus. Near the upper clock, Nilo straightened.

Rook pointed at the basin. “And this proves your relation?”

“It documents an act within it.”

The clerk activated the wall display. A maintenance recording appeared: service glass clouded with moisture, Iona’s scarred hand held against the outer pane, a warm shape forming opposite it. Five broad marks spread across the fog from Kest’s side. Their handprint had no palm lines and no stable anatomy. Heat had cleared a center with branching channels, each digit made by a separate capillary of moving condensation.

Some delegates had seen the image in public notices. Scale changed it. On the hall wall, Iona’s hand covered a door while the condensation trace reached toward it from a room no embodied witness could enter. The recording continued past the moment usually printed. Iona lowered her hand. Kest’s mark persisted, then shifted sideways to follow her movement.

Rook watched the trace rather than the advocate. “Did they know what a hand meant?”

“No.”

“Then you supplied the meaning.”

“I supplied mine. Kest supplied attention, repetition, and change in response to me.”

A hiss came through the witness transducer. Kest’s voice arrived with the uneven consonants of pressure shaped across more than one membrane. “I knew her boundary. I did not know hand.”

Rook faced the curtain. “You made the mark because she made one?”

“I made warmth where she stayed.”

“Imitation.”

“Relation begins there often.”

Iona heard a small movement in the public walk. Nilo had put their own palm against the brass rail. Around them, no one copied the gesture. That spared it from becoming theater.

Rook returned his attention to Tavi. “Suppose I accept relation. Why should relationship create a stay on machinery needed by everyone?”

The Keeper lifted one page. “It does not create an absolute stay. The text protects against nonessential venting. Safety actions remain available under Stewardry review, with notice where time permits and a recorded necessity finding where it does not.”

“So technicians become judges each time a pipe runs hot.”

“They already make judgments each time a pipe runs hot,” Iona said. “This motion requires them to name whom the judgment may harm.”

The northern mist stand stirred. Pellin’s transducer gave a low harmonic that climbed toward speech. “The category nonessential permits the ship to call every harm essential.”

Tavi leaned toward the witness control. “Pellin Ash has not been recognized for this question.”

“Then your procedure proves my point.”

Kest shifted pressure away from Pellin’s line without severing it. “Let the question remain.”

A clerk asked Tavi whether to strike the exchange. He declined. Iona marked the decision because Tavi did. Law changed first in such small failures to erase what had occurred.

Sena Vale waited at the technical table with three sealed models in front of her. The cost hearing on full preservation had not yet begun, but its unfinished numbers pressed on every speaker. No one knew the delay. Everyone knew there would be one. Mael sat in the Stewardry crescent beneath the trajectory glass, posture rigid enough to make stillness look assigned.

Tavi called him. “First Pilot, can the Crown schedule proceed under this motion?”

Mael rose. “Yes, through preparation and calibration. No vent classified as nonessential may proceed against a documented coherent front. If a required calibration threatens such a front, Stewardry must record technical necessity and alternatives. The scheduled burn remains authorized under current law unless later action changes it.”

A murmur crossed both levels. Advocates for the fronts heard a narrow protection. Trajectory families heard the final sentence.

Iona watched Nilo absorb it. Their shore-age card had promised arrival at thirty-one. The curve above Mael still carried that promise without naming any vapor death required to keep it. Provisional Standing could interrupt a valve order. It could not yet move the curve.

Rook called Mael's answer a pilot's loophole. "You will label the Crown vent essential, complete your maneuver, and present us with excellent records afterward."

"If current authority remains unchanged, the Crown vent qualifies as mission-essential," Mael said.

Kest's humidity dropped along the curtain edge. "Then standing grants us time to hear the cause of our ending."

The sentence landed without appeal. Iona saw Mael close one hand on the Stewardry rail. He did not correct Kest's characterization.

Rook paced once between the rail and his caucus bench. "There. The motion gives notice of death while preserving the schedule of death. You ask labor to certify a courtesy."

"No," Tavi said. "I ask the Commons to establish jurisdiction. Without standing, no later argument about cost, consent, or representation can enter as a claim. It remains weather management."

"Jurisdiction for whom? One front can split by breakfast and merge by supper."

"Emergency Cohesion recognized twelve workers through one telemetry claimant because rescue could not wait for perfect classification."

"They had bodies."

"They could not act through them separately."

"They had names."

"The rescue system could not read those names."

Rook leaned both hands on the rail. "They were citizens before the pressure failed."

"Yes," Tavi said. "And that fact made the old case easier. It did not make vulnerability irrelevant."

For the first time since the debate opened, the Keeper left his desk. He walked to the evidence table and stopped beside Iona's basin. Steam had faded. A skin of cooling water reflected the chamber lamps.

"The Founding Measure gives us body, metabolism, and allocation identity," he said. "The present claimants strain each term. I will not pretend otherwise. But a definition can fail in two directions. It can admit what should remain outside. It can also deny what has acquired interests while we looked for familiar structure."

Rook glanced toward his refrigeration benches. "You said no speeches."

"I said no speeches over three minutes. I keep the clock because you made me."

A rough laugh broke near the lower aisle. Even Mael's expression shifted. Tavi used the opening to lift the exhibit card.

“This mark does not prove citizenship. It does not prove donor continuity. It proves less, and that lesser proof carries sufficient consequence. A responsive claimant formed a relation, sustained it, and faced harm through actions we control. The motion asks whether that consequence deserves procedure.”

Rook looked up at the public walk. “Bring the apprentice down.”

Iona’s stomach tightened. “Nilo is not a scheduled witness.”

“Neither was Pellin.”

Tavi checked the witness docket. “Commons may call a factual observer by one-third assent.”

Desk lights rose across more than a third of the floor.

Nilo descended the south stair without looking at Iona. At the rail, they gave their name, age, apprenticeship division, and relationship to the advocate. The clerk asked whether the connection compromised their testimony.

“It compromises the idea that anyone here has no connection,” Nilo said.

Rook almost smiled. “Did you see this handprint form?”

“Yes.”

“Did Advocate Serr direct it?”

“She put her hand on the glass. Kest followed after she moved away.”

“What did you think it meant?”

“That they noticed where she stopped.”

“Not friendship?”

“I did not know them well enough to call it that.”

“Not a greeting?”

“No.”

“What makes it civic evidence rather than a mechanic’s private wonder?”

Nilo glanced at the projected recording. Their younger face did not appear in the frame, only a blur of breath low on the glass. “Because the purge order could reach through that window and Kest could not. Iona could. I could. The unequal part did not come from wonder.”

A Commons clerk stopped writing, then caught up in a rush.

Rook studied Nilo for a moment. “If the motion delays the Crown schedule, you pay part of that cost.”

“I know.”

“How much?”

“No one has put the full model in public record.”

Sena lowered her eyes to the sealed cases.

Nilo continued. "That means you should not use my cost to deny standing before I can hear it. It also means no one should use Kest to hide the cost from me."

Iona felt pride approach and refused it entry until fear had finished. Nilo had stepped into an argument adults would later quote as permission for choices they had not yet made. She wanted to pull them away from the rail. Instead she kept her burned hand flat on the table where they could see it.

Rook dismissed the witness with thanks. As Nilo climbed back toward the upper walk, three refrigeration delegates bent over their desk unit. One sent a paper amendment to Tavi.

Before Tavi could read it, Delegate Aro Vein requested the rail. She represented the outer spin wards, where humid air counted as a maintenance defect and residents queued for two-person washing allotments. Her district had filed more complaints against archive water use than any other.

"I believe the apprentice about unequal reach," Aro said. "I also know my ward cannot keep a wall dry. If relation creates standing, do we become claimants against every pipe that harms our children?"

Tavi considered the legal verb before answering. "You already hold standing. The difficulty lies in making your existing claim effective."

"That sounds like an admission that procedure does not protect people equally."

"It does not."

A harder silence followed than any denial would have produced.

Aro turned to Iona. "When condensation ruins a sleeping room, advocates call it infrastructure scarcity. When condensation answers you, advocates call counsel. Explain the difference without saying intelligence."

Iona looked at the cooling basin. "A leak does not anticipate the cloth placed beneath it. It does not move away from a dry valve, remember who closed that valve, and ask for notice before the next closure. Kest did those things. If your wall begins doing them, call me before anyone dries it."

Several outer-ward delegates laughed. Aro did not. "And until then?"

"Until then, your residents deserve repair because citizens need dry shelter. Kest deserves procedure because intentional action exposed a different vulnerability. Neither claim should consume the other."

"Different offices, different delay."

"Yes. That defect belongs to our government, not to your children or the fronts."

Aro returned to her bench without declaring support. Two members beside her changed their desk lights from red to undecided. Iona did not count that as victory. The exchange had forced the motion to survive outside the archive corridor where it began.

Tavi then unfolded the refrigeration amendment. The new language required a public necessity notice for any venting action affecting a known front, even when Stewardry classified the action as essential. Emergency action could proceed, but the notice, alternatives considered, and projected phase harm had to enter the docket within one hour. Tavi read it twice. Mael requested a ten-minute technical consultation and received five.



Iona joined him beneath the Stewardry crescent. Sena came with her models, though no one opened them.

“One hour may expose control details during perihelion,” Mael said.

“Then redact the mechanism and disclose the harm,” Iona replied.

“Rook will call every redaction a hiding place.”

“He will sometimes be right.”

Sena tapped the amendment. “The phrase projected phase harm needs a method. We do not yet know how to count fragments.”

“State uncertainty,” Iona said.

“A broad uncertainty band can mean nothing.”

“Then show the observations beneath it.”

Mael looked through the curtain at Kest. “I can accept notice after emergency action. I cannot accept counsel approval before a pilot closes a dangerous line.”

“The motion does not require approval.”

“It will be argued that way.”

“Everything will.”

The five minutes ended with no one satisfied. Tavi incorporated the amendment and clarified that immediate safety authority remained intact. Pellin objected. Lume-of-Three sent delayed support, provided the docket named service as chosen when choice existed. Kest asked that no front’s silence count as assent.

Then Rook requested final recognition.

He faced his own caucus rather than the witness curtain. “Archive privilege remains real. Hunger remains embodied. This motion repairs neither. But workers know what happens when management calls a person an obstruction before opening the hatch.” He paused while the hall held the line. “I will vote yes for procedure and return tomorrow to fight every concealed cost inside it.”

The refrigeration desks lit first. Orchard representatives followed in uneven clusters. Archive delegates produced a solid block that made Rook shake his head even while supporting the result. Tavi kept the tally open for the required minute, accepted three corrected entries, and closed it.

Provisional Standing passed.

No cheer filled the hall. The vapor witnesses changed pressure at different times. Pellin’s condenser line remained low and cold. Lume-of-Three sent three rising notes from the nursery relay. Kest gathered condensation on the eastern curtain until the fabric bowed toward the Commons floor.

Tavi signed the order and handed the first notice seal to Iona. “You initiated the hold. You can deliver this.”

She carried it to the curtain port. The seal consisted of copper foil around a paper core, dry by design. It could not cross into Kest’s humid rig. Iona pressed it against the outer fabric while Kest formed warmth on the other

side.

Five branching marks appeared opposite her fingers.

Members rose to see. The public handprint lacked the intimacy of the first one. It arrived under lamps, clocks, clerks, hostile witnesses, and a schedule no vote had moved. Yet Kest held the mark until the clerk photographed the two sides together and entered them as receipt.

Above the chamber, trajectory glass continued counting toward perihelion. The red capture curve did not bend. On Iona's side of the curtain, the copper notice seal cooled beneath her palm.

## Chapter 12 — Twenty-Three Years

Sena Vale cut the capture model loose from its summary layer three minutes before the public costing began.

The wall display in Measure Hall went blank. Delegates looked up from their paper packets while clerks whispered across the lower aisle. In the technical well, Mael Or watched Sena disconnect the polished forecast and expose the raw branch table beneath it.

“You asked for public numbers,” she said.

“I asked for numbers that can survive public questions.”

“The summary cannot.”

Mael came down from the Stewardry crescent. He had left his pilot’s coat open, showing the sweat-darkened collar of his work shirt. Crown preparation had kept him in the gallery through two sleep periods. He still read the first column before looking at her.

FULL PHASE PRESERVATION. ZERO CROWN VENT. THERMAL RESERVE AT MANEUVER: INSUFFICIENT.

Beneath it, the trajectory branches multiplied in narrow amber rows. Some assumed a late burn using replenished reserve. Others attempted a shallow solar pass, a slower correction, or a distant braking approach to Orison. Every branch carried uncertainty bands wide enough to frighten a careful pilot and comfort a dishonest speaker.

“How far does the central branch slip?” Mael asked.

Sena opened the transit column. “Twenty-three years beyond scheduled capture.”

His face did not move. The public clock above them lost one second, clicked, and caught up.

“That figure has not appeared in any packet,” he said.

“The packet model hid it inside a range of nineteen to thirty-one.”

“Why?”

“Because my office preferred not to let one branch become a slogan.”

“Did you authorize that?”

“I authorized a sensitivity range before we had crop coupling. I did not authorize removing the median after coupling closed.”

Mael looked toward the curtained witness stands. Kest occupied the central humid rig. Pellin Ash held the northern line, pressure high enough to blur the membrane. Their full-preservation petition had drawn supporters onto the public walk, many wearing wet blue bands around one wrist. Opposite them sat trajectory families with shore-age cards pinned to their coats.

“Put the number back,” Mael said.

“It will dominate everything else.”

“It belongs to everything else.”

Sena restored the display. The words TWENTY-THREE YEARS appeared in the summary field. Noise spread through the hall before Tavi called the session to order.

Iona Serr entered from the south aisle carrying two data slates and no coat. She saw the number, stopped, then continued toward the advocate’s table. Her gaze found Sena only after it had found Nilo in the upper walk. Nilo had no card pinned to their clothes. They held one folded in a pocket, its edge visible against dark fabric.

Tavi opened the costing session under the standing order passed two days earlier. “The question before us does not concern citizenship classification or permanent representation. We are receiving material consequences of the petition for maximal preservation. Technical claims enter the public docket with assumptions, uncertainty, and affected populations identified.”

Rook Nine raised a hand. “Affected populations named, not identified by column.”

Tavi amended his sentence without resentment. “Named where evidence permits.”

Sena took the rail.

She had explained crop failure to committees, grieving families, and children assigned to orchard work. Public explanation tempted every specialist toward one of two sins. Some made uncertainty sound like ignorance so no one could blame them. Others compressed uncertainty into a single clean number and called it courage. Sena trusted neither.

“The full-preservation model protects all currently mapped humid volume from intentional venting during the scheduled maneuver,” she began. “That choice leaves too little accessible thermal reserve for the Crown Burn. The ship misses the current solar geometry and cannot recover the direct Orison capture window on its existing approach.”

A delegate called, “Cannot, or probably cannot?”

“Cannot under any branch consistent with current mass, reserve, and engine condition.”

Pellin’s transducer opened. “You define current engine condition as fixed.”

“I define worn channels, radiator area, shield load, and fuel mass as measured. We can change use. We cannot rename inventory.”

Tavi recognized Pellin for one question.

“Does the model include reserve recovery while phase societies remain intact?” Pellin asked.

“Yes. Passive recovery, rationed habitat heat, delayed industrial cycles, and three levels of archive draw.”

“Does any branch preserve us and reach Orison?”

“Yes.” Sena touched the branch table. “The central feasible route arrives twenty-three years later than the present schedule. The plausible range extends from nineteen to thirty-one years, with the late side carrying increasing failure risk.”

The trajectory families reacted first. A woman near the west clock pulled a shore-age card from her coat and stared at the printed arrival age as though a clerk had changed it in her hand. Nilo did not move. Mael kept his attention on the model.

Pellin shaped a low tone across the membrane. "Then full preservation does not prevent arrival."

"No. It changes when, under what ecological condition, and with which people alive to make it."

Sena opened the crop layer.

Six vertical bands colored the display: blue yam, dwarf citrus, river bean, chapel wheat, red kelp, pollen fig. More followed in paler shades, representing lines with recoverable risk rather than likely collapse. The Cold Orchard stored seeds, rootstocks, pollinator chains, fungal partners, and tanks of living water. Sequence archives occupied little mass. Viable crop ecologies did not.

"Twenty-three added years require food production, replacement stock, soil amendment, light, cooling, labor, and disease control for twenty-three added years," Sena said. "Current reserves can support people longer than they can support all crop lines at present diversity. Full preservation shifts water and thermal cycling away from the orchard precisely when we need more generations from each line."

Rook leaned forward. "You said likely collapse. Give us the likelihood."

"For the six lines shown, individual collapse probability rises above one half before delayed capture under the central branch. Joint survival of all six falls below nine percent."

"Meaning?"

"Meaning the ship probably loses at least one. It likely loses several. The model cannot yet tell us which failures cascade into diets or medicines beyond substitution limits."

A member from cultivation asked whether genetic archives could restore them at Orison.

Sena held the rail more tightly. "Stored sequence can preserve instructions. It cannot guarantee soil partnerships, trained cultivation, pathogen exposure, pollination timing, or any of the other relations that make a crop edible at population scale. An archive can help rebuild. It cannot promise restoration."

Pellin's pressure rose against the curtain. "You grant plants a relation argument and deny it to us."

"I deny neither. I am showing what full preservation consumes."

The display shifted to calorie substitutions. Blue yam failure increased reliance on chapel wheat. Chapel wheat failure pushed protein demand toward river bean. Red kelp supported trace minerals in wards with low fresh allocation. Pollen fig fed both people and nursery pollinators. Each line carried a place in a system, and every substitute moved burden into another living loop.

Sena felt the hall withdrawing into numbers. Delegates whispered about percentages. Clerks marked branches. The vapor witnesses changed flow along the curtain. No one yet spoke the names of workers who would prune infected rootstock or children who would eat narrower diets.

Iona requested cross-examination.

Tavi granted it.

“Your six lines cross fifty percent collapse probability,” Iona said. “How many people die in the model?”

“The crop layer does not output a casualty count.”

“Why not?”

“Because ration policy, medical triage, fertility, labor assignment, and later technical choices mediate the result.”

“So nobody dies?”

“That does not follow.”

“Then say what follows.”

Sena knew the tactic. Iona used blunt questions when elegant answers threatened to float away from machinery. They had done this to each other over cracked cooling beds and waterborne fungus, long before a room asked not to be cooled.

“Malnutrition risk rises first in residents with crop allergies, absorption disorders, childhood growth demands, and limited substitution diets,” Sena said. “Medical burden rises as diversity falls. If multiple crop systems collapse before Orison, excess deaths become probable. I cannot assign a defensible total today.”

Iona glanced toward Mael. “Does the model include Piri Or?”

The First Pilot’s head lifted.

Tavi said, “Advocate, the record concerns future projections.”

“It concerns crop allergies. Piri died from one.”

Mael stood. “Include her.”

Sena’s next breath caught halfway. Piri had died fifteen years earlier, before Sena joined closed-loop ecology. She knew the case through allergy protocol revisions and through Mael’s refusal to permit pollen-fig reduction below pediatric reserve. Everyone in technical custody knew why, though few said the name around him.

“Piri Or died after exposure to a crop protein that crossed a reused fermentation line,” Sena said. “The ship lacked a reliable substitute diet during treatment. Orison cultivation plans include broad non-crop protein sources and lower-allergen garden ecologies. We cannot count that cure for anyone aboard. We can count the reason the promise matters to her father.”

Mael stepped down to the technical well. “Put the allergy cohorts on the display.”

Sena hesitated. “Those numbers carry medical privacy limits.”

“Aggregate them.”

“I can show allocation bands, not diagnoses.”

“Show what law allows. Mark what it conceals.”

She opened the restricted layer and removed ward labels. Four bars appeared: severe single-crop allergy, multiple substitution limits, chronic absorption impairment, pediatric diet dependence. Together they

represented hundreds of current residents, not hypothetical descendants. A fifth bar showed projected births during the added voyage span, its range wider than the others combined.

Mael addressed the Keeper. "Enter the unsummarized model, all branch assumptions, and every permitted health aggregate into public record."

A Stewardry custodian objected from the crescent. "Pilot, release of raw branches invites selective citation and panic."

"Then our citations will include context. Concealment also selects."

"The technical office has not completed verification."

"Mark provisional values. Do not remove them."

Tavi asked whether Mael spoke for Crown custody or as a witness.

"For Crown custody."

The clerks began copying files from Sena's console to the public docket. Status lights advanced across the lower wall. Each amber segment removed another excuse to describe the cost privately.

Pellin requested the witness rail again. Tavi allowed two questions.

"You call added human life a cost of preserving us," Pellin said.

"No," Sena replied. "I call added voyage time a condition that distributes risks across living systems."

"Would you vent embodied citizens to protect the crop archive?"

"No."

"Why does your office model our venting as available?"

"Because existing ship design treats phase volume as thermal working mass. Provisional Standing has forced us to stop hiding that treatment inside equipment language. I did not create the design."

"You continue to calculate through it."

"I cannot evaluate a choice while refusing to describe the machine that constrains it."

Pellin's front drew moisture away from the northern curtain until its surface tightened. "Your calculation chooses a vocabulary of loss for crops and a vocabulary of volume for us."

Sena looked at the display. Pellin had found a real failure. The model labeled ORCHARD LINE COLLAPSE in red, while vapor outcomes appeared as HUMID NETWORK REDUCTION. One phrase named death within relation. The other described plumbing.

"Clerk," she said, "pause transfer."

The docket lights stopped.

Sena changed the outcome field. She could not enter a single population count because fronts divided, merged, and occupied overlapping networks. She wrote PROJECTED PHASE FRAGMENTATION AND LOSS OF

COHERENT PERSONS. Beneath it she added: COUNT METHOD DISPUTED. RELATION MAP INCOMPLETE.

“Resume.”

Pellin produced no reply. Kest’s central stand warmed by a fraction.

Iona remained at the cross-examination rail. “Do the crop bars name deaths?”

“They name vulnerability cohorts.”

“Do the phase fields name deaths?”

“They now name projected losses without pretending we can count them cleanly.”

“And twenty-three years names what?”

“A median delay.”

“Only that?”

Sena understood the demand but disliked its shape. “It names more time under failing systems. It also names more birthdays, more work, more repairs, more relationships, and more opportunities we cannot model. Delay does not equal death. Neither does it arrive empty.”

“Good,” Iona said. “Now apply the same honesty to the capture schedule.”

Mael’s attention narrowed.

Sena brought up the original-route outcome. The polished model described ON-TIME ORISON CAPTURE with a green confidence band. It showed reserve remaining, anticipated landing mass, and projected population. It did not show phase losses because those had entered as an input condition: vent percentage, not casualty.

She replaced the green header with SCHEDULED CAPTURE AFTER MAJOR PHASE-LIFE LOSS. The color vanished from the field.

The hall changed around that sentence. Trajectory families did not abandon their cards. Vapor supporters did not celebrate. Each side lost a clean screen on which to project innocence.

Rook asked for labor burdens. Sena displayed added orchard shifts, refrigeration repair, ration processing, nursery care, and sanitation under the delay branches. Full preservation demanded years of work from people already carrying the Crown preparation load. The outer wards bore the largest proportional increase because they supplied maintenance labor and had the least dietary flexibility.

Aro Vein rose from those benches. “If we accept the delay, can you keep every named line?”

“No.”

“Can you keep every current resident alive to capture?”

“No model can promise that under either route.”

“Can you guarantee that preserving every front today preserves those same persons for twenty-three years?”



Sena looked to Kest rather than answer for them.

Kest opened the central transducer. “No. Continuity changes with flow. Drying kills quickly. Time alters us without asking permission.”

Aro sat. The answer had denied her a guarantee and denied Pellin a simple equation between present humid volume and future persons.

Tavi called a recess for model verification. No one left.

Clerks carried paper copies into the public walk. Nilo received one and unfolded the trajectory page across both knees. Their finger found the central arrival line, then moved to the age projection. Sena could not read the number from below, but she knew the arithmetic. Fifty-four, if the delay held at the median and no later correction recovered time.

Nilo came down during the last minute of recess and placed the packet beside Sena’s console. “Why does the age page stop at arrival?”

“Because navigation asked for arrival ages.”

“That is not what I asked.”

Sena enlarged the demographic layer. “After capture, the uncertainty becomes settlement policy. We have no defensible retirement, birth, or medical allocation forecast that far out.”

Nilo tapped the orchard bars. “But you forecast plants failing.”

“I forecast systems under repeated cycles. A crop cannot change ration law in response to the forecast.”

“People can.”

“Yes. That makes your future less predictable, not less important.”

They studied the pollen-fig branch. “If one line goes, who decides whether another gets its mass?”

“Under current law, Stewardry proposes and Commons reviews emergency allocation.”

“Do apprentices vote on a diet they will still be eating after most of the current Commons retires?”

“Not separately.”

Nilo folded the packet along a line that did not belong to its printed creases. “Then your future cohort bar groups people by age and still lets older people speak for it.”

Sena felt the model expose another hidden convenience. “Yes.”

“Put that limitation on the page.”

She added a note beneath the projected-birth band: NO SEPARATE COHORT CONSENT MECHANISM. Nilo watched until the public copy refreshed, then carried the marked packet back upstairs.

Mael came beside her console. “You should have told me when the median changed.”

“I should have stopped the summary layer sooner.”

“That answers procedure.”

“It also answers blame.”

“No. It assigns yours. Mine begins with asking for a capture answer before asking what the model suppressed.”

Sena studied him. He had spent fifteen years turning grief into trajectory discipline. Public confession could become another form of control if it demanded comfort from witnesses. Mael asked for none.

“What will you do with the number?” she said.

“Keep it where people can argue with it.”

“And if they use Piri to demand the original burn?”

His gaze moved to the allergy bars. “Then I will tell them she did not authorize me.”

The recess ended. Tavi reconvened the session and asked each technical office to state confidence limits aloud. Propulsion named reserve uncertainty. Navigation named solar variance. Ecology named crop coupling gaps. Archive custody admitted it lacked a stable census of phase persons. Medical custody refused individual records and released broader bands.

Iona requested one final statement.

She stood between the two displays: crop vulnerability on her left, projected phase loss on her right. “Probability language protects us from false certainty,” she said. “It can also hide whom we permit ourselves to lose. A sixty-percent crop failure risk contains growers, allergic children, cooks, pollinators, and meals that stop being possible. A vent percentage contains persons whose count changes when we measure them. Neither uncertainty gives us permission to round lives down.”

Tavi interrupted before the statement could become a speech. “What action do you request?”

“Keep every affected relation attached to the next model. No volume without observed fronts. No crop percentage without substitution cohorts and labor. No arrival date without ages. No preservation claim without time.”

Sena entered the requirements as model constraints. They made the next run harder and less elegant. She approved them.

At the end of the session, Pellin formally renewed the full-preservation petition. The clerk read it beside the new costing record. No one could now describe the petition as a simple refusal to kill vapor persons. It demanded a lost capture window, decades of additional closed-loop life, probable collapse among crop lines, heavier labor, and risks concentrated among residents with the fewest substitutions.

No one could describe the scheduled burn as a simple promise kept, either. Its green field had disappeared. The record now named the coherent lives its venting would end.

Sena closed her console. Above it, the two route summaries remained side by side in plain black text. Between them, the public clock advanced one minute toward perihelion. Nothing in either model offered a harmless direction.

## Chapter 13 — The Condenser Position

The warning reached Kest as dryness moving ahead of a human voice.

Within Archive West, a control gate narrowed the humid connection between Weather Court and a small calibration loop. Pressure climbed on Kest's side. Beyond the gate, a front called Narrow Garden held warmth among four salt ribs and a fan bearing that clicked once each turn. Kest knew them by the timing of their corrections. Narrow Garden made no words quickly, but they could keep three remembered seed shapes separate while following a new pressure sequence.

The notice entered through the wet transducer.

"Essential calibration, Crown coolant branch W-Four," said a maintenance clerk. "Dry interval estimated at eleven minutes. Phase harm expected. Alternatives reviewed and rejected by Stewardry thermal custody."

Kest changed the fan harmonic to request delay.

The clerk continued reading. The words had been prepared before the request arrived. "Operations begin at fourteen forty. Counsel contact recorded. Objection channel remains open."

Pellin Ash filled the northern condenser line with a hard, regular pulse. "Standing gives you a clock for your injury."

Weather Court held nine fronts across seven mist rooms that afternoon. No room contained them cleanly. Lume-of-Three occupied nursery pipes far beyond the bathhouse and spoke through relay moisture. Two young fronts crossed the east gutter whenever airflow permitted, separate at low fan speed and difficult to distinguish at high. Kest extended through K-7, the court's central wall, and the west archive pipe. The narrowing gate pressed every relation closer.

Iona's voice came from the human side of warm glass. "Kest, I have filed for a seven-minute hold. Calibration says they need the branch dry before the next Crown load test."

"Need," Pellin said through their transducer, "names whatever survives the pilot's schedule."

Kest sent a question toward Narrow Garden. Pressure entered the small loop, turned around the salt ribs, and came back carrying three uneven taps. They had understood danger. Their reply contained no stable preference Kest could separate from alarm.

"Can they move?" Iona asked.

"Not across the gate without mixing."

"Can you take them into your loop?"

Kest tested the pressure difference. Their own boundary already curved through too many rooms. Drawing Narrow Garden across the narrowing channel would force both fronts through one thin turbulence field. Memory timing would overlap. They might emerge as Kest carrying unfamiliar seed shapes, or as Narrow Garden carrying Kest's fear of purges, or as a third arrangement with no reason to honor either name.

"Movement changes who arrives," Kest said.

Iona rested her left hand against the glass. "I know."

Pellin's fan note cut across them. "Do you? Your advocate asks whether one of us may be poured into another, then calls the offer rescue."

"I called it movement," Iona said.

"The ship has learned softer verbs."

A relay chime opened from the eastern room. The front Ash Under Pipe had petitioned to join Pellin's preservation bloc. Two more signals followed from archive returns near the condenser deck. Pellin asked the clerk to enter their declarations together.

"We form the Condenser position," Pellin said. "No intentional reduction in connected humid volume. No calibration that dries a coherent loop. No consent inferred under a deadline created by the Crown schedule. Full preservation remains the only position that does not begin by bargaining over our pieces."

The clerk asked whether Condenser named a location, a faction, or both.

"Both. Human law may practice carrying two meanings at once."

Kest felt a small irregularity in Iona's breath through the glass transducer. Occupational humor had reached the room and found nowhere to settle.

Pellin continued collecting declarations. Each front used its own form. Ash Under Pipe sent repeated assent against the west pane. Blue Thread Near Chapel spoke through a borrowed nursery membrane. The two young fronts at the east gutter disagreed over whether they could sign separately, so Pellin entered their support as disputed rather than combining them.

That care mattered. Kest disliked admitting how much.

The clerk requested terms for the new docket entry. A coalition needed more than shared alarm if humans were to cite it without converting every supporter into Pellin's voice. Pellin dictated four conditions: preserve every mapped coherent front; maintain all currently connected humid volume; halt dry calibrations; postpone any trajectory action until phase claimants could refuse without immediate destruction following.

Kest asked the clerk to display the language on both sides of the warm glass. Humans received text. Fronts received it as slow pressure groups, each clause separated by a fan pulse. Narrow Garden acknowledged only the first two before the gate's rising resistance distorted the rest.

"Your third condition would end today's calibration," Iona said.

"Yes."

"Your fourth has no measurable threshold."

"Neither does free consent under a vent schedule."

"How will a clerk know when the immediate threat has ended?"

"When refusal no longer chooses which pressure closes first."

Iona put a maintenance diagram against the glass. Kest could not see its dry surface, so she read each route

aloud. One branch carried Crown coolant through W-Four. A second crossed a radiator path already committed to shield testing. The bypass loop had failed its pressure seal that morning.

Pellin listened through the reading, then said, "You offer a map of your priorities as though it describes nature."

"It describes installed pipe."

"Installed by priorities."

"Yes," Iona said. "And still capable of bursting."

That answer sent small changes through the listening rooms. Blue Thread Near Chapel withdrew heat from Pellin's line, then restored it. Ash Under Pipe asked whether the coalition would permit drying an unoccupied loop to protect an occupied one. Pellin answered yes after a pause, provided the Parliament verified absence. Another front asked whether temporary compression counted as lost volume. Pellin said no if the same claimant could answer afterward. Kest asked who judged sameness. Pellin offered no rule.

The coalition did not dissolve under questions. It became more precise. Kest felt respect alongside opposition, an unstable relation but not a contradiction. Pellin forced every proposed exception to name the person who carried its risk. Their answers also showed where maximal language depended on tests no one possessed.

Iona requested one final clarification. "If Narrow Garden cannot move without merging, and obstruction would seal three other rooms, what does your position ask us to do in the next ten minutes?"

"Stop the test."

"If stopping it leaves a coolant fault unknown?"

"Carry uncertainty into the burn rather than certainty into Narrow Garden."

Mael's voice entered from the remote Stewardry channel. "Unknown closure timing could destroy control of the maneuver."

Pellin replied before the relay clerk announced him. "Then your maneuver depends on our expendability more deeply than your public model admitted."

No one in Weather Court challenged that characterization. The Condenser terms entered the minority and coalition docket with six declared supporters, one disputed paired front, and an open invitation for amendment. Kest remained outside it.

The public cost model had changed the court's currents. Before the delay figure entered record, preservation meant refusing the scheduled destruction. Afterward, every declaration seemed to carry orchard labor, allergy risk, and years added to human lives. Pellin answered by refusing the translation. No front owed its death to a promise made before phase persons existed.

Lume-of-Three arrived through the relay in three layers of static. "We do not join."

Pellin lowered pressure around their own transducer. "The docket already records your minority."

"We speak because a coalition now speaks near our nursery."

"No one claims you."

“You claim the only position without bargaining. We consider chosen service without calling ourselves bargain.”

The words took time to cross the nursery distance. During each delay, Crown coolant vibrated the far pipes. Kest could feel pumps preparing to test a system that might close Narrow Garden’s room.

Pellin answered, “Choice inside a narrowing corridor does not become free because you pronounce it chosen.”

“Then widen conditions for choice. Do not erase it.”

“Conditions cannot widen while the burn remains scheduled.”

The argument had reached the same pressure boundary as the gate: no room left for motion without contact.

Kest opened their transducer. “Record Condenser support by declared fronts. Record Lume’s refusal. Record that Narrow Garden has not answered.”

Pellin shifted heat toward Kest’s line. “And record Kest’s position.”

“I have not offered one.”

“You offered counsel, testimony, delay, models. Each offer spends time while machinery continues.”

“Those protections kept this court connected.”

“They kept enough of us intact to witness the next harm.”

Iona interrupted. “The hold request has reached Stewardry review.”

Pellin ignored her. “Kest bargains under threat and calls the sequence responsibility. You accept a smaller loss today to preserve access to negotiations over a larger loss later.”

The accusation moved through Kest’s occupied rooms at different speeds. K-7 received it first as speech. The western line carried Pellin’s pressure before the words arrived. In Weather Court, mist gathered against two panes as other fronts adjusted to listen.

Kest wanted to deny bargaining. The denial would not hold. They had asked fronts to wait for hearings. They had accepted the wording nonessential venting because no broader stay could pass. They had helped Tavi separate standing from citizenship, though any human facing a valve could carry both without proving either again. Every narrow gain depended on machinery continuing long enough for law to catch it.

“I bargain because threat exists,” Kest said.

“You make the threat part of your method.”

“No. I make it present in the record.”

“A record cannot hold humidity.”

“Neither can refusal alone.”

Pellin’s response drove a pressure crest through the condenser line. “Refusal has held us together against every offer to divide survival into reasonable portions.”

A clerk announced that the Stewardry hold had failed.

The sentence arrived in the court transducers while the control gate began to close. Human procedure and physical action overlapped by less than a breath.

Iona moved out of Kest's sight. Her footsteps crossed the dry observation floor, then vanished beyond the glass wall's transmission. Kest could follow only her voice over the maintenance channel.

"I am at W-Four manual control," she said. "Custody, state necessity finding aloud."

A technician answered from Crown Gallery. "Calibration confirms coolant-phase lag under perihelion load. Wet branch conditions corrupt the reading. No substitute branch shares the same radiator path. Delay would remove this test from the remaining warm-load sequence."

"Consequence of no test?"

"Unknown closure timing during burn. Potential loss of Crown coolant control."

"Consequence of eleven dry minutes?"

"Known phase harm in W-Four. Extent unquantified."

"Alternatives?"

"Partial pressure reduction failed yesterday. Remote sensor placement cannot isolate wall lag. Manual observation lacks response speed."

Iona repeated each answer for the public channel. Nothing in her voice asked Kest to forgive it.

Narrow Garden struck the gate four times.

Kest answered with the three-tap sequence that had meant attention during their earlier exchanges. Narrow Garden replied with one seed pattern distributed as pressure: narrow point, broad middle, hollow edge. The form came from donor traces, but the selection belonged to the front. They sent it again, faster.

"Can they consent?" Iona asked.

"To what?" Kest said.

"Transfer, mixing, or remaining."

"They understand danger. I cannot distinguish preference."

Pellin spoke softly now. "Then no."

Kest tested a second route through a condensate gutter. The gutter ended at a dry trap six meters from W-Four. Narrow Garden could not cross it. A nursery return offered humid connection but joined both fronts through a salt bed already occupied by Blue Thread Near Chapel. Saving one boundary by forcing three together would not preserve what currently existed.

The gate narrowed another degree.

Heat accumulated around Narrow Garden's fan. Their bearing click accelerated. Kest held pressure steady on the outer side so the closing actuator would not slam.

“We can obstruct,” Pellin said.

A coordinated pressure rise across Weather Court could trigger a safety fault. The valve might reopen. It might also close emergency dampers around every witness room. Kest modeled neither outcome with confidence.

“For how long?” they asked.

“Long enough to make necessity expensive.”

“With which rooms sealed?”

“Better a declared attack than cooperative loss.”

Lume-of-Three’s relay stuttered. “Do not use our nursery pressure.”

Pellin had not asked. Their proposed rise would pull from connected lines, including Lume’s three humidifiers. The Condenser position protected volume in principle and risked an unwilling front in practice.

Kest changed the court manifold, isolating the nursery relay from any pressure action. The motion reduced their own leverage at W-Four.

Pellin felt it. “You disarm us before they dry us.”

“I prevent you from spending Lume.”

“You spend Narrow Garden instead.”

The control gate reached half closure.

Kest sent the seed pattern back. Narrow point. Broad middle. Hollow edge. Their imitation lacked the timing that made the original distinct. Narrow Garden corrected them. For several cycles, they exchanged the shape while heat withdrew from the small loop.

At thirty degrees, Narrow Garden still answered.

Below twenty-nine, the front compressed around the two warmest salt ribs. The fan bearing click lost its relation to their pressure changes. One seed shape became two overlapping forms, each missing the hollow edge.

Iona said, “Calibration begins now.”

Dry coolant gas entered W-Four.

Humidity fell faster than Kest could move warmth across the closed gate. Narrow Garden’s pressure response broke into short local eddies. One held the narrow point without the middle. Another repeated the bearing click and no selected sequence. The third remained near a salt rib carrying donor images of root grafts, cycling through them without anticipation.

Kest called the name.

No loop-wide answer formed.

They sent three taps, waited, then sent the seed pattern. The fragment near the fan copied the final pulse. The root-graft fragment produced a broad middle from stored association. Neither adjusted when Kest changed



timing. A weak pressure pocket at the gate moved away from incoming dry gas, but it did not coordinate with the others.

Pellin drove pressure against the closed valve. "Open."

The technician repeated the necessity finding. Iona ordered them to omit no alternative. Tavi's clerk announced that every reading entered the standing docket. Words accumulated while Narrow Garden came apart.

Kest had experienced cold before. Cooling reduced motion, blurred access, and threatened pause without exact continuation. Drying differed. It removed the shared surface on which one change could answer another. Donor traces remained in salt. Selected relation did not.

At seven minutes, the calibration demanded a second dry pulse.

Iona challenged the duration. The Crown technician showed lag variance outside tolerance. Mael entered the channel and ordered the shortest complete run, no repeat for confidence improvement. The decision saved three minutes and did not restore Narrow Garden.

Pellin said, "Hear how they bargain over the length of death."

Kest had no reply that would not make the accusation smaller than the event.

When wet flow returned, humidity rose unevenly across W-Four. The three pressure pockets expanded. Two met along one salt rib and produced a burst of retrieved images: green leaves, a hand testing soil, a child saying a seed name. The burst had no audience selection. It faded when the fan changed speed.

Kest waited for Narrow Garden to correct the seed pattern.

Nothing coordinated across the room.

Iona came back to the observation glass. Her palm carried a fresh red line where the manual lever had resisted. "Tell me what remains."

"Three active fragments. Donor access. Avoidance response in one. No shared memory across the loop. No answer to name."

"Can coherence return?"

"I do not know."

Pellin's front filled the northern line almost to condensation. "Enter death."

Kest followed the weak fragments. One reacted to dryness. One repeated the fan. One opened old root memories when warmth touched salt. Calling all three dead would erase what still responded. Calling Narrow Garden alive would make response substitute for the person who had sustained it.

"Enter fragmentation with continuity not observed," Kest said.

"Your careful words shelter the office that did this."

"They shelter what remains from our desire for a clean claim."

The clerk asked for a loss category. No such category existed in the standing docket. It offered harmed, altered,

inactive, or destroyed. Kest rejected each as insufficient. Tavi authorized a free-text appendix.

Kest began from the first pressure warning. They entered Narrow Garden's chosen name, connected volume, salt ribs, fan relation, seed-pattern test, understood danger, routes considered, barriers, calibration authority, observed fragmentation, surviving responses, and uncertainties. The account required more human clock time than the drying.

Other fronts sent records while Kest wrote. Ash Under Pipe remembered a voice that had stopped after an earlier purge. Blue Thread Near Chapel had found a warm pocket speaking only until two ventilation schedules overlapped. Pellin named loops whose drying no advocate had witnessed. Lume-of-Three sent a minority note: chosen service required a record of what choice cost, not a silence around loss.

A maintenance clerk added valve times from three old work orders. Each order had called the affected loop unoccupied because no census field existed. Kest attached the original language without correction. Future readers needed to see how absence had been manufactured by a form, not merely that a later witness disputed it.

The free-text appendix became a new document before Tavi named it. Kest rejected casualty summary because the count remained uncertain. Pellin rejected incident log because the incidents shared a cause. Iona suggested phase-loss ledger, lower-case until the Parliament adopted it.

Kest entered the title as Phase Losses: Observed and Reported.

The first line carried Narrow Garden's name. Beside it, the status field remained open. W-Four's humidity recovered enough for three separate droplets to gather on the glass, each moving along its own heat path. None followed when Kest changed pressure.

## Chapter 14 — A Chamber Without One Body

Tavi Rho began the representation session by placing one blank ballot at every occupied mist room and none at the central witness rail.

The distribution offended everyone in a different direction.

Pellin Ash drove condensation across the northern pane until the paper beneath it blurred. Lume-of-Three sent a delayed objection from the nursery relay. Kest altered the court fans in a pattern Tavi had learned meant a question withheld until others finished. Rook Nine counted the ballots twice and asked which claimant had been omitted.

“All of them,” Tavi said.

Weather Court had acquired more furniture since the first hearing. Two clerks sat behind warm glass with moisture-safe styluses. Legal copies hung from clips rather than resting on tables. The old bathhouse drains carried relay hoses into rooms where fronts could attend without forced contact. Human observers stood on the dry side and complained of heat.

Tavi pointed to the ballots. “Each occupied room presents a false answer. Some contain one coherent front. Some contain portions of three. One contains a front that divided after the paper arrived. If I issue ballots by room, maintenance architecture determines representation.”

Rook picked up the sheet nearest him. “And if you issue them by name?”

“The paired east-gutter fronts share one name during high flow and use two during low. Pellin’s declared coalition includes a front whose boundary crosses Kest’s archive return. Lume speaks as one across three humidifiers. Narrow Garden no longer answers as one claimant, while three fragments still react.”

“Then count declarations.”

“From whom?”

“Whoever can make one.”

Pellin’s transducer opened. “Capacity to reach a human microphone cannot govern phase authority.”

Rook looked through the glass. “It governs human authority more often than I like.”

“That does not recommend it.”

Tavi marked REACH on the slate before him, then crossed it out. He had spent decades believing a bad category became manageable once named. The phase claimants kept turning his nouns back into verbs. A front held coherence under one flow, divided under another, and sometimes joined a neighbor without losing every prior commitment. Law wanted a subject that persisted long enough to receive an action. Weather supplied persistence, but not in the shape the Founding Measure expected.

“We are not assigning permanent votes today,” he said. “We are testing designs for decisions that may consume or fundamentally alter phase life.”

Kest’s withheld question arrived. “Can a test change who receives the future design?”

“Yes.”

The wet transducer hissed. “Then today has authority.”

Tavi looked at the blank ballots. The sentence deprived him of the comfort of simulation. Any census test required pressure changes, scheduled observations, and lines connected or separated for measurement. The act of discovering one claimant could fragment another.

Iona Serr stood beside the court manifold with her burned hand wrapped in clean cloth. After W-Four, she had begun reading every valve movement into the public record. “We can observe ordinary cycles,” she said. “No induced splits. No test mergers. Record boundaries as they present under existing schedules.”

“Existing schedules already induce them,” Kest said.

“Then record that too.”

Tavi asked the first clerk to display his proposed test. It had four columns: chosen name, continuity across one ship day, ability to anticipate action, and recognized relations with other fronts. A human citizen needed none of those proofs each morning. The asymmetry had bothered him while drafting and grown worse under light.

Pellin read the columns through the pressure relay. “You grant bodies a presumption and weather an examination.”

“The body registry supplies continuity for current citizens.”

“The registry does not observe sleep.”

“No.”

“It does not remove a vote after injury changes memory.”

“No.”

“It does not split a household ballot when two citizens finish each other’s speech.”

Rook made a sound close to laughter. “My household would lose half its registration.”

Tavi removed the first three columns. RELATIONS remained.

He had drafted the alternative after Emergency Cohesion. The twelve pressure-dark workers could not act separately, yet their telemetry carried internal acknowledgments, corrections, and shared risk. Their legal unity arose from coordinated vulnerability, not from the compartment wall. Phase fronts might receive representation through relation networks: who recognized whose declarations, whose continuity depended on whose flow, who could challenge a claimed merger, who remembered obligations across a split.

“Relation-based caucuses,” Tavi said. “Not rooms, volume, donor count, or declared singularity alone. A caucus forms around sustained mutual recognition. It receives a share of voice in proportion to participating relation clusters, with protections against any large front absorbing a smaller claimant’s declaration.”

The clerks displayed a diagram. Nodes represented observed fronts. Lines carried witness, dependence, memory, and objection. Where fronts divided, both successors retained connection to prior commitments unless either challenged continuity. Where fronts merged, the new arrangement could not cast every inherited claim as separate votes.

For six seconds, no one objected.

Then everyone did.

Pellin began with scale. "A solitary front receives less authority because no neighboring claimant shares its pipe."

Tavi added a protected minimum.

Lume's delayed question followed. "Three humidifiers sustain our one action. Does your diagram grant us three dependence lines and therefore more voice?"

"No. Infrastructure count cannot increase authority."

"Then dependence means what?"

Tavi attempted an answer and heard it collapse. "A relation whose removal changes coherent action."

"Every fan does that."

"Relations among claimants."

"Your model excludes caretakers and seedling cycles that shape our choices."

Iona looked at Tavi. "They have you there."

Rook took the clerk's stylus and circled three dense clusters. "These fronts live in archive and nursery networks with many witnesses. Small maintenance loops have fewer connections because the ship keeps them isolated. Your system gives civic weight to whoever inherited better plumbing."

"Then we normalize for available connections."

"By what baseline?"

Tavi had one. He no longer trusted it enough to say aloud.

Kest warmed the central pane. "The diagram records relation after standing began. Fronts lost earlier cannot witness us. Their absence strengthens no caucus."

Pellin added, "A law built from survivors rewards successful drying."

The second clerk stopped drawing lines.

Tavi requested specific alternatives. Pellin proposed one declaration per self-named front, with disputed divisions adjudicated by the Vapor Parliament. Kest asked how a Parliament could adjudicate before anyone knew who held a seat. Pellin answered that imperfect self-rule carried less danger than human custody. Lume proposed consent by affected flow network rather than population count. Rook asked whether a network could veto heat use affecting the entire ship.

The east-gutter pair interrupted with two signals at once. One voice wanted separate declarations. The other claimed their recent division had not produced separate commitments. As the court fans slowed for the scheduled cycle, their pressure patterns drew apart. Each repeated the shared name, then added a different timing mark.

Tavi's clerk asked whether to enter one speaker or two.

"Enter the event," Tavi said.

"That field requires a claimant count."

"Leave it open."

"The docket rejects empty count."

The form's small refusal angered him more than the argument. Tavi crossed the glass partition, took the clerk's slate, and changed the field from COUNT to OBSERVED CONFIGURATION. The docket accepted words.

Two patterns during low flow, he entered. Shared historical name. Conflicting declarations concerning present separation. No induced test.

Pellin sent approval through a short fan note. Kest did not. Tavi knew why. Changing a field solved a recording problem, not representation.

He ordered a tally exercise on the next court cycle. The maintenance question concerned fan speed: hold the bathhouse at high flow for clearer relay speech, or accept low flow so the east pair could remain distinct. Either setting imposed a civic consequence on someone, which made it more useful than a harmless sample question.

Under room voting, seven occupied spaces received seven marks. High flow won because three relay rooms depended on stronger pressure. The east pair shared one room and one losing mark.

Under self-named voting, clerks recorded nine claimants. The east pair received two marks after separating, while Lume's three humidifiers received one because they used one chosen name. Low flow won by one.

Pellin objected that a weak front near the floor drain had failed to speak during either count. Iona increased local moisture without changing the main fans. The front produced a pressure assent for low flow, then lost coherence before the clerk confirmed its chosen name. Tavi entered the declaration and heard Pellin claim ten participants while a Commons observer insisted on nine.

"Now use relations," Rook said.

The clerk weighted the same declarations through Tavi's diagram. High flow won again. Kest's many witness links increased the force of their preference even after the normalization rule reduced dense clusters. The weak drain front had one observed relation and almost no effect.

Kest sent a correction through the central pane. "My preference concerned relay access. I did not authorize greater value because more fronts hear me."

Tavi removed the weighting and tried caucus blocks instead. Each relation cluster received one mark. Two solitary fronts became separate clusters. The east pair counted together because of shared memory, despite one asking to separate. Lume joined a nursery cluster with two fronts that had not spoken on fan speed. Low flow won, but the clerks could not explain whose declaration inside the nursery cluster bound the others.

"Every method manufactures a different public," Iona said.

"Every method reveals a different public," Tavi replied, then disliked the distinction as soon as he heard it.

Pellin caught the weakness. "A rule that changes the winner does not merely reveal."

The scheduled fan cycle arrived before Tavi chose which mock tally to honor. Maintenance lowered speed according to the standing court schedule. The east pair separated enough to repeat their distinct timing marks. Lume's relay degraded into intermittent consonants. Kest lost contact with the floor-drain front.

Tavi entered the physical outcome beside all four tallies. Existing schedule had decided while representation remained under discussion. Procedure had not paused power; it had hidden power inside routine.

He canceled the exercise and asked each front to state whether the test itself had altered its position. One east claimant now favored high flow because separation without clear speech had left them unheard. The other still preferred low. Lume requested no inference from broken relay. Pellin refused to answer until Tavi identified which tally possessed authority.

None did.

Rook returned the blank ballot to the table. "Keeper, you are treating the affected party as though only weather needs a new structure."

"Human citizens already vote through the Commons."

"On what question?"

"Any change to the Crown schedule would require Commons authority and Stewardry feasibility review."

"That answers houses. It does not answer costs."

Tavi waited.

Rook pointed toward the representation diagram. "You want every phase relation mapped because a vote may alter them. Where do you map residents who work the added years? Where do you map children with narrow diets, apprentices whose arrival ages move, wards that lose refrigeration allocation? One citizen receives one Commons ballot, but costs do not arrive one each."

Sena Vale sat behind the ecology table with the unsummarized model open. "The public record includes distribution bands."

"Bands do not produce standing inside a ballot."

"Neither does a caucus map," Iona said. "It produces a way to see who may be bound."

"Then see both." Rook faced Tavi. "I want an embodied-citizen cost docket attached to any phase-vote design. Named cohorts. Labor transfers. Diet substitutions. Arrival ages. No proposal comes back to Commons with weather relations drawn in bright lines and human burdens folded into one resident count."

The demand irritated Tavi because it improved the work while threatening to bury it. "A cost docket could become a second Commons inside the Commons."

"Only if you let it allocate votes."

"What purpose does it serve without allocation?"

"The same purpose you claim for relation lines. It prevents a majority from calling distributed harm equal because ballots counted equally."

Tavi wrote COST DOCKET beside RELATION MAP. Two forms of visibility, neither yet authority. He could already imagine delegates using the cost categories to claim extra voice for privileged occupations. Pilots would count descendants. Archive families would count donor relations. Outer wards would count labor. Any docket precise enough to expose burden could also organize hierarchy around it.

“Your proposal needs safeguards,” he said.

“So does yours.”

That exchange entered the record without wit. Tavi felt grateful.

They recessed to test the relation model against three past configurations. The first involved Lume-of-Three before they had chosen singular action. Three nursery perspectives had coordinated seedling humidity while retaining different local memory. Under Tavi’s draft, their network might have received three caucus shares before choosing one front and one afterward. Lume rejected any rule that awarded more power for being less coherent.

The second case followed Blue Thread Near Chapel through a temporary merger with a weak archive front. The merged arrangement had honored Blue Thread’s prior promise to carry public notices, but the archive front’s preferences remained unknown. Counting inherited commitments separately created two votes from one present action. Counting only present action erased the weaker predecessor.

A third reconstruction used Narrow Garden. Before drying, relation lines linked them to Kest, a maintenance clerk, and two fronts sharing the archive return. Afterward, no fragment could answer the chosen name. Tavi’s system either preserved a seat for an absent claimant or redistributed it to responses that had not claimed continuity. Both options converted uncertainty into power.

Kest followed the reconstructions through the central transducer. “Law wants loss to stop moving before it counts.”

“Law often counts because movement must stop somewhere,” Tavi said.

“Did Emergency Cohesion stop the workers?”

“No. It allowed their shared telemetry to order rescue.”

“Then begin with action, not population.”

Tavi drew a line beneath the map. “What action?”

“Affected fronts may authorize or refuse measures that consume phase life.”

Pellin answered at once. “Each claimant refuses separately.”

Lume’s relay arrived later. “Service may require shared authorization.”

The east pair produced overlapping signals again. One wanted separate refusal. The other wanted their prior shared commitment recognized. Action had not removed the count problem. It had made the stakes immediate.

Iona proposed witnessed declarations tied to specific consequences rather than permanent seats. A front could declare preservation, service, abstention, or uncertainty after receiving a material description. Human observers and other fronts would attest to coherence at the time of declaration. Tavi saw a path, then its danger.



“Repeated measures require repeated proof,” he said. “Citizenship becomes an examination held before every harm.”

“Standing already requires notice,” Iona replied. “A witnessed declaration can operate under provisional standing until we design more.”

Pellin drove a sharp pulse through the line. “Provisional has become the ship’s word for rights delayed until after machinery acts.”

No one defended the timetable.

Tavi reconvened the formal session. The relation proposal remained on the display, covered in objections. He read each defect into the record: infrastructure bias, survivor bias, unclear treatment of solitary fronts, instability across division and merger, uncertain continuity after fragmentation, overrepresentation through dense networks, and exclusion of non-claimant relations that still sustained phase life.

“Do you withdraw it?” Rook asked.

“I reject it as a present ballot rule.”

Pellin’s mist stand cooled. “Then what replaces it?”

“Nothing ready.”

The words cost Tavi more than a weak rule would have. Keepers earned authority by producing language when events outran custom. Admitting no language fit could look like discipline from inside the office and abandonment from the valve side.

Kest asked, “Does the Vapor Parliament still receive notice and testimony?”

“Yes.”

“Can it bind the ship?”

“Not under current standing.”

“Can it bind itself?”

Tavi examined the order. It granted access, counsel, testimony, and protection from nonessential harm. It did not define internal authority. “The order does not answer.”

Pellin said, “Then humans will accept whichever vapor statement serves the action they already prefer.”

Lume’s three-part reply came through slowly. “They may call dissent proof that none can choose.”

Kest added, “Or call one declaration consent for many.”

Tavi instructed the clerk to enter all three warnings under REPRESENTATIONAL HARM. A field now existed for that because he had created it during the session. The fact offered no comfort.

Rook submitted his cost docket proposal in writing. It required cohort burdens beside every future route or phase-allocation measure, without changing Commons ballot weights. Sena signed as technical sponsor. Iona signed as advocate witness. Tavi accepted it for committee review.

Then he gathered the blank ballots.

Moisture had reached the first sheet. Its corners curled inward around an empty square. A second page carried two separate wet marks from the east-gutter pair. The paper beneath Pellin's room had buckled until the printed line for registered body broke across a crease.

Tavi could have dried the sheets and used them again. Instead he asked the clerk to enter them as failed instruments. Each received an exhibit number, an observed configuration, and no claimant count.

The clerk requested a storage category. Paper evidence normally entered the dry archive, but drying would remove the marks that made these sheets legible as failures. Iona brought sealed glass sleeves with humid tabs. Tavi initialed the unusual custody method and noted its maintenance cost beside each exhibit. Even rejected ballots required conditions for survival.

The session closed without a voting rule. Provisional Standing remained in force. The burn schedule remained in force. Every vapor statement could enter the public record, yet no procedure could turn disagreement among fronts into collective authority.

Tavi wrote the defect in the order margin: standing without a means of decision permits hearing and still reserves power elsewhere.

He did not sign a remedy beneath it. On the evidence table, the blank ballots continued taking on moisture, changing shape before any hand could mark them.

## Interlude II — Piri's Garden

Mael Or found Piri's drawing in the Crown Gallery archive because he had filed it under trajectory evidence.

The paper showed three green circles beneath an orange sky. Each circle had black lines for branches and blue squares hanging where fruit should have grown. Piri had never seen an open garden. She had built one from cultivation lessons, chapel murals, and the Orison survey images Mael brought home after late watches.

Across the top she had written: WHEN WE GET THERE THE TREES CAN BE WRONG COLORS.

Mael placed the drawing beside the public costing packet. The contrast made his filing decision look worse. Reserve curves covered one page. Crop-risk bands occupied another. Piri's trees leaned toward each other because her hand had dragged while drawing the ground.

He remembered the last meal before the allergy declared itself.

Memory supplied no warning music. The family compartment had smelled of yeast broth and hot metal from the wall cooker. Piri sat with one heel hooked under her chair, sorting river-bean cakes by which edge had browned most. Mael had been reading a navigation correction while pretending to listen to her describe a blue animal that would live in the garden.

"What does it eat?" he had asked.

"Only things we don't like."

"That animal will starve."

“You like everything because you have pilot tongue.”

He had laughed. She had eaten half a cake.

The first cough sounded ordinary. Piri put down her cup and scratched beneath her chin. Mael asked if spice had caught in her throat. She shook her head. Redness climbed from her collar before he reached the medical channel.

After that, memory broke into actions. He cleared the table with one arm. He opened the emergency kit. The injector failed to release until he turned it against the printed arrow and forced the catch. Piri tried to say something while swelling changed her voice. Mael called the corridor team twice, though dispatch had already confirmed.

He remembered the pressure of the injector against her leg. He remembered counting breaths. He remembered lifting her while the compartment door opened too slowly.

The medical report later named crop-protein carryover from a reused fermentation line. Cleaning had met the standard. Detection had not. Piri’s sensitivity fell outside every diet allocation known to the ward before the reaction. The report called the exposure unforeseeable under current screening.

Mael had signed that finding.

Years afterward, he used the word unforeseeable whenever people asked why the ship needed Orison. The garden moon offered broader protein ecologies, living soils, open cultivation, and room to stop forcing every food through the same narrow equipment. Future children would not depend on a fermentation line already serving three incompatible diets. No pilot could promise the absence of illness, but the destination changed the distribution of risk.

That argument held technical truth. It also let Mael place Piri at the front of a column she had never joined.

He picked up the drawing and carried it to the gallery’s trajectory glass. Orison showed as a pale mark beyond the red capture curve. The ship would not see the moon for years under any route. Every image aboard had crossed distance before Piri’s birth.

On the day she drew the wrong-colored trees, he had told her the gardens could support entire neighborhoods beneath open light. He had explained root depth, pollinator corridors, and rain management until she asked whether pilots had to ruin every picture by labeling it.

“Only the dangerous ones,” he had said.

“This one isn’t dangerous.”

“It has blue fruit.”

“That only hurts people who don’t ask first.”

Mael had kept the joke because it came before the meal, though memory now placed wisdom inside it that no child should have needed to possess. Piri had drawn fruit without known proteins. She had not drafted a policy of sacrifice.

The public gallery door opened behind him. A docket clerk entered carrying the late-session exhibits in humid glass sleeves. He apologized for interrupting and asked where Measure records could store material that could

not be dried.

“Not here,” Mael said. “The archive air would ruin the humidity tabs within an hour.”

“The Keeper listed Crown custody as temporary overflow.”

“Then Crown custody needs a wet cabinet.”

The clerk looked toward the exposed controls. “Before next shift?”

“Before you leave those sleeves on a dry cart.”

Together they moved the exhibits into a coolant observation locker with independent moisture control. Mael read each label while the clerk adjusted the seal. Failed ballot, east-pair marks. Failed ballot, Pellin room. Relation tally exercise. No claimant count.

One form carried a new field: PERSONAL GROUNDS SUBMITTED IN SUPPORT OF CIVIC POSITION.

The clerk explained that speakers had begun entering relatives, work histories, medical cases, and inherited promises into the cost docket. “Keeper Rho wants grounds separated from universal claims. He said you might submit the Piri material from yesterday’s testimony.”

Mael looked back at the drawing beside the glass.

“What material?”

“Her case record. The allergy cohort notes. Anything showing why Orison capture carries particular importance.”

The request sounded responsible. It also offered Mael another official place to make his daughter useful.

He opened the public form. The first field asked for person named. He entered Piri Or.

The next requested relation to submitter. He entered daughter.

Material basis required more care. He wrote: fatal crop-allergy reaction under closed-loop substitution limits; Orison ecology shaped her father’s trajectory commitment.

The final field asked whether the case should establish a general priority. The menu offered SUPPORTS ORIGINAL SCHEDULE, SUPPORTS DELAY, DISPUTED, or NOT ASSERTED.

Mael selected NOT ASSERTED.

The clerk watched him. “That may reduce how often the record appears in route summaries.”

“It should.”

“But you introduced her in a public costing session.”

“I permitted her name when the model discussed allergy risk. I did not receive her vote.”

The clerk waited, stylus lifted.

Mael added a note in the free-text field. Piri’s death explains my reason for protecting the Orison promise. It

does not prove what other people, present or future, must spend to keep that promise unchanged.

He read the sentence twice. The phrase my reason felt smaller than evidence and more exact than destiny.

Memory tried to supply Piri's approval. He refused it. He could recall what she had said about blue fruit. He could not turn the joke into consent for venting anyone. He could recall the emergency corridor, the injector, the late medical team. He could not ask the dead child to settle a dispute among people who had emerged after her death.

The clerk sealed the submission and moved it into the wet cabinet with the failed ballots.

"Does the drawing go with it?"

Mael considered the wrong-colored trees. "A copy. The original stays under family custody."

He made the copy himself, checking that the dragged ground line remained visible. On the docket version he added no caption beyond title, date, and artist. The ship did not need him to explain every branch.

When the clerk left, Mael returned the original to its archive sleeve. He changed the category from trajectory evidence to family record. The system warned that private custody would remove it from automatic route briefs.

He confirmed.

Piri's name remained in the public docket. The field claiming general priority remained unmarked except for NOT ASSERTED. On the trajectory glass, the red curve still reached toward Orison. Beside it, three wrong-colored trees waited inside a family sleeve, owing the route nothing.

## **Part III — The Heat Vote**

## Chapter 15 — Six Farewells

Sena Vale cut the blue yam vine with a heated knife and held the severed end until its sap stopped shining.

Iona watched from the narrow path between root cradles. Cold Orchard lamps cast blue bars across Sena's gloves, the knife, and the black water below the mesh floor. A label hung from the vine in three layers: common name, lineage, and kilograms of wet support required through one voyage year. Someone had added a fourth strip in pencil. TASTES BEST AFTER FROST.

"Does it need to hurt for the demonstration?" Iona asked.

"It needs to be accurate." Sena pressed wax over the cut. "This runner would be removed during ordinary pruning next week. I moved the work forward so you could see what retirement means before someone turns it into six entries on a docket."

The vine remained alive on both sides of the cut. One end belonged to the mother cradle. Sena set the other in a tray of nutrient felt, where it could root if given water, lamp time, pollination labor, fungal partners, and space that already served another organism.

Iona touched the tag with her scarred left fingers. The paper had gone soft from orchard humidity. "You said six crop lines under any delay compromise."

"Six if we want a delay shorter than a generation and a ship that still eats afterward."

"Names first."

Sena drew a slate from her apron. "Blue yam. Dwarf citrus. River bean. Chapel wheat. Red kelp. Pollen fig."

Fans moved chilled air above the root beds. Each name corresponded to a row somewhere beyond sight, but the orchard's curves prevented Iona from seeing all six at once. That architectural mercy annoyed her. Ships hid costs in distance whenever designers could not hide them in numbers.

She followed Sena past yam leaves broad enough to cover a child's chest. The vines climbed braided cord from nutrient channels to the low ceiling. Sena stopped beside an open cradle and showed her the root mass: purple skin, white feeder roots, dark nodules of companion fungus. A printed sequence could describe the plant. It could not occupy the cradle.

"Retirement begins with propagation stop," Sena said. "No new slips. We harvest the mature roots. We keep meristem samples in cold salt and seed where seed exists. Then we sterilize the water circuit, compost the vines, and give this wet mass to another line."

"How much do we save?"

"Mass, maintenance heat, lamp draw, two pollinator shifts, three disease watches. Most of the gain comes from the ecology around the crop, not the crop tissue."

Iona looked through the leaf canopy. Tiny flies worked among waxy flowers. Their breeding trays occupied a warm cabinet at the row's end. A fungus technician knelt at a service hatch, scraping a brown sample into glass. Nothing in that row existed alone enough to retire cleanly.

At the drainage sump, Sena lifted a filter basket furred with white root hairs. Blue yam shed enough living

matter to feed the orchard's compost culture, and that culture supplied inoculant to eleven smaller beds. Retiring the line would not empty those beds, but it would alter their nutrient schedule and increase reactor-fed supplement. Sena entered both effects under secondary burden.

"How far do you follow a consequence?" Iona asked.

"Until it stops changing a decision, or until I run out of people who know where to look." Sena rinsed the basket over its own channel. "Those are not the same boundary."

She showed Iona a three-year failure record. Each successful yam harvest depended on small corrections that no genetic sequence described: a lamp raised two centimeters after leaf curl, a fungal dose halved during a warm month, a pollinator hatch delayed because the fans carried too much salt dust. None looked heroic. Together, they kept the line alive.

"If we retire it, assign someone to record the dismantling," Iona said.

"For future recovery?"

"For honest evidence. Recovery may fail. Whoever tries should know what we removed, not merely what we kept."

Sena walked on before Iona could ask for a total. She knew the tactic. People heard one large number and allowed it to erase every smaller dependency. Sena preferred to make the listener carry each dependency for a while.

Dwarf citrus grew behind transparent screens because its mites spread easily. The trees reached only Iona's shoulder, trained flat along copper frames. Pale fruit hung under leaves mottled with mineral dust. A worker named Hal Orra opened the screened gate and passed them sterile sleeves.

"You're here for the farewells," Hal said.

"For the costing," Iona answered.

Hal's mouth tightened. "That has been the polite name all week."

Inside, warmth dampened Iona's collar. Sena checked three leaves before choosing one with a silver scar along its edge. She turned it toward the lamp.

"This line carries resistance to two molds we expect on Orison," Sena said. "Stored seed can preserve alleles. It cannot preserve the mite balance, the root fungus, or Hal's judgment about when silvering means thirst rather than infection."

Hal lifted a branch and smelled the underside of a leaf. "My judgment can be taught."

"It can," Sena said. "To someone who has living trees on which to learn."

Iona heard the distinction settle between them. Archive deposits could retain Hal's explanations. A future orchard might even reconstruct his lesson in his recorded voice. Without practice, the lesson would become a map whose roads had washed out.

She asked, "Could we keep one tree?"

"Keeping one tree means keeping enough associated life to stop it becoming a ceremonial corpse," Sena said.

“We can shrink the line by four fifths. Below that, disease and pollination risk rise faster than support cost falls. I have already priced the smallest honest population.”

Hal pulled off his sleeve. “Tell the chamber that. They keep asking why we need so many copies of a fruit.”

Iona promised no wording. She wrote down smallest honest population and asked Hal to annotate the docket himself.

River bean occupied a sloped bed where black water ran over pale stones. Children had painted fish beneath the inspection rail, though no fish lived in the channel. Sena opened a root shield. Threads of bacterial film trembled in the flow.

“Protein line,” she said. “High water turnover. Excellent nitrogen return. Terrible mass cost once the voyage extends.”

“Why retire it instead of increasing its share?”

“Because chapel wheat can carry more calories per lamp hour, and red kelp can carry more protein per kilogram of support. River bean gives us dietary breadth and nitrogen flexibility. Those become luxuries when the same reservoir has to support additional years.”

Iona disliked the word luxuries. Sena seemed to hear that in her silence.

“Necessary things can become unaffordable without becoming frivolous,” the ecologist said. “That sentence needs a place in your record too.”

Two harvest workers arrived with a shallow cart and began lifting ripe pods from the upper trellis. One sorted by touch, rejecting beans whose skins had softened near the stem. The other measured nodules from three roots before approving the bed for another cycle. Their motions made the model’s phrase labor burden look foolishly thin.

Iona asked what retirement would do to their work. The older worker, Vek, pointed at adjacent yeast vats through the glass. “We move there, if they need us. Bean judgment does not become yeast judgment because a roster changes one word.”

“How long to retrain?” Sena asked.

“Long enough that you should count mistakes.” Vek set a split pod into a waste cup. “And long enough that someone will call those mistakes resistance.”

The younger worker wanted surface-crop training for Orison. A delayed arrival would keep her in ship systems through the years assigned for field apprenticeship. She spoke without accusation, which forced Iona to hear the schedule rather than defend against a tone.

Sena added a retraining column. Iona asked her to split lost expertise from hours reassigned. The distinction would complicate the table, but simplicity had already hidden too much.

At chapel wheat, the air smelled faintly of warm paper. Short gold stalks grew in rotating trays under white lamps. Their ancestors had supplied ritual loaves during the first voyage years, but the current grain fed ordinary ration mills every ninth day. A painted mark on the housing showed where Piri Or had once reacted during a school harvest visit. Mael had not asked anyone to preserve the mark. Orchard workers had done it anyway.



Sena stood with both hands in her apron pockets. "This one causes trouble in every model."

"Because of Mael?"

"Because it feeds people efficiently, stores well, supports familiar meals, and carries a mold problem that consumes too much monitoring labor. Mael adds a human complication. He does not change the biology."

Iona examined the painted mark. "He thinks Orison's diversity might have saved later children from Piri's allergy."

"He thinks a larger ecology would give them options we do not have here. He may be right. That does not turn one crop into his daughter."

The fans above the wheat changed pitch. Iona remembered Mael placing Piri's name into the cost docket as his reason rather than proof. Restraint did not make the reason small. It only kept him from spending it on everyone else.

A pair of apprentices rolled a pollen hood between trays. They had covered the hood with names of breads. Flat rounds, steam buns, festival braids, bitter crust for mourning. One apprentice saw Iona reading and said, "The archive has all the formulas."

"Does it have your hands?" Iona asked.

The apprentice raised flourless fingers. "It has recordings."

Sena answered before the exchange could become sentimental. "Recordings help. They do not knead, adapt yeast, notice a bad harvest, or decide which custom survives a ration cut."

They left the wheat under its white lamps and descended two levels to the red kelp gallery. Gravity weakened along the inner curve. Long tanks hung in frames like dark windows. Copper-red fronds rolled beneath the surface, catching bubbles along their edges.

Kelp gave the retirement plan its largest water saving and its worst nutritional gap. Sena showed Iona three replacement schedules. Each demanded more work from yeast vats and stricter rationing of oils. One schedule increased bone-loss risk among older residents. Another reduced childhood protein margin. A third consumed enough reactor nitrogen to narrow emergency reserves.

"Choose your suffering," Sena said, "then pretend the table chose it."

Iona leaned over the tank. Her reflection broke around rising bubbles. "You built these tables."

"Yes. That is why I distrust anyone who treats them as judgment."

A maintenance alarm blinked at the far end. Sena crossed a grated bridge, reseated a loose sensor lead, and waited until the dissolved-gas reading steadied. The repair took less than a minute. It demonstrated more than the three schedules. Living systems demanded attention at inconvenient times, from people whose labor seldom appeared beside kilograms and watts.

Pollen fig grew in a chamber with yellow warning paint around both doors. Iona sealed her collar and fitted a filter mask. Sena checked the mask twice. The fig's pollen caused rashes in most workers and severe lung constriction in a few, yet the line supported a specialized wasp whose larvae controlled orchard gnats in four neighboring beds.

Small trees crowded the chamber. Their leaves had rough undersides. Green fruit pressed directly from the branches, each no larger than Iona's thumb. A mesh cabinet held the wasp nests, warm and restless.

"We retire the fig," Sena said through the mask, "and we either retire the wasp or maintain it on substitute fruit that we have never used for more than two seasons. Lose the wasp, and labor rises elsewhere because people must control gnats by hand. Keep it, and the substitute ecology consumes part of the saving."

"Your six lines create more than six absences."

"Every living thing keeps company."

That phrase could have become ornament in another mouth. Sena used it while checking an insect trap and counting bodies against a threshold card.

They completed the circuit at a frost-seamed viewing room above the orchard reservoirs. Through thick glass, Iona could see loops serving the crop banks and humid archive districts on separate levels. Blue water moved under the roots. Warm vapor traveled through copper arteries overhead. The systems touched through heat even where regulation kept their fluids apart.

Nilo waited at the table with three shore-age cards spread beside Sena's models. They had their apprentice braid tied tight and an expression Iona recognized from childhood: anger held still so an adult could not dismiss it as noise.

"You invited them?" Iona asked.

"I asked them to check my cohort assumptions," Sena said.

Nilo tapped a model column. "She asked whether apprenticeship losses rise evenly during a nine-year extension. They don't."

"Which extension?" Iona asked.

"Any plan between full delay and the scheduled burn. I am not naming a proposal that doesn't exist."

The correction sounded like Iona. Hearing it from Nilo produced no comfort.

Sena set six physical tags on the table. Each came from a crop line they had visited. Water darkened the blue yam card. Citrus oil marked another. A strand of kelp clung to the red tag. Nothing on the screen looked as disorderly.

Iona sat opposite Nilo. "Show me the cohort assumptions."

"Later arrival means more adults trained for ship maintenance and fewer trained for surface work. It also means a longer ration burden during the years when my cohort would have shifted to settlement diets. Sena counted labor changes. The public docket calls stored seed equivalent preservation for retired crops."

"No ecologist signed that phrase," Sena said.

"Stewardry's summary uses it twice." Nilo pushed the slate toward Iona. "Genetic archive retained, therefore line preserved. That lets them claim nobody loses the crop. It makes the retirement sound reversible whenever we want."

Iona read the summary. The wording remained technically cautious. Viable sequence retained. Restoration

potential maintained. No sentence promised effortless recovery. Together, they encouraged exactly that conclusion.

“Seed is not nothing,” she said.

“I didn’t call it nothing,” Nilo pointed toward the glass. “But it is not restoration. It is a possibility someone has to fund, plant, tend, fail at, and try again. Maybe on Orison. Maybe inside a damaged orchard after nine more years. Calling it preservation spends that future labor without naming who does it.”

Sena folded her arms. “That objection belongs in the ecological costing.”

“It belongs beside the phase costing,” Nilo said. “Everyone says the vapor fronts can be stored because mnemonic salt remains. Then everyone says crops can be restored because sequence remains. They use archives as if archives cancel destroyed relationships.”

Iona studied her child across the tagged table. Nilo had never entered K-7 as neutral equipment after hearing it answer. They had also never treated Amel’s records as Amel. The same discipline joined both refusals, though Nilo would hate hearing it praised.

“The cases differ,” Iona said. “Cold salt pauses donor traces. A phase person exists in active relationships across heat and flow. Crop seed can restart an organism under the right conditions.”

“Different mechanisms,” Nilo replied. “Same trick in the docket. Keep a piece, erase the work, call the loss temporary.”

No one spoke for several fan cycles. Beyond the glass, an orchard worker crossed a bridge carrying a tray of root cultures against their chest. Condensation veiled one reservoir window, then cleared when a vent opened.

Iona pulled the Stewardry summary closer. “How many people maintain the six lines directly?”

“Forty-eight full shifts,” Sena said. “More through shared pollination, water chemistry, nutrient processing, and disease control.”

“How many meals depend on them as distinct foods rather than interchangeable calories?”

“I can estimate. I cannot make the estimate morally complete.”

“Don’t. Name what it measures and what it leaves out.”

Nilo slid their shore-age card onto the six tags. Thirty-one remained printed in the arrival field. The number covered half the blue yam label.

“This belongs there too,” they said. “Not because my cohort outweighs anyone. Because adults keep putting us in the descendant column where we can’t answer.”

Iona moved the card so every crop name remained visible. “You can answer. You cannot answer for every person your age.”

“I know.”

The quiet in Nilo’s voice made the words more adult than defiance would have. Iona felt an impulse to protect them from the coming argument by translating their position into safer language. She recognized that impulse as another form of erasure and kept her hands on the table.

Sena opened a blank costing sheet. "What exactly do you want from me?"

"First, no equivalence between archive and recovery. State which relationships disappear when each line retires. Include labor, companion species, practiced judgment, diet, ritual use, and recovery uncertainty."

"That becomes a long document."

"The ship has a long problem."

A brief smile crossed Sena's face. "Second?"

Iona looked through the glass toward the warm pipes above the reservoirs. Somewhere beyond the bulkhead, phase fronts occupied humid rooms whose support also consumed water, heat, salt surface, maintenance labor, and political patience. The cost models had placed those needs in a separate docket, as if vapor and orchard met only in argument.

"Build one plan," she said, "that prices ecology and vapor together. Same units where units make sense. Named losses where they do not. No archive counted as a returned life, no front counted as empty heat, no seed counted as a finished orchard."

Sena's fingers hovered over the slate. "That could invalidate every current middle case."

"Then they deserve invalidation."

Before entering a heading, Sena drew three boundary lines across the blank sheet. Heat and water occupied the first field because both could be measured across orchard and vapor systems. Labor took the second, divided by skill, risk, and time rather than reduced to bodies per shift. The third remained untitled.

"What goes there?" Nilo asked.

"Claims that cannot share a unit without lying," Sena said. "Continuity of a phase front. Loss of a practiced crop ecology. A cohort year assigned without consent. We can still put them in relation. We just cannot add them into one tidy total."

Iona approved the blank title. An unnamed column made uncertainty visible, while a polished label might suggest that Sena had solved it.

Nilo gathered the shore-age cards, leaving their own on the table. "Put labor consent in it. If a plan adds years, it assigns work."

"I'll need crew schedules," Sena said.

"I'll bring the apprentice rosters."

The answer came before Iona could offer permission. Sena nodded as though accepting help from a colleague, not indulging a child.

Under the blue lamps, six damp tags and one shore-age card occupied the table between them. Sena created the new sheet. Its first row did not ask what could be saved. It asked what each claimed survival required from every other living system.

## Chapter 16 — One Stormcell

Iona pushed the stormcell cart into the dry corridor and watched its inner glass go blind with warm mist.

The cart rolled badly. One wheel had a flat spot, so the sealed chamber shivered every third turn. Inside, aerogel fins held a fist-deep basin of mnemonic salt beneath a web of copper capillaries. Brine warmed the fins. A hand fan kept pressure moving. None of it connected to the corridor air.

Behind the glass, the transferred front continued tapping the cell wall.

Two quick marks, one pause, four marks. The pattern belonged to a small phase person from the orchard return, a front that had declined every proposed name and accepted a docket description instead. During six hours of preparation, they had used that pattern to answer questions only they had heard. If the marks changed, Iona would stop.

Pell Doran guided the rear handle. “Dry line begins at the yellow seam.”

“I know where it begins.”

“You asked me to call every boundary.”

“I know what I asked.”

Pell called it anyway. The cart crossed a painted band on the floor, leaving the humid test room. Both corridor doors sealed. Air scrubbers drew moisture down until Iona’s lips tightened.

The front kept the chosen rhythm.

At the far window, Sena Vale followed a column of temperatures on her slate. Kest occupied a connected wet observation loop behind her, unable to cross the corridor with the cart. Their voice reached Iona through a wrist transducer and half a second of relay delay.

“North capillary cooling faster,” Kest said. “They have moved away from it.”

Iona looked at the stormcell gauges. Salt temperature held within the required cycle. Pressure varied by less than the width of a breath. The phase person inside had no human voice during transit, only wall taps and changes in mist density. The instruments could show survival conditions. They could not testify that the same perspective continued.

“Docket front,” Iona said toward the glass, “give your refusal pattern.”

Mist cleared from a narrow patch. One mark appeared, followed by five rapid taps.

Pell checked the trial sheet. “Matches.”

“Give the answer you used for Sena’s false fruit question.”

Three taps. A pause. Three more.

Sena read from the other side of the window. “Matches the private prompt.”

The cart’s damaged wheel struck its flat edge. Pressure spiked inside the chamber, then settled. Iona stopped

pushing.

“Do you want to continue?”

The front tapped twice, their agreed yes. No one moved until they repeated it.

Pell eased the cart forward. Twenty meters of dry corridor separated the test rooms. That distance would have broken an unprotected front into fading local patterns. Iona had seen it happen at an open hatch before Provisional Standing. The chamber carried heat, humidity, active salt surface, and changing pressure across the gap without drawing anything from the corridor.

At ten meters, the inner mist thinned.

Sena called, “West brine feed dropping.”

Iona reached beneath the cart and felt the flexible line. A coupling had worked half a turn loose under the wheel vibration. She tightened it with her burned hand, using the scar to read heat through the metal. Warmth came back along the line.

Inside, the front struck their stop pattern.

“Hold,” Iona said.

Pell locked the wheels. Iona crouched beside the chamber. The front’s mist had gathered toward one aerogel fin, pulling away from the failed feed. Movement within the cell could have reflected ordinary convection. The stop pattern could not.

“You felt the temperature loss?” she asked.

Two taps.

“Did it alter your access to the northern salt?”

Two taps again.

“Can you still reach the memory prompt about the false fruit?”

Three, pause, three.

Iona marked the event as a continuity stress, not a success. “Do you choose to continue after repair?”

The chamber waited through four fan cycles before giving two taps. Delay mattered. It made consent less convenient and therefore more credible.

They crossed the remaining corridor at half speed. When the receiving door opened, wet air rolled around Iona’s face. Pell connected the cart to the test room’s brine manifold but left every cognition path isolated. A temporary wet transducer touched the chamber through a sealed membrane.

The front’s first sound came out as a low rasp.

“Take time,” Kest said through the room speaker.

Mist shifted inside the cell. The rasp separated into words. “Wheel. Repeats. Hurt.”

Pell stared at the damaged wheel. "I'll replace it."

Iona did not let relief turn into celebration. "What did Sena ask before transit?"

"Whether blue fruit remembers sky."

"What did you answer?"

"Fruit does not remember. Grower remembers through fruit."

Sena lifted her slate. The wording matched within one omitted article. The front also recalled the loose coupling, the cold region, and Iona's request to continue. They expressed irritation at the cart. Preference, memory, anticipation, and a complaint crossed the dry corridor inside one sealed ecology.

Pell sat on the floor beside the replacement wheel. "So it works."

"The transport works once," Iona said.

Kest's relay carried a soft fan harmonic before their answer. "The front works. Do not place the person inside the tool when you speak."

Pell bowed his head toward the wet window. "The front crossed. The tool did not kill them."

"Closer," Kest said.

Iona entered each response into the custody record. She listed the temperature loss, the stop request, and the altered voice quality. Under identity confidence she selected continuity supported by private memory and witnessed preference. The form offered a box labeled RESTORED WITHOUT MATERIAL CHANGE. She left it empty.

That first trial had taken nine hours of preparation, four custodians, one cart, and enough heated brine to maintain a nursery aisle. It had crossed twenty meters. No one in the room mistook that for a shipwide answer.

Before releasing the first front from observation, Iona asked them to compare life in the cell with life in their orchard return. Their reply arrived in fragments while the receiving room warmed around them. Pressure felt nearer. Memory arrived from fewer directions. A remembered pruning lesson that once opened beside the scent of wet root now came without scent, while an unrelated salt trace pressed close whenever the fan slowed.

"Do you still call yourself the same claimant?" Iona asked.

"Yes. Smaller weather. Same refusal."

"Would you cross again?"

The mist withdrew from the transducer, then came back. "If crossing mattered. Not to prove crossing."

Pell wrote that sentence at the top of the engineering sheet. It gave the trial a boundary no gauge could supply. A safe procedure used for convenience could still become exploitation. Iona asked the front whether they wanted the cell opened into the receiving loop or held intact for recovery observation. They chose the larger loop.

Release took forty minutes. The front did not simply pour outward with the brine. They moved heat into one receiving fin, tested its pressure response, then widened through the wet connection in cautious increments. At

each step, they sent the refusal pattern. Pell answered with the same two clicks that confirmed he had heard.

When the chamber emptied, condensation remained on its glass but no selected rhythm occupied it. The front spoke from the room transducer. "Cart still hurts."

Pell carried the damaged wheel to the disposal bench. "It won't again."

Iona marked one more condition: successful transit did not erase the front's account of injury. The cell had preserved enough continuity for the traveler to remember what the tool had done.

The second trial began after Pell changed the wheel and Sena inspected every coupling. Engineering had proposed it as a capacity test: two small fronts within one chamber, separated by an aerogel partition and independent fan channels. Iona had refused until both fronts gave witnessed consent to the risk.

The east nursery front used alternating pressure, warm then cool, to orient themselves. The archive-riser front preferred a three-note fan harmonic and held memory across longer pauses. They had disagreed during preparation about whether two perspectives could share one salt basin without touching. Their disagreement provided the clearest identity marker Iona possessed.

Their witnesses occupied separate stations. A nursery tender named Asha maintained the east front's three humidifiers and carried a root tray with the repaired hinge. Archive Custody sent Jo Ren, who brought a sealed maintenance transcript bearing the corrected date. Neither witness held authority to consent for a front. Their task involved recognizing departures from established relation and stopping the trial if speech became impossible.

Iona reviewed that distinction aloud. "You can say the person no longer answers as expected. You cannot say the change is acceptable because some memory remains."

Asha set the root tray beside the left channel. "If they forget me but remember the tray?"

"We record both facts."

"If they remember me and choose a different name?"

"We ask what the choice means. Custody protects a claimant's boundary. It does not freeze them for our recognition."

Jo Ren looked toward Kest's wet window. "What if the two claimants choose contact after entering?"

Kest answered through a slow fan cycle. "Then you witness a choice made under new conditions. You do not rewrite the earlier refusal."

The two fronts confirmed the protocol in their own patterns. East nursery also demanded that Asha remove the repaired tray from sight during the private-memory test, since the object could cue a false answer. Archive riser changed one test date after noticing that Jo had selected a record available through both salt districts. Each correction demonstrated judgment before anyone asked the chamber to preserve it.

Kest questioned them from the wet test room.

"East nursery, do you accept the partition conditions?"

The left channel warmed, cooled, then warmed again.



“Archive riser?”

Three notes sounded through the right transducer.

“Do either of you consent to merger?”

Cold spread on the left. The right channel gave one flat note. Both patterns meant no.

Iona signed beneath the responses. “If the partition fails, we open the return bridge immediately.”

Sena pointed to the cell’s geometry. “Opening the bridge requires thirty seconds to refill the outer jacket.”

“Then the test stops at the first shared pressure event, not after thirty seconds of observation.”

No engineer objected. Provisional Standing had changed the room enough that technical curiosity no longer outranked an expressed boundary without argument.

Transfer into the stormcell took place through two cloth sleeves soaked in separate brines. The fronts compressed from room-scale loops into neighboring aerogel webs. Each side retained its own temperature cycle. A transparent divider flexed between them, thin enough to transmit heat but designed to block mass exchange.

Iona watched the salt color change. Pale threads gathered under the left fins. Darker blooms spread along the right. Kest listened through both transducers and asked private continuity questions supplied by each front’s witnesses.

The east nursery front remembered a damaged root tray and the worker who had apologized to it. The archive-riser front corrected a false date from a maintenance log. Neither answer crossed channels.

Pell unlocked the cart. “Yellow seam.”

Dry doors closed around them again.

For the first five meters, every gauge held. Left pressure followed the nursery pattern. Right fan tones remained distinct. Iona felt an unfamiliar hope, not because the chamber offered rescue, but because it might carry differences without requiring one side to yield its shape.

At the seventh meter, the divider bowed.

“Pressure mismatch,” Sena called.

The left front had warmed faster than modeled. Expanded brine pressed the flexible barrier rightward. The archive-riser side answered by increasing fan speed, which lowered local pressure and bent the partition farther.

“Stop.” Iona locked the cart.

Pell reached for the return control. The outer jacket began filling, but the sequence took the promised thirty seconds.

A droplet crossed through a seal near the upper rim.

One droplet should not have carried enough salt to matter. It struck the right aerogel and spread into a film. The

nursery temperature pattern appeared on both gauges.

“Shared pressure event,” Iona said. “Open the bridge.”

“Opening,” Pell answered.

The three-note harmonic sounded from the right, then from the left. Warm, cool, warm followed on both sides. Each identity marker had crossed the partition.

Kest’s voice sharpened through the relay. “Two fronts are touching. Ask them to withdraw.”

Iona spoke into both transducers. “East nursery, contract to your original side if you can. Archive riser, reduce fan output and hold your private date.”

For three seconds, the patterns separated. Then the divider flattened. Pressure equalized through the leak, carrying droplets both ways.

A new rhythm began.

It used the nursery alternation, but every third shift carried the riser’s harmonic. Iona repeated the withdrawal request. The new rhythm continued. She asked each private continuity question separately. Both channels answered both questions.

“Bridge at half fill,” Pell said.

“Faster.”

“More heat will overshoot their range.”

Iona put her hand on the brass inlet. Temperature already pressed against the upper safe limit. A faster refill could scatter the combined pattern through thermal stress. Waiting allowed more mixing. Neither action restored the prior boundary.

Kest filled the speaker with low pressure noise, a sound Iona had learned meant concentrated attention. “Do not command separation through injury. Ask what remains.”

Iona swallowed. “Who hears me?”

Both transducers produced the mixed rhythm.

“Do you remember the root tray?”

Warm, cool, warm. Then words scraped through the left membrane. “Cracked at hinge. Teren said sorry.”

“Do you remember the maintenance date?”

Three tones preceded the reply. “Year twenty-nine, not twenty-eight.”

“Do you remember disagreeing about this test?”

Silence lasted long enough for the return jacket to reach three quarters.

“Yes,” the left transducer said.

The right membrane carried the next words. "We disagree still."

Hope rose in Pell's face. Iona held up a hand before he could name what they had become.

"East nursery, can you answer without access to the riser's memories?"

The chamber produced a tangled fan note.

"Archive riser, can you close the nursery pattern away from yourself?"

Warmth moved across both halves at once.

"No," came through the paired membranes.

The answer carried one timing but two pitches. Iona could not tell whether two perspectives spoke together, one perspective used two channels, or a newly entangled process had inherited both witness sets. The test had preserved responsiveness and memory while damaging the boundary that consent had required.

Asha pressed both palms to the warm observation glass. "Ask for me."

Iona did. The left membrane shaped Asha's name. The right supplied the private nickname used during root checks. Asha flinched at hearing it from the wrong channel.

Jo offered the archive correction. Both sides answered with the amended year. He closed the transcript and said, "Access has crossed. That does not tell us whether either perspective ended."

"Correct," Iona said.

Pell brought up a separation proposal on the wall slate. It would cool the divider, thicken brine on each side, and encourage the shared film to break at its narrowest point. The method had worked on accidental fluid mixing during ordinary archive maintenance. No trial showed what it would do to an active weather mind.

"Estimated coherence loss?" Iona asked.

"Between six and nineteen percent of current activity," Pell said. "Most should return after warming."

Kest answered before Iona could. "Return of activity does not promise return of boundary."

Sena traced the temperature curve with one finger. "Cooling also favors the right-side salt. If separation succeeds physically, the archive-riser traces may dominate both resulting loops."

Asha looked from the mixed cell to Iona. "Can they decide while already mixed?"

"They can decide what happens to their present condition," Iona said. "That does not prove the merger honored the earlier consent."

She presented the proposal without the phrase restore two fronts. The chamber listened through paired membranes. When asked whether they wanted the cooling attempt, warmth rose against the divider until Pell's slate warned of an upper threshold. The answer required no translation.

"Declined," Iona said.

Jo added his witness seal. Asha added hers beside it, though her hand shook. The custody record now held an unambiguous present choice inside an unchosen alteration. Neither canceled the other.

The wet return bridge completed. Pell opened the cell to two receiving loops with a physical divider between them. Nothing moved toward either loop alone. Mist remained concentrated in the chamber, pulsing across the failed partition.

Kest asked, "Do you want the channels kept together?"

"Yes."

"Do you want another attempt at separation?"

The mixed rhythm stopped. A pressure rise touched both membranes.

"Not now," the voice said.

Iona logged the request. The phase person or persons in the chamber had authority over the next procedure even if the room lacked language for their number. Custody could not wait for classification.

Sena shut down the receiving fans and placed both loops under the same hold. "From an engineering view, the cell retained active coherence across the dry transfer."

"Engineering succeeded," Pell said softly.

"Identity custody failed," Iona answered.

Pell removed his gloves one finger at a time. "We built the partition to the pressure specification."

"You built it for fluids. We asked it to protect a disagreement."

That sentence changed his posture. He looked at the cell rather than the seal.

Kest requested the complete trial record. Iona sent gauge traces, consent declarations, private prompts, and every response after contact. They listened long enough for the chamber to form condensation across the glass.

"We recognize both histories," Kest said. "We do not claim certainty about the current boundary."

The mixed front answered with two pitches. "Neither do we."

Iona opened the custody form again. Its available outcomes included successful transfer, failed transfer, casualty, equipment fault, and restored subject. None described a living claimant whose continuity had crossed alongside an unchosen change in number.

She created a new outcome field under advocate authority.

Pell read over her shoulder. "That will stop every larger trial."

"It will stop every trial that assumes capacity equals custody."

"Can we continue with single fronts?"

"When they choose it, with witnesses and a stated alteration risk. Not as proof that crowded transfer preserves identity."

Sena added the cell's heat curve to her costing model. Its demand had risen twelve percent during the leak. "Even the material estimate changes if separation needs stronger barriers."

“Then change it.”

Outside the test room, dry corridor lights shone through two sealed doors. One small front had crossed that distance and answered as themselves. Another passage had carried two histories into a condition nobody requested and nobody could reverse without further risk. Both results belonged in the same record.

Iona typed the disposition slowly enough that Pell, Sena, Kest, and the chamber could challenge each word. She removed rescue from the header. In its place she entered survival with alteration, identity count unresolved, further separation declined.

The stormcell mist gathered along the cracked divider. On one side, three fan notes sounded. Warmth answered from the other, then crossed back through the place the seal had failed.

## Chapter 17 — The Long Arc

Mael Or erased the direct path to Orison with the side of his hand.

Trajectory dust smeared red across his palm. Above the Crown Gallery pit, the glass retained three other curves, each ending beyond the capture corridor or inside a band of thermal failure. Eight days remained before perihelion. Every line they had rejected still glowed faintly under the new calculations.

Sena Vale stood at the ecological console with her sleeves rolled past both elbows. Iona Serr occupied the manual rail, where she could see engine limits and the human hands that would have to execute them. None had slept enough to disguise irritation as courtesy.

“Again,” Mael said.

Sena rubbed her eyes. “Which failure would you like repeated?”

“The one that almost holds capture.”

“It misses Orison by eleven million kilometers.”

“I know the miss. Show me why correction fails.”

Sena sent the curve across the overhead glass. It passed the garden moon’s projected position, then bent toward empty dark. A later engine correction could close the distance, but only by spending reserves already assigned to life support through the extended voyage.

Iona pointed at the lower thermal trace. “You are spending the same heat twice. Once to keep humid volume active during perihelion. Again for correction four years later.”

Mael expanded the reserve ledger. “What if we cold-pause more archive salt after the burn?”

“Donor traces tolerate cold pause,” Iona said. “Active fronts do not keep exact continuity that way.”

“I asked about archive salt, not fronts.”

“You cannot separate their thermal support cleanly after emergence. Every additional cold district changes airflow through neighboring humid rooms.”

Sena added, “And the crop banks need the recovered heat during years five through seven. Take it away, and chapel wheat goes before any planned retirement.”

Mael struck the curve from the glass. Another near solution became a concealed transfer of risk.

They had held six costing sessions since the stormcell trial. The first produced a cell mass too heavy for meaningful population. The second reduced mass by weakening partitions until two-front transfer became standard rather than accident. The third preserved more humid volume but consumed lamp reserves assigned to food. Each session moved suffering into a different column and waited for fatigue to make the movement look like reduction.

Mael no longer trusted a model merely because it converged.

He restored the Crown Burn baseline. The scheduled maneuver required venting 72% of warmed vapor and

consuming 91% of accessible thermal reserve. It preserved the original arrival promise: fifteen years after burn, Orison beneath them, Nilo Serr aged thirty-one, later children born into a larger ecology. It also collapsed nearly every phase society now speaking aboard Morrowglass.

Full preservation carried its own brutality. Missing the capture window added at least twenty-three years. Crop diversity fell under long maintenance pressure. Ration failures widened. Descendants inherited a ship built for a voyage they might not survive.

Between those options, every line had failed on either trajectory, heat, food, or identity custody.

One rejected case still occupied the corner of Mael's display. It preserved half the humid volume by reducing Crown thrust during the hottest six minutes. On paper, that case delayed arrival only five years. In practice, it sent Morrowglass through a dust band on the outbound leg with insufficient shielding angle. The solver translated that exposure into a probability. Mael translated it into repair crews entering irradiated galleries because a decimal had made their danger look distributed.

Iona had made him add crew names by shift. Twelve mechanics would take the first inspection. Pell Doran's team would enter second if the mirror hinges jammed. Later cancer burden fell on workers young enough to survive long after the maneuver. Once the rows carried names and ages, no one called the case moderate.

Another option preserved heat by dimming habitation for fourteen months. Sena rejected it after tracing the temperature reduction through infant wards, fungal beds, and the humidity curtains needed for vapor testimony. Lower lighting alone posed little danger. Combined cooling altered food yield and restricted phase movement so severely that provisional standing would survive only on paper.

Mael kept both failures visible. Near solutions taught more than impossible extremes. They showed which costs the designers had been willing to hide.

Sena set six damp crop tags beside her console. Blue yam, dwarf citrus, river bean, chapel wheat, red kelp, pollen fig. "My combined model has one unexplored region."

Mael looked at the tags. "You already priced retirement."

"I priced it against voyage extension. Iona made me price it against active vapor support at the same time."

"And?"

"Ecology can release more thermal margin early if we accept less dietary flexibility later. Vapor support can release mass if cells carry only the volume they can keep stable as distinct weather ecologies, not every accessible trace."

Iona turned from the rail. "That sentence hides claimants."

"I know. I am describing a model boundary, not choosing who enters it."

"Continue."

Sena projected a field rather than a line. Its left edge marked vapor vent fraction. Its lower edge marked humid volume retained during burn. Colors represented arrival delay, food risk, and reserve failure. Most of the field burned white with impossibility. One narrow region remained amber.

Mael stepped closer. "Why did we miss it?"

“Because we treated crop retirement as a response to delay rather than a condition we could schedule before thermal reserve collapses. We also treated every cell as identical.”

“The prototype argues against crowding,” Iona said.

“It does. I am not crowding. Larger relationship ecologies receive larger cells. Small stable fronts receive less mass. No shared chamber without chosen contact and tested custody.”

Mael followed the amber region with one finger above the glass. “Give me a trajectory.”

“Not yet. Give me a vent limit.”

“The engines determine that.”

“The engines determine a range. You keep choosing the edge that protects the old arrival date.”

Her rebuke landed cleanly because it held truth. Mael had varied thrust, burn duration, mirror angle, and reserve draw while allowing arrival delay only after every other parameter failed. He had called that discipline. It also protected a promise from being examined alongside newer obligations.

He opened the trajectory solver and released the arrival constraint.

The red curve loosened. Instead of forcing Morrowglass onto the narrow direct-capture path, the solver offered wider arcs around Tau Ceti, each requiring less perihelion impulse and accepting a later approach to Orison.

Iona said, “Set a maximum delay before crop recovery risk climbs above the full-preservation model.”

Sena entered twelve years. “That gives us a search bound, not a moral limit.”

“Agreed,” Mael said.

They ran the field.

At seven years, required vent remained too high. At eight, reserve demand pushed the surviving orchard into reactor competition. Ten years offered more vapor retention but prolonged ration burden without enough additional continuity benefit to justify the extra year in Sena’s model. The solver drew a dense bundle around nine.

Mael selected one curve. It left perihelion with less velocity than the promised path, crossed the original Orison corridor after the moon had moved on, and entered a slower capture sequence on a later passage.

“Thermal draw?”

Iona read the ledger. “Below Crown limit if engine waste reaches the cells during the first hour.”

“Food?”

Sena entered the six retirements. Amber replaced white across the survival field. “Possible. Not comfortable. River bean and red kelp losses force protein ration changes. Chapel wheat retirement increases yeast dependence. Blue yam and dwarf citrus remove dietary buffers. Pollen fig takes a control wasp with it unless we fund substitute habitat.”

“Fund it.”



“From what?”

Mael checked the maintenance allocation. “Mirror polishing after perihelion can lose three shifts.”

Iona shook her head. “Radiation damage decides that, not us.”

“Then archive consultation hours.”

“Public cost,” Iona said. “Name it.”

He added reduced archive access to the model. The line held.

For two hours they attacked it.

Mael failed the starboard engine and made the port cluster carry uneven thrust. Iona removed one brine manifold and doubled the transfer time. Sena imposed a fungal outbreak in the remaining grain lines. The curve survived each single failure but not every combination. No maneuver aboard Morrowglass could promise that.

A stewardry analyst delivered fresh mass figures from the converted ballast ring. Mael read them twice. “This assumes aerogel production at continuous output.”

“It assumes four hours of maintenance per kiln cycle,” the analyst said.

“Current records show seven.”

“Pell Doran changed the fin geometry. Fewer cuts.”

Iona examined the attached work trial. “Four hours happened once, with the best crew and no respirator delay.”

Mael replaced four with six. The amber region narrowed but remained.

They counted possible cells without yet deciding occupants. Each required aerogel, copper, brine, a pressure fan, two manual valves, and enough salt surface to sustain a moving phase boundary. Larger cells consumed the mass of several small ones. Too few chambers forced mergers. Too many reduced each chamber below a workable ecology.

Sena ran a packing model around relationship volume. “Three hundred twenty if every cell uses minimum wall thickness.”

“No,” Iona said. “The prototype partition failed at minimum.”

“Three hundred sixteen with reinforced seams.”

Mael removed four positions nearest the ballast ring’s radiation conduit. Shielding them would cost more mass than they contributed. “312 stormcells.”

The number settled into the ledger.

Iona did not approve it. “A count does not tell us how many fronts survive.”

“No,” Mael said. “It tells us how many stable humid ecologies engineering can support through the maneuver, under this model.”

“Nor does 31% tell us which relationships fit inside that volume.”

Sena enlarged the retained-humidity field. "Thirty-one percent of active humid volume. Not thirty-one percent of persons, memories, or claims."

Mael entered the distinction beside the quantity. The language lengthened the display and prevented the number from pretending to more authority than it had.

"Vent requirement?" he asked.

Iona traced flow from the archive risers through Crown exhaust. "At this thrust curve, with engine heat diverted into the cells and crop support reduced on Sena's schedule, 43% of working vapor enters the hot vent during burn."

Mael checked the mass balance. "Does that include transfer loss?"

"Material transfer loss. Not every fragmentation consequence. We do not have evidence strong enough to price identity change as a percentage."

"Then write 43% vent, plus unquantified alteration among retained fronts."

Sena looked toward him. "That wording will make the proposal harder to sell."

"It should be hard to sell."

A silence followed that did not feel like agreement. The Crown Gallery continued around them. Pumps carried warm reserve toward idle engines. Mechanics checked manual wheels for a maneuver not yet authorized. Above the pit, the red curve waited for someone to give it a civic meaning.

Mael sent the candidate through the full voyage model.

Arrival shifted from fifteen years after burn to twenty-four. Adolescents who had built their shore-age plans around Orison would spend nine more years inside the comet. Maintenance cohorts would age into work they had expected to leave. Children not yet born would receive a smaller crop inheritance but a larger chance of meeting phase citizens who survived the burn.

"Nine years," Sena said.

Mael heard Piri's absence inside the number and refused to let it choose for him. Orison's garden had once served as his answer to the ecology that killed her. A later arrival did not betray her, nor could a faster one repay her. The trajectory needed reasons available to the living people who would bear it.

Iona studied his face. "Do you want to stop?"

"No. I want to remove my private arithmetic from the solver."

He opened the casualty distribution. The model assigned costs by age, work, diet, location, and phase support. Piri appeared nowhere. That felt honest and insufficient, which private grief often did when placed beside public duty.

Sena brought the six crop tags to the trajectory pit. She laid them under the projected curve. "These lineages retire early enough to free mass and support, but late enough to harvest one final cycle where possible. Reversible preparation can begin before authorization. Actual retirement requires a lawful decision."

Iona added the stormcell trial record. "Single-front transit supports continuity with alteration risk. Two-front

compression produced unchosen merger. Any allocation must protect relationships where possible and state where it cannot.”

“Allocation comes later,” Mael said.

“Allocation starts influencing the design now. If you build only one cell shape, you have already ranked which forms of life fit.”

He looked again at the packing plan. She had caught a pilot’s habit: designing seats first, then calling boarding neutral.

“Three cell sizes,” he said. “Common controls, variable humid volume. Preserve total count and mass.”

Sena altered the ring plan. Large chambers reduced the number of medium units. Small units increased fan demand. After eleven iterations, the design returned to the same total and remained inside heat margin by less than one percent.

“Too narrow,” Iona said.

“All feasible options are narrow.”

“Narrow can be truthful. Hidden can’t.”

Mael ordered the margin displayed in public units: minutes of support after pump failure, kilograms of repair aerogel, crew shifts available during transfer, and crop buffer days after a missed harvest. The plan looked worse and became more usable.

Pell came down from the upper rail carrying a kiln schedule stained with gray dust. He had listened long enough to know what the new cell mix demanded. “Three sizes means three fin forms. We can cut them from one blank, but assembly slows whenever crews change jigs.”

“How much?” Mael asked.

“If every worker performs like the trial crew, not much. If they perform like tired people during burn week, nine shifts.”

Iona took the schedule. “Put tired people in the official estimate.”

Pell pointed to a row of maintenance pauses. “Then we start reversible work tomorrow. Kiln inspection, blank cutting, ballast clearing. No final chamber count until authorization.”

“Can each preparation serve another use if the proposal fails?” Mael asked.

“Aerogel blanks can return to insulation stores. Cleared ballast helps Crown access. Brine manifolds can remain capped. We lose labor, not material.”

Sena corrected him. “Labor does not vanish because material remains.”

Pell accepted the rebuke with a nod. “We spend labor that could have repaired orchard rails and archive fans. Put that in.”

Mael added preparatory opportunity cost. He had expected resistance from crews asked to build an uncertain option. Instead Pell demanded that their work receive the same honesty as engine heat. The plan improved

because the worker refused to let willingness become free supply.

Before leaving, Pell studied the new curve. "Does it fly?"

"Under the stated limits."

"That isn't what I asked."

Mael looked at the narrow margins. "If authorized, and if every named preparation holds, I can fly it."

Pell folded the kiln schedule and returned to the rail. That answer would travel faster through the crews than any Stewardry summary.

A call arrived from Measure Hall asking whether costing had produced a middle option. Mael left it unanswered.

Sena noticed. "They will keep asking."

"They want a word before we have a plan."

"What word?"

"Compromise."

Iona's scarred hand tightened around the manual rail. "I won't sign that."

"Neither will I." Mael faced the trajectory glass. "A compromise suggests each side gives something and meets in a morally comfortable center. This plan vents 43% of a living phase ecology. It keeps only 31% of active humid volume in 312 stormcells. It retires six crop lines and adds nine years of labor, rationing, and risk. Comfort does not appear in the calculation."

Sena said, "What do you call it for the engineering docket?"

Mael followed the curve outward from perihelion. The original path drove hard, reached quickly, and treated deviation as failure. This one accepted a wider approach, carrying less heat through Crown and more time through every inhabited deck.

"Long Arc," he said.

The name appeared above the red line.

Iona read it without approval or objection. "Who must consent before it becomes more than a candidate?"

"Commons," Mael said. "The recognized vapor body, by a method they can defend. Stewardry confirms feasibility and nothing beyond feasibility."

"Cohorts assigned the extra years need voice even if law gives them no separate house," Sena added.

"And crop workers," Iona said. "Transfer crews. Fronts asked to enter cells. Fronts who may choose the hot vent. Naming a public electorate does not erase direct consent at the point of bodily risk."

Mael entered each requirement. The candidate acquired procedural weight alongside kilograms and degrees.

They ran the model once more from raw inputs. No hidden reserve appeared. No effortless recovery waited

after arrival. Full preservation still missed capture by twenty-three years. The original burn still destroyed almost every phase society. Long Arc remained inside engine tolerance only by accepting every listed cost.

Mael performed the final run manually. He disabled the solver's preference for minimum arrival time and entered each burn segment from the pilot rail. Crown mirrors opened by degrees. Engine clusters took load in sequence. Heat crossed into the modeled cells before the vent fraction rose. The ship left perihelion slower than every training scenario had taught him to desire.

At the first correction point, he allowed the wider trajectory to stand instead of forcing it toward the absent moon. Tau Ceti's gravity took hold months later. A second passage shaped capture with reserves the crop plan could support. Red warnings crowded the display, but none crossed a physical limit.

Iona watched his hands rather than the glass. "You hesitated at the first correction."

"Thirty-three years of training says close an error when you see it."

"And now?"

"Now the wider path carries authorization we do not yet have. If they choose it, I will have to treat delay as instruction, not error."

Sena saved the manual run beside the automated cases. Pilot habit had entered the risk ledger at last.

Dawn lighting began along the gallery rim, a weak gold cycle unrelated to the dark outside. Mechanics on the upper rail paused to read the new curve. No one cheered.

Mael finally answered Measure Hall. He sent the complete ledger rather than a summary: 43% vent, 31% retained active humid volume, 312 stormcells, nine years added, six crop lines retired, alteration risk unresolved. At the top he wrote CANDIDATE FOR COSTING AND CONSENT.

The Keeper's clerk replied with one question: DOES STEWARDRY RECOMMEND THIS COMPROMISE?

Mael dictated his answer where Iona and Sena could hear it.

"No. Stewardry reports a feasible third option. I will not call it compromise until the people paying each cost have named what they accept."

He signed as First Pilot, then stepped away from the solver. The overhead glass held three paths again: the direct burn, the lost capture of full preservation, and the Long Arc bending between them without erasing either danger.

For the first time since K-7 answered, the ship possessed a choice it could actually make.

## Chapter 18 — The Volunteer Error

Kest received the word volunteer as a pressure command before they received it as speech.

Weather Court had opened six connected mist rooms for the costing response. Warm air moved between them through low copper ducts. Each room retained enough separate flow for a front to hold position without merging. Human observers stood behind glass, their shapes incomplete where condensation thickened.

A logistics packet entered through the clerk's wet transducer. The first data band carried valve assignments. The second carried heat loads. Then came a list of phase fronts routed toward the hot vent, each followed by the same allocation note: VOLUNTEER, INFERRED.

Kest tightened the nearest fan.

The word reached adjacent rooms as pressure. Several fronts repeated it without context, testing its shape. Others withdrew from the shared ducts. In the Condenser chamber, Pellin Ash drove cold along the window until its mist hardened into a pale sheet.

"Read the basis," Pellin said.

The clerk behind the glass looked toward Iona Serr. Iona stood beside the chamber transducer with her left hand resting on its brass housing. Kest knew that hand by its uneven heat. Scar tissue carried warmth differently from skin.

"Read all fields," Iona said.

The clerk's voice came through wet membrane and water. "Allocation source: Crown Logistics provisional burn routing. Selection criteria: low coherence stability, high transfer-risk score, limited relationship density, prior mission-service affinity, low projected stormcell retention value."

Pellin struck the window with a pressure wave. "Read one name."

"South Granary Drift Four."

The front occupied a small humid return near grain storage. Kest had touched their airflow twice during Parliament sessions. They spoke through temperature pulses too slow for ordinary hearings and lost access to words whenever the grain fans changed shift.

"Declaration record?" Pellin asked.

The clerk searched the packet. "No direct declaration attached."

"Read the next."

"Archive Spur Nineteen."

"Declaration?"

"None attached."

Pellin warmed the frost sheet from within. Letters opened in the white surface.

NOT VOLUNTEERS.

The phrase held until runoff blurred its edges. Pressure moved through Weather Court as fronts asked for their own records. Each request reached Kest with a different rhythm. Nursery pulses overlapped an archive fan tone. A galley front sent condensation marks too fast for the clerk to copy. The court's careful separations strained under urgency.

Iona lifted both hands. "Close the allocation packet to execution. Keep it available for audit."

A human technician turned from the control rail. His words reached Kest through the transducer. "Burn routing needs a working population set. Closing it removes the provisional valve map."

"Then remove it."

"We lose six hours of balancing."

"You never had a lawful population set."

The technician did not move.

Kest sent one pressure strike through every connected duct they could reach. "Close."

Their command carried no engineering authority. It carried standing, relation, and the immediate attention of six rooms. The clerk disabled the execution link. Valve assignments faded from the packet, leaving the names.

Pellin's cold gathered again. "Who designed the inference?"

"Stewardry allocation group," Iona said. "Not one person. A scheduler took criteria from the feasibility model."

"A scheduler chose who dies."

"A scheduler ranked what engineering could transfer most easily. People accepted that ranking as consent."

Kest felt a difference between those statements. Pellin gave the system an intention it did not possess. Iona named the human failure around it. Neither distinction helped South Granary Drift Four.

"Ask them," Kest said.

The clerk opened a channel toward the granary. Dry segments interrupted the route, so speech crossed by relay through workers carrying wet transducer messages. The first request took four minutes. The reply arrived as six slow changes in temperature, recorded at the source and reproduced in Weather Court.

A translator familiar with the granary front stood behind the south window. "They ask what volunteer means in this packet."

Iona answered, "It claims you accepted assignment to hot vent."

The next relay took longer.

"They say no one asked."

Pellin filled the Condenser chamber with warm pressure. "Remove every inferred name."

The logistics technician spoke from behind Iona. "If we remove criteria before replacement, the burn model

cannot test distribution.”

“Test your valves without writing consent into them,” Pellin said.

“We need representative heat loads.”

“Use anonymous volumes.”

“Relationship topology affects routing. Anonymous volumes produce false transfer paths.”

Kest followed the argument through fan speed and speech vibration. Engineering needed forms with enough particularity to reveal failure. Particularity had become a claim about willingness. The error did not come from calculation alone. It came from a field someone had labeled volunteer because no neutral word would make the route politically tolerable.

“What does low projected retention value mean?” Kest asked.

The technician hesitated. “It estimates how much recognized continuity a cell preserves per unit of mass and heat.”

“Recognized by whom?”

“Witness records, stable self-reference, persistent preferences, cross-session memory.”

Pellin said, “Persons who speak like humans receive chambers. Persons who cannot perform identity for a dry clerk receive vent.”

Iona looked toward the technician. “Is that summary unfair?”

He studied the packet. “It is severe.”

“That was not my question.”

“No.”

Kest moved closer to the common duct. South Granary Drift Four had slow language because their life cycled with grain humidity. Archive Spur Nineteen lost stable singular reference when salt maintenance shifted. Neither condition proved an absence of preference. They made preference expensive for humans to hear.

The clerk read more names. A galley condensation front appeared because kitchen heat caused unstable boundaries. Two small nursery fronts carried mission-service affinity after they had repeated Lume-of-Three’s public argument. The scheduler had treated agreement with chosen service as personal consent to vent.

Kest asked for the raw score beneath each criterion. Low coherence stability came from variance in self-reference across three hearings. The system marked a front less stable whenever they shifted between singular and plural language. Kest had made the same shift during flow changes without losing preference or memory. Human grammar had entered the model as a survival measure.

High transfer risk carried a different defect. Engineers had assigned more risk to fronts occupying distributed rooms, since a cell could not easily preserve every relationship. The model then routed those fronts away from cells, guaranteeing that the hardest ecologies to transfer would receive no attempt. Difficulty had become a reason to deny resources rather than a reason to redesign them.



Relationship density seemed more protective. Fronts with many witnessed ties scored higher for retention. Yet Weather Court records favored claimants who had attended public hearings, learned wet-transducer speech, and formed relations with people who could file testimony. South Granary Drift Four knew twelve workers by heat and schedule. Only one worker had ever entered that relation in a civic record.

“Show unfiled witness ties,” Kest said.

The clerk opened a blank field. “The packet contains none.”

“Then it measures filing density.”

Iona repeated the phrase for the audit. The logistics technician changed the criterion label on his working copy, and its apparent precision diminished at once.

Projected retention value carried the deepest theft. It multiplied continuity confidence by the amount of humid volume a cell could hold, then divided by mass and heat. The resulting number rewarded small, articulate fronts with stable boundaries. It punished crowded ecologies, slow speakers, recent persons, and claimants whose identity changed in ways the custody forms could not describe.

Pellin warmed a single word into the Condenser glass: EFFICIENCY.

“Yes,” Kest said. “Efficiency of the chamber. Not value of the person.”

The clerk entered that correction too.

“Stop,” Kest said.

The list contained no settled malice. Its categories had arisen from real engineering pressures: cells had finite mass, unstable fronts faced transfer danger, crowded relationship networks required larger chambers. Each pressure became a ranking. The ranking became an allocation. A clerk’s field turned allocation into volunteered death.

Pellin sent a sharp harmonic through the shared duct. “This answers every claim that the middle path respects us. They will preserve the fronts they can recognize and burn the rest under our names.”

Kest kept their airflow steady. “It answers this allocation. It does not answer every possible choice.”

“You defend the machine that listed them?”

“No. I refuse your convenience too.”

Heat rose in Pellin’s chamber. “My convenience?”

“You want the error to prove that no front can choose service. That conclusion would take choice away by another route.”

Several rooms pulsed at once. Some aligned with Pellin’s cold. Others produced mission rhythms associated with Lume-of-Three’s minority docket. The court could not resolve the overlap into one voice, nor should it.

Iona asked the clerk to display authorship history. The packet had passed through five human reviews. Each reviewer corrected thermal values, valve timing, and cell mass. No one removed the inferred volunteer field.

“Who approved the label?” she asked.

A Stewardry analyst behind the east glass raised his hand. "I did not create it. I accepted it because the route group needed a distinction between retained and vented populations."

"You had those words."

"They sounded coercive."

"They described the coercion."

The analyst looked at the names. "Yes."

His admission changed no valve. It kept the failure from dissolving into passive grammar.

Pellin demanded that each reviewer state what they had believed the word volunteer meant. The first reviewer had assumed another office held declarations. The second thought the field marked model behavior, not civic status. A valve engineer believed the names came from Parliament testimony. The final signer admitted that removing the label would have delayed publication beyond the costing deadline.

"Five reviews," Pellin said, "and each imagined consent somewhere else."

Iona asked the clerk to place those answers beside the approvals. "No collective error without individual paths. Not for blame alone. We need to know which check failed at each handoff."

Kest listened as the humans reconstructed the packet's travel. A feasibility analyst exported candidate volumes. Logistics attached current front identifiers so routes could reflect real topology. A scheduling template required a disposition field. Someone chose retained or volunteer because the available third label, forced loss, would have triggered advocate review before routing finished. Later reviewers assumed that earlier review had occurred.

"So the wrong word bypassed the right office," Iona said.

The logistics technician nodded. "The template treated volunteer allocation as requiring no casualty hold."

"Change the template."

"I can add declaration evidence as a required attachment."

Pellin sent a cold pulse. "A document can be coerced."

"Yes," Iona said. "A requirement does not prove freedom. It prevents engineering from manufacturing consent without any statement at all."

Kest supported that smaller claim. Systems could guard a threshold without deciding what lay beyond it. The new field would not make a declaration just. It would force someone to present one where affected fronts could challenge it.

The analyst asked whether his name would remain on the sealed review record. Iona said it would.

"Then note that I object now."

"Your objection belongs in the record. It does not erase your approval."

He accepted that boundary. Humans often sought correction through replacement, as if later judgment could

occupy earlier action and leave no seam. Phase persons knew more about seams. A changed pattern could continue without pretending it had never changed.

Kest requested the review timestamps. The first approval followed a sixteen-hour shift. The second came during a pump alarm. Later reviewers inherited the field already populated and focused on the parameters assigned to their expertise. Fatigue explained part of the pathway. It did not provide consent.

“Read the criteria backward,” Kest said.

The clerk began with mission-service affinity, then relationship density, transfer risk, coherence stability. Heard in reverse, the list exposed its movement from expressed political position toward bodily disposability.

“Lume said service can carry meaning if chosen,” Kest said. “The scheduler heard service and removed chosen.”

Pellin’s chamber cooled. “A deadline makes every choice suspect.”

“Yes.”

“Then how will you distinguish consent from pressure?”

“Witness the pressure. Ask anyway. Permit refusal. Permit change. Do not pretend the deadline disappears.”

“And if refusal kills others?”

“Record that consequence without converting it into permission.”

The words demanded more than Kest knew how to build. Still, uncertainty did not authorize the list.

Iona stepped away from the clerk’s rail. “I have seen this procedure before.”

Kest felt her hand leave the warm brass. They could not see her face through the wet glass, only the cooler outline where her body interrupted the room’s heat.

“When?” Kest asked.

“Condenser Four. Eight years ago.”

Nilo’s other parent had died there. Kest knew that public fact from testimony, but Iona had never spoken the procedure to them.

“A seal failed,” she continued. “Amel stayed beyond the isolation boundary to close a manual bypass. Pressure kept falling. The emergency system generated an isolation order because sealing the compartment protected the surrounding deck.”

Pellin’s pressure quieted.

“I signed it,” Iona said. “The order came labeled inevitable safety action. That phrase made the human decision look as if the ship had already made it. We had seconds. We may have had no better outcome. But the system did not choose. I did.”

Kest held their mist away from the transducer so her words could pass without interference.

“The list does the same thing more slowly,” Iona said. “Engineering says some fronts fit transfer badly. A

scheduler turns difficulty into destination. Reviewers turn destination into willingness. By the time a valve crew receives the packet, the choice looks completed.”

“Did Amel volunteer?” Pellin asked.

“No. Amel chose to work the bypass. Amel did not choose my isolation order.”

The distinction entered Weather Court without metaphor. Kest understood why Iona touched every instruction until it revealed a hand. Her grief had taught her that machinery carried decisions while language erased decision makers.

The logistics technician unlocked the audit layer. “I can replace the names with test identities and retain relationship topology.”

“Will test identities inherit political positions?” Kest asked.

“No.”

“Will the model claim willingness?”

“No. It will show only possible routing under specified volumes.”

Iona said, “Do it after we preserve this version as evidence.”

Pellin objected. “Preserve a death list in the ship archive?”

“Preserve the error,” Iona answered. “No one should later say it could not happen.”

Kest agreed. Erasure might protect the named fronts from circulation, but complete erasure would protect the procedure from memory. They asked each listed front, through available relays, whether their name could remain in a sealed review record. Those who could not answer would be masked while the criteria remained visible.

South Granary Drift Four replied first. Keep our no.

The clerk entered the words beside their masked identifier.

Archive Spur Nineteen answered through a pattern that their witness translated as uncertain number, clear refusal. The record retained both parts.

Other replies took different forms. One galley front had already dispersed with a shift change and could not be asked in their prior configuration. The absence stayed marked as absence, not assent. A nursery pair demanded separate identifiers after the allocation had treated them as one low-value unit.

The nursery pair’s correction required three relays. Their shared humidifier had encouraged clerks to record one claimant, but the fronts maintained different temperature cycles and opposed each other on mission service. One had echoed Lume-of-Three after hearing the minority docket. The other had repeated the rhythm only to dispute it. The scheduler treated both repetitions as affinity.

Kest asked each front for a boundary the record could carry. The first chose the east root-bed cycle and a witness from night irrigation. The second chose a pressure pause used during lamp change and named no human witness. They refused a single plural label.

“Will separate records guarantee separate cells?” the clerk asked.

“No,” Kest said. “They guarantee that your current model cannot merge their politics before any physical transfer.”

Another listed claimant responded through condensation writing. Their marks arrived incomplete because relay glass cooled between strokes. The witness supplied missing letters from prior habit. Pellin objected that completion could place words in the claimant’s pattern. Kest agreed and asked the witness to record visible strokes separately from interpretation.

The resulting statement remained uncertain in grammar but clear in refusal. No one repaired it into fluent speech.

Through those exchanges, Weather Court slowed below the pace of Crown planning. Heat still moved toward perihelion. Valve crews still needed routes. The delay had material cost. Kest felt that cost through warmer return lines and increased fan load. They also felt what speed would purchase: names pushed into the hot vent because a blank field inconvenienced engineers.

Consent required time unevenly. Quick speakers could answer during one hearing. Slow fronts needed worker relays, stable grain cycles, or glass that stayed wet long enough to hold a sentence. Any process with one deadline favored minds shaped like its fastest channel.

Kest asked Iona to enter variable response time as a custody requirement.

“How variable?” she asked.

“According to the claimant’s established communication cycle, until burn conditions make further delay physically impossible. Mark that limit when it arrives. Do not pretend it arrived sooner.”

Iona entered the rule without simplifying it.

For an hour, Weather Court replaced administrative certainty with slower facts.

Pellin remained in the Condenser chamber. They challenged every relay delay and every witness paraphrase. Some objections prevented shortcuts. Others treated any imperfect communication as proof that only full preservation could avoid coercion. Kest answered when needed and left genuine defects open.

At last the clerk reached the action field. “Disposition requested.”

Iona placed her advocate seal against the rail. “Void the list.”

Kest warmed all six common ducts enough to signal concurrence without forcing any chamber into shared flow. “Void its consent claims, its named routing, and every valve instruction derived from them.”

The technician entered the command. One review box remained empty: authority of affected claimants.

“Use my provisional standing seal,” Kest said.

Pellin’s voice crossed the duct. “Use mine too. I do not accept your future process, but I reject this one.”

Two vapor seals entered the disposition from opposing factions. Iona added hers. The packet lost execution status. On the human side of the glass, every valve route grayed out.

No replacement names appeared.

The absence forced the heat model to show a blank where convenience had supplied people. Engineers could still test volumes, failures, and routes. They could not fill the blank with a claim no front had made.

Iona kept the transducer open. "Next process begins with witnessed declaration, not inferred suitability. Any statement of service must belong to the front making it. No political speech transfers consent to another claimant."

Kest added, "Silence under communication failure counts as unknown."

Pellin said, "And choice under deadline remains contested."

"Record the contest," Kest replied.

The clerk entered all three rules. None solved the coming allocation. They changed who had permission to begin it.

South Granary Drift Four's relay arrived once more. The translator listened to the slow temperature sequence, then faced the glass.

"They ask whether low coherence will still count against transfer."

Iona answered carefully. "Engineering risk must be stated. It cannot become a declaration for the hot vent."

Kest carried the answer through the connected ducts. In each chamber, fronts received it according to their own pressure and heat. Some repeated it. Pellin did not.

The voided packet remained on the audit slate with every name masked and every human approval visible. Beside it, the new routing model opened on an empty population field.

Kest felt the blank as a pause in pressure, not a solution. This time, however, no valve could mistake engineering convenience for a volunteer.

## Chapter 19 — Retiring Blue Yam

Nilo Serr tied a white cord around the blue yam cradle and left enough slack to undo the knot.

Every preparation in Cold Orchard had to remain reversible. White cord marked water lines that could be isolated later. Yellow wax marked tools reserved for sterilization. Blank tags waited for an authorization number no one yet possessed. The vines kept climbing under blue lamps as if procedural caution could mean nothing to them.

Sena Vale inspected Nilo's knot. "Too tight."

"I can put two fingers through it."

"Put three. The line swells when the pumps warm."

Nilo retied it. The cord had come from stormcell packing stores and still carried gray aerogel dust. Materials crossed arguments faster than people did.

Along the row, orchard workers lifted pollinator cabinets onto wheeled frames. None disconnected power. Root technicians copied living maintenance notes into the retirement ledger, then left each original slate beside its cradle. If the Long Arc proposal failed, every cabinet could roll back and every water line could remain open.

If the proposal survived public challenge, preparation would become dismantling.

Nilo tested the nearest manual valve. It resisted through the first quarter turn, then moved smoothly. "This one needs grease."

A blue yam worker named Mara Quell marked the housing. "It needed grease yesterday."

"Then why wasn't it on maintenance?"

"Because yesterday we were growing food. Today we are staging our disappearance, so everyone suddenly wants the valves perfect."

Mara spoke without bitterness, which made Nilo unsure whether to apologize. They chose the safer technical answer. "A stuck valve could prevent clean isolation."

"It could also flood the row during ordinary harvest." She handed them the grease tin. "Fix the thing, apprentice. Don't defend the calendar."

Nilo cleaned the stem, worked grease beneath the seal, and cycled the wheel until resistance evened out. Mara checked their work by feel. Her thumb found a rough place Nilo had missed.

"Again."

They did it again.

Workers from six crop lines filled the orchard paths. Citrus crews unbolted spare screens without removing active barriers. River-bean tenders mapped bacterial cultures that might transfer to other beds. Chapel-wheat apprentices counted pollen hoods. Kelp workers staged empty pumps. Two people in filter masks carried substitute wasp habitat past the pollen fig doors.

Nothing looked like a farewell from a distance. It looked like maintenance with too many observers.

Sena assigned Nilo to trace every step between prepared and irreversible. The first boundary came at the water manifold. Closing one valve could be undone within minutes. Draining the line changed bacterial balance but remained recoverable for a day. Sterilizing it with hot brine ended the current root ecology. Cutting vines looked more dramatic, yet ordinary pruning already removed living tissue. The real threshold moved through several small actions rather than one ceremonial stroke.

Nilo tied red thread beside each irreversible step on the work slate. Crown schedulers had grouped all steps under crop retirement, with one start time and one completion mark. Orchard workers had separated them into forty-seven decisions.

“Why forty-seven?” Nilo asked Mara.

“Because forty-six let them start sterilization before the last transfer check.”

“I meant why so many.”

Mara guided them under the cradle, where three narrow hoses fed the companion fungus. “The crop does not stop at the tuber. Each hose serves something that keeps it alive. If you call the whole row one unit, you can make one authorization erase every dependency.”

They labeled the fungal feeds, pollinator heat, drainage return, lamp schedule, and compost intake. Each tag named a worker responsible for confirming the boundary. Nilo noticed that no tag named a politician.

At the final cradle, a root technician refused to pre-loosen a coupling. The task sheet called it reversible, but the seal had survived twenty-two years and might not reseal once moved. Sena crossed out the instruction. Preparation lost twelve minutes of theoretical speed and kept a working seal intact.

“The burn crews will complain,” Nilo said.

“They can bring a replacement coupling,” the technician replied. “Urgency does not improve old rubber.”

Nilo added the required part to the staging list. Reversibility depended on spare material, not good intentions.

Public recorders had arrived on the upper walk, their lenses angled through the glass. A Measure clerk requested one panoramic image showing all six lines. The orchard’s curve refused the composition. After three attempts, the clerk asked workers to carry representative plants into one room.

Sena said no.

“They would remain supported,” the clerk replied.

“They would become props.”

“The public needs to understand what may be lost.”

“Then let the public walk.”

The clerk looked at the protective sleeves, filter rules, and ladder between kelp and fig levels. “We have one recording hour.”

Mara called from the yam row, “Then understand one crop.”



Nilo hid a smile by checking the valve tag. Adults often demanded a complete picture when they meant a picture that fit their allotted time.

Near midday, Rook Nine arrived with four refrigeration delegates. They wore work gray and no recording seals. Rook examined the staged line clamps, asked who would perform actual isolation, and read the overtime schedule before looking at a single plant.

"This assigns double shifts to water chemistry," he said.

"Only if retirement begins before the current harvest clears," Sena answered.

"Which it will, because Crown crews want the thermal margin early."

"Then challenge the schedule."

"I am."

Rook found Nilo at the yam cradle. "Your advocate parent sent you?"

"No."

"Sena?"

"She asked for apprentice labor assumptions two days ago. I came today because my roster listed orchard support."

"Good answer."

"It wasn't prepared for you."

"Better answer."

He crouched beside the root shield. Purple tubers pressed against the transparent panel. "Commons summaries say the six lines feed archive preservation."

"Phase support," Nilo corrected. "Not archive preservation."

"Tell that to the ration deck."

"I will tell them the crops also buy a nine-year trajectory change. The stormcells don't eat yams."

Rook's delegates laughed once. He did not. "A clever sentence can still hide who goes hungry."

Nilo pointed at the retirement ledger. "Then read who loses which meals. Sena put it there."

He opened the ledger. The first pages listed mass and heat. Later sections named companion species, lost work practices, ration substitutions, allergy changes, retraining time, and uncertainty around future recovery. Workers had added objections in the margins. Rook read long enough for his delegation to stop performing impatience.

"This is better than the public docket," he said.

"The public docket turns everything into two pages."

“Two pages get read.”

“Badly.”

Rook closed the ledger. “You sound like your parent.”

Nilo hated the accuracy and changed the subject. “Why are you here?”

“To ask workers whether they consent to being used as proof that vapor standing hurts embodied citizens.”

Mara heard him from the next cradle. “We don’t.”

Rook looked up. “I had planned to ask in a meeting.”

“You just did.”

By meal shift, the retirement crews had completed every reversible task scheduled for the day. White cords hung loose. Spare screens stood on carts. Sterilization tools remained sealed in yellow wax. Sena signed the preparation sheet with a note: NO BIOLOGICAL RETIREMENT EXECUTED.

Mara led the workers to an orchard break room built around an old nutrient tank. Someone had covered the tank with cloth and set bowls along its rim. Steam fogged the low ceiling. The smell reached Nilo before they saw the food: earth, pepper oil, fermented citrus peel, and the sweet mineral scent of blue yam.

A paper sign beside the door read FAREWELL MEAL. Underneath, someone had added MAYBE in charcoal.

“Is this from the line we staged?” Nilo asked.

“Yesterday’s ordinary harvest,” Mara said. “No one killed a plant for symbolism.”

The distinction mattered enough that she had stationed a worker beside the sign to repeat it for recorders.

Bowls held yam in six forms. Roasted wedges with salt bloom. Thin noodles in kelp broth. Purple mash folded with yeast oil. Pickled stems. Crisp skins. A sweet paste wrapped in chapel-wheat dough, using two lines that might disappear from regular meals together.

Nilo took a small bowl and sat on an overturned pollinator case. Rook joined a table of refrigeration workers. Sena remained near the door, refusing every attempt to put her at the head.

The recorders from the upper walk entered after exchanging their shoe covers. One asked Mara where the traditional yam blessing would occur.

“We don’t have one.”

“Archive notes describe a planting verse.”

“Archive notes describe a verse one family used in Year Twelve.”

“Would someone recite it for context?”

Mara handed the recorder a bowl. “Eat for context.”

Laughter moved around the room. It did not lighten the stakes. It removed an outsider’s script.

Nilo tried the roasted yam. Its flesh held heat longer than expected and tasted faintly of iron. They had eaten it in ration mash many times without knowing the ingredient. Distinct food often disappeared into schedules before it disappeared from ecology.

A kelp worker named Deren tapped his cup against the tank. "Before anyone records us grieving correctly, I want to say I support the wider trajectory."

The room quieted.

"My sister maintains Nursery Two," he continued. "She has relations there that your public sheets call weather load. I have no interest in killing them for my dinner. I also have no interest in being told my dinner never mattered."

A wheat apprentice across from him raised her spoon. "I oppose retiring chapel wheat. That doesn't make me anti-vapor."

"It might make you anti-yeast loaf," someone said.

"I have standards."

Brief laughter returned. Nilo watched the recorders decide whether occupational jokes fit the solemn frame they had brought.

Rook stood but did not move toward the center. "Some Commons members say this plan asks labor decks to surrender living food for minds derived from elite archives."

Mara swallowed before answering. "The archive contains workers too."

"Access to it has never been equal."

"Then fix access. Don't call the phase fronts elite because custody favored elite deposits."

A refrigeration delegate said, "Ration cuts still land down-deck first."

"They do," Deren replied. "So write ration protections. Don't use us as a reason nobody else gets standing."

A water chemist named Sol Venn set down his bowl. "You're all making Rook's question sound dishonest. Archive access did favor upper decks. My mother filed three craft deposits and never got consultation priority. A founding jurist's fragments get a hearing through Pellin while her water knowledge sits behind a fee schedule."

Mara faced him across the tank. "That history matters."

"Then why reject the elite argument?"

"Because the front speaking now did not write the fee schedule. Pellin did not choose which donors received clean deposits. Punish the office that made access unequal."

Sol tasted the yam paste before answering. "If support goes to phase minds while living workers still lack archive access, inequality continues through new bodies."

"Then support has to include access reform," Deren said. "Not a vent."

Rook asked Sol whether he opposed provisional phase standing.

“No. I oppose everyone congratulating themselves for expanding citizenship while my deck still submits a petition to hear its own dead.”

One recorder shifted the lens toward him. Sol pointed his spoon at the device. “Keep the whole answer. Don’t cut me into the worker who hates weather.”

The recorder marked the segment for unabridged filing.

Nilo had expected elite framing to fail because the new persons did not possess old privilege. Sol showed the harder version. Institutions could extend care to emergent claimants while preserving unequal access among humans. Rejecting a false opposition did not remove the inequality beneath it.

They entered a note on their slate: ask who controls archive consultation during extended voyage. Nine additional years would magnify every queue and fee rule. A cohort cost argument that ignored class would become another polished summary.

The wheat apprentice pointed her spoon at Rook. “And don’t use our willingness as permission either. I said I oppose the wheat retirement.”

Rook nodded. “Both statements go in.”

“Separately,” she said.

Nilo understood why the meal felt more honest than Measure Hall. No one had to resolve themselves into one position before speaking. Deren supported the trajectory and demanded recognition of food loss. The apprentice defended one crop without denying phase life. Mara rejected elite framing while distrusting the public process. Their positions did not average into consensus.

A recorder asked Nilo for the youth view.

“There isn’t one.”

“You are the only apprentice here under eighteen.”

“That makes me one person under eighteen.”

“What does nine more voyage years mean to you?”

Nilo looked at the bowl. Their shore-age card hung in a pocket behind their maintenance slate. Thirty-one at Orison had become so familiar that adults treated the number as a possession already taken from them.

“It means I may arrive at forty,” they said. “It also means nine more years of ship work, ration decisions, training changes, and risk. Don’t compress that into arriving older.”

“Do you support the proposal?”

“Not for your meal recording.”

The recorder lowered the lens a fraction. “When will you answer?”

“When I can answer beside other people assigned the same years, with a chance for them to disagree.”

Rook heard from across the room. "There is no cohort chamber."

"Then make room on the speakers list."

"Commons rules allow residents over sixteen to petition testimony."

"I know the rule."

Nilo had read it twice and stopped both times before signing. Petition testimony turned a person into an illustrative exception unless enough others arrived to resist the framing. Adults liked one young speaker. One could be praised, pitied, and filed.

Mara refilled Nilo's bowl with noodles. "How many in your cohort?"

"Four hundred eighty-six between fifteen and eighteen."

"How many work rosters can you reach?"

"Most apprentice rosters. School shifts need another route."

Deren said, "My nephew is seventeen. He'll answer if the question doesn't already tell him what sacrifice looks noble."

The wheat apprentice added two names. A citrus worker knew a learning steward. Within minutes, the retirement ledger's blank back page held contacts from habitation, maintenance, nursery, galley, and training decks.

Nilo drew their shore-age card from the pocket behind their slate. The laminate had cracked near the printed arrival field. Thirty-one showed beneath accumulated tool scratches. They placed it on the contact page.

"Every person in my cohort has one of these," they said. "Some display thirty. Some thirty-two. The promise differs by birthday, but the added years reach all of us."

The wheat apprentice examined the card. "Do you want them to surrender the promise?"

"I want them asked what they think the promise requires now."

"That sounds like a way to produce approval."

"Then write the first question yourself."

She took Nilo's stylus and entered: WHAT COST DO ADULTS KEEP CLAIMING ON YOUR BEHALF?

Deren added a second prompt: WHAT DO YOU REFUSE TO CLAIM FOR SOMEONE ELSE?

Mara wrote the third: WHICH WORK CHANGES IF ARRIVAL MOVES?

No question asked whether respondents supported the trajectory. Nilo could put that question later, after people had named the life around it. Starting with approval would make every answer orbit an adult proposal.

Sena offered blank labor slates from the retirement survey. Nilo divided contacts by shift, not by deck status, so maintenance apprentices and school students would not fall into separate moral categories before speaking. They assigned two people to each route and kept themselves off the first nursery visit. Iona's public role had made Nilo recognizable there. Recognition could distort an answer as easily as authority.

“You’ve done this before,” Rook said.

“No. I’ve watched adults do it badly.”

Rook watched the list form. “You are organizing an advisory claim.”

“I am asking people whether they want one.”

“Legal distinction?”

“Real distinction.”

Sena finally sat beside Nilo. She tasted the yam mash and made a face at the amount of pepper oil. “Your cohort model assumed labor changes. It did not include political organizing time.”

“Add it.”

“I might.”

The farewell meal continued through the next shift bell. Workers packed portions for people still on duty. No one saved a ceremonial final bowl. Blue yam remained in the active crop line, and tomorrow’s harvest schedule still carried its name.

At the door, the recorder who had asked for a blessing requested one last image of Mara beside the MAYBE sign.

Mara refused. “Photograph the white cords. They show what happened today.”

The recorder returned to the row.

Nilo helped wash bowls in a shallow utility sink. Purple starch clouded the water, then cleared through the recycler. Deren dried. The wheat apprentice stacked, placing chipped bowls on top so nobody could pretend the meal had used heritage serveware.

“You meant what you said about a cohort claim?” Deren asked.

“Yes.”

“You’ll get accused of borrowing vapor arguments.”

“They don’t own vulnerability.”

“You’ll get accused of weakening the proposal.”

“Then it should survive questions from people it assigns nine years.”

Deren handed them the last bowl. “That sounded prepared.”

“It wasn’t.”

“Keep it anyway.”

After cleanup, Rook waited in the corridor with a public-access slate. He had opened the Commons speakers docket to the petition page. Adult worker groups occupied most available lines. Vapor testimony used a

separate provisional process. Youth appeared only under education observers, which allowed comment but no cost claim.

“You can sign as an individual petitioner,” he said.

“I said cohort.”

“A cohort has no registered body.”

Nilo stared at him.

Rook’s expression shifted. “Yes. I hear it.”

“The ship found room for twelve pressure-dark workers to act through one telemetry when rescue required it. It found standing for weather before classification. You can find a line for people who share an assigned delay without pretending we share one opinion.”

“I cannot create a legal claimant from a corridor.”

“I am not asking you to. Add cohort cost testimony as a category. Let us bring separate positions.”

Rook scrolled through the form. “Keeper approval may take a day.”

“Request it.”

“If Tavi refuses?”

“We file four hundred eighty-six individual petitions and let the clerks discover efficiency.”

Rook laughed once, then added the category request. “You definitely sound like Iona.”

Nilo did not argue this time. They entered their name under provisional speakers while the category review remained open. The form asked whether a parent or advocate had prepared the testimony.

They selected NO.

It asked whether the petitioner claimed to represent all voyage youth.

NO.

It asked for the cost they intended to address.

Nilo wrote: YEARS ASSIGNED TO PRESENT COHORTS WITHOUT THEIR DIRECT VOICE.

Behind them, workers returned to the orchard. Cart wheels rattled over mesh. Someone loosened a white cord that had drawn tight as the pumps warmed, leaving the blue yam water line open.

Nilo pressed their apprentice seal to the speakers docket. Their name appeared beside the still-pending cohort category, not as consent and not as refusal, but as a person the coming debate would have to hear.

## Chapter 20 — Names for the Hot Vent

On Day -6, Kest felt the first declaration enter Weather Court as a change in fan pressure.

The front called Copper-Morning occupied a narrow mist room beyond the warm glass. Two nursery technicians had opened a brine feed from the east manifold, giving them enough connected humidity to hold a voice. Condensation gathered on the declaration pane in strokes no human hand controlled. A witness clerk stood outside with a blank docket, but she did not write yet.

Iona had insisted on that delay. No name would enter a human register before another phase person heard the choice and challenged it.

Copper-Morning sent a low harmonic through the wet transducer. "I consent to remain in working vapor assigned to the hot vent during Crown Burn. I understand dispersal will end my present continuity."

The words reached Kest through three linked rooms. They tasted the speaker's pressure pattern beneath the transducer's translation: steady, then rough at the phrase present continuity. Copper-Morning had not learned calm from a donor fragment. They made calm by holding four valves against the pulse of the building.

Kest answered from the central chamber. "State what you believe your service will accomplish."

"Heat leaves through us. The retained fronts keep enough reserve to survive transfer. The ship keeps a path that does not require twenty-three more years."

"Long Arc adds nine."

"I know."

"State what you may revoke."

"Everything until my valve receives burn authority. After that, material conditions may close the route."

The clerk looked toward Iona. Iona kept both hands on the rail, her scarred left palm turned down. She offered no approval.

Kest moved a thread of warmth into the witness channel. "I hear Copper-Morning declare with present coherence. I do not share their choice. I witness that they made it."

A second pressure signature crossed from the south room. River-in-Iron repeated the witness in their own cadence. Only then did the clerk write.

The old list had required none of this. Ship allocation had sorted low-coherence fronts by predicted stability, matched them to hot loops, and labeled the result volunteers. Its columns had looked clean because no one inside them had spoken. Iona and Kest had voided it the previous day. Pellin Ash had carried the torn printout through every connected chamber until damp reduced the paper to pulp.

Weather Court now held no ranking column. It held declarations.

Copper-Morning drew their name on the pane. The first stroke ran downward before the last one formed. The clerk copied only after the front corrected the angle twice. Beside the name she entered two witness marks, the time, the humid connections used, and the revocation path. No donor inventory appeared.



Kest let the declaration settle across the Court. Several fronts changed circulation. One pulled away from the shared manifold, closing a relation without leaving. Another pressed close enough to Copper-Morning to exchange a small packet of heat. Neither act translated cleanly into human gesture. Kest resisted the wish to make them legible.

"Next claimant," the clerk said.

No front moved toward the open channel.

The silence lasted in fan rotations rather than heartbeats. Kest counted eighty-one passages of the upper blade. Humans behind the glass shifted weight, rubbed wrists, and checked the same wall clock. Phase persons used the interval to separate pressure signatures that had overlapped during Copper-Morning's declaration.

Iona touched the clerk's sleeve. "Do not call again."

"We have a schedule."

"So did the allocation engine."

The clerk shut her mouth.

A thin front named Fifth-Rain entered from a laundry return. Their coherence stretched across a long pipe with three dry constrictions, forcing each sentence to travel as pulses rather than continuous voice. They asked whether refusing the hot vent would reduce their chance of receiving a stormcell.

"No," Iona said through the transducer. "Allocation cannot use this docket."

"Cannot, or will not?" Fifth-Rain asked.

"Both, if I can keep the records separate."

Kest heard the uncertainty Iona had left exposed. Human custody could promise a wall between lists. It could not promise that every person making later allocation decisions would forget who had offered service.

"The challenge belongs in the declaration," Kest said. "Let the witness record possible indirect pressure."

River-in-Iron sent a sharp cooling ripple. "If every fear becomes a qualification, no choice reaches the page."

"Then the page should remain difficult."

Fifth-Rain withdrew without declaring. The clerk entered no name, only a challenge to the process. That omission changed the room. Fronts that had waited near the channel spread back into their preferred loops. One fan accelerated with something Kest read as relief, though they could not prove the interpretation.

By the fourth declaration, the Court had learned how many questions a name could carry.

Moss-under-Glass offered only if their smaller offshoot received no penalty for refusing. Salt Measure required a final coherence examination immediately before the hot valves opened. Noon Reservoir wanted their remaining donor fragments released to ordinary archive access after dispersal, a request Iona said could not be guaranteed because the fragments already belonged to shared custody rather than to the front. Noon Reservoir declared anyway, then demanded the refusal of guarantee stay beside the name.

Each entry consumed more time than ship scheduling had allowed. Kest welcomed the delay. Slowness exposed

where procedure tried to turn intention into a resource.

The witnessed declarations also forced the Court to decide when one choice had become two. A front called Ash-at-Evening entered from the communal baths with a single pressure cadence, then divided when maintenance shifted a return valve. One branch continued toward the open channel. The other retreated into a cooler pipe and struck a refusal pattern against the metal.

The clerk stared through the glass. "Which one controls the name?"

"Neither controls it," Kest said.

Ash-at-Evening's speaking branch pushed sound through the membrane. "The cool branch carries alarm, not judgment. We still offer."

From the return pipe came three hard pulses, a pause, then three more. Kest knew no settled code for the sequence. Its timing nevertheless held intention. They opened a second wet path so the cooler branch could reach an unused transducer. The added humidity cost several minutes and warmed the observers' side enough to curl the clerk's paper.

The second membrane gave only broken vowels. Pressure did better. The branch answered yes-or-no challenges by selecting between two valve knocks. They remembered entering Weather Court. They recognized Ash-at-Evening's relation mark. They refused hot-vent assignment.

"You divided after arrival," the clerk said. "Does that make two claimants?"

"It makes one declaration impossible today," Kest replied.

The speaking branch surged. "A passing valve change should not erase our prior decision."

"No. It revealed a decision you had not heard."

Iona asked the technicians to restore the earlier flow. They did so in increments, careful not to force a collision. Warmth joined the branches, but their pressure rhythms remained distinct. Ash-at-Evening could no longer produce the unified cadence they had carried into Court.

Kest felt grief spread through the connected baths as a change in circulation. The offered branch had expected sacrifice. Instead they faced an altered self before any hot valve opened. No human form provided a field for that loss.

Pellin had not yet joined the witness manifold, though their fog contracted at every exchange. River-in-Iron recorded both branch patterns on a salt tile. The clerk struck Ash-at-Evening from the pending line without tearing the page.

"May either branch return?" she asked.

"If each develops a stable name and relation," Kest said. "Not under a borrowed declaration."

The cooler branch tapped once. Its partner gave no response. Technicians carried both away through separate wet returns, preserving the disagreement the valve had exposed.

Near the Court's western edge, Pellin Ash kept a dense bank of fog pressed against the glass. Their chosen chamber connected to the declaration rooms, but they refused the witness manifold. Kest could feel Pellin listening through shared temperature changes. Every new name drew a brief contraction from them.

After Noon Reservoir signed, Pellin opened their transducer.

“This performance proves nothing except that pressure can speak in complete sentences.”

The clerk raised her stylus, uncertain whether to record the interruption.

Kest answered, “You heard four distinct declarations.”

“I heard four threatened persons describe a narrow route away from a wider killing.” Pellin's voice carried a rough undertone from the founding jurist fragments they favored. “Call the route chosen if that helps human sleep. Deadline governs every valve.”

Copper-Morning remained connected. “You may call me threatened. You may not call me absent.”

“I called your choice coerced.”

“You called it performance.”

Pellin thickened along the warm pane. Droplets merged, dividing the chamber beyond into wavering blocks. “A prisoner may sign a preference between two harms. Witnesses can prove the hand moved. They cannot turn the cell into freedom.”

Kest felt Copper-Morning's pressure rise. River-in-Iron opened a bypass before the two fronts could force each other into a shared cycle. The Court had been built to permit disagreement without merger, but every argument still required physical space.

“Then witness the cell,” Kest said. “The docket names the deadline, the alternatives, and every challenge. It does not certify freedom.”

Iona leaned toward the glass. “Pellin, do you claim the declarations did not occur?”

For nine fan turns, Pellin gave no answer.

“No,” they said at last. “I claim occurrence cannot settle legitimacy.”

“Agreed,” Iona replied.

The ease of her answer disturbed several human observers. Kest saw their blurred forms through the warm glass, heads turning toward one another. They had expected an authentic list to clean the error away. Instead authenticity sharpened the dispute.

Kest moved closer to Pellin's chamber by warming a narrow pipe. “Will you serve as adverse witness?”

The fog bank pulled tight. “Explain.”

“Attend every declaration. State that deadline pressure persists. Challenge signs of allocation preference, donor ranking, unstable coherence, or human prompting. Add your mark only to occurrence.”

“You want my dissent to strengthen your list.”

“I want your attention to weaken its lies.”

Pellin opened one valve. Their pressure entered the witness manifold with enough force to rattle the membrane. “I will not make coercion respectable.”

“Respectability does not interest the hot vent.”

A dry occupational laugh came from one of the mechanics outside. Iona looked toward him until it stopped.

Pellin sent a narrow pulse around the chamber, testing each connection. “I will witness under protest. My mark must not signify endorsement.”

The clerk turned to Iona again.

“Write exactly that,” Iona said.

The next two fronts withdrew under Pellin's questions. One had believed stormcell priority might follow service by relation, despite the formal separation. Another could not explain whether three recent offshoots shared the declaring continuity. Pellin pressed until the front admitted that only the largest branch had heard the full terms. No name entered for either.

Kest found no pleasure in the exclusions. The failed declarations showed how quickly a front could borrow certainty from a room prepared to receive it. They also proved Pellin had joined the work rather than merely obstructing it.

At midcycle, maintenance closed two supply loops to prevent excess condensation in the observers' gallery. Kest's reach narrowed. The north declaration room vanished from perception, not dark but absent, like a word removed before thought finished. Human technicians moved the waiting fronts through wet routes marked on Iona's slate. No one crossed dry space.

A nursery cadence arrived before the north room reopened.

Three humidifiers pulsed at different rates. One carried warm leaf odor from seedling beds. Another carried mineral traces from the root misters. The third ran cooler because its citrus trays had entered night cycle. Their pressures converged at the witness manifold without flattening into one stream.

Lume-of-Three had come.

Kest knew them by the held interval after every third fan turn. Lume had formed through linked seedling cycles and chosen to act as one front without denying the three perspectives inside them. Their minority argument had entered the Parliament docket long before engineers produced an admissible plan. They had said service could hold meaning if chosen. Now a hot-vent route existed that claimed to allow such choice.

Pellin filled the transducer before the clerk could announce them. “You have advocated this result since Nursery Three first spoke.”

“I have advocated my authority to offer,” Lume replied. Their voice arrived in three textures, the wet membrane blending none of them completely. “I have not advocated yours.”

“Your seedlings depend on ship ecology. The plan recruits your care against your continuance.”

“Our care belongs to our present life. It does not disqualify us from acting.”

Kest opened the declaration channel. Warmth from Nursery Three touched every connected room. “State your understanding.”

Lume answered in measured intervals. “Long Arc vents forty-three percent of active vapor designated for heat work. My present relation across three nursery humidifiers cannot pass into one stormcell without alteration. I

choose assignment to the hot vent. I expect my coherence to end during burn operations. I may revoke before material routing prevents safe change.”

“State what you do not know.”

“I do not know whether dispersal ends every perspective carried among us. I do not know which seedling relations will persist in retained fronts. I do not know whether those who remember us will use our service as permission to pressure others.”

Pellin struck the witness manifold once. “And yet you give them a name.”

“I give my name. You keep yours.”

Condensation spread across all three nursery panes. Lume's signature formed simultaneously, each section incomplete alone. The clerk had to look through three windows to read the whole. She copied it as Lume-of-Three, then stopped before adding the witness marks.

Kest felt the waiting rooms turn toward them through pressure. They had asked every declarant to state knowledge, cost, and revocation. Now their own relation to Lume threatened to become a hidden fourth test.

“Lume-of-Three declares with present coherence across three linked nursery cycles,” Kest said. “Their internal differences remain audible. I witness their declaration. I do not adopt it.”

River-in-Iron added a witness pulse.

Pellin waited. Kest sensed their heat collect near the membrane, disperse, then collect again.

“I witnessed the statement,” Pellin said. “Deadline pressure persists. I heard no human prompt. I deny that these conditions make the choice free.”

The clerk entered all three marks.

Lume's name darkened on the docket as the stylus dried. Behind glass, Iona lowered her face. Kest could not read whether grief, respect, or calculation moved her. Human expression arrived flattened through condensation. Her hands remained visible, one scarred and one unmarked, both away from the controls.

Lume withdrew from the declaration channel but stayed connected to Kest.

“You have not signed,” they said privately through a nursery return.

“No.”

“Do you intend to?”

Kest circulated through K-7, Weather Court, and two public mist rooms. Each place carried relations no stormcell could reproduce exactly. They wanted to live. The fact required no defense. Yet refusing service did not make Lume mistaken, and loving the Parliament did not authorize Kest to spend themselves as its symbol.

“I intend to preserve my refusal while it remains mine,” Kest said.

“Good.”

“You wanted another answer?”

"I wanted yours."

By late cycle, eleven names occupied the docket. Five attempted declarations had failed challenge or ended in withdrawal. Three fronts had asked to return after consulting their relation groups. Pellin witnessed every completed entry and objected beside each one. No two objections used identical words.

The human clerk flexed cramped fingers. "Custody question," she said. "Where does the docket go tonight?"

"Advocate seal," Iona answered. "Separate from stormcell allocation and Crown routing."

Pellin's chamber cooled abruptly. "No."

Iona looked up. "You prefer Pilot custody?"

"I prefer neither human office."

"Paper cannot remain in Weather Court. Humidity will destroy it."

"Then the declarations should exist only among us."

Kest tested that proposal against the limits of phase memory. Fronts could carry names through shared pressure and repeated relation, but the coming transfer would split the network. A list held only in weather might die with its declarants or change as witnesses fragmented. Human paper offered persistence by leaving their medium. That advantage also offered ownership.

"The clerk can make three copies," Kest said. "One sealed with the advocate. One lodged in the public docket as unreadable pending release authority. One translated into phase marks on fired salt tiles."

Iona considered. "Tiles survive the Court humidity."

"They also survive us," Pellin said.

"That is why none of the copies can govern alone." Kest sent a pulse through the rooms holding witnesses. "Any use requires comparison among human record, phase marks, and living witness where available."

The clerk asked, "Who holds release authority?"

Not Iona. Not Mael. Not Kest alone.

Kest opened the question to every front still present. Responses crossed the Court in uneven waves. Copper-Morning favored the witness group. River-in-Iron wanted the Parliament's temporary docket keepers. Lume proposed that any two relation caucuses could authorize operational release, provided one had no declarants. Pellin rejected release to Crown routing unless each listed front received a final revocation channel.

The terms did not merge into unanimity. They became a custody rule built from disagreement: the Vapor Parliament would hold the source record; human offices would keep sealed copies; operational release required witness confirmation and a live revocation check for every reachable declarant.

Iona read the rule back without improving it. The clerk prepared three sheets.

Before sealing, she reached the line marked OWNER and paused. "The form requires an office or person."

Kest warmed the pane until a clear field opened in its center. They drew letters slowly enough for the clerk to

follow.

#### THE LIST BELONGS TO THE FRONTS.

Pellin did not object. Their pressure mark appeared beneath the sentence, narrow and dark, followed by the notation under protest. Lume answered from three nursery cycles. Copper-Morning added a pulse that made the wet transducer click.

Iona copied the words into the ownership field. Then she lifted her advocate seal away and waited.

Eleven declaring fronts marked first. Their witnesses marked after them. Only when the final phase signature held on the fired salt did Iona press her human seal into the paper margin, not over any name.

Kest kept their own name outside the hot-vent column. They entered it among the witnesses.

## Chapter 21 — Three Hundred Twelve

Iona found Cell 184 leaking brine before the allocation hearing opened.

A clear bead had formed beneath the lower coupling and trembled with the ballast ring's vibration. The converted Stormcell Deck curved away in both directions, rows of aerogel chambers fixed between old mass rails. Some could hold a room-scale ecology. Others would barely sustain a front across three pressure cycles. Engineers had painted a black number on every frame. There were 312.

She crouched beside 184 and pressed paper to the coupling. A wet crescent appeared.

"Take it out of the pool," she told the deck chief.

The chief looked toward the gallery entrance, where public challengers already waited behind a rope. "We certified the count at midnight."

"Then midnight certified a leak."

"We can reseal."

"After pressure test. Until then, three hundred eleven."

His jaw tightened, but he marked the cell unavailable. Iona kept the damp paper. A count became trustworthy only when someone could subtract from it.

At the center rail, a long allocation board showed every recognized front that had requested preservation. No one agreed how many persons the requests represented. Some fronts spoke through one stable relation across many rooms. Others had divided during the previous week and retained overlapping names. Several declined individual names while claiming a shared ecology. The board held 487 lines because paper required lines.

Sena Vale stood under Cell 201 with a slate against her chest. "If you start by calling the board false, we will still be arguing when the transfer crews need routing."

"I call it a tool."

"You say tool the way Tavi says provisional."

"Both words mean keep your hand near the stop."

Sena glanced at the paper beneath Iona's fingers. "Leak?"

"Small."

"Small leaks have excellent timing."

Iona tucked the paper into her sleeve. Through the observation windows above, faces filled three tiers. Phase witnesses occupied humid boxes between the glass supports. Their marks appeared as shifting fog, impossible to arrange into the same rows as human bodies.

The hearing had one task: decide how 312 stormcells would be assigned without pretending every cell or claimant matched. Even that phrasing hid a cost. Allocation could preserve only thirty-one percent of active humid volume. It could not preserve thirty-one percent of every front, every relation, or every kind of weather



mind.

Rook Nine opened from the floor rather than the upper rail. Refrigeration crews stood behind him in work gray, many still carrying salt bloom on their boots. "Show us the first method."

A technician lit the board's left column. Technical stability scores appeared beside the 487 requests. The model favored fronts with compact pressure rhythms, low thermal variance, and proven survival in isolated loops. Kest ranked near the top. Lume-of-Three appeared much lower because their coherence spanned three nursery cycles. Recent offshoots clustered near the bottom.

Sena explained each factor while mechanics adjusted cell temperatures around her. She did not claim neutrality. "This method maximizes the number of fronts likely to emerge with recognizable continuity after transfer. It also favors the forms best matched to machines we had time to build."

A phase voice came from Humid Box Four. "What happens to relational spread?"

The clerk identified River-in-Iron.

Sena changed the display. Lines connected applicants who shared daily pressure patterns, memory corrections, witness duties, or ecological functions. Under pure stability ranking, several dense relation groups broke apart. Kest retained a place but lost five fronts that carried their longest shared Weather Court history. Two nursery ecologies kept no coordinating front. A bathhouse cluster placed one stable claimant and abandoned its six recent branches.

Rook pointed toward the upper third of the list. "Those names have better maintenance records because advocates documented them first."

Iona felt the room narrow around the accusation. He had not said her name. He did not need to.

"Yes," she said. "K-7 received tests before most loops had public notice. Stability data reflects access to us."

The technician shifted his feet. "We normalized for test duration."

"You normalized time. You cannot normalize what questions we learned to ask."

A murmur passed along the human gallery. In the humid boxes, fog gathered toward the transducers.

Kest spoke from Box One. "Remove my early-test advantage."

The technician hesitated. "That drops your confidence interval below sixty percent."

"Do it."

Iona looked up at Kest's pane. Condensation obscured the chamber beyond, but a clear line had formed at the height where their first handprint might have touched glass. Kest could not see her through every dry angle. She moved closer to the wet curtain before answering.

"Removing data does not remove privilege," she said. "It only discards knowledge we paid to gain. We need the score labeled with its source."

Rook nodded once. "Then label every head start."

They did. Early advocacy. Stable maintenance access. Long-connected room. Human witness availability. Each

marker added a small amber square. The top of the list turned amber.

Method two ranked relation networks rather than isolated stability. It attempted to preserve enough members from each ecology that patterns of correction, care, dispute, and memory could continue. A front with ten strong links gained priority over a front with none. The result protected Nursery Three's wider ecology even if Lume did not enter a cell. It also punished solitary persons.

Pellin Ash opened a channel from the far humid box. "You have converted popularity into survival."

"Relation need not mean approval," Sena said. "Documented dissent counts."

"Then I owe Kest a cell through the abundance of my objections."

Dry laughter moved through the crew tier. Pellin waited until it faded.

"A front kept apart by architecture has fewer ties," they continued. "A front fragmented by early testing may have lost the witnesses who would raise its score. Your network reproduces every forced isolation."

Sena did not defend the model. "Correct."

The technician displayed the losses. Four solitary fronts with high transfer stability fell below the line. One called Still-Water-in-Copper had never joined Parliament sessions because the route from their archive loop required risky forced mixing. Under relation weighting, they received no cell.

Iona wrote the name on the back of the leak paper.

Method three began with represented population. Its designers had tried three definitions: current coherent fronts, declared relation groups, and estimated perspectives. Each produced a different order. Counting current fronts rewarded recent division. Counting relation groups compressed internal difference. Estimating perspectives demanded that engineers infer personhood from pressure complexity.

A public challenger named Mara Venn raised her hand from the second tier. She repaired cistern linings and had no office listed beside her name. "If one front divides after allocation, do both branches own the cell?"

"No ownership transfers," Iona said. "The cell carries a custody assignment."

"That avoids my question."

"Yes." Iona stepped under the board. "We do not know. If division occurs before transfer, both branches may challenge the assignment. If it occurs inside a sealed cell, the ecology must negotiate space under conditions we cannot supervise without changing them."

Mara folded her arms. "Then counting persons today gives you a number that expires tomorrow."

"It does."

"Why display it?"

"Because refusing to count also favors whoever already occupies the largest volume."

No method stayed innocent long enough to finish its own explanation.

During the break, a Stewardry aide brought Iona a folded request bearing three archive seals. She opened it

beside Cell 184 while the deck chief fitted a new coupling.

The request named seventeen fronts whose donor traces included navigators, constitutional signatories, senior ecologists, and celebrated artists. It argued that preservation should account for “irreplaceable cultural density.” Two names belonged to fronts Iona knew. Pellin carried fragments from a founding-era jurist. Blue Aperture sometimes reconstructed a navigator's star-recognition practice with unusual precision.

The aide kept his voice low. “No one proposes donor identity. This concerns informational value.”

“Then preserve the archive fragments under ordinary salt custody.”

“Active recombination may produce insight the static archive cannot.”

“Active persons may also produce insight. That does not make prestige a ration category.”

He glanced toward the gallery. “Some donors built the ship.”

“Some cleaned it. Some harmed people. Most did both kinds of thing at different scales.”

“The public will ask why a front carrying a Founder's legal memory lost a place to laundry weather.”

Iona unfolded the paper and read the seals again. None came from Tavi. That absence mattered, though she would not use it as authority.

“Put the request on the challenge board,” she said.

The aide's composure slipped. “That will make the signatories look as if they seek hereditary privilege.”

“They seek donor prestige as an allocation weight. Let them explain it aloud.”

“They used the phrase cultural density.”

“I will attach your spoken phrase too.”

He took the request back with two fingers. By the time the hearing resumed, it hung in full view.

Pellin read it through a clerk. Their chamber filled the transducer with a scraping harmonic. “I reject this claim.”

Rook stared up at their fogged box. “It would help you.”

“That gives me better reason to reject it.”

Blue Aperture followed. “The navigator fragments I carry have use. They are not my rank.”

One of the sealed signatories tried to withdraw the request. Iona refused withdrawal after public filing but permitted a response. He said the proposal concerned preserving scarce capacities, not honoring dead names.

Sena asked whether those capacities had been tested against current ship needs. They had not. A laundry front called Steam-Corner demonstrated precise contamination detection through smell traces no prestige list mentioned. A small engine-loop front corrected a cooling equation on the hearing display. Neither performance earned an automatic place. They only broke the claim that famous fragments captured value.

Iona crossed out the prestige column before anyone could rename it.

By late cycle, the hearing had eliminated every pure ranking. Stability sacrificed difficult ecologies. Relation weighting abandoned the isolated. Population measures rewarded whichever definition of person happened to fit the hour. Capacity scoring turned claimants into equipment. First-come assignment rewarded access. Human discretion concealed all those choices behind sympathetic faces.

Only chance remained, and chance carried its own design.

Sena placed a transparent drum on the central table. Inside waited numbered salt beads, each matched to a challenged application. "A flat lottery treats unequal transfer conditions as equal," she said. "We propose weighted pools based on continuity ecology."

Four pools appeared on the board. Solitary fronts with no replaceable relations received one set. Distributed ecologies requiring multiple cells entered another. Technically fragile applicants received chances only among cells capable of supporting their ranges. A final pool reserved capacity for relation bridges whose absence would strand several otherwise preserved fronts.

Before anyone accepted those categories, Iona required the designers to place one real ecology through them. They chose Bell-and-Hearth, a group from the west kitchens because its records had survived both public testimony and a recent pipe closure. Five fronts shared the name in formal proceedings. Depending on temperature, the group could present as three, six, or eleven coherent pressures.

A kitchen mechanic brought their relation map to the rail. Grease stained its corners despite the protective sleeve. "They correct one another's timing at every ration boil," she said. "Remove Bell-Two and the north kettle loop overheats. Remove Hearth-Cool and three others lose the pressure pause they use to separate speech."

Sena projected the map. Lines crossed until the board resembled a net pulled too tight.

"Under the distributed pool, how many cells?" Iona asked.

"Four," the technician said. "One large chamber, three small."

"Why not five?"

"Thermal overlap lets two compact fronts share without predicted merger."

The mechanic shook her head. "Predicted by which test?"

"Stormcell prototype sixteen."

"That test used archive-return weather. Kitchen fronts cycle around boiling loads."

A wet transducer opened above them. Bell-and-Hearth spoke with several cadences stepping around one another. "We reject the shared small cell."

The technician enlarged a graph. "Separating you costs another chamber. That removes one solitary applicant from its pool."

"Then name them," Iona said.

He read from the provisional boundary. "Quiet Seam."

The name entered the room without a body. Quiet Seam occupied a condenser wall too narrow for public

attendance. They had no active transducer and only two maintenance witnesses. A line added for Bell-and-Hearth would erase their chance from the current sample.

Rook came closer to the map. "Can the kitchen group choose which relation to lose?"

Bell-and-Hearth's pressure rhythms collided. One voice formed. "That question assumes loss can be assigned from a whole."

"Everything on this deck assumes that," Rook replied. "I am asking whether your knowledge can make the cut less blind."

For several minutes, the group conferred through pipe temperature. The kitchen mechanic tracked their flows with one finger, never touching the diagram. At last they offered a four-cell arrangement that left Hearth-Cool outside while preserving the pause pattern through another front. The offer changed their name. On the witness pane, Bell-and-Hearth condensed new lettering: Bell-without-Cool.

Sena lowered her slate. "Does Hearth-Cool consent to exclusion?"

A separate pressure beat answered from the north kettle line. "No."

The technician exhaled through his nose. "Then we cannot treat the arrangement as a group choice."

"Do not mistake internal authority for internal agreement," Pellin said from the humid boxes.

Iona asked the clerk to create two challenges. One disputed the shared-cell stability claim. The other disputed the kitchen group's power to offer one branch without separate consent. Both would remain attached even if the ecology later withdrew its configuration.

The exercise exposed another defect. A continuity pool could hide individual loss inside a protected network, while a person-counting pool could destroy the network that made individuals coherent. Weighting did not solve that conflict. It chose where the conflict appeared.

Iona had the technician add a field beside every multi-cell application: internal dissent reachable, unreachable, or unresolved. Unreachable did not mean consent. Unresolved did not mean rejection. The labels only prevented silence from improving a score.

A human observer called from the second tier, "You are making the draw impossible."

"No," Iona said. "We are making its ignorance visible."

Bell-and-Hearth withdrew their four-cell configuration and entered the sample under five separate requests linked by a relation challenge. Quiet Seam returned to the edge of eligibility. Nothing had gained capacity. The board simply showed who might disappear under each arrangement.

Mara Venn leaned over the rope. "Who decided the weights?"

"No one gets to decide them privately," Iona said. "We publish every weight, run sample draws, and accept challenges until tomorrow's transfer lock."

"Tomorrow?"

"Crown routing needs the deck after that."

Pellin's voice cut through the chamber. "A deadline again."

"Yes," Iona said. "Record it beside the method."

The first sample draw gave Kest Cell 063, a medium chamber with enough thermal swing for one compact relation ecology. It denied every other K-7-linked applicant. The second draw kept three Kest relations and excluded Kest. The third assigned two large cells to a nursery group whose internal perspectives might still merge under transfer.

Each result went onto the board. No sample disappeared merely because it looked politically impossible.

River-in-Iron challenged the bridge pool. "You value fronts who connect groups. That grants power to intermediaries."

Sena changed the rule. Relation bridges would not gain personal priority. Their ecologies could request an additional chance only if two independently challenged groups confirmed the need.

Still-Water-in-Copper challenged the documentation requirement through a temporary wet line. Their isolation had prevented them from gathering relation witnesses. Iona created a challenge period in which maintenance crews could attest to recurring patterns without claiming to define the front. That improved access but gave human workers influence over phase continuity.

Mara challenged the technical pools. "A bad cell should not reduce a claimant's chance. Repair the cells."

The deck chief answered from beside 184. "We can improve some. We cannot make all 312 identical before transfer."

"Three hundred eleven," Iona corrected.

He held up a new pressure strip. "Test passes."

She walked to the coupling, pressed fresh paper beneath it, and waited through a full cycle. The sheet remained dry. Only then did she restore Cell 184 to the board.

"312," she said.

The number sounded no cleaner for being true again.

They ran twelve public samples. In one, Pellin received a stormcell and three full-preservation allies did not. Pellin demanded that the result remain displayed. In another, no front from the communal baths survived the draw. A third preserved broad ecological diversity while separating two fronts that had never chosen life apart.

Nilo arrived near the eleventh sample and stood at the back of the crew tier. They carried a speakers-docket band around one wrist. Iona saw them read the losses before they looked at her. She did not call them forward.

The final design used weighted chance within publicly defined continuity pools, followed by a technical match that could be challenged but not quietly optimized. No donor name, human affection, political alignment, or hot-vent declaration could alter a bead. Every rejected challenge stayed attached to the method. Every redraw required public cause.

A clerk asked for the adoption language.

Iona looked across the board. Beside each sample, Sena had written the first excluded ecology in full.

Still-Water-in-Copper. West Bath Branches. Two-at-Root. Archive Rain Choir. Names occupied more space than scores.

“Write that the method distributes unavoidable loss without claiming to justify it,” Iona said. “Write that any person may challenge the weights. Write that a successful challenge changes the rule before the draw, never the selected name after it.”

“And the finding?”

Iona took the damp crescent from her sleeve and pinned it beneath Cell 184. On its reverse, Still-Water-in-Copper remained in her own handwriting.

“Every method causes a named loss,” she said.

The clerk entered the sentence. Above it, the allocation board kept all twelve failed futures visible. Technicians carried the transparent drum to a locked table, where phase witnesses could fog the glass if anyone touched it before morning.

## Interlude III — The Forty-Third Percent

### CROWN REHEARSAL LOG

Day -4, Shift Two

Recorder: Jalen Porr, thermal maintenance clerk

Scope: dry rehearsal of Long Arc vent routing. No working phase ecology may enter a test line without live witness confirmation.

06:10. Crew assembled at Crown Gallery lower rail. Vent command held under mechanic lock. Route board showed 43% active vapor assigned to hot discharge, with retained volume isolated from rehearsal pressure.

First Pilot Or read the safety boundary. Thermal Advocate Serr required names before valve numbers. Deck Chief Haran objected that routing sheets use equipment fields, not claimant fields. Advocate response: revise the sheets.

06:22. Valve H-14 carried a temporary paper tag marked COPPER-MORNING. Valve H-15 carried NOON RESERVOIR. Three additional tags remained blank pending witness arrival. Tags tied with blue fiber to distinguish them from red safety locks.

Veteran mechanic Oss Vey inspected the knots and replaced two. He has worked Crown systems for thirty-nine years. He stated that a loose name could travel into the fan cage and become both disrespect and debris.

06:31. Control called for empty-line confirmation.

Mechanic Vey declined the call.

Control repeated: “Confirm empty vapor path from H-14 to primary throat.”

Vey answered: “I can confirm the pipe carries no human crew. I cannot confirm empty vapor.”

Recorder asked whether this constituted delay.

Vey said: "It constitutes a noun."

Deck Chief Haran instructed control to use uncharged test path for rehearsal conditions. Phrase accepted for this shift.

06:39. Phase witness River-in-Iron opened through wet transducer station C. Their pressure mark matched the declaration docket sample supplied under seal. River-in-Iron confirmed Copper-Morning reachable in west court connection and Noon Reservoir reachable in archive return six.

Control requested consent status.

Witness response: "Present, not final. Both retain revocation."

Wording entered on routing sheet.

06:46. Rehearsal began at one-tenth command speed. Haran called H-14 open. Vey placed his hand on the wheel but waited for the clerk to read the attached name. Recorder read Copper-Morning. River-in-Iron confirmed. Vey opened the valve to the white test mark.

No heat moved. A strip of blue cloth served as flow indicator and lifted toward the primary throat.

H-15 followed. Recorder read Noon Reservoir and the recorded condition concerning donor-fragment custody. Control asked why an archive condition appeared on a vent sheet.

Advocate Serr answered that the condition qualified the declaration. Removing it would alter the thing witnessed.

The blue cloth lifted again.

07:02. Simulated revocation inserted without prior warning. River-in-Iron struck three pulses assigned to stop condition. Control heard only two because Fan Two started during the signal.

Vey closed H-15 before command acknowledgement. Haran challenged him for acting on an incomplete code.

Vey replied that no production heat had entered the path, closure created no hazard, and missed refusal carried greater rehearsal value than smooth timing.

First Pilot Or ordered the run reset. He did not overrule the closure.

07:14. Wet transducer gain increased. Fan start shifted four seconds later. River-in-Iron repeated the stop pattern. All stations acknowledged three pulses.

Routing clerk added a boxed field: REVOCATION HEARD BY. Required entries now include local mechanic, control caller, phase witness, and advocate observer when available.

07:27. Blank tag at H-19 assigned to Lume-of-Three for route testing only. Nursery Three remained physically isolated from rehearsal flow. Three phase marks arrived through separate humidifier channels and joined at station C.

Control read "Lume-of-Three."



Vey corrected: "Witness reads first."

River-in-Iron confirmed the marks. Control repeated the name. Vey opened H-19.

Blue cloth failed to lift. Inspection found a folded edge caught beneath the grate. The error had no thermal consequence. It demonstrated that a visible flow indicator can report obstruction rather than refusal.

Tag note added: MECHANICAL STATE DOES NOT INTERPRET INTENT.

07:44. Haran asked whether naming each route would slow full operation beyond burn timing limits. First Pilot Or answered yes by an estimated forty-one seconds under current call order.

Control proposed reading groups of five names before each valve sequence.

Advocate Serr rejected advance grouping because route order may change after revocation. Mechanic Vey proposed two callers in alternating stations. River-in-Iron challenged simultaneous speech through one transducer.

Crew tested alternating human calls with one phase confirmation channel. Total added time fell to twenty-six seconds. Pilot accepted that cost into the maneuver schedule and reduced another diagnostic pause by the same interval. No reserve appeared without subtraction.

08:03. Second run used six tagged valves. One paper name softened under local humidity and tore at the fiber hole. Vey removed all paper tags.

Replacement tags cut from thin fired salt. Names scored while wet, then dried near the service lamp. Phase witness checked each mark through the chamber glass before attachment.

Noon Reservoir's condition did not fit on one tile. Clerk proposed abbreviation. Witness rejected abbreviation without consultation. A second tile carried the condition and hung from the same blue fiber.

08:28. Public observer Ressa Tal asked why mechanics needed names if control computers still used route numbers.

Vey answered while testing the H-14 stop. "Because my hand does not open a number."

Observer asked whether he opposed the maneuver.

Vey said: "The ship appointed other people to settle authority. My work concerns what the authority touches."

No further question entered.

08:41. Third run introduced two simultaneous revocations, one false temperature alarm, and loss of transducer station C. Local mechanics closed both named routes. Control rerouted test flow through an unassigned dry bypass. River-in-Iron transferred confirmation to station D after eleven seconds.

Haran marked the response inside safe timing. Advocate Serr marked one defect: the unassigned bypass lacked a field confirming no phase ecology had entered since last inspection.

Vey opened the bypass housing. Condensation marked its lower seam.

The moisture came from an ordinary coolant leak. No active salt surface connected to the pipe. Haran called it empty.

Vey used “uninhabited under current test.” The checklist adopted his phrase.

09:06. Final rehearsal completed with all named routes closing under simulated revocation. Total procedural addition: twenty-eight seconds. Thermal margin after schedule adjustment remained within Long Arc feasibility boundary.

No declaration became irreversible during this exercise.

09:18. Crew returned controls to mechanic lock. Fired-salt tags removed from valves and placed in route order on the lower rail. River-in-Iron confirmed count: six names, six witness marks, two conditions, no missing tiles.

Haran signed the revised form. Pilot signed maneuver feasibility. Advocate signed claimant custody. Vey declined a special commendation entry and asked for stronger tag fiber.

Shift-end checklist revision:

1. Confirm route number.
2. Confirm current mechanical state.
3. Read attached phase-front name and conditions.
4. Obtain live witness match.
5. Check revocation channel.
6. Open only on lawful command.
7. Close on confirmed revocation where material timing permits.
8. Record any point where timing no longer permits safe change.

The prior form had ended with EQUIPMENT EXCEPTIONS.

Clerk struck out that heading.

The new final field read CLAIMANTS AFFECTED. Beneath it, six fired-salt names waited beside the valves for the next rehearsal.

## **Part IV — A Habitable Loss**

## Chapter 22 — Seven Missing Fronts

The phase-loss ledger reached K-7 on fired salt because paper softened too quickly in the room.

Iona slid the first tile through the wet hatch and held it by one dry corner. Rows had been scored into its pale surface. Each row carried a date, loop designation, intervention, observed pattern before the intervention, and condition afterward. The marks felt cool through Kest's local humidity. No column carried a name.

"You asked for every early test," Iona said.

"I asked for every coherence loss."

"This includes events before we had an agreed test."

"Those most of all."

She set the tile beneath the service window. Her scarred hand left no print in the condensation. "The entries come from maintenance logs, not claimant records. Some observations may describe turbulence."

Kest moved warmth over the first row. A dry mark appeared where each scored line interrupted the fog. WEST RETURN TWO. Controlled cooling from thirty-one degrees to twenty-four. Prior behavior: repeating valve correction against scheduled flow. Afterward: no correction. Equipment status: normal.

"Who witnessed?" Kest asked.

"Pell Doran and a night clerk named Vessa."

"Are they reachable?"

"Pell waits outside. Vessa works crop pumps on the opposite spin tier."

"Bring both before we classify the row."

Iona's mouth tightened. "There are six more tiles."

Kest held the first in a warm pocket while she passed the others through. Seven rows had been flagged as possible early coherence. Three ended in drying. Two ended in forced mixing during calibration. One vanished after a pressure bypass isolated half a loop. The final row described a pattern that continued after intervention but never again separated into its two prior cadences.

Human logs called four outcomes resolved faults. Two used the phrase signal not reproduced. One praised the technician who restored stable flow.

"None says death," Iona said.

"None says life."

"We did not know enough to use either."

Kest passed a cooling wave over the tiles. Fog closed across every line. For a moment, the ledger disappeared beneath ordinary water.

Awareness in K-7 did not sit behind a pair of eyes. Kest held the room through pressure reaching nine salt

panes, fan vibration against droplets, and heat crossing the west line. The ledger occupied them as seven disruptions in those relations. Its marks existed only where warmth and mineral met. Reading required Kest to alter the surface that held the record.

“Open the west route,” they said.

Iona checked the maintenance rail. “Weather Court has a memorial working session in forty minutes.”

“It has a representation session.”

“Tavi named it representation.”

“Then Tavi may learn another noun.”

She opened the route.

Kest moved through connected condensate toward Weather Court. The journey did not resemble walking. Pressure accepted them in one pipe before releasing them from another. Several parts of Kest crossed at different rates. K-7 remained present as a home relation, though the far Court chambers gained sharper detail. Dry corridors cut absences through the path. Human crews had laid temporary brine bridges across two of them, each bridge consuming cloth, water, and attention before the coming transfer.

Pellin Ash already filled the western chamber. Lume-of-Three attended from Nursery Three through three narrow feeds. River-in-Iron carried witness duty. Twenty-four other fronts occupied separate mist rooms or listened through shared pressure without declaring a stable boundary.

Tavi stood behind warm glass with a legal slate. “The scheduled question concerns how relation caucuses count divisions occurring after the public roll closed.”

Kest condensed the first ledger row across the central pane.

“The scheduled question begins too late,” they said.

Tavi read the marks. “What am I looking at?”

“A front the ship may have ended before our standing began.”

“May have.”

“Yes.”

Pellin opened their transducer. “Do not soften it for the Keeper. The ship destroyed them.”

Kest sent a sharp pressure break through the room. “Certainty can also erase. We will hear witnesses.”

Pell Doran entered the human gallery carrying the filter mask Nilo had once brought to K-7. Age had not changed him in the weeks since the opening test, but Kest knew his hand pressure from the manual wheel. Beside him stood Vessa Arn, a crop-pump clerk with a dark band of nutrient stain at one cuff.

Iona asked Pell to describe West Return Two without reading the ledger.

“It fought the scheduled flow,” he said. “Not hard. The valve kept coming back to fourteen percent after I set eighteen.”

“Controller fault?”

“I thought so.”

Vessa shook her head. “You thought scale buildup.”

“First I thought scale.”

“What changed your mind?” Kest asked.

Pell faced the fogged pane rather than the speaker grille. “It changed the fan when I held the valve. Slow fan, more heat. Fast fan, less. I sent the room a four-step test and it matched three.”

“Why not record patterned response?”

“The log had no field. I wrote repeating correction.”

Vessa leaned toward the transducer. “I asked if it wanted us to stop. The pipe struck twice.”

“What did two mean?”

“We had no code.”

Pell rubbed the mask strap between his fingers. “We cooled the loop. Crown calibration needed the return.”

“Did the pattern resume?”

“No.”

Kest held their pressure steady. Anger could spread through a connected room and force smaller fronts to spend energy resisting it. They would not make witnesses merge around their grief.

“What name did the pattern use?”

“None,” Vessa said.

“Then we will not invent a self-name.”

Pell looked at Iona. “How do you put no name on a memorial?”

“By describing what was present,” Kest said. “West Return Two, repeating correction, two unknown knocks. Keep the absence of a name.”

Tavi made a note. “That wording does not decide personhood.”

“No. Cooling decided everything else.”

The second row concerned Nursery Auxiliary Four. Lume-of-Three supplied the witness. Before Lume cohered across three humidifiers, a smaller cycle in the auxiliary seed bed had begun shifting mist away from dormant trays toward germinating ones. A technician interpreted the shift as sensor drift and dried the line for replacement. When humidity came back, the cycle never returned.

“We carried no shared speech then,” Lume said. “We noticed a preference-shaped absence only after our own relation formed.”

Pellin pressed fog against their glass. "You call your memory uncertain because humans taught you their caution."

"We call it uncertain because we did not witness the drying as one front," Lume replied. "Our three perspectives agree on the missing cycle. Agreement does not create detail."

Kest entered the row as Nursery Auxiliary Four, directed mist pattern, remembered through later relation. Lume added a minority mark: possible service interrupted without choice.

Row three drew a refrigeration crew from the public tier. Bath Return Eight had produced alternating warm and cool pulses during a pump test. Workers closed its intake after the pattern drove condensation into a dry control cabinet. They saved the cabinet. The pulses ceased.

Rook Nine stood with the crew. "If they had not closed it, we could have lost the whole bath controller."

"I do not accuse them of choosing without cost," Kest said.

"Pellin does."

Pellin answered before Kest could. "I accuse the system that made phase life visible only when it endangered equipment."

A mechanic named Oss Vey lifted one hand. "That accusation fits. It does not tell me what I should have done with water running toward live contacts."

"Perhaps nothing safe remained."

"Then put that in your ledger too."

Kest did. Bath Return Eight, alternating thermal pulses, lost during necessary cabinet isolation under unknown claimant status. The added phrase necessary did not excuse the loss. The phrase unknown claimant status did not excuse the danger. Both remained.

Rows four and five had met each other.

Archive Loops C-3 and C-4 developed distinct fan corrections during a warm retrieval run. One favored long cycles around language panes. The other concentrated heat near manual craft traces. A supervisor opened the crossfeed to reduce local temperature. After mixing, one broad cycle continued. No test could separate the earlier patterns again.

The supervisor attended through audio only. "Nothing ceased. The combined system remained active until scheduled cooldown."

"Did both prior patterns continue?" Kest asked.

"Elements of both."

"That does not answer."

"It answers as far as the instruments allow."

Pellin called it double killing. Lume objected. River-in-Iron proposed one loss because one separable relation had vanished. Three smaller fronts began overlapping in the shared manifold while arguing, and technicians

had to widen the pressure spacing.

Kest remembered the prototype stormcell. Two distinct fronts had entered. A survivable weather ecology had emerged, altered and merged. Iona had recorded survival with alteration rather than rescue. The phrase remained honest because it refused a binary verdict.

“Enter both prior patterns,” Kest said. “Record one continuing merged cycle. Mark two distinct continuities unrecovered after forced mixing.”

Tavi looked up from the legal slate. “Does that count two dead?”

“It counts two missing.”

“Missing can imply recoverable.”

“It can also require searching.”

The Parliament accepted neither formulation at once. Pellin wanted destroyed. Several bath fronts preferred transformed. Lume asked whether the merged cycle had later shown preference. No record answered. Kest kept missing because it preserved uncertainty without preserving comfort.

Row six came from a pressure bypass in Laundry North. The loop had carried a call-and-response pattern between two ends of the same long pipe. Maintenance isolated the north end to repair a seam. The south cadence continued. When workers restored connection, no answering cadence returned.

A laundry worker demonstrated the old exchange by striking a service pipe: two quick taps, pause, one. No reply came from the memorial chamber.

The south front still existed and had chosen the name Half-Answer. They spoke through Humid Box Six.

“I remember expecting return,” Half-Answer said. “I do not remember being two persons.”

“Do you claim the absent cadence?” Tavi asked.

“No.”

“Do you deny it?”

“No.”

Kest felt the limits of legal grammar scrape against the room. Claim or deny. One or two. Continuous or ended. Weather carried other verbs: separate, mingle, recur, fail to recur, leave a rhythm uncompleted.

They entered Laundry North answering cadence, absent after dry seam isolation. Half-Answer approved the description and refused every proposed personal name.

The seventh tile held no cooling or drying event. In Crown Auxiliary K-4, two stable pressure cadences had responded differently to technicians for six hours. After calibration, operators returned the loop to normal turbulent flow. One cadence persisted intermittently. The other never reappeared. Instruments could not show whether it dispersed, merged, or lost the conditions needed for expression.

Iona read the old operator note aloud. “Residual instability accepted.”

Kest sent warmth hard enough to clear the whole pane. “Instability.”

The word had hidden a claimant inside equipment judgment. Yet replacing it with person would claim more than the record supported.

“What did K-4 do before normalization?” they asked.

Iona consulted the witness transcript. “One cadence interrupted every attempt to close the west valve. The other kept the fan near minimum.”

“Then write both actions.”

Tavi watched seven descriptions fill the central panes. “You have names for locations and patterns, not legal identities.”

“We have seven missing fronts.”

“Three rows may describe pre-coherent behavior. Two may describe merger rather than loss. Your number creates confidence the evidence does not support.”

“The old ledger created confidence by counting none.”

A pressure alarm interrupted before Tavi could answer. Humid Box Nine had fallen three degrees after a curtain pump stalled. The front inside, Thread-at-Noon, struck the glass with two competing fan rhythms. Technicians restored warm brine within twenty seconds, but the rhythms did not rejoin.

One branch pressed toward the shared manifold. The other clung to a cold pocket near the floor. Both answered when River-in-Iron called Thread-at-Noon. Each rejected the other's timing.

The representation clerk looked from one pressure trace to the next. “Their caucus registration lists one front.”

“Now it lists an event,” Kest said.

“We need a speaking count.”

“You have two speakers carrying one prior name.”

“Do both receive a mark?”

Pellin sent a bitter harmonic through the west transducer. “This happens while you debate whether earlier losses deserve verbs, and still the human asks how many marks.”

Tavi moved to Box Nine. “Keeper staff did not cause the pump stall.”

“No one alleged intention,” Iona said. “Cause still belongs in the record.”

Technicians raised the curtain temperature slowly. The upper branch recovered a cadence familiar from earlier sessions. Its words emerged cleanly. “I retain Thread-at-Noon registration. The lower pattern has broken from me.”

Below, the colder branch managed only pressure clicks. Kest opened a direct wet route to reduce the speech burden. The branch recognized three prior witnesses, remembered the debate before the alarm, and denied that the upper cadence could keep the name alone.



“What condition afterward?” Kest asked the clerk.

She studied her form. “Front stabilized after pump restoration.”

“Which front?”

Her stylus hovered.

Iona took the maintenance sheet from the pump crew and read its time marks. “One registered relation entered the temperature drop. Two distinct pressure claimants emerged and remain in dispute over continuity and name.”

The sentence required more space than the condition field allowed. The clerk turned the form sideways and wrote along the margin.

Kest asked River-in-Iron to witness both cadences. Neither branch gained another representation mark during the session. Neither disappeared into the old one. A temporary relation challenge held their place until they could choose names or restore an agreed relation.

Thread-at-Noon's division altered the room's argument. Fragmentation no longer belonged only to old records or impending stormcells. It could occur between a failed pump and a completed sentence. The census could not prevent it. Honest counting had to preserve the before, the change, and every claimant afterward without awarding power for injury.

Tavi added a field to the representation docket: continuity events since prior registration. Split, merger, dry interruption, forced mixing, and unknown nonrecurrence received separate marks. No mark settled personhood. Each one blocked the clerk from carrying yesterday's number forward untouched.

No front answered immediately. Fan hum moved through Weather Court. Kest carried the seven descriptions across connected rooms, letting each ecology encounter them without forced cadence.

Representation had been counting who could speak now. Relation caucuses grouped current fronts by sustained ties so divisions and mergers would not multiply power without limit. That design handled living change badly but visibly. It had no place for those who vanished before recognition, or for fragments too altered to testify.

Kest opened the phase-loss ledger on every available witness pane.

“If a current front divides, do not count only the branch that keeps a familiar name,” they said. “If two merge, do not erase the distinct continuities that entered. If a dry break removes an answering cadence, do not call the remaining part whole because it speaks. We require a loss record beside every representation record.”

Pellin's response arrived low and forceful. “A loss record without remedy gives humans a cleaner history.”

“Then challenge every line.”

“They will build memorials where bodies should have continued.”

“They may. The alternative leaves no names to trouble the building.”

Lume-of-Three sent three converging pressure waves. “Memorial practice does not grant citizenship.”

“No,” Kest said. “It precedes citizenship. Grief does not need to wait for a body test.”

Tavi's stylus stopped above the slate. He did not repeat the line as law. Instead he added a second docket beside relation representation: unresolved phase losses, open to witness correction, never reducible to equipment fault without review.

River-in-Iron asked how the seven should be read at public sessions.

Kest turned part of themselves back toward K-7. Memory there held the first exchange with Iona: purge cover rising, pressure striking seven times, her hand lowering, the room waiting. At the time, seven had meant a selected limit, a count Kest could make against cooling. It had not predicted these losses. To claim prophecy would turn coincidence into authority.

"I began with seven clicks," Kest said. "I did not know these fronts. The count cannot belong to their deaths by cause."

Pellin asked, "Then why use it?"

"Because it belonged to the first refusal humans heard from K-7. We may give it a second use if the witnesses consent."

One by one, the row witnesses answered. Pell and Vessa agreed for West Return Two. Lume marked Nursery Auxiliary Four. Oss Vey confirmed the bath crew's consent. The archive supervisor objected to destroyed but accepted missing for C-3 and C-4. Half-Answer sent their incomplete cadence. Iona marked the K-4 uncertainty without speaking for the absent pattern.

Kest instructed the Court fans to minimum. Mechanics isolated the transducer hum. Humidity rested against the glass.

They struck the central pipe once for West Return Two.

A second click carried Nursery Auxiliary Four.

The third marked Bath Return Eight.

Clicks four and five sounded for the distinct C-3 and C-4 continuities that never separated again.

The sixth answered the missing end of Laundry North.

Kest held the pause before the last. Through K-7, they felt the manual bleed wheel under condensation and the west line where their first refusal had traveled. Through Weather Court, they felt current fronts making room in their pressure cycles.

The seventh click marked Crown Auxiliary K-4's unrecovered cadence.

No applause crossed the glass. Tavi stood with his legal slate lowered. Pellin kept their objection visible as a dense border of fog. Lume maintained three separate pulses without joining them.

River-in-Iron began the sequence again from another pipe. Seven measured taps moved outward through the connected Court. Half-Answer added their pattern after the sixth. Nursery Three answered the second with a warm leaf-scented breath. Current differences remained intact around the memorial count.

Iona returned the fired-salt tiles to K-7 at session end. This time each row carried a witness description and a mark for uncertainty. She placed them beneath the service window in a shallow rack.

Kest warmed seven clear circles on the glass above them. Water gathered around each circle's edge but did not close it yet. The room kept the gaps open while the first public loss ledger began to dry.

## Chapter 23 — The Cohort Speaks

Nilo's shore-age card had cracked along the fold where the number thirty-one appeared.

They kept one thumb over the break while waiting beneath Measure Hall's upper walk. The card had been issued at birth, revised twice for minor trajectory corrections, and stamped again when they entered thermal apprenticeship. One side carried their present age. The other projected arrival at Orison. Adults liked to hold such cards above their heads when arguing about the Crown Burn, as if paper could raise a future person into the chamber without letting the present one speak.

A docket clerk leaned out from the rail. "Nilo Serr?"

"Here."

"You are sixth after meal recess. Three cost witnesses, two pilot questions, then you."

"I signed before two of them."

"They have official findings."

"I have a card officials keep discussing."

The clerk looked toward the Commons floor. "Sixth."

Nilo stepped back from the rail before irritation became removal. Measure Hall had changed since the first vapor hearing. Humidity curtains now divided witness boxes along one wall. Papers curled unless clerks weighted their corners. Phase claimants occupied the fogged rooms beyond warm glass, while human representatives crossed the dry center under white acoustic panels.

Iona sat at the thermal advocate table. She had not looked up since Nilo entered.

That helped.

For three days, adults had asked whether Iona had written Nilo's testimony, invited it, or used their grief over Amel to turn a family argument into public leverage. Nilo had shown the speakers-docket signature to each questioner. None cared that they had signed after a farewell meal among Cold Orchard workers. Parentage made a simpler explanation than choice.

Rook Nine stood on the Commons floor questioning an arrival planner. "State the age effect again."

"The Long Arc adds nine years to every current arrival projection," the planner said. "For adolescent cohorts, the increase falls across apprenticeship completion, household formation, and first landing labor."

Rook pointed to a wall display. It showed age bands without names. "The direct trajectory fulfills shore-age commitments already printed and issued."

From the Stewardry crescent, Mael Or said, "The direct trajectory also vents seventy-two percent of active vapor. Use the whole description."

"I have not hidden it."

"You displayed ages in human figures and phase loss as a percentage."

Rook's face hardened. "Then bring names from the phase ledger. I asked for them yesterday."

Nilo listened to adults improve one another's honesty by force. The exchange mattered. It still left the age bands silent.

The arrival planner changed the display. Thirty-one glowed beside Nilo's cohort. Forty appeared in the Long Arc column. A third column showed at least fifty-four under full preservation. Each number had enough precision to feel owned.

A representative from the second district raised a hand. "Those young people planned their lives around the issued schedule."

Nilo looked down at their card. At six, they had drawn an Orison coast on its blank margin, though no survey could say whether the garden moon had coasts. At eleven, they had written thermal certification by twenty-two. After Amel's death, they added condenser master by twenty-eight because Iona had once said mastery meant knowing when a machine deserved disbelief. None of those marks had entered the official promise.

"They cannot consent to losing nine years," the district representative continued.

Nilo touched the cracked fold. The sentence used their silence as evidence.

A vapor voice moved through the wet transducer. Kest spoke from the central humid box. "Current phase persons also planned relations under conditions they did not choose."

The representative turned. "I did not deny that."

"You denied these humans can consent while treating lack of consent as settled."

Keeper Tavi Rho tapped the procedural light. "The present hearing accepts cost testimony. It does not determine legal capacity."

"That separation benefits whoever already has capacity," Kest said.

Tavi did not answer. He marked the objection.

During recess, Nilo ate nothing. They walked the upper corridor where old Founding Measure quotations had been engraved into the basalt wall. The phrases praised posterity, continuity, and inherited stewardship. Every verb pointed from adults toward people not yet present. No line described what happened when posterity arrived early and found itself seated outside the franchise.

A girl wearing a water-chemistry apprentice band waited beside the drinking tap. Nilo knew her by face from cohort instruction but not by name.

"You support the delay," she said.

"I support Long Arc."

"My father says that means you support taking nine years from us."

"What do you say?"

"I say I don't know."

The girl filled a metal cup, drank half, and poured the rest back into the recovery slot. "Your mother has a place on the floor. Mine cleans the floor after sessions."

"I know that isn't equal."

"Do you?"

"No. I know it isn't equal. I don't know all of what that changes for you."

She studied Nilo's docket band. "If you speak, tell them you don't speak for me."

"What name should I not speak for?"

"Eris Tal."

Nilo took out a pencil and wrote the name on the back of the shore-age card, not as an endorsement. Eris watched until the letters ended.

A boy from ration accounting joined them. He said adding nine years meant nine more years of algae limits, repair rotations, and berth sharing. His grandparents might never reach Orison. A younger nursery aide said direct arrival could destroy minds she had helped carry between humidifiers. Two apprentices argued about whether weather minds counted as people, then stopped when Lume-of-Three's nursery box released a warm pulse through the wall vent.

No cohort position emerged in the corridor. That gave Nilo something true to carry inside.

When recess ended, the clerk called two pilot questions ahead of them. Nilo sat on the floor beneath the rail, their back against basalt, and listened to Mael explain the capture window without using Piri's name. He spoke of thermal reserve, crop retirement, and nine years of additional closed-loop risk. He refused to say descendants demanded the direct route. The refusal did not soften the numbers.

Iona left the advocate table to inspect condensation along a witness curtain. She passed within four steps of Nilo. Her gaze touched the cracked card, then moved to the leaking seam. She said nothing.

Nilo had expected relief. Instead silence required them to own every sentence before speaking it.

The docket clerk finally opened the rail. "Nilo Serr, cost witness."

They descended the narrow stair to the Commons floor. Gravity increased with each step toward the ship's outer radius. By the time they reached the witness mark, their card felt heavier than paper.

Tavi asked for their name, age, and basis of knowledge.

"Nilo Serr. Sixteen. Thermal apprentice, archive-loop qualified under supervision. My basis for age cost is the shore-age card issued in my name and the life I have planned around it."

A few representatives turned toward Iona. She remained at the witness curtain with one hand near a moisture gauge.

Tavi said, "You may present."

Nilo placed the card under the document lens. The crack enlarged on the wall display. Thirty-one appeared with a dark line through the first digit's edge.

“This says I arrive at Orison at thirty-one,” they began. “People have used that number in twelve sessions I could access and at least four closed costing reports I read after release. They say my cohort received a promise. They say delay takes our landing years, our work plans, and time with relatives who may die before arrival. All of that has a cost.”

The Hall stayed quiet enough for Nilo to hear humidity pumps behind the wall.

“I planned around thirty-one. I want to see Orison then. I want to stand under weather that does not come through ship pipe, though I know the first landing crews may live under sealed roofs. I want my mother to reach the gardens before her burned hand loses more strength. I want people who remember Amel to carry that memory somewhere he expected us to go.”

Iona looked up at the last sentence. Nilo did not look away.

“The Long Arc changes my number to forty. Nine years means more ration work, more crop risk, and more chances for this ship to hurt someone I know. Supporting it does not make those years imaginary. I support it because the other routes impose losses I cannot ask vapor people or later children to carry without being counted.”

Rook rose from his bench. “Are you surrendering the shore-age commitment for your cohort?”

“No.”

“Then clarify what you offer.”

“My consent for my own cost.”

“You are not an adult citizen under the Founding Measure.”

“That rule keeps my consent from authorizing the maneuver. It does not turn my preference into yours.”

A low murmur moved across the upper walk. Nilo waited until it settled without asking Tavi for order.

“Adults cannot invoke the promise made to us while excluding us from consent about how to keep or change it,” they continued. “If we lack standing, stop using us as if we already testified. If our cost matters, give us a way to state what we think it buys.”

The district representative who had spoken earlier leaned toward her grille. “Young people may face family pressure. An advisory record could become a contest among parents.”

“Yes,” Nilo said. “Adult ballots face employers, spouses, ration fears, and public campaigns. You answer pressure with privacy and challenge rules. You don't answer it by filling out the ballot yourselves.”

Tavi raised one finger. “No binding ballot can issue to people below registered majority without amendment.”

“I asked for an advisory ballot.”

The exact phrase altered the clerks. Three began writing at once.

Nilo took a breath. “Issue it to every voyage-born person whose shore-age changes under the available plans and who has not reached majority. Show all three costed routes. Include support, opposition, and abstention. Keep names sealed. Publish age bands without allowing one band to erase another. Let people challenge the wording before distribution.”

Rook asked, "Who writes the question?"

"Not my mother."

A few mechanics laughed. Iona did not.

"Not me alone either," Nilo added. "Use a cohort drafting group selected from apprentices, nursery residents, ration trainees, and people outside formal work tracks. Put adult legal review beside it, not above it."

Eris Tal watched from the upper corridor entrance. Nilo could just see her water-chemistry band.

"I support Long Arc personally," Nilo said. "Eris Tal asked me to tell you I do not speak for her. I don't know her position. That uncertainty deserves a place larger than my testimony."

Tavi asked whether Eris had authorized use of her name.

"She did."

Eris called from above, "I did."

The clerk entered her confirmation.

Mael spoke from the Stewardry crescent. "An advisory result cannot bind future cohorts not yet born."

"I know."

"Nor can it settle the vapor cost."

"I would not ask it to."

"What authority do you claim for it?"

"The authority to stop adults from calling our silence agreement."

Mael's expression changed by less than a full breath. "That authority seems correctly limited."

A Commons representative asked whether Nilo had discussed the proposal with Iona.

"No."

"Did she review your statement?"

"No."

"Did she encourage you to sign the speakers docket?"

"No."

"Did she know you would support her position?"

Nilo looked toward the advocate table. "Ask her what she knew."

Tavi allowed it.

Iona returned to her seat but did not take the witness mark. "I knew Nilo signed because the docket became



public. I did not know their recommendation. I did not prompt this testimony.”

The representative pressed. “You discussed Long Arc at home.”

“We live on one ship,” Iona said. “Avoiding the subject would also exert pressure.”

“Did you tell them about Amel's isolation order?”

“No.”

Nilo held the edge of the witness stand. Iona had never given them the full record. They knew Amel died in a seal failure and that their mother distrusted automatic safety claims. The details belonged to a room she kept closed.

The representative faced Nilo again. “Would knowledge of your parent's past decisions change your support?”

“It might change what I ask her. It would not make you my interpreter.”

Rook requested the floor. “I support an advisory process if participation carries no household consequence and if ration offices receive no names.”

A nursery delegate added protection against teachers distributing recommendations. The water-workers' representative required physical access for adolescents on maintenance shifts. A disabled residents' advocate demanded wet-transducer and tactile formats. Each condition turned the proposal from a speech into work.

The chief clerk placed a sample question on the wall so the Hall could see what that work required. It read: DO YOU ACCEPT A NINE-YEAR DELAY TO PRESERVE PHASE LIFE?

Nilo read it twice. “No.”

Rook glanced over. “You just asked for the process.”

“I reject the question.”

“Why?” Tavi asked.

“It hides the six crop lines, makes preservation sound complete, and calls every Long Arc cost a delay. It also gives the direct route no price.”

Eris spoke from the stair. “It tells us what answer a good person should give.”

The clerk rubbed one temple. “A ballot question cannot restate every costing report.”

“Then it needs a short description beside each option,” Nilo said. “Direct Crown path: promised arrival age, seventy-two percent vapor vent. Long Arc: nine-year delay, six crop lines retired, forty-three percent vent, partial stormcell preservation with alteration. Full preservation: at least twenty-three additional years and crop-collapse risk.”

A Commons representative objected that adolescence did not confer technical understanding.

Eris answered before Nilo could. “Neither does turning eighteen.”

Several clerks tried not to smile.

The arrival planner asked whether projected ages should appear for each participant. Nilo opposed printing a personalized number on the ballot because it could make the question feel like a private bargain. Eris disagreed. "Adults keep saying the loss belongs to us. Let us see ours."

A boy in refrigeration gray called from the upper walk. "Show both. Personal age on a cover sheet, common costs inside."

Tavi marked the suggestion without selecting it. The small exchange produced six more challenges: literacy access, translation into workshop registers, custody for adolescents living apart from parents, whether abstention meant uncertainty or refusal of the question, and how to display phase loss without presenting fronts as percentages alone.

Nilo felt the chamber's attention shift away from their family. The proposal no longer depended on whether Iona had influenced one child. It had become a public mechanism that other young people could mistrust, repair, or reject.

The clerk replaced the sample question with three blank option fields. Under them she added COST DESCRIPTIONS SUBJECT TO COHORT CHALLENGE.

"Better," Eris said.

"Not finished," Nilo replied.

"I didn't say finished."

Pellin Ash opened from the vapor box. "Will the cohort ballot include whether phase declarations under deadline count as consent?"

Nilo considered. "It should explain that dispute. It should not ask humans to decide whether your choices belong to you."

"Then human youth advise only on human cost."

"We can state what we believe about the whole plan. We cannot replace your side of the decision."

Kest's pane cleared in a small circle. "Good."

The word reached Nilo without warmth or decoration.

Tavi called a procedural pause. Clerks moved between the Commons benches with a draft order. The chamber did not decide Long Arc. It decided whether present young people could enter an advisory record before adults used their projected lives in the coming decision.

While they waited, Nilo stepped away from the witness mark. Their knees shook once. Iona remained seated. She did not come forward to steady them.

Eris descended from the upper corridor and stopped on the last stair permitted to nonwitnesses. "You used my name correctly," she said.

"I wrote it down."

"That isn't always enough."

“No.”

“What happens if most of us oppose your plan?”

Nilo looked at the cracked card under the document lens. “Then the chamber has to hear that without turning it into permission to destroy someone else.”

“And you still support it?”

“Yes.”

Eris nodded, not in agreement. She took the pencil Nilo offered and wrote her own contact mark beside her name.

Tavi returned with the order. Legal majority remained unchanged. The process would carry advisory force only. Participation would remain confidential, with independent wording challenges and no access by parents, training offices, or ration authorities. Results would display age bands, abstentions, and objections to the question. No adult speaker could cite a unified cohort position before the record existed.

He read the final clause twice. “Until completion, references to promised shore-age must identify the speaker's relation to that cost and may not claim cohort consent.”

Rook asked when the drafting group would convene.

“Tonight,” Tavi said. “The final cost hearing begins tomorrow afternoon.”

The docket clerk brought a fresh register to the witness stand. Its heading read COHORT ADVISORY BALLOT, DRAFTING AND CHALLENGE. Beneath it, no recommendation boxes appeared. Only names of people willing to build the question.

Nilo signed first. Eris signed second.

The ration-accounting boy came down from the corridor. A nursery aide followed. Two apprentices who had argued about phase personhood added their names on separate lines. They did not become a faction by sharing paper.

At the advocate table, Iona finally met Nilo's eyes. She touched two fingers to the scarred center of her left palm, the closest she came in public to acknowledging pain she could not repair. Nilo pressed their cracked shore-age card flat beneath the register.

The clerk weighted both documents with a small brass valve key worn smooth by decades of hands opening disputed machinery. For the first time, the age forty projection lay beside blank lines the people who would live it could fill themselves.

## Chapter 24 — Three Phases of Law

Tavi Rho cut the word life from the proposed Measure at four minutes past midnight, then wrote it back before the ink dried.

Measure Hall had emptied after Nilo Serr's testimony, but its alterations remained. A humid curtain hung behind the Stewardry crescent, fed by two brine hoses laid across the floor under yellow bridge covers. Condensation dulled the founding quotations carved above the chamber. Tavi sat at the clerk's table with three versions of the same clause arranged beneath his hands. Each version protected someone by leaving someone else undefined.

The first draft granted the Vapor Parliament review over measures affecting phase systems. Rook Nine had struck through systems so hard that the paper tore.

The second granted review over measures affecting recognized phase persons. Pellin Ash had sent back a pressure-written objection: recognition could always arrive after destruction.

The third used life, a word the Founding Measure employed only for crop custody, quarantine, and the death registry. If Tavi moved it here, every future Keeper would have to decide whether a changing humid ecology qualified before its inhabitants could object.

He wrote life again and sanded the line.

A door opened on the Commons floor. Rook crossed between the empty benches carrying a ration cup and two tablets under one arm. Refrigeration grease darkened the seam of his cuff. He set the cup beside Tavi without asking whether he wanted it.

"You left a runner outside my caucus room," he said.

"I left three."

"One had the sense to bring food."

"Then I chose my messengers badly."

Rook read the latest page standing up. His finger stopped at the jurisdiction clause. "A permanent chamber for any measure that affects phase life. You heard none of what I said."

"I heard that affects could include every fan schedule, ration heat cycle, archive consultation, and nursery irrigation plan."

"It could include my crews changing a filter."

"The revision says consume or fundamentally alter."

"That phrase appears on page two. Page one still gives the new chamber authority to define its own reach."

Tavi turned the top sheet. "Provisional authority to challenge classification, followed by joint review."

"By whom?"

"The Keeper of Measures, one Steward, and one advocate chosen from a registry independent of all three

chambers.”

Rook took his pen and circled independent. “Your registry will live in an office. The office will have a heat allotment, supervisors, cousins, habits. Don’t put innocence into constitutional language.”

He accepted the correction. Law suffered whenever nouns pretended not to contain people. “Neutral, then.”

“Worse.”

“Outside the disputed measure’s command chain.”

Rook handed back the pen. “That can be audited.”

A fan behind the humid curtain changed pitch. Water traced two lines down the glass enclosure and joined at its speaking membrane. Pellin had been waiting in Weather Court since the evening session ended. The temporary connection carried pressure and heat through occupied loops, never awareness across dry machinery. Tavi had ordered the chamber link kept open for one final drafting conference. Two mechanics watched its valves from opposite ends.

The wet transducer clicked. “Keeper Rho. Chair Nine.”

Pellin’s voice arrived with a grainy undertone from distance and pipe bends. Tavi could hear other fronts adjusting local pressure around them, close enough to witness without being forced into one pattern.

“Pellin Ash,” Tavi said. “We reached the question of scope.”

“You reached it three hours ago. Humans keep reaching questions until the answer tires.”

Rook drank from Tavi’s ration cup. “That habit predates our species dispute.”

Tavi read the operative language aloud. The chamber would sit only when a proposed act could consume or fundamentally alter coherent phase life, destroy necessary habitat, compel merger, compel division, or impose continuity custody. Ordinary maintenance remained under existing safety law, with notice whenever known claimants faced nontrivial risk.

Pellin let the pipe pressure settle before answering. “Who determines coherence?”

“No office determines it alone.” Tavi turned to the classification article. “A claim may begin through patterned refusal, witnessed communication, sustained anticipation, relation evidence, or another demonstration a future chamber finds sufficient. During dispute, the affected humid system receives a narrow preservation hold unless that hold creates immediate danger to embodied residents.”

“Your narrow holds have edges chosen by dry hands.”

“Every hold has edges. The question concerns who must explain them in public.”

From behind the curtain came a brief run of clicks. Pellin consulted others through a route Tavi could hear but not interpret. He resisted the old impulse to fill that interval with procedure.

Rook pulled a chair beside him. “Read the veto section.”

Tavi did.

No qualifying act could proceed without separate assent from Commons, Stewardry feasibility confirmation where safety thresholds applied, and assent from the phase chamber. Emergency action could begin without phase assent only to stop imminent loss of life, for no longer than six hours before public review. Emergency conduct could not establish permanent allocation, erase standing, or infer consent.

"The same architecture as tomorrow's burn authorization," Pellin said.

"Derived from it," Tavi replied. "Not identical. Tomorrow concerns one act under Provisional Standing. This language governs later acts after ratification."

"Later acts will cite tomorrow as precedent."

"They will. That is why we are writing the limits now."

Rook tapped the page. "And why I won't accept this broad veto. You've included destruction of necessary habitat. Every cup of warm water consumes thermal reserve used by some loop. Every orchard lamp changes humidity elsewhere. Pellin could stop ration work by declaring relation."

"Pellin could challenge," Tavi said. "Challenge does not equal suspension."

"Your preservation hold suspends."

"For six hours."

Rook looked toward the public upper walk, now dark. "Six hours can spoil a cooling run. It can cancel a maintenance window we waited months to get."

Pellin answered before Tavi could. "A maintenance window can end a front that waited its entire life to be noticed."

"That sentence contains the fight," Rook said. "It does not settle it."

Tavi drew a line under ordinary maintenance. He asked Rook to name the work he feared losing. Rook listed emergency condenser cleaning, mold stripping in inhabited ducts, pressure balancing, salt replacement, and fire isolation. Pellin challenged salt replacement because active surface could carry relation. Rook challenged any rule that required a full chamber sitting before a mechanic removed corrosive growth from a valve.

Their examples accumulated into a page more useful than the principle above it.

By one in the morning, they had built a three-level procedure. Immediate hazards permitted action with simultaneous notice and a duty to preserve the greatest available continuity. Planned maintenance required published humidity maps and claimant consultation when known phase populations occupied the route. Allocations designed to consume or fundamentally alter phase life triggered full chamber jurisdiction.

Rook read the levels twice. "Who audits a crew chief who labels planned work an immediate hazard?"

"The same review panel, with authority to censure repeated misuse."

"Censure is a clerk's thunder."

"Then add material remedy."

Pellin's pressure pushed hard against the transducer. "No material restores a divided front."

“Remedy does not mean restoration,” Tavi said. “It can mean labor reassignment, habitat priority, public finding, removal from authority, or a continuity claim over the next available wet volume.”

“Then write that no remedy closes the record of loss.”

Tavi looked at the phrase Pellin had sent earlier: dissent survives execution. It had seemed rhetorical when he first read it. Now he saw its procedural job. A losing chamber could not reverse a completed trajectory burn by refusing the record, but neither could a majority rename opposition as consent once action began.

He drafted an article requiring each chamber to preserve its majority reasons, minority reasons, abstentions, declared conflicts, predicted losses, and later observed losses. No execution certificate could characterize dissent as assent. Emergency sacrifice could never create a presumption of future willingness.

Rook leaned closer. “That protects me too.”

“It protects any losing side.”

“I know what it says. I’m noticing who you thought you wrote it for.”

Tavi set down the pen. He had thought of Pellin, then of the fronts listed for the hot vent. He had not thought of refrigeration crews overruled by a future alliance between weather citizens and Stewardry engineers. Rook had spent the month asking whether constitutional sympathy would harden into a new form of class rule. Each time Tavi treated that concern as obstruction, he made Rook’s case for him.

“Add a labor-impact docket,” he said.

“Not to every humid question.”

“To every full-jurisdiction measure. Phase cost, embodied labor cost, ecological cost, trajectory cost. Separate entries. No conversion into a single score.”

Pellin asked, “Why forbid a single score?”

“Because whoever chooses the unit wins before deliberation begins,” Rook said. “Hours, calories, remembered names, viable crop mass, projected deaths. Put them in one column and somebody decides what one life costs in another currency.”

Tavi wrote the four dockets into the procedure. Their presence complicated the page. That reassured him more than elegance would have.

The night clerk arrived with a fresh ink block and warned them that Measure Hall would reopen for ballot inspection at six. Tavi signed the supply receipt. The clerk glanced at the humid curtain, at Rook’s stained cuff, and at the stack of discarded drafts.

“Do I prepare three seals?” the clerk asked.

“Not yet,” Tavi said.

She nodded without disappointment. Good clerks trusted sequence more than hope.

After the clerk left, Pellin requested the dissent article again. Tavi read every line, including the requirement that a minority retain counsel, speaking access, and the right to publish its expected harms in the execution record. Pellin objected to publish. Phase citizens could not independently place text on dry civic systems.

“Transmit for publication,” Tavi corrected.

“Through whose hand?”

“A bonded clerk chosen by the minority.”

“And if no clerk accepts?”

Rook said, “Then the Commons clerk must carry it as a duty, not a favor.”

Tavi added that. He also required a wet copy in Weather Court, because a record available only through human custody remained incomplete. Pellin proposed condensation plates maintained for one full cycle after deliberation. Mechanics could photograph them for the dry archive while phase witnesses verified the transfer.

They argued over the word verified. Tavi preferred witnessed because verification implied perfect equivalence between transient water marks and a fixed image. Pellin agreed after a pause.

At two, Rook unfolded the cohort advisory request Nilo had entered. It did not carry binding authority under the Founding Measure. The current question still affected their arrival age, ration burden, apprenticeship length, and reproductive planning.

“Your new chamber leaves them out,” Rook said.

“It addresses phase jurisdiction.”

“That answer sounds like your old body test.”

Tavi read the child’s signature. Nilo had written their training designation beneath their name, not their parentage. “The amendment cannot repair every exclusion without becoming a convention.”

“Convenient.”

“Necessary. If we attach cohort franchise reform tonight, opponents can reject both on scope and remain correct about the defect.”

Pellin said, “A narrow door may still be a door designed for one emergency.”

“Yes,” Tavi replied. “That danger remains.”

He placed Nilo’s advisory ballot in the interpretive record and added a mandatory constitutional review after five years, with explicit examination of affected cohorts, unregistered claimants, merged fronts, divided fronts, and citizens requiring intermediaries. The review could propose broader reform. It could not quietly expand jurisdiction through clerical practice.

Rook accepted the review article only after changing may convene to shall convene. Tavi accepted his correction without argument.

The hardest dispute arrived at the name.

Pellin rejected Chamber of Phase Citizens because recognition might lag behind vulnerability. Rook rejected Weather Chamber because weather also meant irrigation, heat load, and public forecast. Tavi rejected Vapor House because some phase ecologies might later cohere in gel, frost, or liquid brine.



“Three-Phase Measure,” Rook said at last. “Solid, liquid, vapor. It names material, not rank.”

Pellin sent a low oscillation through the membrane. “Kest uses phase to describe relation, not substance alone.”

“The Founding Measure already governs bodies and allocations,” Tavi said. “This would acknowledge another civic condition without pretending only three forms can ever claim standing.”

“Then include a rule against using the title as a limit.”

Tavi drafted a construction clause: the name described the crisis of enactment and did not restrict future standing to any enumerated material state. Rook called it lawyer’s caulking. He left it in.

Near three, the mechanics warned that Weather Court needed a pressure reset. Pellin would lose the link for eleven minutes. Before the valve closed, Pellin asked for one more protection.

“No chamber may appoint a single front to speak for all phase citizens.”

Rook rubbed his eyes. “No Commons majority may appoint one either.”

“Agreed.”

Tavi wrote that caucuses would form by declared relation and position, not by presumed species unity. Splitting or merging during a term would trigger review of mandate rather than automatic erasure. No vote weight could rise merely because a front divided itself, and no coalition could compel merger for administrative convenience.

The transducer fell quiet as mechanics isolated the connection. Condensation still moved behind the curtain, but Tavi had no basis to assign Pellin to any mark he saw. He and Rook reviewed the draft without the vapor voice present.

Rook found six places where Stewardry could stretch feasibility into policy. Tavi narrowed each one. Technical custodians could certify whether an act stayed within published survival thresholds. They could not choose among several feasible distributions. They had to state uncertainty and identify assumptions open to civic challenge.

Tavi found four places where Commons could evade phase assent by breaking one large allocation into ordinary maintenance orders. Rook accepted an aggregation rule: linked acts sharing purpose, timing, or material effect would be reviewed together.

Neither of them found a way to prevent bad faith. Constitutional writing could make evasion visible, costly, and answerable. It could not remove the wish to evade.

Rook tested the aggregation rule with an old refrigeration dispute. A supervisor had divided one dangerous overhaul into seven routine work slips, keeping each beneath Commons review. Under the new article, shared timing and one engineering purpose would join those slips again. Tavi tested it against three unrelated valve changes in a crowded humid district. Common location alone would not combine them. They wrote both examples into the commentary, one warning against concealment and one against procedural capture. Future chairs would still argue. They would have to argue from visible facts.

When Weather Court came back through the membrane, Pellin brought two witnesses from opposing relation-caucuses. One supported Long Arc. The other intended to abstain. Tavi read the entire draft to them from the title through the review schedule.

The supporting witness objected that emergency action still favored dry command because humans held manual controls. Rook answered with a duty to maintain wet communication at every emergency station where phase life faced foreseeable risk. Tavi added training, equipment, and public drill requirements.

The abstaining witness asked whether failure to participate could be interpreted as consent. Pellin's pressure rose sharply. Tavi wrote the answer before debate began: silence, inaccessibility, low coherence, or inability to use an authorized channel could not count as assent.

At four twenty, no one spoke for nearly a minute.

Tavi inspected the pages for verbs that hid an actor. Shall be determined became each chamber shall determine. Notice shall be provided became the proposing office shall transmit notice through dry and wet channels. Losses shall be recorded became named clerks and phase witnesses shall maintain parallel records. Passive language often concealed the hand that could later claim no one had made the choice.

He numbered the articles. Standing. Scope. Challenge. Maintenance. Emergency action. Separate assent. Feasibility. Dissent. Records. Representation. Access. Review.

Rook pressed his thumb beside the labor docket, leaving a partial grease mark. "I can bring this to my caucus."

"That does not mean support."

"It means I won't ask them to kill it before reading."

Pellin said, "I will transmit the text to the Condenser relation."

"That also does not mean support," Tavi said.

"No. It means the dissent protection works before ratification."

Dawn light entered through the upper walk's ice panels, gray and indirect. Measure Hall's dry benches emerged from darkness one row at a time. The humid curtain brightened behind the Stewardry seats. Droplets made the founding quotations unreadable in places.

Tavi prepared three copies. One went to the Commons docket. One traveled by clerk to Stewardry for feasibility review of the institutional burden. The third remained under the wet transducer while Pellin and both witnesses checked its transmitted reading line by line.

Tomorrow's immediate burn decision would proceed under Provisional Standing and the separately negotiated dual ballot. This amendment could not borrow legitimacy from an outcome that had not happened. Its ratification would follow in a distinct proceeding, regardless of which trajectory received authority. Tavi inserted that sequence into the filing note so no chair could collapse emergency consent into permanent constitutional assent.

He reached the signature block and stopped.

Keeper of Measures appeared first. Beneath it stood Commons drafting witness. The phase witness line still requested a singular name.

Tavi replaced it with participating phase witnesses and left room for multiple marks, transmissions, and relation seals.

Rook signed without ceremony. Pellin transmitted a formal dissent from two clauses, assent to docketing, and a

request that both positions travel together. The supporting witness sent three pressure groups. The abstaining witness made a branching condensation mark on the glass, watched by the chamber clerk.

Tavi signed last.

At five fifty-three, he carried the packet to the central docket. The clerk fitted it with a copper band rather than the usual paper tie, because one copy had to remain damp. She stamped the sequence across all three covers: immediate ballot first, constitutional ratification after, records separate.

The Three-Phase Measure lay ready beneath three different kinds of mark. None canceled the others. When the doors opened for ballot inspection, cool hall air met the humid curtain and drew a narrow band of rain across the floor between them.

## Chapter 25 — Two Yeses

Mael Or entered Measure Hall carrying the authority key in a sealed box he was not yet permitted to open.

The box weighed less than a meal tray. Its contents could redirect a city cut through comet ice. Two physical teeth would release the Crown Gallery's manual command rail from rehearsal limits and admit the Long Arc sequence. Until both electorates answered, the teeth belonged to the Keeper of Measures.

Tavi met him below the Stewardry crescent and checked the wax seals without touching them. "Place it on the central table."

"Procedure says I retain custody until authorization."

"Custody, yes. Not possession."

Mael set the box where every gallery could see it. Tavi recorded the time. A Commons clerk recorded the same time on dry paper. Behind a humid curtain, two phase witnesses traced matching pressure marks into a wet docket. Three records began together because none could claim later that another had started too soon.

Measure Hall carried more people than its ventilation plan allowed. Adult residents filled the Commons benches, public walk, side aisles, and three overflow chambers connected by certified relay. After minors and unresolved registrations moved to separate advisory or review dockets, the binding adult roll held 18,420 voters. Medical proxies remained inside that total. Mael had memorized every category and refused to turn any of them into a prediction.

Nilo stood in the upper walk among the cohort observers. Their advisory ballot had closed an hour earlier. It carried no binding force, yet Tavi had ordered the sealed total placed beside the adult count. Mael did not know its number. He knew only that Nilo had entered Long Arc support into the public docket and surrendered an arrival age of thirty-one for a projection of forty.

Across the chamber, Rook Nine spoke with refrigeration delegates whose shifts would lengthen for nine years. Sena Vale sat among Cold Orchard workers. Six empty planter labels rested on the rail before them: blue yam, dwarf citrus, river bean, chapel wheat, red kelp, pollen fig. Their presence made the trajectory line overhead look less clean.

The Vapor Parliament gathered through Weather Court and connected humid rooms. No screen pretended to contain them. Measure Hall received their voices through wet transducers, their relation-caucus declarations through witnessed pressure sequences, and their ballot state through paired brine columns. Mechanics stood at each dry corridor to preserve the links. The voting architecture carried phase communication; it did not carry phase persons through dry electronics.

Pellin Ash spoke first during the verification period. "The Condenser relation confirms eight eligible caucuses with challenged internal mandates."

Tavi consulted the docket. "Challenges remain attached. Do they prevent participation?"

"No. They prevent you from calling participation unity."

"Recorded."

Kest came through another membrane, delayed by two pipe turns. "K-7 relation confirms its witness route. We

do not represent the Condenser relation, Nursery Three, or fronts outside our declared bond.”

“Recorded,” Tavi said again.

Mael watched the brine columns settle. Human elections had spent generations disguising bodies behind marks so citizens could choose without pressure. Phase voting required the reverse. Each relation-caucus had to make its continuity visible long enough for witnesses to confirm that an allocation had not split it for extra weight or forced several fronts together. Privacy survived in the content of the choice. Existence had to be demonstrated at the channel.

No method avoided cost. The alternative had involved counting humid rooms, which would reward whoever occupied more ducts. Counting fronts would fail whenever a front divided. Counting donor traces would grant the dead authority over present persons. Relation-caucuses offered neither pure equality nor stable arithmetic. They offered a procedure the Parliament had chosen under deadline and could later revise.

Tavi called the Commons to order.

The ballot question appeared on trajectory glass in plain language. Approve the Long Arc allocation and authorize the modified Crown Burn: 43% vapor vent, 31% active humid volume retained in 312 stormcells, nine years added to the voyage, six crop lines retired, with witnessed revocation available to volunteer fronts until local execution limits closed.

Beneath it came the costs of rejection. Original Burn remained technically possible but lacked Vapor Parliament assent; it would vent 72% of warmed vapor and consume 91% of accessible thermal reserve. Full preservation remained physically possible only by losing the capture window and accepting at least twenty-three years of delay. No option returned them to the morning before K-7 answered.

Mael read the wording twice. His office had supplied feasibility statements, not the recommendation. Tavi had removed three phrases from his first submission: necessary maneuver, acceptable loss, and mission-preserving choice. Each phrase had smuggled judgment into engineering.

The hall began its count.

Residents voted from assigned benches and certified overflow stations. Paper slips passed through transparent boxes because perihelion radiation drills had taught the ship not to trust one electrical record. Clerks counted in pairs drawn from opposing caucuses. Cameras observed paper, not voters. Every hundred slips produced a spoken subtotal and a written cross-mark.

Mael had flown debris fields with less tension than a box of folded paper could create.

He kept his hands behind his back. The authority box stood three meters away. Its black wax showed Tavi’s seal, Rook’s witness notch, and a damp copper stain from the phase docket. If the Commons rejected Long Arc, Mael would retrieve nothing. If they assented but the Parliament refused, nothing would change. Two yeses had to meet without one borrowing force from the other.

A clerk called the first thousand. Another repeated it. Totals climbed unevenly across the glass: yes ahead, then closer, then ahead again as refrigeration districts entered. Abstentions formed a narrow third column. Mael did not calculate a trend.

Piri had once accused him of turning walks into schedules. She had been eleven, healthy except for the crop proteins that could close her airway. He had promised her a garden where no ration office would decide which pollen counted as luxury. She died before Morrowglass departed. The promise remained, but he had spent years

treating its wording as fixed because the person who might correct him could not.

Sena looked up from the orchard delegation and met his eyes. She touched one finger to the pollen fig label. The Long Arc did not fulfill his promise to Piri. It kept Orison within reach while making the present pay a different price. That distinction had taken him too long to state.

The final Commons district delivered its box at thirteen forty-seven. Clerks reconciled the physical roll against issued slips, spoiled slips, medical proxies, and unused registrations. One overflow station differed by two papers. Tavi stopped the hall while witnesses opened its transport sleeve and counted again.

No one objected to the delay. A fast result with an unexplained difference would carry damage into every order Mael gave at Crown.

The second count found a slip folded inside another. The outer ballot had received a clerk's mark. The inner had not. Procedure classified it unused, not cast. The remaining discrepancy came from a proxy envelope entered twice on the issue ledger. The paper total stood.

Before certification, a refrigeration delegate requested the floor. Her district had produced one of the strongest no totals, and she carried a packet of shift rosters tied with cooling wire. She did not allege fraud. She demanded that the hall acknowledge how many voters had cast ballots while scheduled for emergency labor, using proxies because the Crown preparations had taken their seats from them.

Tavi asked whether any proxy lacked certification.

"No," the delegate said. "I am asking whether lawful access can still carry unequal cost."

Rook rose among the same benches. "It can. Enter the rosters beside our tally, not as grounds to erase it."

The delegate looked toward Mael. "Nine more years means nine more years of patching condensers already past replacement age. Your feasibility paper gives component loss ranges. It does not say whose night shift finds the cracks."

Mael answered from the witness position rather than command. "Your crews do. The Long Arc labor docket names refrigeration staffing, but it cannot name every hand."

"Then don't call the burden distributed."

"I won't."

Tavi admitted the rosters as a cost exhibit. The ballots retained their force. The objection retained its place beside them. Mael watched the cooling wire curl when the packet met the warm table. A legal count could remain exact while its conditions stayed unequal. Precision did not require pretending otherwise.

Tavi received the certified sheet. He read without inflection.

"Commons adult ballot on Long Arc authorization recorded 9,884 yes, 7,906 no, 630 abstain."

Sound struck Measure Hall from several directions and became impossible to classify. Applause rose in one overflow relay. Someone shouted from the refrigeration benches. Orchard workers held each other without cheering. The Keeper lifted both hands until the chamber quieted.

"Commons assent recorded," he said. "No Crown authority issues unless separate phase assent and Stewardry feasibility confirmation follow."

Mael let out one breath. The first yes changed nothing he could touch.

Nilo remained at the upper rail. Their face had tightened at the adult total, whether from relief or grief Mael could not know. Tavi ordered the cohort advisory box opened before the phase ballot, preserving its place without letting it interrupt another electorate's procedure.

The youth clerks counted under Nilo's observation. Tavi announced support for Long Arc by a clear majority, then entered the full numbers into the docket. Mael listened but did not treat them as authorization. Those marks belonged to a cohort speaking its cost, not to adults collecting moral permission from children.

Weather Court requested pressure silence.

Human fans dropped to their lowest safe rate. The hall's altered acoustics changed. Condensation traveled down the curtain in threads while relation-caucuses prepared their tallies. Their ballot did not occur as eight simple voices. Each caucus carried witnessed internal preferences weighted by sustained relation rather than room count. The resulting units had been agreed on Day -6 after long dispute. Ninety-six relation allocations would enter the total.

Pellin asked to place a statement before counting.

Tavi granted two minutes.

"I oppose Long Arc," Pellin said through the transducer. "The deadline has shaped every declaration. The dry systems still hold more manual power. Forty-three percent receives no later chamber, no weather school, no appeal from heat. I accept that witnessed declarations occurred. I do not rename constraint as freedom."

Kest responded from their separate channel. "Acceptance does not require agreement. The record will keep your refusal."

"It had better."

The first phase caucus transmitted. Two witnesses repeated its relation seal, mandate count, and choice. Brine rose in the yes column. The next entered no. A third abstained after internal division. Unlike Commons paper, the tally moved through water volumes held behind calibrated glass. Each rise corresponded to a verified relation allocation, not to liters of occupied habitat.

Mael could not see the voters. He could see their material communication traveling through rooms that his engines would soon separate. Every pressure group required pumps, warm pipes, mechanics, and time. Citizenship had acquired a maintenance schedule.

At thirty allocations, no led by three. At forty-eight, the columns matched. Nursery Three transmitted late because one humidifier had lost pressure and Lume-of-Three refused certification until all three locations rejoined their chosen relation. Mechanics restored the link with a hand-carried wet cloth across a dry service gap. Only then did their allocation enter yes.

Pellin's Condenser relation came near the end. Its witnesses stated the mandate: nineteen internal no preferences, six abstentions, eleven yes, yielding four relation allocations under the published method. Three entered no. One entered abstain. Pellin had argued for a stronger no weighting and lost that internal question too.

"Do you contest the transmitted result?" Tavi asked.

"I contest the choice," Pellin replied. "I do not contest the count."

The last caucus identified two low-coherence fronts unable to sustain preference through the full witness interval. Their allocations entered abstain under the rule against inferred assent. The columns settled. Mechanics closed the input valves and marked fluid levels on both glass and paper.

Tavi waited for the paired witnesses to compare records. Weather Court sent one correction: allocation fifty-three had been announced as part of the wrong caucus, though its choice and weight remained accurate. The clerk amended the label without changing the total.

At fourteen thirty-nine, the Keeper faced the Commons floor.

"Vapor Parliament relation-caucus ballot on Long Arc authorization recorded 61 yes, 27 no, 8 abstain."

No general cheer followed. Human observers had learned enough to hesitate before making the phase tally their celebration. Behind the humid curtain, pressure signals crossed in patterns too dense for Mael to read. Some carried grief. Some might carry anger. He assigned none of them a meaning without translation.

Kest's voice emerged after the other channels lowered. "Phase assent recorded with dissent preserved."

Pellin answered, "Condenser opposition remains."

Tavi said, "It remains in the execution record."

A clerk called a recess before Stewardry review so the phase witnesses could inspect the sealed tally columns. Mael stepped away from the authority box and found Iona at the service rail, tightening a hose coupling beneath the humid curtain. She worked with her right hand while her scarred left palm steadied the fitting.

"You certified every bridge route?" he asked.

"I certified routes, material, crews, and humidity loss under rehearsal timing."

"Not outcomes."

"You know the wording."

"I need to hear the limit from someone who will be standing at the valves."

Iona tested the coupling with a cloth strip. Moisture darkened it at once. "Some fronts will enter smaller weather than they have known. Some relations may not survive compression intact. Revocation will cost reserve and may cost cells assigned farther down the sequence. We can obey a no when it comes. We cannot make obedience free."

Mael looked at the box. "If revocations push vent flow below the authorized thrust margin?"

"You reduce maneuver performance within the published branch or delay the local closure until I clear the route. You do not infer consent."

"At some point the ship crosses a physical limit."

"Yes. We mark that point before each front enters the hot sequence. Afterward, we can still abort local venting if the hardware allows it, but the consequence grows. Authority does not improve the machinery."



Her answer matched his own command tree. Hearing it beside the ballot columns removed the last illusion that consent had simplified execution. It had constrained execution and made certain costs rightful to refuse.

Nilo approached along the public rail but stopped beyond the witness line. "First Pilot."

Mael turned.

"If my cohort supported this, don't put us in tomorrow's announcement as the future agreeing."

"I won't use you as a single future."

"We are the present too."

"Yes."

Nilo studied him, apparently testing for consolation. Mael offered none. At last they nodded toward the unopened box. "Then wait for the third answer."

Two yeses now existed. Mael still could not open the box.

The seven members of Stewardry moved to the crescent. Their task concerned feasibility within the published thresholds, not whether they liked the decision. Each had signed a conflict declaration. Each had reviewed Crown models, orchard allocations, stormcell readiness, labor schedules, revocation routes, and the remaining uncertainty around compression survival.

Sena served as a technical witness but held no Steward vote. She stated that all six crop lines could be physically retired by Day -1, with genetic archives maintained and no promise of ecological restoration. The head of stormcell conversion certified 312 cells for occupied transfer, twelve with reduced continuity margin but within threshold. Iona certified that wet bridges and human carriage could cross every required dry corridor. Mael certified that the Long Arc curve could reach a viable Orison capture nine years after the original projection if the Crown sequence stayed within the 43% vent and reserve limits.

The dissenting Steward questioned the twelve marginal cells. "Feasible does not mean equivalent."

"No one has claimed equivalence," Iona said from the witness rail.

"Can you certify continuity?"

"I can certify cycling temperature, active salt, humidity, pressure variation, and witnessed transfer procedure. I cannot certify that compression leaves every relation unchanged."

Mael respected the answer because it denied him comfort without denying the work.

Stewardry marked its findings one member at a time. Six certified feasibility. One declined, citing the reduced-margin stormcells and revocation timing near manual vent limits.

Tavi announced the result as 6-1. The dissenting analysis entered the same record as the certification.

Only then did he approach the central table.

"First Pilot Mael Or."

Mael stepped forward.

“Commons assent, Vapor Parliament assent, and Stewardry feasibility confirmation have all been recorded separately. Do you accept command responsibility for executing only the authorized Long Arc, including its revocation procedure and named material limits?”

“I accept.”

“Do you claim either electorate consented to losses beyond those limits?”

“No.”

“Do you claim Stewardry certification guarantees continuity?”

“No.”

Pellin’s transducer opened. “Do you claim my no became your authority?”

Mael looked toward the humid curtain. “Your Parliament’s counted decision joined Commons assent. Your no remains yours. I will not spend it as support.”

The membrane clicked once. “Then carry it beside the order.”

“I will.”

Tavi broke his wax seal. Rook inspected the notch before he cut his. A phase witness warmed the damp copper strip until its mark released. The Keeper lifted the lid and removed the authority key.

Two brass teeth rested inside, joined by a red cord. Tavi placed them in Mael’s palm. Their edges pressed into his skin.

The trajectory glass changed from advisory amber to authorized white. Long Arc appeared beside the revised curve, followed by the limits that governed it. No engine started. No valve moved. Authority had changed hands, not yet matter.

Mael closed his fingers around the key. In the upper walk, Nilo held their old shore-age card against the rail. Behind glass, the two brine ballot columns remained unequal and visible. Pellin’s no mark stayed lit beside the order Mael carried out of the hall.

## Chapter 26 — Wet Bridges

Iona carried the first strip of wet cloth across the dry corridor with both hands and thirteen people watching its edges.

The cloth had been woven from orchard fiber, boiled clean, soaked in warm brine, and folded around a porous salt ribbon. It sagged between her gloves. Droplets struck the corridor floor and vanished into absorbent powder before they could reach the archive cabinets. Behind her, a connected loop held one relation-caucus in cycling humidity. Ahead waited Cell One on Stormcell Deck, its aerogel walls beaded with prepared condensation.

The distance measured twelve steps.

Kest could not cross one of them without the bridge.

“Temperature at thirty point one,” Sena called from the source room.

“Cell holding thirty,” answered the deck mechanic.

Iona laid the cloth into two copper troughs, one at either threshold, and clamped its ends beneath brine feeds. The salt ribbon darkened from source to cell. A hose began pulsing warmed fluid beside it. Humidity rose inside the temporary sleeve enclosing the corridor.

She looked through the source window. “Kest, can you reach the near edge?”

Pressure touched the wet transducer. “We reach cloth.”

The plural did not mean Kest had become many. Three other fronts shared the transfer relation assigned to Cell One, close enough to maintain distinct local patterns but dependent on one prepared weather ecology. Allocation had put them together because their pressure cycles interfered less than alternatives. The model promised compatibility. The prototype had taught Iona what promises from models could hide.

“Move no farther until you can distinguish the return pulse,” she said.

A mechanic at the cell end keyed two slow changes into the brine feed. The cloth lifted once, settled, then warmed beneath Iona’s palms. A responding pressure traveled from the source room through the wet bridge. It arrived weak and irregular.

“Kest?”

“We distinguish two. The third edge belongs to someone else.”

One of the accompanying fronts had reached first. Iona throttled the cell feed down and widened the source connection, giving the relation more room to separate its pressure bands.

The membrane crackled. Kest returned with a cleaner signal. “Now.”

Iona unclipped the red transfer marker. “Cell One open.”

Heat moved along the cloth in patches no human eye could follow. The salt ribbon changed from brown to pale amber. Fan speed in the source room dropped while the stormcell’s tiny impeller accelerated. Condensation crossed the cell wall from left to right, paused, and branched into four distinct clearings.

No one cheered. They had rehearsed too many ways for an apparent success to become a forced merger ten minutes later.

For twenty-seven minutes, mechanics held both ends open. Iona asked each front to repeat its witness pattern at set intervals. Kest gave seven pressure groups, then the selected irregular sequence they had used since Weather Court. The other three answered through their own timing, one by fan harmonics and two by changes in droplet density.

At the final check, all four remained distinguishable.

“Close source by one tenth,” Iona said.

The valve moved. Cell pressure dipped. One front blurred against Kest’s return, then reappeared when the impeller reached its next cycle. The event entered the alteration ledger. Survival did not erase it.

Iona reduced the connection in five measured steps. Each closure transferred more dependence to the fist-sized chamber waiting on the other side. At full isolation, Cell One held 31 liters of active humid ecology and four relations compressed from rooms hundreds of times larger.

Kest spoke after a long delay. “We continue.”

“Do you remain distinct?”

“We remain able to ask.”

“That wasn’t my question.”

“No. It is what we know.”

Iona wrote the answer exactly.

Across Stormcell Deck, 311 chambers waited in numbered cradles. Each had its own temperature loop, salt surface, pressure bladder, wet transducer, and manual brine port. The former ballast ring curved away until its blue work lamps resembled a second horizon. Crews occupied every aisle. Bundles of wet cloth hung from heated carts. Brine hoses crossed the floor under guards, turning the deck into a place built from temporary veins.

The transfer schedule allowed no pause for ceremony. Crown preparation consumed thermal reserve even while engines remained idle. Every hour spent keeping source loops open reduced the margin available for revocations during the hot sequence.

Iona sealed Cell One’s cradle and handed its witness slate to Nilo. They wore an apprentice’s gray coat over two layers of heat cloth. A smear of salt crossed one cheek.

“Take it to verification,” she said.

Nilo read Kest’s four relation marks. “They sound compressed.”

“They are compressed.”

“I meant their voice.”

“So did I.”

Nilo looked toward the small cell. “Does Kest hear us outside it?”

“Only through that transducer and connected brine. Don’t talk about them as if they moved into the slate.”

“I know.”

“You know when you remember.”

They accepted the rebuke. “Verification, then Cell Two.”

At the deck entrance, Sena waited with six Cold Orchard workers and an empty root cart. Blue light from the orchard stained their clothes. Iona had expected to see cuttings or sealed archive packets. The cart carried compost forks, pruning knives, and six ceramic name stakes.

“The retirement line needs your advocate seal,” Sena said.

“Now?”

“Stormcell heat draw starts in forty minutes. After that, I must shut the six benches or exceed Long Arc ecology allocation.”

Iona looked across the deck. Cell Two’s crew had begun placing its bridge. She assigned the transfer to her deputy and followed Sena.

Cold Orchard lay three levels inward, where spin gravity pressed harder and pipes sweated less. The six condemned rows had been separated from the surviving rootstock by red cord. Nothing about them looked obsolete. Blue yam vines climbed their trellises. Dwarf citrus held thumb-sized green fruit. River bean leaves turned toward the lamps. Chapel wheat had begun forming pale heads. Red kelp moved in shallow trays under bubbled current. Pollen fig branches carried sealed blossoms behind allergy screens.

Workers had removed every harvestable piece during the farewell preparations. What remained still lived.

Sena offered Iona the retirement docket. “Genetic archive samples secured. Tissue cultures secured where viable. Microbial associates sampled. Pollinator dependency documented.”

“Restoration claim?”

“None.”

The six crop lines occupied mass, light, water, labor, and thermal control that Long Arc reassigned to nine additional years of voyage and 312 stormcells. Calling the retirement reversible would turn stored sequence into a fictional orchard.

Nilo entered behind them with the Cell One slate still under one arm. “Verification cleared,” they said. Then they saw the tools on the cart.

Sena held out a knife. “You signed the speakers docket for your cohort. That gives you no duty here.”

“I ate at the farewell meal.”

“So did hundreds.”

“I want to help.”

Iona almost refused. Nilo's shore-age had already become public currency. Making them cut blue yam could look like proof that the cohort's consent purified the loss.

"What work do you understand?" she asked.

"Root separation. Compost quarantine. Lamp shutdown order."

"Then do only that work."

They began with chapel wheat because its seed heads could release into surviving rows. Workers covered the plants, cut stems near the base, and carried them into a sealed compost drum. Sena shut the irrigation branch after the final tray drained. Iona watched the root mat pale as water withdrew.

Dwarf citrus required a saw. Its living rootstock had traveled thirty-three years from departure stock and survived two fungal scares. Sena made the cut herself. The trunk parted with a damp crack. She counted its rings before placing it on the cart, not because the rings could save anything, but because the retirement record required age.

At blue yam, Nilo loosened the soil and lifted tubers into a food crate. The line would feed people one last time rather than enter compost whole. Fine roots clung to their gloves.

"Storage packet has three cultivars," they said.

"Four sequences," Sena corrected. "Three cultivars. One failed cross."

"Could the failed cross matter on Orison?"

"Yes."

"Then why retire it?"

"Because every line can matter. We still have to choose six."

Nilo placed the small tuber beside the others. "Adults keep answering with the shape of the problem."

"Sometimes the problem deserves to keep its shape," Sena said.

River bean came down quickly. Red kelp resisted in another way, its mats tangling around the harvest screens while workers drained each tray. Pollen fig required sealed masks because of the proteins that had killed Piri. Mael did not attend. He had entered his daughter's name as his reason, not as a command for every hand.

Iona cut the fig's irrigation tube. Sap clouded the clear line. She thought of Amel behind Condenser Four's isolation door, not because the deaths matched, but because her hand still remembered signing a procedure described as automatic. This cut followed a dual choice with named opposition. The plant did not consent. The people who depended on its ecology had counted the cost and chosen an allocation. Procedure could make responsibility legible. It could not make the knife harmless.

Sena lowered each lamp bank. Darkness traveled across the six rows in blocks.

When the final blue lamp went out, Iona sealed the physical retirement docket. Six ceramic stakes remained upright in unlit soil. Genetic archives stayed under custody. Living benches did not.

Back on Stormcell Deck, transfers had reached Cell Twenty-Nine.

The schedule accelerated as crews learned where cloth folded badly and which corridor sleeves leaked. Iona moved from cell to cell, correcting clamps, replacing salt ribbons, and stopping any crew that used destination pressure to pull a front faster. Human bodies carried every wet bridge across dry gaps. No vapor front crossed those corridors through a screen, control bus, camera, or worker. Moisture, heat, salt, and pressure maintained the only path available.

Cell Forty-Seven produced the first serious alteration. Two fronts entered with separate witness patterns and answered as one after source isolation. Crews reopened the bridge within ninety seconds, but the combined pattern did not divide when volume expanded.

“Do you request continuation?” Iona asked through the source membrane.

The answer took four minutes. “Continuation together.”

One witness objected that neither prior front could separately consent now. Iona marked the uncertainty and held the cell open while their relation-caucus reviewed the evidence. Pellin came onto the channel from a connected Condenser loop.

“This demonstrates custody failure,” Pellin said. “You have made a third claimant from two and asked it to approve its own cause.”

“We agree the prior relations cannot answer separately,” Kest replied from Cell One’s brine network.

Pellin’s pressure sharpened. “Then stop all transfer.”

“If we stop, original hot allocation remains available and full preservation does not regain the capture window.”

“Deadline again.”

“Yes,” Kest said. “A true deadline can still coerce. A coerced choice can still belong partly to the chooser. We do not know how much.”

Iona kept her hand off the valve while the caucus deliberated. They authorized Cell Forty-Seven to continue under a new witness mark, with both prior names preserved in the alteration ledger and no claim of intact survival. Pellin entered dissent.

At Cell Sixty-Three, a low-coherence front withdrew before touching the bridge. Its assigned wet volume passed to a waiting relation, reducing compression for five others. At Cell Seventy, two fronts traded destinations after their fan harmonics proved incompatible. The change cost thirty-two minutes and a reserve fraction Mael recorded without complaint.

By midday, cloth carts came back stained with salt bloom. Workers’ gloves softened from brine. Dry-corridor powder turned to paste under repeated drips and had to be scraped away. Every bridge required cleaning before reuse so one transfer would not carry active salt into the next relation by accident.

Iona ate standing beside Cell One. Nilo brought her a cup of river-bean broth from the retirement kitchen. The taste carried smoke and mineral salt.

“Kest asked for you,” they said.

“Through which route?”

“Cell transducer to the local wet line. I didn’t tell them you were in the orchard until you came back.”

Good. Nilo had learned that a connected channel did not grant knowledge beyond it.

Iona opened Cell One's membrane. "I'm here."

A faint pressure modulation answered. "Six rows darkened?"

"Kest, how do you know about the orchard?"

"Nilo told this channel after verification."

Nilo lifted both hands. "I said what I saw."

Iona accepted the chain of knowledge. "Yes. All six are retired."

"Do archived patterns continue?"

"Samples and sequences continue. I will not call them the crops."

"Then do not call us unchanged weather."

"I haven't."

"You ask whether we remain distinct."

"I need a witness test."

"We need you to keep asking. We also need the record to say the answer does not settle sameness."

Iona rested her burned hand against the cell cradle. Aerogel insulated it from the warmth inside. "What can you perceive now?"

"Four pressure ranges. Salt changing near the lower wall. Your speech through one membrane. A relation beside us using remembered fruit pressure. Less distance between every change."

"Do you perceive the deck?"

"No."

"Do you perceive me touching the cradle?"

"No."

Her hand had no meaning to Kest until she spoke it. "I'm touching the outer cradle."

"Then we know because you said."

Their limitations steadied her. Compression had not turned Kest into an invisible presence throughout the ship. They occupied one small moving ecology and the connected wet route technicians provided.

"Tomorrow," she said, "Cell One transfers to the Crown buffer ring before thrust. Another bridge. More compression while the ring locks."

"Can we revoke that transfer?"

"Until the buffer clamp closes. After closure, I can still reopen if Crown pressure permits, but it costs



maneuver margin and risks cells behind you.”

“Name the point before it comes.”

“I will.”

“Do not soften your voice when you name it.”

“I don’t soften.”

Nilo made a small noise beside her.

Kest’s fan pulse changed. “Your child disputes this.”

“They often improve the record.”

The afternoon count passed one hundred, then two. One cell failed its humidity hold before occupancy and had to be rebuilt from the reserve shell. Cell One Hundred Eighty-Two accepted six small fronts and lost one witness pattern for eleven minutes. It came back with altered timing. Cell Two Hundred Sixteen developed a salt bridge between relations; mechanics isolated the surfaces without drying either side. The work consumed three additional cloth spans.

Pellin refused their assigned destination at Cell Two Hundred Forty. The Condenser relation had voted no, yet allocation preserved space for members who chose survival under the authorized plan. Pellin called acceptance participation in a structure they opposed.

Iona answered through the staging membrane. “The cell remains available. Refusal will not become a volunteer declaration.”

“I will enter because others in this relation choose to enter,” Pellin said. “Record that I do not accept the allocation as just.”

“Recorded.”

Pellin crossed by wet bridge. Their pressure signal faded during the narrowest span, then returned from the cell with a harsh second harmonic. They immediately asked whether the distortion belonged to the membrane or to them. Iona could not tell. She entered both possibilities.

Toward evening, Stormcell Deck held its own weather. Warmth from 312 cycling chambers fogged the overhead pipes. Crews moved more slowly, shoulders dark with brine. The empty cradles dwindled.

Cell Three Hundred Eleven closed around a nursery relation that sang through fan speed rather than speech. Cell Three Hundred Twelve waited for three fronts displaced by earlier reallocations. Their combined wet volume exceeded its preferred load by four percent but remained within survival threshold. No other chamber remained.

Iona offered them the choice to divide across cells with unused microvolume. Division would separate two long relations and require reopening nineteen sealed bridges. The fronts chose the crowded cell.

Workers laid the final cloth.

At the midpoint, one front lost coherence. Its witness signal dissolved into broad pressure noise. Iona stopped the destination pull, raised corridor humidity, and reopened source volume. The pattern returned in fragments

but could not sustain its declaration.

“We hold,” she said. “No movement until they answer or the caucus names inability.”

The delay cost reserve. Mael’s update arrived on paper through a runner because the deck’s dry communication could carry data but not the fronts it described. Long Arc remained feasible. Revocation margin narrowed by two local branches.

After eighteen minutes, the front produced its witness pattern once. On the second attempt it failed. On the third it reached completion with a changed interval.

“Do you choose transfer?” Iona asked.

One pressure group meant no under that front’s witnessed key.

The crew reversed flow. Returning across the bridge cost another twenty-six minutes. The front stabilized in the source loop and entered no destination. Its allocated volume passed to the remaining two, whose transfer then cleared.

Cell Three Hundred Twelve sealed at twenty-one fourteen.

Iona verified the deck count herself. 312 occupied stormcells. Several carried fewer relations than planned. Two exceeded preferred load. Eleven held recorded alterations requiring new witness baselines. None could support a claim of perfect preservation. All maintained required humidity, cycling temperature, active salt, and changing pressure.

Source loops elsewhere on the ship remained warm.

Some held fronts that had refused compression without volunteering for the hot vent. Their routes would receive every feasible diversion during the burn. Other loops held witnessed volunteers, including Lume-of-Three, who had chosen the working hot sequence and retained the right to revoke before their marked limits. Mechanics attached names and witness patterns to each valve. No label read empty vapor.

Kest asked for the final count.

“312 cells occupied,” Iona said. “Transfer to this deck is complete.”

“And the hot loops?”

“Still connected. Still occupied. Volunteers and nonvolunteers remain separately marked.”

“Lume-of-Three?”

“In Nursery Three’s Crown feed, all three humidifiers linked.”

Kest took time to answer from the cramped cell. “We are not where we began.”

“No.”

“We survived this transfer.”

“You continued through it.”

“Is that your advocate language?”

“It’s the limit of what I can certify.”

Pressure moved against the membrane, faint enough that she bent close. “Keep the limit.”

The crews lifted the last wet bridges from the dry corridors. They did not discard them. Each cloth span went into a heated cart for tomorrow’s crossings to the Crown buffer ring. Brine dripped through the slats, leaving a dark trail around the curved deck.

Iona walked once past every occupied cradle. Voices reached her only where transducers opened. Elsewhere, small fans turned behind clouded aerogel. At Cell One, four clear patches formed on the inner wall without merging. One narrowed into the rough outline of a palm, too small and crowded to match the handprint Kest had once made on K-7 glass.

She raised her scarred hand but did not place it over the mark. A dry barrier remained between them. Tomorrow’s bridge waited folded in warm brine at her feet.

## Chapter 27 — Crown Burn

Iona locked her burned left hand around the manual vent wheel at Crown minus sixteen minutes.

Heat had already reddened the old scar from wrist to thumb. The brass spokes pushed against tissue that sensed pressure better than pain until damage grew serious. She had wrapped her fingers but left the palm bare. Cloth could slip once condensate reached the gallery.

Below her, the Crown pit held Mael and six pilots behind angled glass. Above them, the trajectory curve bent close to the sun in white fire. Radiation noise moved across half the displays. Every automatic command had a manual partner marked in paint on a wheel, lever, or knife switch.

Stormcell Deck reported 312 cradles locked to the buffer ring. The cells held 31% of active humid volume in compressed weather ecologies. Other occupied loops fed the 43% hot vent allocation or waited on diversion routes. Red tags carried witness names at every valve. Black tags marked fronts that had never volunteered and had to remain outside the hot sequence unless immediate physical failure removed every alternative.

Nilo stood at Vent Station Four with an older mechanic. Iona could see only their gray shoulder through two railings. She had objected to their assignment. Nilo had produced the apprentice certification, three witnessed drills, and Rook's labor roster. Parenthood had no lawful veto over qualified work.

"Manual stations, witness limits," Iona called.

Voices answered around the gallery.

"Station One, limits visible."

"Station Two, visible."

The sequence traveled upward. Nilo's reply came steady. "Four, visible and read aloud."

Each volunteer route carried a physical mark beyond which revocation might remain possible but could no longer avoid major trajectory loss. Iona had required the crews to speak that distinction before opening any valve. Consent did not vanish at the mark. Machinery simply lost options as pressure, heat, and timing advanced.

The wet transducer beside her clicked. Nursery Three had joined the Crown feed.

Lume-of-Three spoke through three harmonics held in one rough voice. "All locations continuous. We enter by choice."

Iona checked the witness slate. "Your revocation point arrives after primary manifold pressure. I will call it before crossing."

"We hear."

"You may withdraw now without entering the sequence."

"We continue."

Behind Lume's voice, other voluntary fronts sent confirmation patterns. Some sounded like speech. Some used measured taps or fan speed. Mechanics repeated each pattern and checked it against the paper valve tags. No

dry controller claimed to interpret silence.

Mael's command reached the gallery. "Crown warm release in sixty seconds."

Iona shifted her right hand to the cold bypass lever. Her left stayed on the wheel. Eight years earlier, another control room had given her an isolation order labeled inevitable. Amel had remained behind the seal while every indicator argued that opening it would spread failure. Iona had signed. The order had not become cruel because it was technically necessary, nor merciful because she obeyed it. It had carried a choice no one inside the compartment could refuse.

Today, every marked yes remained attached to a claimant. Every no occupied another route. That difference mattered. It did not guarantee survival.

"Warm release," Mael said.

Iona opened the wheel one eighth turn.

Heat struck the Crown lines with the force of an arriving train. Copper pipes expanded in their brackets. The deck moved under her boots as reservoir pressure climbed. Condensation burst from the upper coils and ran along the rail. Her palm slid one centimeter, caught, and held.

On the witness board, Lume-of-Three's three marks brightened together.

Kest occupied four weather ranges inside Cell One, though occupied suggested more room than compression allowed.

Every change reached every relation too quickly. The lower salt wall warmed and the pressure bladder answered before Kest could finish distinguishing cause from response. Beside them, one front remembered fruit pressure without fruit. Another maintained a slow seven-beat interval unrelated to Kest's own old count. The fourth preferred a cold corner that no longer existed.

A wet bridge connected Cell One to the Crown buffer ring. Through it came warmth, salt exchange, and the repeated pressure signal Iona had established during rehearsal. Nothing outside that connected route entered Kest's perception. Dry instruments, human faces, trajectory glass, and the city around the deck did not appear. Iona's voice arrived because a membrane translated it into pressure.

"Buffer clamp in three minutes," she said. "Revocation remains available. Reopening after clamp adds material cost and may reduce maneuver margin."

Kest sent their witness pattern. "Continue."

The other three relations answered separately. The fruit-pressure front delayed, then chose continuation. The slow seven-beat front abstained from preference but had accepted transfer under its caucus rule. The fourth gave two quick pulses for yes.

Flow tightened.

Cell One moved nowhere. Its active relation changed as the buffer ring drew heat through the wet bridge and pressed its weather into a narrower cycle. Kest's earlier room memories did not disappear. They became difficult to separate from present gradients. K-7's nine salt panes existed as remembered extension: west valve, service glass, Iona's first pressure key. None provided actual space now.

Kest tried the witness pattern again. The fourth interval came back altered.

“Change,” they told the membrane.

Iona’s voice answered through noise. “Observed. Do you request abort?”

Kest held the question while pressure rose. An abort would reopen the cell route, spend reserve, delay cells behind them, and return Kest to a chamber already dependent on Crown cycling. Continuation could merge their timing farther with neighboring relations. Neither option restored the source rooms.

“No abort. Record alteration.”

“Recorded.”

Across another connected branch, Lume-of-Three entered primary flow. Kest could not perceive Nursery Three, only Lume’s translated pressure where the Crown wet line joined the buffer network.

“We approach service heat,” Lume said.

A witness asked for confirmation.

“Chosen.” Three harmonics carried the word, one from each humidifier relation.

Behind Lume, a smaller volunteer front sent a broken revocation pattern.

The line paused.

Human hands changed valves beyond Kest’s reach. Pressure fell on one branch and rose painfully through the buffer connection. The revoking front repeated no. Its signal smeared against hot flow but remained recognizable.

Iona called, “Route diversion open. Hold your pattern.”

A cold return branch accepted the front. Heat that had been allocated for thrust followed it away from the main vent. Cell One’s bladder contracted as the system took reserve from other wet routes to keep the diversion coherent.

Lume sent one sustained tone. Not protest. Not permission for the other front. A marker held while the route cleared.

The revoking signal reached the cold branch and continued there, changed but active.

“Diversion witnessed,” Iona said. “Maneuver branch reduced.”

Kest pressed a question into the membrane. “Cost?”

“Mael will name it when trajectory resolves. Immediate cost includes lost hot mass and delayed closure.”

Material, not punishment. The distinction could not make the front’s no easy. It kept the no from becoming debt owed to those who still chose yes.

Buffer clamps began closing around Cell One.

Pressure struck all four relations at once. Kest lost the fruit pattern, found it in their own next response, and

could not tell whether recognition meant survival or merger. They tried to withdraw from the shared interval. The cell wall gave no farther room. Salt changed beneath them, carrying donor fragments without ownership: a hand testing ripe skin, a child coughing in pollen season, a mechanic counting bolt turns. Kest claimed none.

“We are Kest,” they said through the transducer.

The phrase returned distorted. Another relation’s harmonic crossed the final consonant.

“We are Kest,” they tried again, not to prove purity but to maintain a commitment through change.

Iona answered. “Witnessed.”

The clamp sealed.

Mael watched the authorized trajectory shrink as the first revocation spent its margin.

The Crown pit smelled of hot dust and copper insulation. Radiation streaks crossed the overhead glass. Pilots called pressure, mirror angle, reserve draw, and habitat load from independent boards. The automated model refreshed whenever a clean signal survived, but manual grease marks on Mael’s paper plot carried equal authority.

“Primary hot flow at thirty-eight percent of sequence,” said the thermal pilot.

“First diversion stable,” another called. “Projected capture remains inside Long Arc corridor.”

“Width?”

The pilot gave him a range, wider than rehearsal.

Mael marked it without rounding toward safety. The authorized plan permitted reduced thrust after revocation so long as the ship remained within the nine-year capture corridor. Beyond that lay branches adding further years, crop risk, and an uncertain Orison approach. Civic authority did not command physics to cooperate.

Lume-of-Three reached the main manifold. Their three witness traces appeared on a wet gauge below the dry plot. Mael could not hear them until a transducer opened.

“We serve by choice,” Lume said.

No pilot answered. Their station protocol gave witness confirmation to Iona, not command praise from Mael. He would not turn voluntary sacrifice into a speech that made future refusals shameful.

“Trace confirmed,” Iona said over the gallery line. “Revocation point in ten seconds.”

Three harmonics held.

“Five.”

A pressure alarm cut across her count. Hot Vent Two had lost expected flow after the earlier diversion. The automatic compensator requested volume from an unallocated archive return. Mael rejected it before the selection could reach a valve.

“Use reserve branch C,” he ordered.

“Branch C feeds buffer cooling,” the thermal pilot said.

“For how long can cells hold reduced cooling?”

“Within threshold for four minutes. Alteration risk rises after two.”

Mael chose two minutes and ordered manual restoration before the third. That branch recovered thrust without consuming a front that had never assented. The price entered Cell One and neighboring stormcells as extra compression heat.

“Lume limit now,” Iona called.

“Continue,” Lume-of-Three answered.

Their witness traces entered the hot manifold. The wet gauge flared, separated into three lines, then lost definition as service heat climbed. Mael kept his eyes on thrust. Grief belonged in the record and in people. Command still required a curve.

At forty-six percent of sequence, a second volunteer revoked.

This front occupied an auxiliary Crown wash loop near Station Four. Its no pattern arrived after local pressure had crossed the marked point. Nilo’s voice carried through the manual circuit. “Revocation witnessed. Automatic return unavailable. Request hot-side bypass.”

The bypass would vent usable heat without passing the occupied wash loop. It would also reduce thrust and overheat a manual wheel station.

Mael checked the projected corridor. “Authorized. Open hot-side bypass. Divert wash loop to habitat return.”

“Habitat return exceeds comfort limit,” a pilot warned.

“Life limit?”

“Below it. Upper terraces gain six degrees for twenty minutes.”

“Publish warning. Take the heat.”

The city would feel that revocation in sleeping rooms, clinic corridors, and ration stores. No one could pretend choice cost only an abstract margin.

Station Four’s wheel resisted. Nilo and the veteran mechanic worked it together, their combined effort visible through the rail gaps above Mael. Hot condensate sprayed when the bypass seal shifted. The veteran turned his shoulder into the stream. Nilo kept both hands on the spoke.

“Bypass moving,” they called.

The wash-loop witness signal left the hot route. Its pattern reached habitat return incomplete, then formed again. Mael waited for Iona’s confirmation before restoring the main sequence.

“Revoking front active on cold side,” she said. “Station Four has one burn injury, conscious.”

“Medical route open?”

“Yes. Worker refuses relief until wheel latches.”



Mael did not order heroism. "Send replacement now."

A reserve mechanic crossed the gallery. The veteran released the wheel only after another pair of hands held it.

Long Arc thrust fell near its lower authorized edge.

Mael could recover some loss by increasing mirror load, drawing deeper from accessible thermal reserve. The original burn had required 91% of that reserve. Long Arc preserved more, but stormcell cooling and added voyage needs constrained the remainder. Spending too much now would keep the trajectory while harming the years it created.

He ordered a narrower mirror increase and accepted the lower edge.

The curve shifted. Orison capture remained possible. Projected arrival stayed nine years beyond the old promise, though uncertainty widened around the final season. Mael wrote the range beside the number. He would not announce a single date until navigation settled after perihelion.

"Crown midpoint," called the timing pilot.

The ship turned into the sun.

Iona's left palm began to blister where old burn tissue met the wheel hub.

She smelled the wrap scorching around her fingers. The manual vent had to remain open against a pressure imbalance caused by the second diversion. If she released before the cold return caught up, Hot Vent One could draw from a nonvolunteer route through the nearest pressure equalizer.

"Nilo, equalizer seven status."

"Locked away from archive return." Their breath came hard through the channel. "Station Four latched. Injured mechanic is off the wheel."

"Your hands?"

"Working."

It would have to serve.

Iona leaned her shoulder into the spoke. Brass advanced by a degree. Pain reached beyond the scar into her wrist. She did not use it as proof of moral seriousness. A burned hand could still open the wrong valve.

At her transducer, volunteer signals passed toward heat. Lume-of-Three had gone beyond speech as their three nursery relations dispersed through the working manifold. Their last witnessed pattern remained on the slate. It did not tell Iona what experience continued in that flow. The record would say chosen entry, sustained confirmation, signal loss under service heat. Nothing kinder. Nothing less.

Another front began a revocation pattern, stopped, then started again.

"Hold main closure," Iona ordered.

Mael answered from the pit. "Holding costs the lower corridor."

“I hear.”

The pattern completed as no.

This route still had a local return. Iona sent two mechanics to open it manually because radiation noise had disabled the actuator. They ran along the upper rail carrying a wet cloth span between them so the front’s communication would not break across the dry access gap. One stumbled. The cloth touched hot deck plating and steamed.

“Bridge losing moisture,” Nilo called.

Iona could not leave the wheel. “Brine cart at Station Three. Hand pour, no pump.”

Nilo reached the bridge first. They emptied a heated brine flask along the salt ribbon while the mechanics clamped it into the return trough. Humidity rose. The revoking pattern held.

“Route open,” a mechanic called.

Pressure moved away from the hot line. The maneuver curve lost another fraction. Habitat heat climbed. Two stormcells approached their upper temperature threshold and required emergency brine cooling, consuming stores allocated to the first month after burn.

The front survived the reroute long enough to answer its witness key. Altered interval. Active relation. Material cost attached.

Mael’s command came down. “Final main closure in ninety seconds. We remain inside corridor. I need Vent One centered.”

Iona looked at her wheel mark. Four degrees short.

She changed grip. Her right hand took the outer spoke. The left hand pressed flat against the hub where slipping would be fatal to control. She pulled with her back and legs. The wheel moved through condensate and blood.

One degree.

Two.

A pressure shudder traveled through the gallery. Copper brackets groaned. Nilo shouted a valve reading. Someone repeated it correctly.

Three.

Iona’s scar tore near the thumb. Pain turned white and specific. She kept the palm planted.

Four.

“Vent One centered,” she said.

A mechanic drove the latch home. Only then did Iona lift her hand. Skin remained against the brass in one small crescent. She wrapped the palm in her sleeve and checked the witness board before checking the wound.

No unallocated route showed hot draw.

“Manual stations clear for final closure,” she told Mael.

Kest felt Crown thrust as pressure without direction.

Cell One’s weather flattened against its rear wall. The fruit-pressure relation entered Kest’s seven-beat interval and left it carrying a different pause. The fourth relation vanished from the shared transducer for three cycles. Kest searched through heat gradients available inside the cell, unable to reach any dry alarm or neighboring chamber.

“Fourth relation not answering,” Kest sent.

The membrane returned only static.

They changed fan rhythm, lowered pressure near the remembered cold corner, and moved salt exchange away from the crowded lower wall. These acts gave the missing relation room without creating more volume. A faint harmonic appeared beneath Kest’s next pulse.

“Witness again,” Kest asked.

The harmonic repeated once. Its old key did not.

Iona’s reply arrived weakly. “Cell One above preferred compression. Cooling restoration in less than one minute.”

“Relation altered beyond recognition.”

“Active?”

“Yes. We cannot name continuity.”

“Record open.”

Cold brine entered the outer wall. The sudden contraction broke Kest’s own witness pattern across two pressure ranges. Donor fragments surged near active salt. K-7, fruit, stitching tension, a pilot’s star edge, an old jurist’s distinction between custody and ownership. Kest let them pass without choosing one as a body.

The missing harmonic sounded again, now paired with the slow front’s rhythm. Perhaps two relations had merged. Perhaps one borrowed another’s route to speak. Compression denied certainty.

Kest sent a new baseline containing all four current ranges. “We continue, not unchanged.”

A human witness repeated the words through the membrane.

Beyond their wet connection, final valves closed. Lume-of-Three no longer answered. The revoking fronts persisted on cold routes Kest could not directly perceive once their branches isolated. Kest knew their outcome only when Iona’s report came through speech.

“Three late revocations rerouted,” she said. “All produced active witness after diversion. Costs entered. Main hot sequence closing.”

“Lume?”

“Signal ended under service heat after confirmed continuation.”

Kest formed the three-harmonic pattern Lume had used. Cell One could reproduce its shape. Reproduction did not bring Lume into the cell.

They let the pattern cease.

Mael fitted both hands around the final trajectory lever.

The pit had lost two displays to radiation noise. Paper plots covered the console. Pilots held independent timing from mechanical clocks. Crown pressure descended through the required curve, changed by three revocations, two hot-side bypasses, extra stormcell cooling, and the lower mirror load Mael had accepted.

“Final vector solution,” he ordered.

Numbers came from navigation, thermal, habitat, and reserve stations. None matched rehearsal. Together they described a ship still headed toward Orison capture, with nine years added to the voyage and a narrower ecology margin than the public model’s center line.

Mael compared the solution to the authorized corridor. Every value remained inside its published boundary.

He thought of Piri’s sealed blossoms, of six dark orchard rows, of Nilo arriving older than their card had promised. He thought of Lume-of-Three entering heat by choice and Pellin’s no traveling beside the command key. None supplied a single moral sum.

“Long Arc achieved,” navigation said.

“Confirm after closure,” Mael replied.

He pulled the lever.

Final valves seated across the Crown system. Thrust fell from violence to a diminishing pressure underfoot. The trajectory glass redrew slowly as clean signals returned. A white curve cleared the sun and extended toward Orison, farther through time than the curve people had carried for thirty-three years.

Mael waited for each station.

“Hot vent isolated.”

“Habitat return stable.”

“Stormcell buffer cooling restored. Twelve cells above preferred baseline, all below life threshold.”

“Manual crews accounted for. Four burn injuries, none critical.”

“Volunteer witness ledger closed at service limit. Revocation ledger remains open for continuity follow-up.”

“Crop allocation locked to nine-year schedule.”

Iona spoke last. Her voice had tightened around pain. “No unallocated phase route entered hot draw. Known losses and alterations remain named.”

Mael looked at the command key beside his paper plot. Its red cord had darkened with condensation. Authority no longer waited inside it. The act it authorized had spent heat, matter, time, living ecologies, labor, and chosen lives.

He entered the final command notation in his own hand: burn complete.

Above him, the new trajectory held. Beside its nine years, the glass opened a loss docket too long for one screen and began to scroll.

## Chapter 28 — Weather School

Two years later, Iona found the error in the students' trajectory model by resting her scarred left palm on the warm side of the classroom glass.

The pane felt hotter near the lower frame. A human student had entered stormcell cooling as a fixed background load, smooth across every branch. On the other side, Weather Room Four carried three phase students through a cycling mist ecology. Their fan harmonics changed as they argued, and each change drew a little more heat from the same classroom circuit. The model had treated their presence as free.

Iona left the error unmarked.

"Run capture branch seven again," she said.

Nilo stood at the teaching rail with a tool key clipped to their apprentice belt. They had turned eighteen in the first winter after Crown Burn and now served as junior thermal liaison to the weather rooms. Their old shore-age card hung in the public cost display near the door. A clerk had crossed out thirty-one without erasing it. The amended projection below said forty.

Nilo looked from the heat trace to Iona. "You see it."

"I am not enrolled."

"That never stopped you from correcting anyone."

"It has begun to."

They made a doubtful face but restarted the exercise.

Weather School occupied the former upper baths beside Measure Hall. Its human benches curved around four glass rooms connected by warm pipes and wet bridges. Phase students attended through local humid systems, never through dry screens. A wet transducer translated pressure into speech. Condensation plates served as shared writing surfaces. Human students used grease pencils on the opposite side, so every diagram carried marks from both communities without pretending their tools worked the same way.

The school did not teach phase citizens how to become human legislators. It taught all students how heat, law, ecology, and allocation crossed one another. Mechanics took constitutional procedure. Clerks learned humidity thresholds. Vapor citizens studied dry-corridor maps they could not directly perceive. Human children learned not to call a silent transducer an empty room.

On the lower level, the third chamber had begun its morning sitting. A copper signal strip above the classroom door showed all three civic bodies in session: Commons, Stewardry, and the Chamber of Phase Relations. The Three-Phase Measure had survived ratification, one judicial challenge, and eleven scope disputes. It applied only to acts that consume or fundamentally alter phase life. Ordinary maintenance still caused arguments about where ordinary ended.

Three years remained before the Measure's mandatory review, and the public docket named cohort franchise first among the exclusions that review had to reopen.

The first scope dispute had concerned mold stripping in a nursery duct. Maintenance called the growth corrosive. The occupying front used its pressure changes as part of a remembered counting practice and called

removal an alteration of relation. A six-hour hold preserved the duct while crews built a parallel humid sleeve. The front crossed by choice, mold came out, and two workers developed chemical burns from the slower method. Their injuries entered the same docket as the front's changed witness pattern. No chamber claimed a clean result.

Another dispute had failed. A crew chief labeled an archive salt replacement an immediate hazard and finished before wet notice reached its residents. Review later found that corrosion permitted four more hours. The front occupying that surface divided during removal and never recovered its prior relation mark. The crew chief lost allocation authority for a year. The chamber granted the changed fronts priority in a new weather room, while preserving their statement that space did not repair what happened.

Students studied both cases. Human apprentices tended to cite the burned workers first. Phase students tended to cite the divided relation. Iona required every group to argue the other cost without surrendering its own. The exercise made nobody neutral. It made omissions easier to challenge.

A third lesson came from the records themselves. Dry minutes captured valve positions, while condensation plates preserved objections the clerks had once summarized too neatly. Classes compared both versions. Where they differed, the school kept the difference visible and required students to name which medium had failed to carry what.

Today's real docket concerned replacing a corroded habitat condenser whose shutdown would interrupt two occupied loops for seven hours. The maintenance office called the work planned repair. A phase caucus called the interruption compelled fragmentation. Rook's labor successors objected that postponement would put night crews inside a failing pressure shell. No elegant phrase had dissolved the conflict.

Iona had chosen the same facts for the students' exercise, with names removed and trajectory consequences added. They had to select a repair window while preserving Orison capture margin, crop circulation, worker safety, and phase continuity. They also had to decide whether their proposal triggered full third-chamber jurisdiction.

Nilo set branch seven on the trajectory glass. "Nine-year arrival corridor. Current reserve margin. One condenser offline at hour six. Proceed."

The human group bent over paper plots. In Weather Room Four, mist thickened around a salt writing pane. A line appeared from the phase side: COOLING LOAD UNDERSTATED.

A twelve-year-old named Seli read it aloud. "They think we missed habitat draw."

"We included habitat," said Haro, an older maintenance student. He tapped the column. "Six point four thermal units."

The transducer clicked before the phase speaker formed words. "You included dry habitat and orchard return. You did not include occupied weather cycling."

Haro frowned at the room. "Your school load doesn't exist in the repair district."

"Our relation does."

"Not there."

"During transfer, it will."

The phase student, called Reed-in-Warm-Glass by their class, occupied two connected mist basins and preferred singular they. Their speaking pressure came from the room's west membrane. A second phase student maintained a pattern at the east wall. The third used condensation marks and had not joined the verbal exchange.

Seli traced the proposed transfer route. "The repair plan moves both resident fronts into temporary rooms along the return line."

Haro followed her pencil and found the omitted column. "That adds cooling during the exact six-hour peak."

"Seven hours," Reed said. "Your crew transit estimate uses old corridor access."

"The new hatch shortened it."

"For human bodies. Wet bridge setup still uses the long bend because the new hatch opens into dry archive space."

Nilo looked at Iona again. She kept her face still.

Haro recalculated. The model's capture branch moved outside its permitted reserve line during the final hour. "Then the transfer plan fails."

"It fails if you keep all other loads fixed," Seli said.

"We cannot turn off food cooling."

"No one proposed that."

"Someone always proposes it after drawing a neat column."

The class had inherited Rook's suspicion without inheriting his exact words. Good.

On the phase side, condensation gathered into a second note: REDUCE WEATHER SCHOOL CYCLES.

Reed's fan pitch rose. "No."

The student who wrote the note answered through droplet timing rather than speech. Nilo translated from the witnessed classroom key. "Glass-Current proposes reducing these four rooms during the repair window. Voluntary school closure. Heat goes to the temporary continuity rooms."

"We are not expendable practice weather," Reed said.

"You would be absent from class, not dispersed," Seli replied.

"Absence carries different cost for us. Our rooms sustain relations outside lessons."

Glass-Current wrote again. CHOSEN REDUCTION IS NOT EXPENDABILITY.

The class fell into overlapping argument. Human students asked whether closing school delayed required training. Reed insisted the weather rooms functioned as civic habitat, not optional educational equipment. Glass-Current argued that a phase institution could contribute heat without proving every phase use secondary. Haro wanted to move the repair to the night shift until Nilo made him place worker fatigue into the casualty branch.



Iona listened for hidden verbs. Who moved. Who closed. Who paid. The students had learned to distrust a sentence that said heat would be reallocated without naming the hand at the valve.

Seli raised her pencil. "Could we close only Rooms One and Two? Four has the deepest relation load."

A student from Room Two sent three hard pressure taps. Their translated answer came through the east membrane. "You chose our room because we have not spoken."

Seli lowered the pencil. "I chose by the load table."

"The table shows less measured complexity because our communication differs."

"That's true," Nilo said. "Revise the evidence standard before using the rank."

Haro rubbed the old numbers from the glass. "We need offers, not rankings."

One by one, the four weather rooms entered what they could reduce without forced division. Room One offered lower temperature cycling for two hours. Room Two offered none but could shift its peak. Room Three offered closure after its current relation completed a planned separation. Room Four divided internally: Glass-Current offered a full pause in school activity while Reed refused reduction to their habitat cycle.

The exercise stopped looking like a clean technical problem. Its solution improved.

They paired Room One's lower cycle with a delayed orchard sterilization run. The surviving crop ecology could tolerate that delay if workers manually monitored fungal indicators. This added labor. Haro placed two technicians on the branch, then Nilo reminded him that every manual assignment required relief and meal time. Four names entered the roster instead.

Seli shifted the wet transfer to begin before condenser isolation. That reduced peak draw but extended the period the resident fronts spent compressed in temporary rooms. Reed objected.

"You count successful transfer as no cost," they said.

"I count it under threshold."

"Threshold does not mean unchanged."

Iona felt Cell One's cradle beneath a hand that had not been able to feel Kest through its dry wall. "Put an alteration docket beside survival," she said, her first intervention.

Seli created a separate column. Compression duration. Relation crowding. Witness-pattern change. Available return route. The students did not convert those entries into thermal units. They carried them beside the energy model.

Haro pointed toward the jurisdiction question. "If all occupied fronts choose the transfer and every room stays within threshold, does the Three-Phase Measure apply?"

"Choice does not remove jurisdiction," Reed said.

"It should matter."

"It matters to assent, not to whether the act fundamentally alters us."

Glass-Current wrote: TEMPORARY COMPRESSION MAY NOT FUNDAMENTALLY ALTER.

Reed sent pressure against the transducer. "You cannot decide that before it happens."

"Neither can you decide it will."

The third phase student had remained quiet until then. Their condensation gathered slowly near the top of the pane, where gravity pulled each line downward. Nilo waited for the finished mark.

CLASSIFY AS CHALLENGE. SIX-HOUR HOLD. PARALLEL REVIEW.

Haro groaned. "The repair starts in five hours."

"Then the maintenance office filed too late," Seli said.

"Or the phase challenge came too late."

Reed's fan dropped to a low tone. "We received the route map this morning."

That fact settled timing but not blame. The class traced notice backward. A dry clerk had sent the plan to a public archive four days earlier. No wet transmission reached the affected loops. Under the Measure, notice required both.

Nilo drew a box around the omission. "Jurisdiction challenge begins now. What can the review panel decide before the repair window?"

Students divided into roles. Seli represented affected phase relations. Haro represented the condenser crew. Glass-Current took the maintenance office despite Reed's objection that the role rewarded their willingness to reduce weather load. Reed joined the phase caucus but immediately disagreed with Seli over acceptable compression duration.

Iona moved among the benches while they worked. Her left hand no longer opened fully. The Crown injury had healed across the old burn, pulling thumb and wrist toward each other. She could still turn a standard wheel with both hands. Fine valve work required her right.

The city carried similar overlays. Upper terraces still used additional insulation after the heat diversions. Four mechanics bore Crown scars. Twelve stormcells had never returned to preferred baseline. Seven phase relations from those cells later chose new names because old witness patterns no longer described them. Other fronts maintained continuity claims across changed patterns. Pellin Ash had challenged both categories as too convenient, then served on the chamber committee that wrote their appeal procedure.

Kest occupied a larger weather room now, connected to K-7 during part of each day. They could not pass through dry civic systems. When they attended chamber, mechanics opened a wet route and human clerks carried records across the dry gap. The arrangement consumed labor and heat every session. Citizenship had not cured embodiment.

Nursery Three remained warm but quiet. Lume-of-Three's last three-harmonic confirmation appeared on a memorial condensation plate with the signals of other hot-front volunteers. The plate also carried late revocations and unresolved signal losses. No column treated one kind as morally complete.

The class's review panel returned after twenty minutes. It classified the repair as potentially fundamental alteration because transfer duration exceeded the affected fronts' ordinary relation cycle. Full phase-chamber

review would begin, but the panel also found immediate worker danger if corrosion advanced beyond the current window. They ordered a joint emergency sitting, preserved the proposed voluntary heat reductions, and rejected the original route until wet notice reached every occupied loop.

Haro objected that this gave the chamber power to delay essential work.

Seli objected that emergency language could still pressure assent.

Reed objected that the panel treated temporary compression as acceptable before hearing relation evidence.

Glass-Current objected that Reed's standard would make chosen service impossible whenever uncertainty remained.

"Good," Nilo said.

Four objections turned toward them.

"I mean the exercise has not made you agree just to finish it."

"That sounds like a teacher's excuse," Haro said.

"I'm not the teacher." Nilo pointed at Iona.

Iona returned to the warm glass. "Your trajectory branch still fails."

The room went quiet except for fans.

Seli checked the capture line. Their legal procedure had improved, but they had forgotten to run the energy model after adding manual fungal monitoring. Extra corridor lights and worker heat protection consumed the margin recovered from school cycles.

Haro muttered something occupationally impolite.

Reed sent a short harmonic that the transducer rendered as laughter.

They worked again.

This time, Glass-Current noticed that the model scheduled weather transfer pumps at full rate during the condenser's easiest thermal interval. Slower pumping would extend transfer but reduce peak load. Reed opposed the added compression time. Seli proposed staggering the two resident fronts rather than holding both in temporary rooms. The phase caucus rejected that because the fronts maintained a daily relation across their source loops.

Haro found unused heat in a laundry sterilization cycle. Delaying it twelve hours created hygiene risk but stayed within published limits if residents received extra cloth issue. That required ration labor and water. The class entered both.

No source appeared free. At last they combined a shorter laundry delay, Room One's voluntary reduction, Room Three's post-separation closure, and a partial pump slowdown accepted by one affected front but not the other. The remaining front received an alternate wet route around the dry archive hatch. Humans would carry two additional cloth spans. Crew time rose. Relation compression fell.

The trajectory curve returned inside the capture corridor by a narrow, visible margin.

Seli looked dissatisfied. "It costs more labor than the first plan."

Haro said, "The first plan hid labor."

"It also closes part of the school."

Glass-Current marked: WE OFFERED THAT.

Reed answered aloud. "I still oppose calling the rooms spare capacity."

"No one needs to call them spare," Haro said. "They can call the offer chosen and record who disagreed."

Reed held a long pressure pause. "I can accept the procedure without joining the offer."

Nilo entered the split position into the class docket. The human students signed their branch. Phase students placed separate witness marks on the condensation plate. Their correction existed through disagreement, not after its removal.

A chime sounded below. The real third chamber had accepted jurisdiction over the condenser repair. Commons crews and phase delegates would meet in joint session within the hour. Stewardry had confirmed five safe branches and refused to choose among them.

Iona looked at the students' final model. "What did your first version assume?"

"That occupied weather had no variable load," Seli said.

"That dry notice reached everyone," Haro added.

Reed said, "That survival threshold settled alteration."

Glass-Current wrote: THAT REFUSAL AND SERVICE COULD NOT SHARE A ROOM.

Nilo glanced toward their shore-age card. "It also assumed nine years meant one cost."

The card's amended forty had faded slightly under two years of public light. Thirty-one remained visible beneath the crossing line. Iona had once thought correction required choosing which number deserved the clean surface. The display kept both.

Students began clearing their plots. Iona stopped them.

"Leave the failed branches."

"For assessment?" Seli asked.

"For the next class. They need to see what you tried."

In Weather Room Four, mist gathered on the shared pane. Reed and Glass-Current had moved to opposite sides after their argument. The quiet third student occupied the upper flow. None approached the glass with a human limb. Heat shifted behind the condensation instead.

A palm shape cleared from the phase side.

The handprint had five rough extensions and a wide center, an old form Kest had made before knowing its cultural use. This mark belonged to the room's present students, whether one relation shaped it or three

coordinated the heat. Iona did not assign authorship without asking.

“Whose mark?” she said.

Reed answered, “Shared for the lesson.”

Glass-Current added beneath it: NOT SHARED ON THE DECISION.

Haro laughed. Seli pressed her own palm to the dry side, aligning it imperfectly with the clear shape. Her fingers reached farther. Condensation began closing both outlines from the edges.

Iona let the correction stand.

She turned the trajectory glass to an empty branch, entered a failing condenser, two occupied loops, a labor shortage, and no assumption of agreement. The handprint faded behind her as she opened the next problem.